

SYSTEM SEGMENT SPECIFICATION

SEA SW OBB - Light Signals / Eulynx

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1 Scope

1.1 Identification

SEA is also called SEC in international context, and is also called EEC in product context; EEC, SEC and SEA are synonyms. This document keeps using "SEA".

SEA consists of several components of hardware and software. All software artifacts of SEA are using the naming prefix SEA SW to distinguish them from hardware-focused documents.

The term "SEA SW system" identifies the sum of all software components working together to fulfill the SEA SW functionality. It does not address any hardware issues.

The "Light Signals SW" identifies the subsystem of all software within the SEA SW system, which are working together to fulfill the functionality for a signal controlling module.

A SEA SW system consists of

- the generic library SEA SW GEN:
This implements the basic, country independent functionality common for the SEA SW systems in different country specific implementations.
- functional submodules for the generic system which are defined in own documents, such as point
- country-specific extensions:
Any specific SEA implementation using the generic software library has to fulfill the requirements listed in this document. Nevertheless it is allowed to redefine certain functional requirements to meet the needs of the specific implementation. For redefined requirements it is recommended to add a reference to the corresponding generic requirement to allow easy cross checking.

This SSS provides the generic system requirements for the functional submodule Light Signal element derived from TODO (reference high-level docs - feature list, SEMP etc...)

The architecture and design of system SEA SW GEN is part of the System Subsystem Design Description document SEA SW GEN [3], the architecture and design of the signal element subsystem is part of the Software Design Description document SEA SW Light Signals [5].

1.2 System Overview

1.2.1 EEC: Overall System

EEC (ELEKTRA 4 element controlling) is a system to control trackside (outdoor) railway equipment. The main architectural approach of EEC is to locate processing units, the outdoor object controllers (OCs), next to the track to control all elements nearby. The OCs are connected to the operating indoor interface host of ELEKTRA interlocking, the element controller multiplexer (EC-MUX), through an IP based data connection over a LAN or WAN network.

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The EEC implementation is done according to the Eulynx standard reference architecture. To connect the indoor EC-MUX and outdoor OC components the standardized SW protocol set of Eulynx is used. The OCs represent the Eulynx subsystems of type outdoor elements. Because the Eulynx reference architecture defines a separation of command and control interface (SCI, standardized command interface, implemented in SIL4 context) from diagnosis and maintenance (SDI, standardized diagnosis interface, and SMI, standardized maintenance interface), just the SCI protocol is of interest for the EC-MUX.

EEC EC-MUX is intended to be executed in an virtualized environment within a data center with possibly high distance to the OCs / field elements. All connectivity to other components is therefore IP based.

EEC is not a stand-alone product, it is designed to be integrated into a superior control logic, which determines commands for the outdoor elements and evaluates state information sent back. EEC is able to receive commands, forward them to the outdoor elements and send back state and diagnostic information.

Each EEC SW system consists of the two logically and physically separated components EC-MUX and the OCs. An object controller is of a specific type according to the Eulynx subsystems:

- OC-P - Point subsystem
- OC-LS - Light signaling subsystem
- OC-TDS - Train detection subsystem
- OC-LC - LevelCrossing subsystem
- OC-IO - Generic IO subsystem

Each system can have one or more EC-MUXes controlling several OCs:

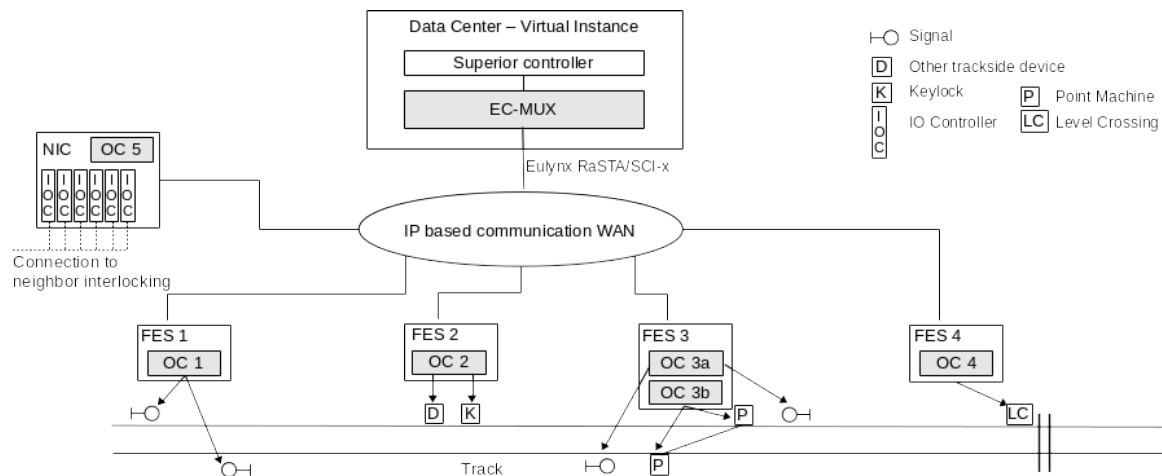


Figure 1: EEC system

- EC-MUX is the element multiplexer. It is an indoor component located in the control center and able to handle many processing units (OCs) along the track. It has the task to provide an interface for receiving commands and sending state and failure information back to the superior controlling instance.

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- OCs are the object controllers inside the FES (field element station, the cabinet). It is located next to the track operating the elements nearby. The OCs represent Eulynx outdoor subsystems and are connected to the EC-MUX via Eulynx SCI protocols.

1.2.2 EEC SW GEN: EEC generic software libraries

The software of an EEC system always consists of some generic parts (EEC SW GEN libraries), which are common for all country specific implementations, and a country specific part extending the generic base libraries. Within this document, the generic parts of EEC software components are in focus. There are no site or country specific issues considered.

EEC SW GEN is the generic software library, which defines a basic system to operate OCs by implementing generic versions of the indoor component EC-MUX and implementing interfaces to the OCs.

This basic system provides all country independent functionality for EEC SW specific solutions:

- common functionality
- base communication
- diagnosis system
- startup and shutdown mechanisms
- build mechanisms
- virtual machine support
- safety concept

EEC SW GEN is deployed as some Java software libraries, which are supposed to be included in country specific EEC SW implementations. It is assumed, that the generic libraries are always embedded into a country specific solution adding specific functionality.

EEC SW GEN implements the following generic software modules and supports the following interfaces:

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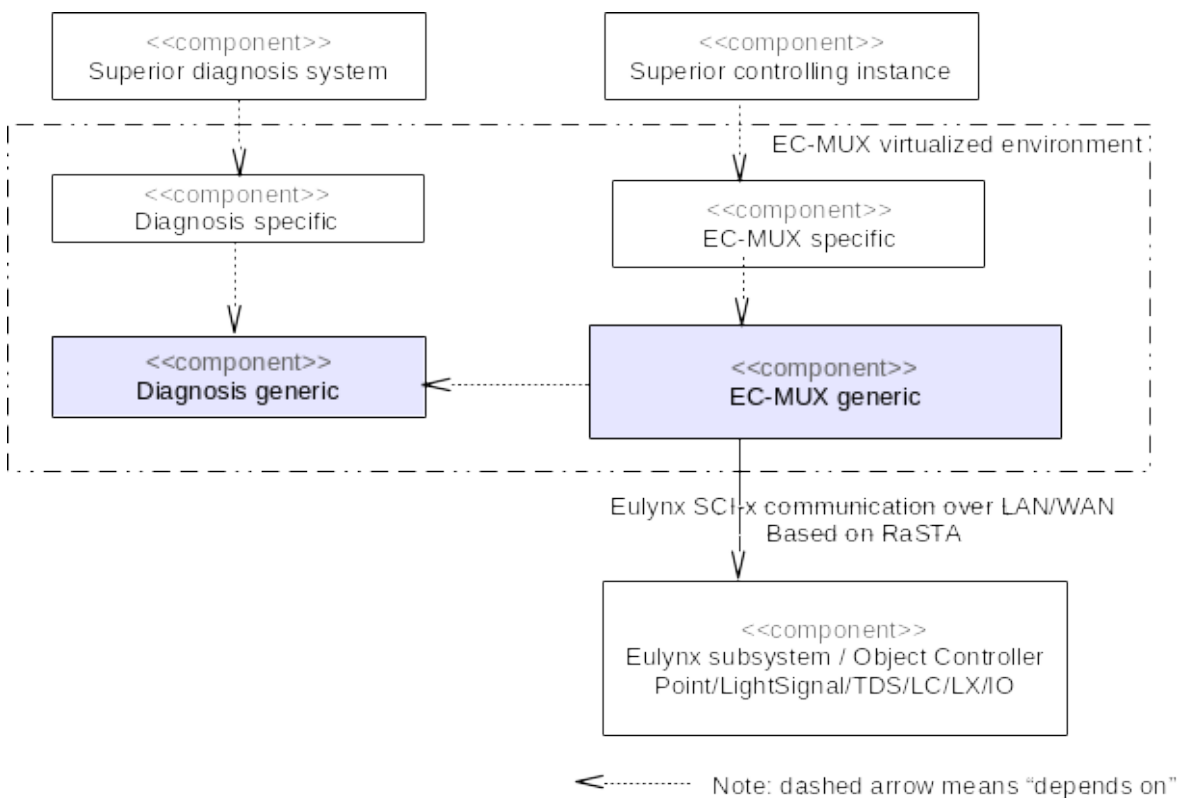


Figure 2: EEC SW GEN: main software modules and interfaces

- EEC command interface: the EC-MUX provides a country independent software interface to allow any controller using the EEC SW system to operate trackside elements and receive state and failure information
- Interface Eulynx SCI-x - IP based communication over WAN/LAN / single-message-safe principle using RaSTA: the protocol SCI-x is the Eulynx application layer interface for communication between EC-MUX and OCs. On transportation layer, according to Eulynx it is sent via the RaSTA [10] protocol
- Module EC-MUX generic: software channels for the element controller multiplexer - operates multiple OCs, provides generic functionality to send/receive command/state information to/from the correct OCs
- Module Diagnosis generic: common instance to collect all generated diagnosis data from the whole EEC SW system

The components supposed to extend EEC SW GEN in any country specific implementation are:

- EC-MUX specific: implementation of the EEC command interface, translation of incoming commands from the superior controller to EEC commands, vice-versa for element states and failures sent back
- Diagnosis specific modules: implementation of specific translation of messages to/from a superior diagnosis system

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1.2.3 Light Signals Overview

The Light Signals element controls and supervises the following signals:

- Main Signal
- Protection Signal
- Shunting Signal Standalone
- Shunting Signal Combined
- Advance Signal Standalone
- Advance Signal Combined
- Speed Indicator Standalone
- Speed Indicator Combined
- Signal Repeater Standalone
- Signal Repeater Combined
- Subsidiary Signal
- Subsidiary Signal Repeater
- Advance Speed Indicator

Not supported in current SCI-LS development phase:

- Caution Signal

For details about the specific signal types see Chapter Requirements.

PDI-version

The PDI-version of the SCI-LS as described in EULYNX Interface specification SCI-LS [7] is 0x05.

Protocol type

The protocol type used for the SCI-LS is 0x30.

Message type

The following table shows permitted subsystem specific message types for the SCI-LS.PDI. All of them are supported by SEA Eulynx Signalling.

Message type	Value
<i>command</i> Indicate Signal Aspect	0x0001
<i>message</i> Indicated Signal Aspect	0x0003
<i>command</i> Set Luminosity	0x0002
<i>message</i> Set Luminosity	0x0004

Table 1: Subsystem specific message types for the SCI-LS.PDI

1.2.3.1 Eulynx Signal Mast configuration

The following table describes the used bytes in the OC-LS message according to SCI-LS [7].

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Remark: For the signal type that is not configured on the mast, the value “Not equipped = 0xFF” shall be used for the corresponding command Byte. This is applicable for Bytes 43 - 48 within Cmd LS Indicate Signal Aspect in OBB context.

Remark: The Byte 47 (direction indicators) and Byte 48 (direction indicator announcements) are removed from the table Eulynx Signal Mast configuration because these are UNUSED.

Element Mast	Signal	Byte 43 = code for basic aspect types	Byte 44 = code for extension of basic aspect types	Byte 45 = code for speed indicators	Byte 46 = speed indicator announcements
Main Signal Mast	Main Signal	For the exact used values, see sub-chapter Main Signal	UNUSED	UNUSED	UNUSED
	Shunting Signal Combined	For the exact used values, see sub-chapter Shunting Signal Combined	UNUSED	UNUSED	UNUSED
	Subsidiary / Caution Signal	UNUSED	For the exact used values, see sub-chapter Subsidiary / Caution Signal	UNUSED	UNUSED
	Advance Signal Combined	UNUSED	For the exact used values, see sub-chapter Advance Signal Combined	UNUSED	UNUSED
	Speed Indicator Combined	UNUSED	UNUSED	For the exact used values, see sub-chapter Speed Indicator	UNUSED
	Advance Speed Indicator	UNUSED	UNUSED	UNUSED	For the exact used values, see sub-chapter Advance Speed Indicator

Table 2: Eulynx Signal Mast configuration - Main Signal Mast

Eulynx Main Signal Mast is in SEA realized as shown in the following figure.

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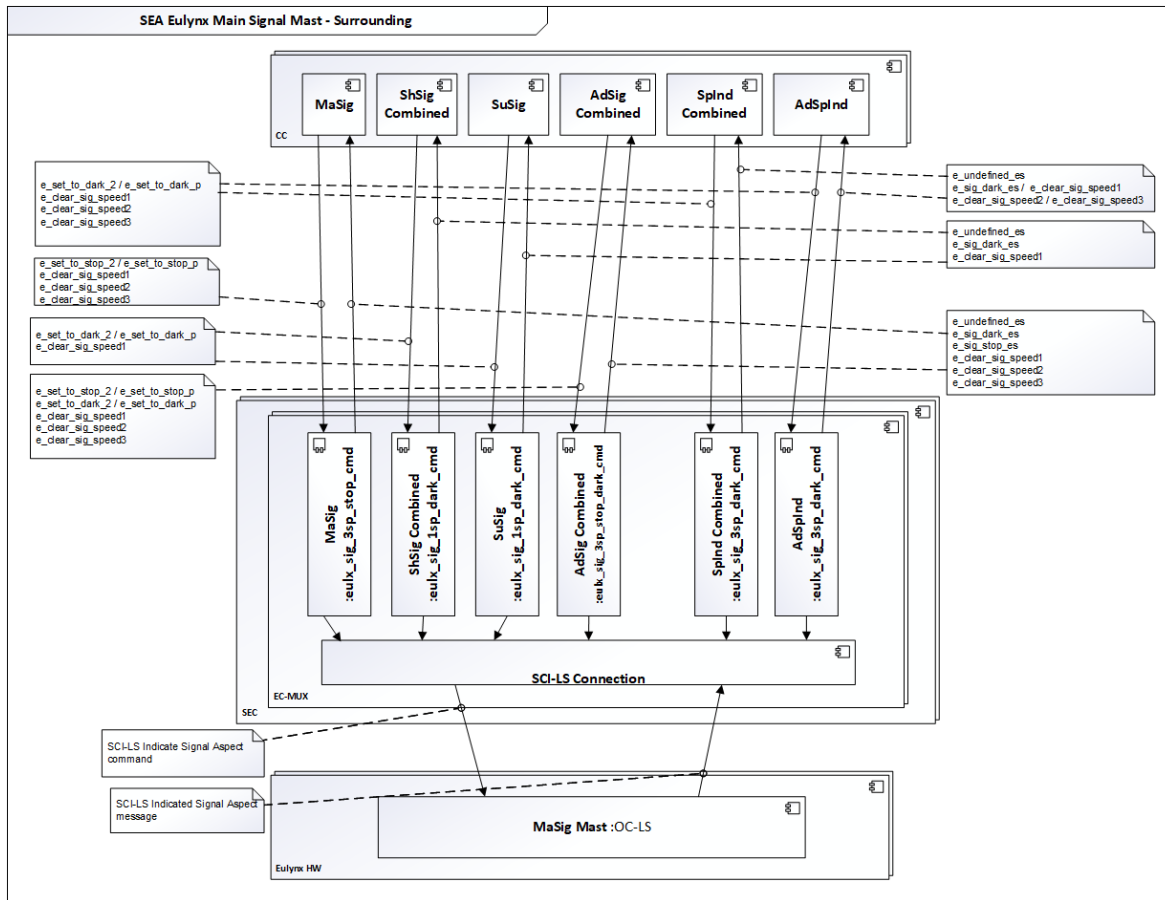


Figure 3: SEA Eulynx Main Signal Mast - Surrounding

Element Mast	Signal	Byte 43 = code for basic aspect types	Byte 44 = code for extension of basic aspect types	Byte 45 = code for speed indicators	Byte 46 = speed indicator announcements
	Protection Signal	For the exact used values, see sub-chapter Protection Signal	UNUSED	UNUSED	UNUSED
	Shunting Signal Combined	For the exact used values, see sub-chapter Shunting Signal Combined	UNUSED	UNUSED	UNUSED
	Subsidiary / Caution Signal	UNUSED	For the exact used values, see sub-chapter Subsidiary / Caution Signal	UNUSED	UNUSED

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Protection Signal Mast	Advance Signal Combined	UNUSED	For the exact used values, see sub- chapter Advance Signal Combined	UNUSED	UNUSED
	Signal Repeater Combined	UNUSED	For the exact used values, see sub- chapter Signal Repeater Combined	UNUSED	UNUSED
	Speed Indicator Combined	UNUSED	UNUSED	For the exact used values, see sub- chapter Speed Indicator	UNUSED
	Advance Speed Indicator	UNUSED	UNUSED	UNUSED	For the exact used values, see sub-chapter Advance Speed Indicator

Table 3: Eulynx Signal Mast configuration - Protection Signal Mast

Eulynx Protection Signal Mast is in SEA realized as shown in the following figure.

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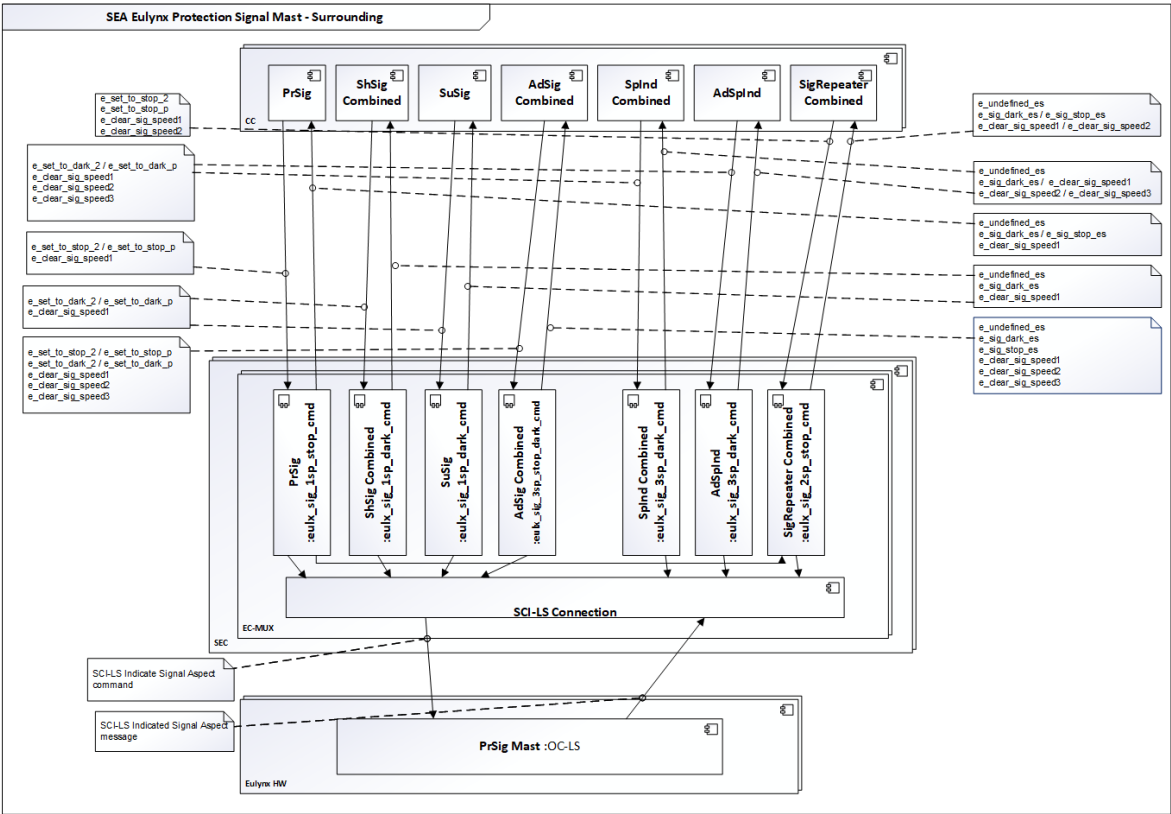


Figure 4: SEA Eulynx Protection Signal Mast - Surrounding

Element Mast	Signal	Byte 43 = code for basic aspect types	Byte 44 = code for extension of basic aspect types	Byte 45 = code for speed indicators	Byte 46 = speed indicator announcements
Shunting Signal Mast	Shunting Signal Standalone	For the exact used values, see sub-chapter Shunting Signal Standalone	UNUSED	UNUSED	UNUSED

Table 4: Eulynx Signal Mast configuration - Shunting Signal Mast

Eulynx Shunting Signal Mast is in SEA realized as shown in the following figure.

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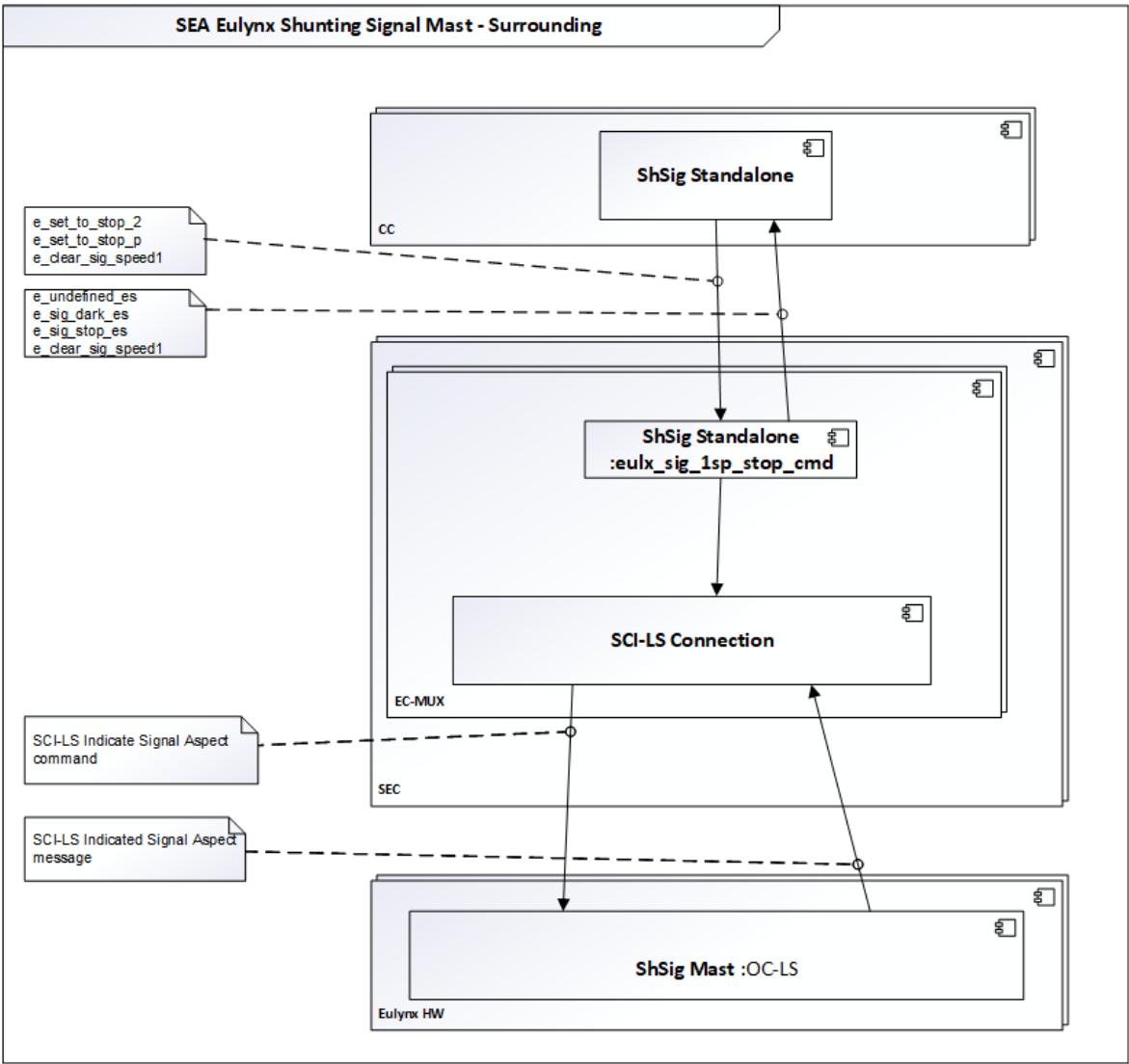


Figure 5: SEA Eulynx Shunting Signal Mast - Surrounding

Element Mast	Signal	Byte 43 = code for basic aspect types	Byte 44 = code for extension of basic aspect types	Byte 45 = code for speed indicators	Byte 46 = speed indicator announcements
Advance Signal Mast	Advance Signal Standalone	For the exact used values, see sub-chapter Advance Signal Standalone	UNUSED	UNUSED	UNUSED
	Advance Speed	UNUSED	UNUSED	UNUSED	For the exact used values, see sub-chapter Advance

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	Indicator				Speed Indicator
--	-----------	--	--	--	-----------------

Table 5: Eulynx Signal Mast configuration - Advance Signal Mast

Eulynx Advance Signal Mast is in SEA realized as shown in the following figure.

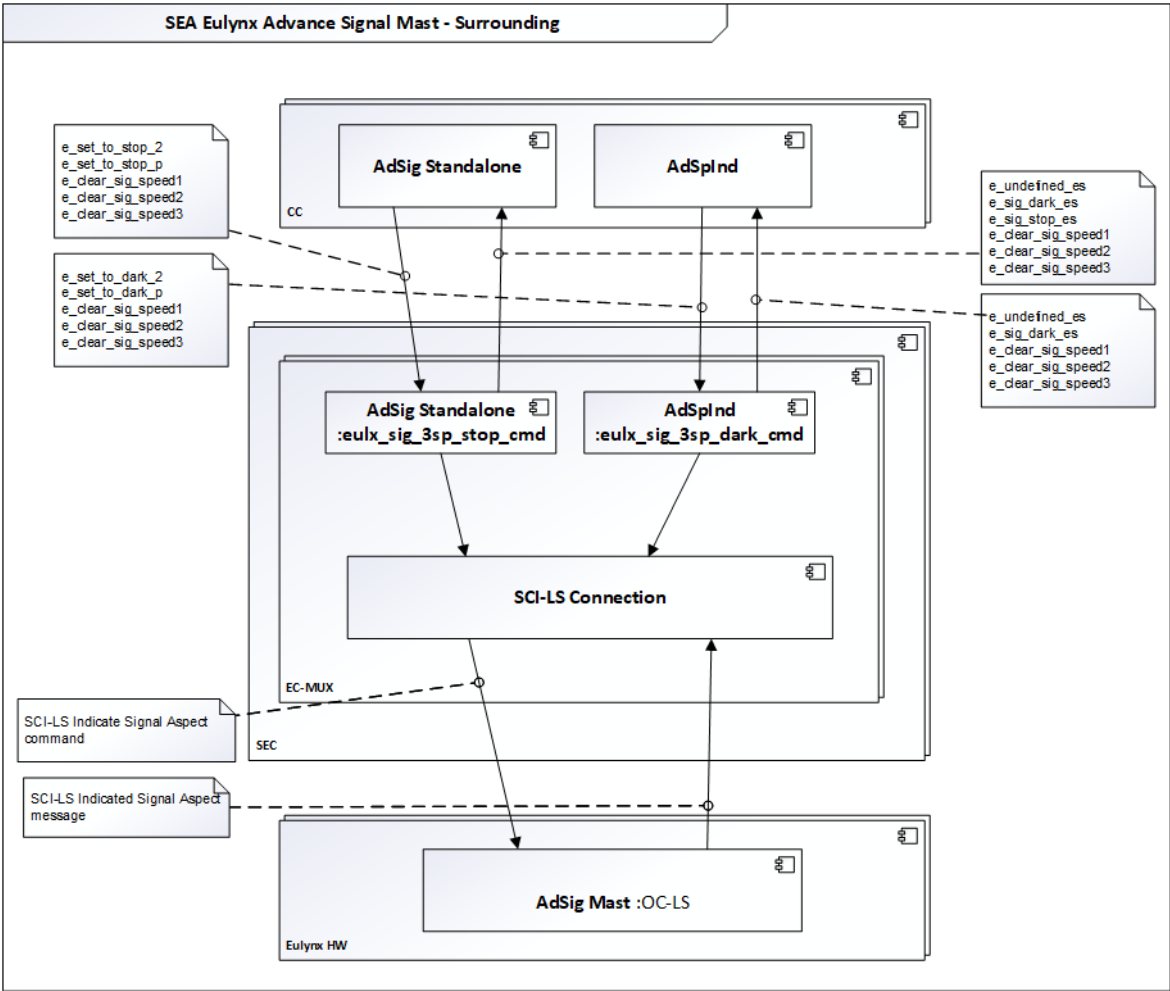


Figure 6: SEA Eulynx Advance Signal Mast - Surrounding

Element Mast	Signal	Byte 43 = code for basic aspect types	Byte 44 = code for extension of basic aspect types	Byte 45 = code for speed indicators	Byte 46 = speed indicator announcements
Speed Indicator Mast	Speed Indicator Standalone	Not equiped = 0xFF	UNUSED	For the exact used values, see sub-chapter Speed Indicator	UNUSED

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Table 6: Eulynx Signal Mast configuration - Speed Indicator Mast

Eulynx Speed Indicator Mast is in SEA realized as shown in the following figure.

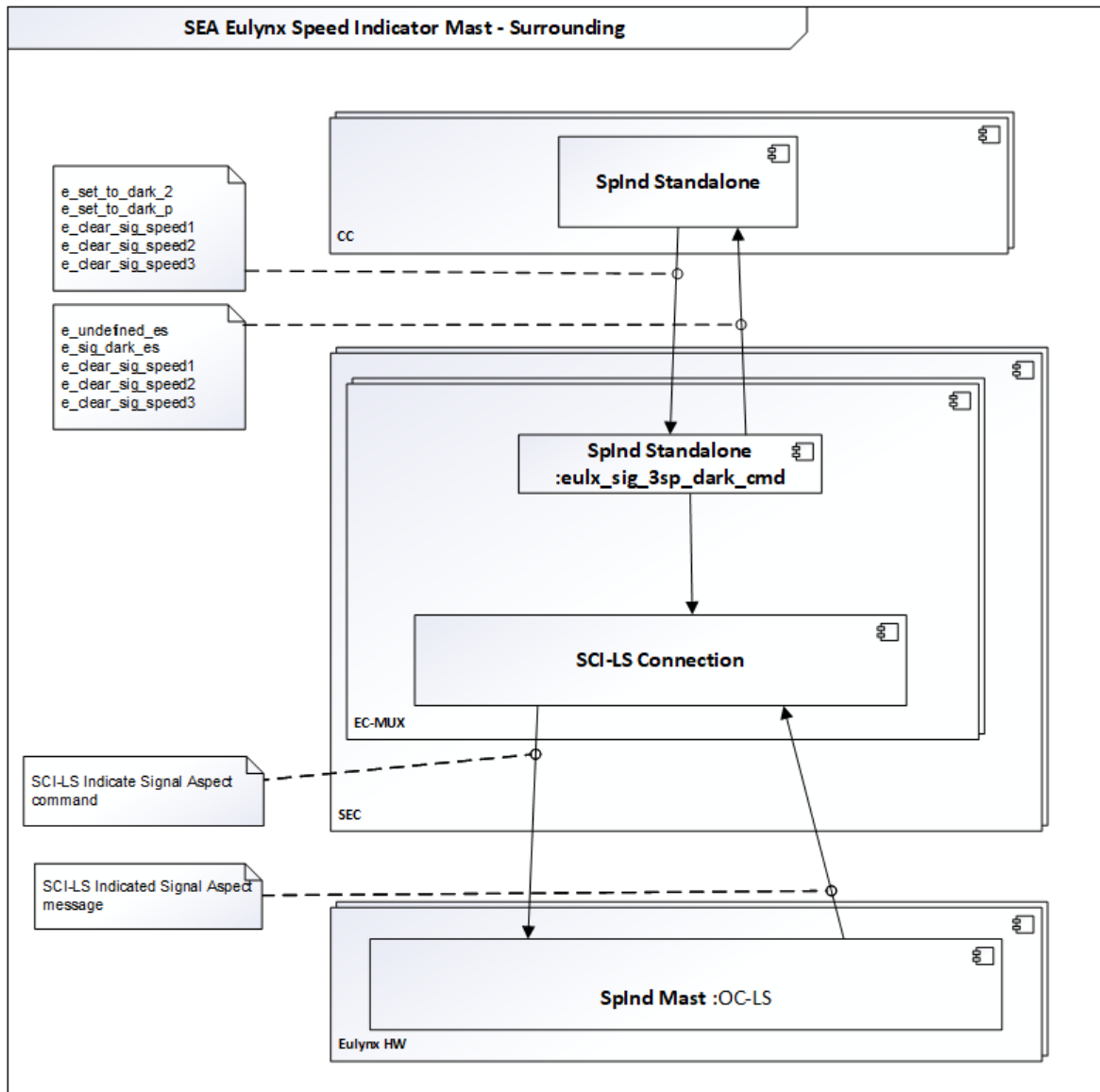


Figure 7: SEA Eulynx Speed Indicator Mast - Surrounding

Element Mast	Signal	Byte 43 = code for basic aspect types	Byte 44 = code for extension of basic aspect types	Byte 45 = code for speed indicators	Byte 46 = speed indicator announcements
	Signal	For the exact used			

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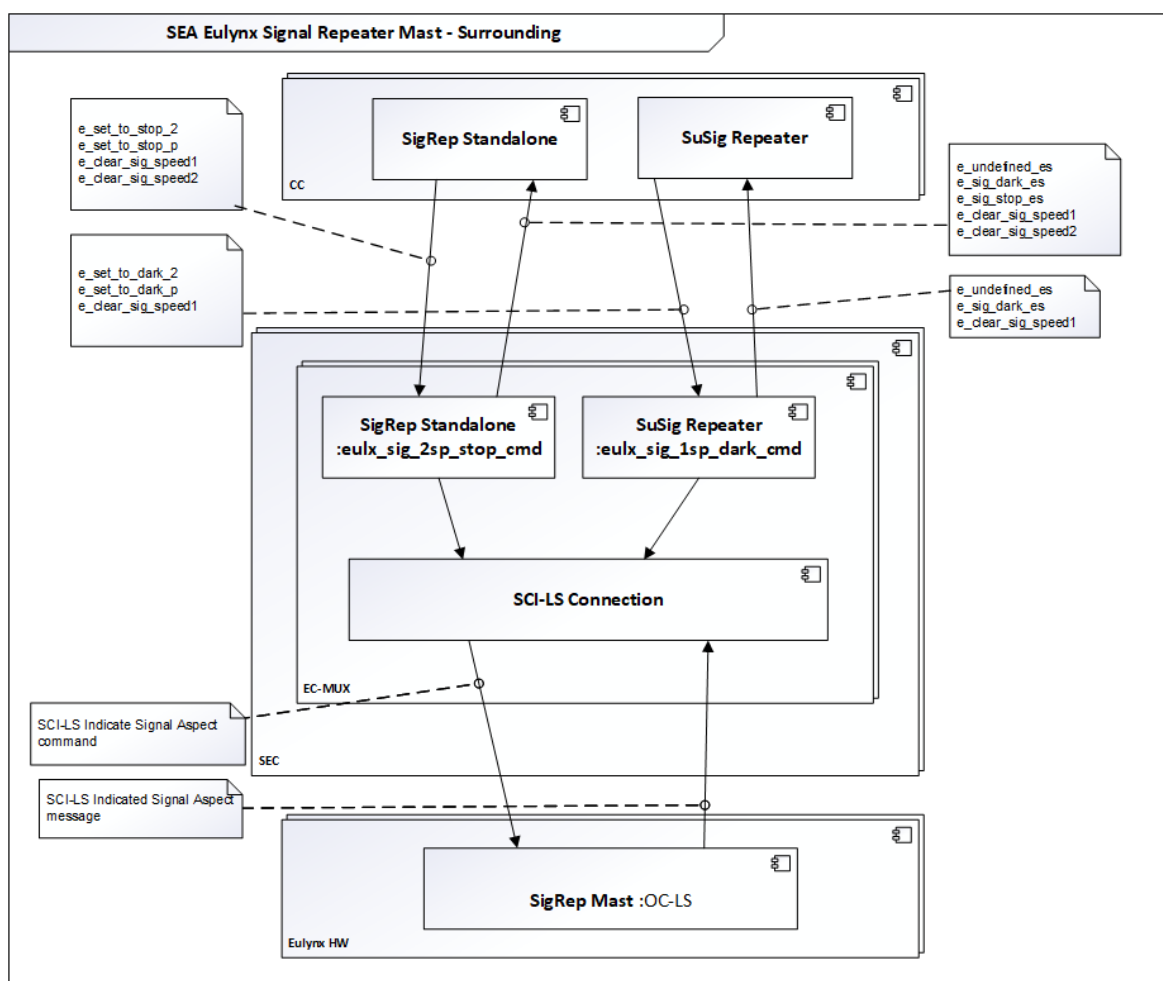
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Signal Repeater Mast	Repeater Standalone	values, see sub-chapter Signal Repeater Standalone	UNUSED	UNUSED	UNUSED
	Subsidiary Signal Repeater	UNUSED	For the exact used values, see sub-chapter Subsidiary Signal Repeater	UNUSED	UNUSED

Table 7: Eulynx Signal Mast configuration - Signal Repeater Mast

Eulynx Signal Repeater Mast is in SEA realized as shown in the following figure.


Figure 8: SEA Eulynx Signal Repeater Mast - Surrounding

Remark: In the SCI-LS [7] there are currently no details defined for the usage of command Bytes 49 - 60 within Cmd LS Indicate Signal Aspect in OBB context. Using 0xFE / 0xFF is just an assumption.

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Command byte	Command parameter	Default value
49	downgrade information	Downgrade information not applicable = 0xFF
50	route information	Route information not applicable = 0xFF
51	intentionally dark	Intentionally dark not applicable = 0xFF
52 - 60	national specified	No information = 0xFE

Table 8: Eulynx Signal Command Byte not specified in SCI-LS

Note that any of the signals on the mast, except the primary, may or may not be configured.

1.3 Document Overview

This document is the System Subsystem Specification for the Light Signals SW subsystem of the SEA OBB software library.

Chapter 1 defines the scope of the document, identifies the system and explains SEA and the Light Signals subsystem.

Chapter 2 lists all references, terms, abbreviations and requirement language.

Chapter 3 of this document lists the requirements for the generic Light Signals functionality:

- Subchapter "Operational" contains all the requirements for operational scenarios (all commands received in all states, spontaneous notifications and notifications as command answer)
- Subchapter "Technical" contains all the requirements for technical scenarios such as booting, connection loss and diagnosis.

At the end there is an index of all requirements listed in chapter Requirements.

1.4 Intended Audience

The target audience for this document is:

- Product Design Authority (PDA)
- Solution Engineering Manager (SEM)
- Safety Assurance Manager (SAM)
- Verifier (VER)
- Validator (VAL)
- Integration Verification, Validation, Qualification Manager (IVVQM)
- Architect (ARC)
- Requirements Manager (RM)
- Product Line Manager (PLM)
- Quality Assurance Manager (QAM)

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1.5 Evolution of the document

The document is written iteratively according to the Agile Process defined in the SEA Team Charter [1]. This means that several document editions can be released before the functionality is completely described/defined. For the details on what is included in which edition of the document please see the section History below.

After the final edition containing all information for the planned functionality is released, it will be updated accordingly in case of requests for changes of the system. For type and reasons of changes please refer to the section History below.

1.5.1 History

1.5.1.1 Edition 01: Initial edition

This document edition contains requirements that define the features of the Light Signals.

1.5.2 Open issues

- Light Signal functionality as a result of reverse engineering and based on EULYNX Interface specification SCI-LS [7] and SRS SEA OBB [4] is covered by the current edition of this document. Still, as no high-level documents - feature list, SEMP - are available it cannot be guaranteed that everything needed is covered. References to these documents need to be specified and are currently marked with a "TODO". Editions of the referenced documents in Chapter 2. *References and Terms* need to be updated.
- An occurrence / repair of a signal lamp fault shall be reported to the CC by the sum_fault / sum_fault_rep signals. Indication of these fault events via SCI-LS is not fully specified yet, clarification needed. For this reason reporting sum_fault / sum_fault_rep is not considered yet in the requirements defined in this document and will be added in a later project phase.
- Consolidate term usage SEA/SEC/EEC in the document, currently Chapter 1.2 *System Overview* uses EEC while the rest of the document uses SEA
- Add a reference to finally used versions of DCA3.x (and DCA2.x) to *System Overview*
- There are scenarios for Signal Repeater Combined where it is not clear how the OC-LS shall work, especially with regard to lamp faulty. These scenarios need to be clarified (SEA-5760).
- The CC connection loss requirements for Signal Mast are not yet clarified and not included in this document and will be added in a later project phase.
- Traceability of system layer requirements (due to its non-availability) or to Eulynx specifications is not included in the document for now

1.5.3 Distribution

A new version (proposal) of this document has to be distributed to all functions listed in the section Intended Audience. A new edition of this document has to be archived (online archive and conventional archive). Due to possible large numbers of involved parties the online archival of proposals is recommended. A copy for information can be downloaded from online archive by everyone in the project, who has read privilege.

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1.5.4 Configuration Management

This document is generated by the Doc Maven Plugin. It is composed of textual building blocks; the content structure and metadata are defined in a "document definition" XML file. These source files are archived in the Git repository of GTS Austria, where all changes are tracked. The generated document is published on the project's website as XHTML and PDF.
Formal review proposals and released versions are archived in the GTS Austria central document repository in PDF format.

Git URL: <https://bitbucket.de.gts/scm/sea/sea>
Git commit: aef0f61772
Document definition file: /doc-obbb/src/doc-definition/sss_ls.xml

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2 References and Terms

2.1 SEA Generic Documents

[1]	Team Charter SEA, 3BU 61400 0137 QMAPA, Ed. 03
[2]	Acronyms and Abbreviations; ELEKTRA Evolution, 3BU 15000 0081 DCAPA, Ed. 06

2.2 SEA Design Documents

[3]	SEA SW GEN SSDD - System Subsystem Design Description, 3BU 61400 0119 DSAPA, Ed. 08
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2.3 SEA Specification Documents

[4]	System Requirement Specification, SEA OBB, 3BU 61400 0141 DTAPA, Ed. 05 - TODO: To be extended for LS
[5]	SEA SW OBB Light Signals, Software Design Description, 3BU 61400 0182 PEAPA, Ed. 01

2.4 Elektra Documents

[6]	IDD CC/DGP - EC-MUX, 3BU 15000 9130 PBAPA, Ed. 02
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2.5 External documents

[7]]	[EULYNX Interface specification for subsystem Signals], EULYNX Interface specification SCI-LS, Eu.Doc.33, Baseline 4.2 (0.A)
[8]]	[EULYNX Requirements specification for subsystem Signals], EULYNX Requirements specification SCI-LS, Eu.Doc.32, Baseline 4.3 (0.A)
[9]]	[EULYNX Signal aspect table], EULYNX Signal Aspect Table, Eu.Doc.37, Baseline 2.2 (0.E), EULYNX Baseline Set 4

2.6 Protocols

[10]	RaSTA, Rail Safe Transport Application, V03.03, DIN VDE V 0831-200, 3BU 01500 0006 PBAPA, Ed. 01
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2.7 Structure of Requirements Definition

This subsection defines the structure of one single requirement. The keywords and their explanations are shown below:

- **Identifier:** Unique identifier of the requirement
- **Type:** A classification of the different requirement types; the following keywords are defined:

CSR Customer Specific Requirement
 DAT Data Requirement
 DOR Document Requirement
 DPR Dependability Requirement
 ECR Environmental Conservation Requirement
 FUR Functional Requirement
 GCR General Constraint Requirement
 MCR Management and Commercial Requirement
 MDR Maintenance and Diagnosis Requirement
 MIR Man-Machine Interface Requirement
 MPR Management and Production Requirement
 OAR Operational Aliveness Requirement
 PER Physical/Environmental Requirement
 PHR Prevention of Hazards Requirement
 PRR Performance and Real-Time Requirement
 PSR Product Startup Requirement
 RSR Resource Support and Consumption Requirement
 SAR Safety Requirement
 SCR Supplier to Customer Requirement
 SER Security Requirement
 TIR Technical Interface Requirement
 TVR Testing and Validation Requirement

- **Synopsis:** A brief and significant description of the requirement.
- **Description:** Complete description of the requirement. Includes, when needed, layouts (screen layouts in principle - no screen dumps), formats, and protocols for interfaces.
- **Satisfies (Origin):** Source of the requirement (feature list, reference to other requirements or internal). It is a precise backward reference of this requirement to its origin. Per convention, references to the Feature List SEA are marked FR:SEA:xxx.
- **Status:** The current status of the requirement, with the following keywords defined:
 - in work: new requirement which is not yet described completely
 - draft: requirement is written down and ready to be reviewed, it is not yet approved
 - approved: requirement was reviewed by all affected groups and is confirmed
 - descoped: Requirement delayed to a later release
 - to be descoped: requirement is planned to be descoped, no formal approval for changing the state yet
 - to be deleted: Requirement is planned to be deleted. No formal approval for deleting by CCB yet
 - deleted: Requirement is no longer valid and will not be implemented at all. Has to be agreed by CCB

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- **Stability:**
 - stable: will probably not change during the life time of the product
 - volatile: subject to change
 - instable: subject to change frequently
- **Target Release:** Scheduled for which release
- **Priority:** Relative importance of fulfilling this requirement from a functional point of view (used for incremental deliveries). The following keywords are defined:
 - **Mandatory:** Requirement required by applications. If requirement is not provided, the fact and the consequences have to be documented in the validation report. The application has to decide if the product is accepted or not in absence of this requirement.
 - **Optional:** Enhancement of the product may be implemented in this or a later release. Absence does not make the product unacceptable.
- **Originator:** "Equipment Practice", "Product Management", "Quality", "RAM", "Safety", "Standards /Legal", "Technics (R&D)"
- **Allocation:** Allocates requirements to the system, subsystem, component, or product environment as described in a product breakdown structure. The appropriate keywords have to be defined by the project.
- **Verification Method:** "Analysis", "Demonstration", "Inspection/Review", "Test"
- **Verification Step:** The following verification steps are defined:

Review
 Code Reading
 Analysis
 Module Test
 SubIt Test
 Integration Test
 System Test
 Performance Test
 Qualification Test
 Acceptance Test
 Field Test
 Tool Test
 Safety Approval

- **Comment:** Any additional information if necessary. (e.g. reason for deleting a requirement)

2.8 Requirement Language

This subsection defines the language used in requirements and SUBIT test cases. Note that all actions are specific to the element for which the requirement or test case is written, e.g. by received a command it is meant the command received for the element in question, by reacting or not reacting to an input it is meant that SEC is doing an action on behalf of the element in question, without considering any other elements, unless this is explicitly mentioned.

The element state mentioned is the SEA internal element state as defined in State Machine Diagram.

General:

- "SEA shall do nothing":

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SEA shall do nothing in regard to the element in question:

- the state of the element is not changed
- element specific timers are not changed
- no command is sent to the HW interface or to another internal SEA component
- no message is sent on the CC and DGP interface
- "Action within t ":

Action is performed within the specified time interval from the last action.

- "Action after t ":

Action is performed specified amount of time after the last action.

- "Action after t from *ReferencePoint*":

Action is performed specified amount of time after the reference point.

- "Action when reaching state *State*":

Action is performed when the SEA internal element state has changed to the specified state.

- "SEA shall set the current element state to *State*":

SEA shall set the SEA internal element state to *State*.

- "SEA shall switch to *State*":

SEA shall

- set the SEA internal element state to *State*
- command the output or check-back port(s) corresponding to the element state *State* (as defined in the mapping table)
- "the current element state is *State*":

SEA internal element state is *State*.

- "the element state has changed to *State*":

SEA internal element state changed.

- "SEA shall execute EC-MUX safety reaction *SafetyReaction*"

SEA shall execute one of the safety reactions defined in SSDD SEA GEN [ref:SSDD_SEA_GEN], Table "EC-MUX Safety Reactions".

- "an *Element* configured as *Type*":

The element configuration corresponds as defined for the specified type.

- "initializing the element"

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At SEA start-up, one of the performed actions is to read the xml database file containing station data and setup the element repository. Each element contained in the station data shall take on an initial state according to its element type configuration.

EC-MUX booting:

- "shutting down EC-MUX"

Shut down the EC.MUX software by causing a reboot of the hardware.

- "starting up EC-MUX"

Wait until the EC-MUX hardware is booted up again. This includes also establishing the connection between EC-MUX and HW (OEC, OC...).

Internal communication between SEA elements:

- "changing the state of *Element*":

The element (for which the requirement is written) detects a state change of another element.

CC Interface connection and fault handling:

- "SEA shall report the *Fault* fault to CC"

Send the fault information for the element from EC-MUX to CC via the s_epa_gcx_C20B_sum_fault/s_epb_gcx_D30B_sum_fault CHILL signal.

- "SEA shall report the *Fault* fault repair to CC"

Send the fault information for the element from EC-MUX to CC via the s_epa_gcx_C20C_sum_fault_rep/s_epb_gcx_D30C_sum_fault_rep CHILL signal.

- "establishing the connection to CC"

This includes successful completion of all phases required for an operational connection, like version handshake, update_history phase and general inquiry. See [6] for details.

- "losing the connection to CC"

EC-MUX does no longer see the CCx. This may either be caused by network connectivity loss, or by CCx reboot. Implicitly, while there is no connection to the CC, no communication (receiving commands or reporting) will be possible.

Connection loss to CC is detected by at least one of the following:

- CCx does not respond to EC-MUX heartbeats
- CCx does not respond to EC-MUX reactivity supervision signals
- EC-MUX gets an "ep_is_down" event
- "CC connection state is [CONNECTED|NOT_CONNECTED]":

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CC connection state **CONNECTED** means that the connection between EC-MUX and CC is fully operational, the connection procedure as explained under "(re)establishing the connection to CC" has been performed.

CC connection state **NOT_CONNECTED** means that there is no connection between EC-MUX and CC, the CC connection was never available or the disconnection procedure as explained under "losing the connection to CC" has been performed.

- "the connection to CC has been lost"

EC-MUX lost the connection to the CC. See also "losing the connection to CC".

DGP Interface:

- "SEA shall report the diagnosis aspect message *Aspect* to DGP":

Send `e_sea_dgp_elem_aspect_reply` with the given aspect (corresponding to the element state) from SEA to DGP via DGP signal `s_elx_dgp_01b2_diagnose`.

- "SEA shall report the *Fault* diagnosis device fault message for the *Device* device to DGP":

Send `e_sea_dgp_elem_hw_fault` for the element with the given faulty device (e.g. IOC) from SEA to DGP via DGP signal `s_elx_dgp_01b2_diagnose`.

- "SEA shall report the *Fault* diagnosis device fault message to DGP":

Send `e_sea_dgp_elem_hw_fault` for the element (no device specified) from EC-MUX to DGP via DGP signal `s_elx_dgp_01b2_diagnose`.

- "SEA shall report the *Fault* diagnosis device fault repair message for the *Device* device to DGP":

Send `e_sea_dgp_elem_hw_repair` for the element with the given faulty device (e.g. IOC) from SEA to DGP via DGP signal.

- "SEA shall report the *Fault* diagnosis device fault repair message to DGP":

Send `e_sea_dgp_elem_hw_repair` for the element (no device specified) from EC-MUX to DGP via DGP signal `s_elx_dgp_01b2_diagnose`.

- "SEA shall report the rejected command diagnosis to DGP":

Send `e_pc_dgp_elem_cmd_info` from EC-MUX to DGP via DGP signal `s_elx_dgp_01b2_diagnose`.

- "When receiving an aspect query from DGP":

refers to receiving the `e_dgp_sea_elem_aspect` for the element from the DGP. SEA responds by one or more `e_sea_dgp_elem_aspect_reply` messages for the addressed element.

Related to SubIT:

- "the element configuration is completed":

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This step is intended to be used only in testing at SubIT level. It represents the moment when OIF and CC filters are created and General Inquiry is sent. It is needed for identifying the specific elements when multiple elements are part of the environment setup.

Interface EC-MUX - OC connection:

- "losing the connection between EC-MUX and OC":

This OC is a placeholder and it means the step is applicable for OC-P, OC-LS, OC-TDS, OC-IO, OC-LC, OC-IO Eulynx subsystems. Eulynx connection between the EC-MUX and OC is lost, EC-MUX sets all elements controlled by this OC to the default state and executes safe commands for all elements.

- "establishing the connection between EC-MUX and OC":

This OC is a placeholder and it means the step is applicable for OC-P, OC-LS, OC-TDS, OC-IO, OC-LC, OC-IO Eulynx subsystems. Eulynx connection between the EC-MUX and OC is established, all phases required for an operational connection like version handshake and general inquiry are successfully completed.

- "SEA shall execute EC-MUX safety reaction OC Disconnection":

This step includes disconnect the connection between EC-MUX and OC, set element state to default state and report and diagnose state change and report and diagnose the disconnect.

- "SEA shall execute EC-MUX safety reaction OC-SCI Disconnection":

This step includes disconnect the connection between SCI and OC and diagnose the disconnect.

OC-LS Interface:

SCI-LS commands and messages referred in the following steps are defined in SCI-LS [7]. These steps are applicable for several signal types. For this reason particular values to be used for the referred command / message parameter and (default) values of every other command / message parameter are not stated here, but in the signal type related functional description and requirements.

- "SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to <SeaSigAspect> to OC-LS"

SEA sends the command LS Indicate Signal Aspect with Code corresponding to the SEA Eulynx Signal Aspect as defined in mapping tables for the given Signal Type.

E.g. for Shunting Signal Standalone this mapping can be found in sub-chapter Shunting Signal Standalone - "Shunting Signal Standalone - Mapping CC command (CC to EC-MUX) to SCI-LS Cmd Indicate Signal Aspect (EC-MUX to OC-LS)" Table.

- "receiving the LS Indicated Signal Aspect message with Code corresponding to <SeaSigAspect> from OC-LS"

SEA receives the message LS Indicated Signal Aspect with Code corresponding to the SEA

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Eulynx Signal Aspect as defined in mapping tables for the given Signal Type.

E.g. for Shunting Signal Standalone this mapping can be found in sub-chapter Shunting Signal Standalone - "Shunting Signal Standalone - Mapping SCI-LS Msg Indicated Signal Aspect (OC-LS to EC-MUX) to gen_elem_stat (EC-MUX to CC)" Table.

- "the current element state is CHANGING_TO_SPEEDn"

This step indicates that any of the e_clear_sig_speedx command is received from the initial state STOP/DARK. The exact CHANGING_TO_SPEEDx depends on the received e_clear_sig_speedx command.

When referring to CHANGING_TO_SPEEDn it means any of the CHANGING_TO_SPEEDx, x={1; 2;3}. The value of x in CHANGING_TO_SPEEDx is equal to the value of x from the received e_clear_sig_speedx command. In the requirements both CHANGING_TO_SPEEDn and CHANGING_TO_SPEEDx are used to clearly specify the signal behavior.

- "the last stable element state is <LastStable>"

Every element state except CHANGING_TO_SPEEDn / CHANGING_TO_STOP / CHANGING_TO_DARK is considered as stable.

This step is used when the last stable element state is relevant for processing commands from CC, messages from OC-LS, aspect diagnosis query from DGP or connection establishment with CC.

Current element state	Last stable element state	Current OC-LS state	Scenario
CHANGING_TO_SPEEDx	STOP /DARK	STOP /DARK	cmd SPEEDx executed in STOP/DARK
CHANGING_TO_DARK /STOP	SPEEDx	SPEEDx	cmd DARK/STOP executed in SPEEDx
CHANGING_TO_DARK /STOP	DARK /STOP	DARK /STOP	cmd DARK/STOP executed in CHANGING_TO_SPEEDx
CHANGING_TO_DARK /STOP	DARK /STOP	SPEEDx	cmd DARK/STOP executed in CHANGING_TO_SPEEDx after which SPEEDx is received from OC-LS

Table 9: Details explaining the step *the last stable element state is LastStable*

- "SEA shall forward the command LS Set Luminosity with luminosity <Luminosity> to each connected OC-LS"

LS Set Luminosity command with luminosity value LUMINOSITY_FOR_DAY / LUMINOSITY_FOR_NIGHT shall be distributed to each Signal Mast configured for this EC-MUX, i. e. each connected OC-LS (there is one OC-LS connection configured per Signal Mast). LS Set Luminosity command is addressed to the signal type represented by the "basic aspect" when commanding an aspect by the Cmd Indicate Signal Aspect.

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- “SEA shall forward the command LS Set Luminosity with luminosity <Luminosity> to this OC-LS”

LS Set Luminosity command with luminosity value LUMINOSITY_FOR_DAY / LUMINOSITY_FOR_NIGHT shall be sent to the OC-LS that is currently connecting to this EC-MUX.

- “receiving the LS Set Luminosity message with luminosity <Luminosity> from OC-LS”

SEA shall receive the LS Set Luminosity message with luminosity value LUMINOSITY_FOR_DAY / LUMINOSITY_FOR_NIGHT / INTENTIONALLY_DELETED.

- “SEA shall set the stored command to STOP”

Applicable for Signal Repeater Combined: even though upon CC connection loss SEA shall forward the DARK command, internally SEA shall set the stored command to STOP and it shall be used if no command is received from CC between the moment starting with the event of CC connection loss, continuing with CC connection establish and until event of Protection Signal aspect change to SPEED1. While CC connection is CONNECTED, if SEA receives e_clear_sig_speed1 or e_clear_sig_speed2 from CC, then this shall be the last stored command. When Protection Signal aspect changes to SPEED1, SEA shall execute the last stored command. This step is also used at initialization of the element when SEA shall set the stored command to STOP, i.e. the first command to be executed when Protection Signal reaches state SPEED1 shall be STOP (if no other command has been received). Note that commands rejected by the element process shall never be stored, otherwise all commands received from CC are stored.

- “the last stored command is <Command>”

Applicable for Signal Repeater Combined: the last stored command can be e_clear_sig_speed1 or e_clear_sig_speed2 if received from CC, otherwise the command is e_set_to_stop. More details in previous entry.

- “the current element state is DARK reached from CHANGING_TO_STOP /CHANGING_TO_SPEEDx”

Applicable for Signal Repeater Combined: We usually have “Given the current element state is <ChangingState> reached from <StableState>” However DARK can be reached from STOP in different ways, and in case of a timeout it needs to be specified which Unstable State was reached before.

- “the current element state is DARK reached from STOP”

Applicable for Signal Repeater Combined: We usually have “Given the current element state is <ChangingState> reached from <StableState>” However DARK can be reached from STOP in different ways, in this case it is reached from STOP directly when a msg for DARK is received. DARK is usually reached on start up.

CC Interface:

- “receiving the command <Command> from CC”

SEA receives a command from CC via the defined restraint-safe s_gca_gpx_2C80_peri_out /s_gcb_gpx_3D80_peri_out CHILL signal or via the defined execution-safe s_gca_gpx_2c81_2chan_out/s_gcb_gpx_3d81_2chan_out CHILL signal and process it.

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- “SEA shall report the element state <CCState> with command reply flag <Flag> to CC”
SEA sends the signal element state <CCState> to CC via the s_epa_gcx_C20A_gen_elem_stat /s_epb_gcx_D30A_gen_elem_stat CHILL signal. Besides the element state itself it contains also <Flag>, indicating whether the state is reported as command reply or a spontaneous state change. This is represented as mode m_logical (true / false).

- “receiving the luminosity command with <Goal> from CC”

SEA receives the luminosity command s_gca_gpx_2C87_day_state from CC with Goal parameter value TRUE / FALSE, corresponding to commanding the luminosity to DAY / NIGHT.

DGP Interface:

- “SEA shall report the plain text diagnosis message MsgText to DGP”
Send e_sea_dgp_string containing the message text MsgText from EC-MUX to DGP via DGP signal s_elx_dgp_01b2_diagnose.

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2.9 Abbreviations and Acronyms

CC	Central Controller
DCA	Distributed Control Architecture
DGP	DiaGnostic Processor , diagnosis component of ELEKTRA
EC-MUX	Element Controller MULTipleXer
EEC	Elektra 4.0 Element Controlling
HW	Hardware
IO	Input / Output
IOC	Input Output Controller , an IFB-based CAN device
IP	Internet Protocol
LAN	Local area network
RaSTA	Rail Safe Transport Application
ÖBB	Oesterreichische Bundes Bahnen (Austrian Federal Railways)
OC-IO	Object Controller - Input Output
OC-LC	Object Controller - Level Crossing
OC-LS	Object Controller - Light Signaling
OC-P	Object Controller - Point
OC-TDS	Object Controller - Train Detection System
OEC	Outdoor Element Controller
OIF	Outdoor InterFace
PDF	Portable Document Format
REQ	REQUIREment
SEA	Sichere Elektronische Aussenelementansteuerung (Engl.: safe electronic outdoor element control)
SEC	Single Element Controller
SEMP	System Engineering Management Plan
SIL4	Safety Integrity Level 4
SSS	System Subsystem Specification
SW	Software
WAN	Wide Area Network

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XML	EX tensible M arkup L anguage
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Additional Terms, Abbreviations and Acronyms can be found in [2].

2.10 Definitions

2.10.1 Terms

- “Generic Eulynx Signal Type” or “Eulynx Signal” refer to a SEA Signal implementation based on Eulynx specification.

2.10.2 Definitions

1. Configuration data
any information or file needed by the application software when executed (station data and software configuration files)
2. Configuration partition
a partition on the flash card, which is supposed to store any configuration data
3. Software configuration files
files, which are loaded by the application at startup to read parameters for executing the software; not site specific
4. Station data
the site specific preparation data, which describes the present elements of a component; output from Orest in XML format
5. Application software (files)
the executables (files), which are executed when the application is started
6. Element process
The EC-MUX contains the logic corresponding to the behavior of each configured element, internally divided into a set of Java classes. This logic is referred to as “element process”.
7. Initial element state
The initial state is the state of the element upon startup when DCA is disconnected.
8. Default element state
The default state is the one the element shall try to recover to when an error occurs.
9. Fault
Defect, mistake or inaccuracy which could result in failure or in a deviation from the intended performance or behavior
10. Failure
Unacceptable difference between required and observed performance.
11. Light Signal
A railway light signal is a visual display device that conveys instructions regarding the train driver's authority to proceed. The train driver interprets the signal's indication and acts accordingly. Typically, a signal might instruct the driver to stop, or inform the driver of the speed at which the train is allowed to proceed. A railway signal usually consists of several light points of different color. For the complete list see chapter *Light Signals Overview*.

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12. Standalone Signal

A signal which is called standalone exists as a primary signal (or sometimes called superior signal) on a mast. Note that there may be other signals on the same mast, but a standalone signal's state does not depend on the other signals on the mast.

13. Combined Signal

A combined signal has a primary signal (or sometimes called superior signal) on its mast and the state of the combined signal may depend on the current primary signal's aspect. For example, an Advance Signal Combined will be DARK when its primary signal (a Main Signal) shows STOP. Note that a primary signal's state does not depend on a combined signal's state.

To describe parts of code, the following terms are used:

- **Class**
(usually) small part of executable code, encapsulating its own data in object-oriented context.
Example: "at.thales.sea.oec.builders.LampDeviceCfg"
- **Package**
group of classes contextually fitting together (e.g. depending on each other). Example: "at.thales.sea.oec.builders". Grouping may be applied recursively; a package may contain sub-packages.
- **Module**
A Maven module. Consists of one or more packages. This is the smallest unit that can be built on its own. Example: "ecbase".
- **Component**
Software part that serves a defined task. Consists of one or more modules. Note that modules may be used in multiple components, so in general components share the same modules.
Example: "EC-MUX generic". Note further, that this is not the definition of the term component as stated in CENELEC standard, but a term used in SW context.
- **Deliverable**
Software part that may be deployed on its own. This includes binaries, scripts, and configuration items like resource files or station data.

2.10.3 Telegram Overview

The following table briefly describes the commands used in SEA Eulynx Signals. More details can be found in table *Generic Eulynx Signal Type* - see the links in column *Specific Signal Types* and column *Sub-Chapter*.

CC Command	Function	SCI-LS command
e_set_to_stop_2 OR e_set_to_stop_p	Command Signal Aspect to STOP	Command LS Indicate Signal Aspect
e_clear_sig_speed1	Command Signal Aspect to SPEED 1	
e_clear_sig_speed2	Command Signal Aspect to SPEED 2	
e_clear_sig_speed3	Command Signal Aspect to SPEED 3	
e_set_to_dark_2 OR e_set_to_dark_p	Command Signal Aspect to DARK	
s_gca_gpx_2C87_day_state	Command Luminosity DAY / NIGHT on Mast	Command LS Set Luminosity

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Table 10: Mapping CC command (CC to EC-MUX) to SCI-LS Commands (EC-MUX to OC-LS)

The used SCI-LS notifications and CC state reporting can be found in table *Generic Eulynx Signal Type* - see the links in column *Sub-Chapter*.

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3 Requirements

3.1 Light Signals

SEA Eulynx Signal is an implementation of Signal element logic based on SCI-LS [7].

Note that terms “speed”, “clear” and “free” are used as synonyms for Signal aspects by different systems. In ELEKTRA context the term “speed” is used in the command value applicable for commanding the signal to an aspect (e.g. e_clear_sig_speed1) and it has been kept also for the SEA internal aspects and element states (e.g. SPEED1).

CC checks strictly the command reply flag true / false contained in the gen_elem_stat message used for the signal aspect reporting. Even a rejected command (in stable states) needs to be replied by repeating the current signal aspect. The information about the particular value of the command reply flag is contained in the requirements defined in the features of this chapter.

SEA Eulynx Signal element process has no knowledge about the faults of particular lamps and their reason. When a lamp fault occurs it will be made obvious by OC-LS only by reporting a downgraded / fallback aspect, e.g. reporting the STOP aspect if a lamp needed for the SPEED aspect is faulty or the DARK aspect if a lamp needed for the STOP aspect is faulty. The downgraded / fallback aspect shall be commanded by the OC-LS controller itself, not by the element process.

In order to report the changed signal aspect after forwarding the command to OC-LS, the aspect indication of the commanded / downgraded aspect (DARK in case of failed STOP command) from OC-LS is required. OC-LS command shall be time supervised, the timer duration will be configured at SEA system level in order to have the possibility to adjust the value to the system needs. Two different timers are defined as shown in the following table.

Timer	Functionality	Default Value
WaitForSafeState	signal commanded to a STOP / DARK aspect, no answer received yet from OC-LS	1000 ms
WaitForSpeed	signal commanded to a SPEED aspect, no answer received yet from OC-LS	2000 ms

Table 11: Eulynx Signal - Command Reply Timers

Spontaneous indication of STOP / DARK shall always be accepted. Spontaneous indication of a SPEEDx aspect or command reply by another SPEEDx aspect than the commanded one cannot be accepted and it shall cause the Safety Reaction OC Disconnection.

SEA Eulynx Signals support following commands / messages defined in SCI-LS [7]:

- Cmd LS Indicate Signal Aspect
- Msg LS Indicated Signal Aspect

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For explicit CC to SCI-LS state / command mapping see the specific Signal configurations in following chapters of this document.

Since in SCI-LS context, not the particular lamps are commanded / indicated according to the desired speed aspect, but the speed aspect itself is commanded / indicated, signals with same number of speed aspects and safe commands (Stop / Dark) can be grouped together: they have the same behavior, only different command and state mappings in Eulynx context.

Within one signal type, the lamp configurations are irrelevant, e.g. for Shunting Signal Standalone it does not matter if the configuration with 2 or 3 white lamps is used, the element process behavior is always the same.

Based on the possible speed aspects and safe commands (Stop / Dark), the signals in SEA are grouped into following Generic Eulynx Signal types:

Generic Eulynx Signal Type	Specific Signal Types	Sub-Chapter
Signal with One Speed Aspect and Stop command :eulx_sig_1sp_stop_cmd	Protection Signal Shunting Signal Standalone	see Signal with One Speed Aspect and Stop command
Signal with Two Speed Aspects and Stop command :eulx_sig_2sp_stop_cmd	Signal Repeater Combined (*) Signal Repeater Standalone	see Signal with Two Speed Aspects and Stop command
Signal with Three Speed Aspects and Stop command :eulx_sig_3sp_stop_cmd	Advance Signal Standalone Main Signal	see Signal with Three Speed Aspects and Stop command
Signal with One Speed Aspect and Dark command :eulx_sig_1sp_dark_cmd	Subsidiary / Caution Signal Subsidiary Signal Repeater	see Signal with One Speed Aspect and Dark command
Signal with Three Speed Aspects and Dark command :eulx_sig_3sp_dark_cmd	Speed Indicator Standalone Speed Indicator Combined Advance Speed Indicator	see Signal with Three Speed Aspects and Dark command
Signal with Three Speed Aspects and Stop and Dark commands :eulx_sig_3sp_stop_dark_cmd	Advance Signal Combined	see Signal with Three Speed Aspects and Stop and Dark commands

Table 12: Eulynx Signal Types

Additionally, there are specific implementations for:

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- Shunting Signal Combined - see sub-chapter *Shunting Signal Combined*
- Signal Repeater Combined on the mast of the superior Protection Signal - see sub-chapter *Signal Repeater Combined on the mast of the superior PRSIG*.

Only one set of requirements is defined per Generic Eulynx Signal Type element process.

3.1.1 Signal with One Speed Aspect and Stop command

SEA Eulynx Signal with one speed aspect and Stop command can reach following regular signal aspects via CC command:

- Speed1
- Stop

Following signal aspect can be reached only due to the lamp fault:

- Dark

This generic implementation can be configured to respond to specific CC commands and OC-LS messages.

The state machine of the SEA Eulynx Signal with one speed aspect and STOP command is shown in the following figure:

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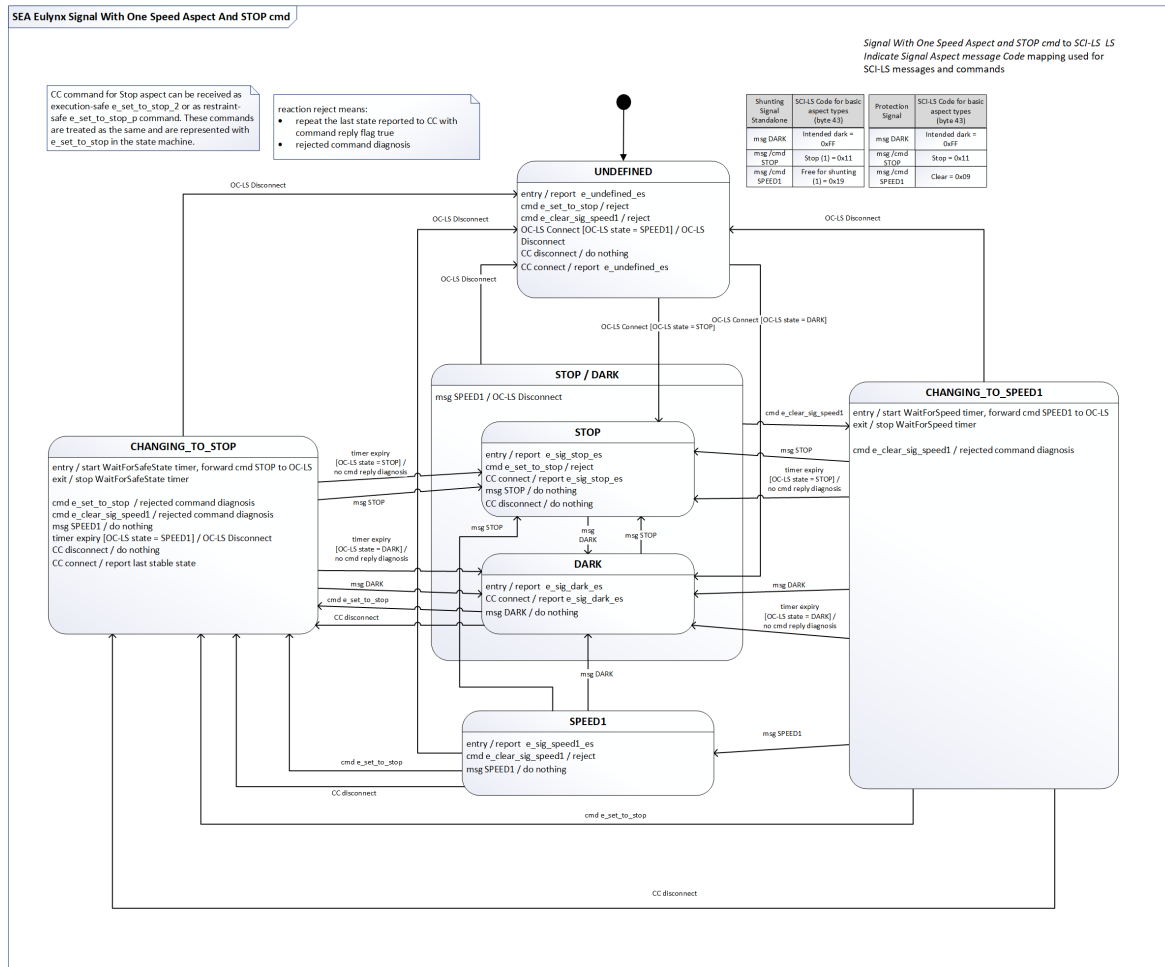


Figure 9: SEA Eulynx Signal with one speed aspect and STOP cmd - State Machine

Note To keep the STM applicable for several Signal types, names based on SEA Signal Aspects are used for SCI-LS messages and commands. A table with mappings to the SCI-LS codes has been provided for each Signal Type.

Following internal element states are defined for the SEA Eulynx Signal with one speed aspect and STOP cmd:

Element State	Functionality	CC State	Faulty
UNDEFINED	no aspect information given, initial / default state	e_undefined_es	yes
STOP	indication for STOP aspect received, as spontaneous msg or after cmd execution	e_sig_stop_es	??
	signal commanded to SPEED1,		

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SPEED1	corresponding aspect indication received already	e_sig_speed1_es	??
DARK	indication for DARK aspect received; as spontaneous msg caused by lamp fault	e_sig_dark_es	yes
CHANGING_TO_SPEED1	command for SPEED1 aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history
CHANGING_TO_STOP	command for STOP aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history

Table 13: SEA Eulynx Signal with one speed aspect and STOP cmd - Internal Element States
Behavior in case of connection loss

When losing the connection to CC, SEA shall execute the safe command, i.e. command the signal to the STOP aspect if the last command forwarded to OC-LS was a different one.

3.1.1.1 Shunting Signal Standalone

The Shunting Signal Standalone in SEA is a configuration of the SEA Eulynx Signal with one speed aspect and Stop command.

Used Cmd LS Indicate Signal Aspect and Msg LS Indicated Signal Aspect shall be addressed to the element identified by the Shunting Signal Standalone operational identifier. CC command and state mappings to SCI-LS telegrams are defined in following tables.

CC Command	SEA Signal with one speed aspect	Command byte / parameter	Parameter value	OBB aspect
e_set_to_stop_2 OR e_set_to_stop_p	Stop	43 = code for basic aspect types	Stop (1) = 0x11	Verschubverbot
e_clear_sig_speed1	Speed1	43 = code for basic aspect types	Free for shunting (1) = 0x19	Verschubverbot aufgehoben

Table 14: Shunting Signal Standalone - Mapping CC command (CC to EC-MUX) to SCI-LS Cmd Indicate Signal Aspect (EC-MUX to OC-LS)

The SCI-LS Cmd Indicate Signal Aspect irrelevant and relevant parameters for the Shunting Signal Standalone are listed in the table *Eulynx Signal Mast configuration*.

Message byte /		SEA Signal with one		
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parameter	Parameter value	speed aspect	OBB aspect	CC State
43 = code for basic aspect types	Stop (1) = 0x11	Stop	Verschubverbot	e_sig_stop_es
43 = code for basic aspect types	Free for shunting (1) = 0x19	Speed 1	Verschubverbot aufgehoben	e_sig_speed1_es
43 = code for basic aspect types	Intended dark = 0xFF	Dark	(*)	e_sig_dark_es

Table 15: Shunting Signal Standalone - Mapping SCI-LS Msg Indicated Signal Aspect (OC-LS to EC-MUX) to gen_elem_stat (EC-MUX to CC)

Remark: (*) The aspect defined in Signal aspect table [9] as "Intended dark" cannot be commanded, it can be reached only due to a lamp fault.

For default values of the remaining parameters (byte 44 - 60) of the Msg Indicated Signal Aspect see table for default values of the corresponding command parameters.

3.1.1.2 Protection Signal

The Protection Signal in SEA is a configuration of the SEA Eulynx Signal with one speed aspect and Stop command.

In OBB context a new and an old variant of the Protection Signal is distinguished. The new variant hosts lamps for shunting on the same mast, the old one does not. From an element process point of view, there is no difference between them; all requirements concerning the Protection Signal are valid for both variants.

Used Cmd LS Indicate Signal Aspect and Msg LS Indicated Signal Aspect shall be addressed to the element identified by the Protection Signal operational identifier. CC command and state mappings to SCI-LS telegrams are defined in following tables.

CC Command	SEA Signal with one speed aspect	Command byte / parameter	Parameter value	OBB aspect
e_set_to_stop_2 OR e_set_to_stop_p	Stop	43 = code for basic aspect types	Stop = 0x11	Fahrverbot
e_clear_sig_speed1	Speed1	43 = code for basic aspect types	Clear = 0x09	Fahrverbot aufgehoben

Table 16: Protection Signal - Mapping CC command (CC to EC-MUX) to SCI-LS Cmd Indicate Signal Aspect (EC-MUX to OC-LS)

The SCI-LS Cmd Indicate Signal Aspect irrelevant and relevant parameters for the Protection Signal are listed in the table *Eulynx Signal Mast configuration*.

Message byte /	Parameter	SEA Signal with one		
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parameter	value	speed aspect	OBB aspect	CC State
43 = code for basic aspect types	Stop = 0x11	Stop	Fahrverbot	e_sig_stop_es
43 = code for basic aspect types	Clear = 0x09	Speed 1	Fahrverbot aufgehoben	e_sig_speed1_es
43 = code for basic aspect types	Intended dark = 0xFF	Dark	(*)	e_sig_dark_es

Table 17: Protection Signal - Mapping SCI-LS Msg Indicated Signal Aspect (OC-LS to EC-MUX) to gen_elem_stat (EC-MUX to CC)

Remark: (*) The aspect defined in Signal aspect table [9] as "Intended dark" cannot be commanded, it can be reached only due to a lamp fault.

For default values of the remaining parameters (byte 44 - 60) of the Msg Indicated Signal Aspect see table for default values of the corresponding command parameters.

3.1.1.3 Operational - Signal With One Speed Aspect and Stop command

In order to control and supervise a Signal With One Speed Aspect and Stop command
As a SEA

I want to be able to receive commands from CC, translate them into SCI-LS commands and forward them to OC-LS and to receive SCI-LS state messages from OC-LS, translate them into signal state changes and report them to CC.

For requirement language see Chapter 2.8 "Requirement Language".

CHANGING_TO_SPEED1 and CHANGING_TO_STOP element states are not considered in the mapping between element state and CC State because when reaching these states, nothing is reported to CC.

WaitForSpeed and WaitForSafeState are configurable timers for receiving a command reply from OC-LS. WaitForSpeed is used when waiting for a reply to a SPEED command (e_clear_sig_speed1), its default value is 2000ms.

WaitForSafeState is used when waiting for a reply to a STOP command (e_set_to_stop_2, e_set_to_stop_p), its default value is 1000ms.

Given a Signal With One Speed Aspect and Stop command

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a STOP command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_stop_es
SPEED1	e_sig_speed1_es

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State	CC State
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_ UNDEFINED	0
SIG_ DARK	12
SIG_ STOP	13
SIG_ SPEED_ 1	16

EC-MUX-SIG:SSS: REQ:3000	Signal With One Speed Aspect and Stop command - Initial state		
When initializing the element			
Then SEA shall set the current element state to UNDEFINED			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3003	Signal With One Speed Aspect and Stop command - reject commands in UNDEFINED state
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED When receiving the command <Command> from CC	

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Then SEA shall report the element state e_undefined_es with command reply flag true to CC
And SEA shall report the rejected command diagnosis to DGP

Command
e_set_to_stop_p
e_set_to_stop_2
e_clear_sig_speed1

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS:
REQ:3004

Signal With One Speed Aspect and Stop command - execute e_clear_sig_speed1 command in STOP / DARK

Given the current element state is <State>

When receiving the command e_clear_sig_speed1 from CC

Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to SPEED1 to OC-LS

And SEA shall set the current element state to CHANGING_TO_SPEED1

State
STOP
DARK

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX

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Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3005	Signal With One Speed Aspect and Stop command - reject STOP commands in STOP					
Given the current element state is STOP When receiving the command <Command> from CC Then SEA shall report the element state e_sig_stop_es with command reply flag true to CC And SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_set_to_stop_2</td></tr><tr><td>e_set_to_stop_p</td></tr></table>				Command	e_set_to_stop_2	e_set_to_stop_p
Command						
e_set_to_stop_2						
e_set_to_stop_p						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:	CC shall not send the STOP commands if the stop aspect is already given.					
Satisfies:						

EC-MUX-SIG:SSS: REQ:3011	Signal With One Speed Aspect and Stop command - execute STOP commands in DARK
Given the current element state is DARK When receiving the command <Command> from CC	

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Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS						
And SEA shall set the current element state to CHANGING_TO_STOP						
<table><tr><th>Command</th></tr><tr><td>e_set_to_stop_2</td></tr><tr><td>e_set_to_stop_p</td></tr></table>				Command	e_set_to_stop_2	e_set_to_stop_p
Command						
e_set_to_stop_2						
e_set_to_stop_p						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

EC-MUX-SIG:SSS: REQ:3006	Signal With One Speed Aspect and Stop command - spontaneous transition between STOP / DARK														
Given the current element state is <State> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag false to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP															
<table><tr><th>State</th><th>NewState</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>STOP</td><td>DARK</td><td>e_sig_dark_es</td><td>SIG_DARK</td></tr><tr><td>DARK</td><td>STOP</td><td>e_sig_stop_es</td><td>SIG_STOP</td></tr></table>				State	NewState	CCState	DGPMMessage	STOP	DARK	e_sig_dark_es	SIG_DARK	DARK	STOP	e_sig_stop_es	SIG_STOP
State	NewState	CCState	DGPMMessage												
STOP	DARK	e_sig_dark_es	SIG_DARK												
DARK	STOP	e_sig_stop_es	SIG_STOP												
Type:	FUR - Functional Requirement														

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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3007	Signal With One Speed Aspect and Stop command - execute STOP command in SPEED1					
Given the current element state is SPEED1 When receiving the command <Command> from CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS And SEA shall set the current element state to CHANGING_TO_STOP <table><tr><th>Command</th></tr><tr><td>e_set_to_stop_p</td></tr><tr><td>e_set_to_stop_2</td></tr></table>				Command	e_set_to_stop_p	e_set_to_stop_2
Command						
e_set_to_stop_p						
e_set_to_stop_2						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

EC-MUX-SIG:SSS:	Signal With One Speed Aspect and Stop command - reject
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REQ:3008	e_clear_sig_speed1 command in SPEED1		
Given the current element state is SPEED1			
When receiving the command e_clear_sig_speed1 from CC			
Then SEA shall report the element state e_sig_speed1_es with command reply flag true to CC			
And SEA shall report the rejected command diagnosis to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	CC shall not send the command e_clear_sig_speed1, if the speed1 aspect is already given.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3009	Signal With One Speed Aspect and Stop command - spontaneous transition from SPEED1 to STOP / DARK											
Given the current element state is SPEED1 When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS Then SEA shall set the current element state to <State> And SEA shall report the element state <CCState> with command reply flag false to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>State</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>STOP</td><td>e_sig_stop_es</td><td>SIG_STOP</td></tr><tr><td>DARK</td><td>e_sig_dark_es</td><td>SIG_DARK</td></tr></table>				State	CCState	DGPMMessage	STOP	e_sig_stop_es	SIG_STOP	DARK	e_sig_dark_es	SIG_DARK
State	CCState	DGPMMessage										
STOP	e_sig_stop_es	SIG_STOP										
DARK	e_sig_dark_es	SIG_DARK										
Type:	FUR - Functional Requirement											
Status:	draft	Stability:	stable									
Target Release:	SEA SW GEN 3.0	Priority:	mandatory									

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Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3010	Signal With One Speed Aspect and Stop command - transition from CHANGING_TO_SPEED1 to SPEED1		
Given the current element state is CHANGING_TO_SPEED1			
When receiving the LS Indicated Signal Aspect message with Code corresponding to SPEED1 from OC-LS within WaitForSpeed ms			
Then SEA shall set the current element state to SPEED1			
And SEA shall report the element state e_sig_speed1_es with command reply flag true to CC			
And SEA shall report the diagnosis aspect message SIG_SPEED_1 to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Successful command execution - commanded aspect reached.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3012	Signal With One Speed Aspect and Stop command - reject e_clear_sig_speed1 command in CHANGING_TO_SPEED1		
Given the current element state is CHANGING_TO_SPEED1 When receiving the command e_clear_sig_speed1 from CC within WaitForSpeed ms Then SEA shall report the rejected command diagnosis to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	CC shall not repeat the command e_clear_sig_speed1. The WaitForSpeed timer shall not be re-started due to the rejected command.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3013	Signal With One Speed Aspect and Stop command - CHANGING_TO_SPEED1 - return to STOP / DARK											
Given the current element state is CHANGING_TO_SPEED1 And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS within WaitForSpeed ms Then SEA shall set the current element state to <State> And SEA shall report the element state <CCState> with command reply flag true to CC <table><tr><th>LastStable</th><th>State</th><th>CCState</th></tr><tr><td>STOP</td><td>STOP</td><td>e_sig_stop_es</td></tr><tr><td>DARK</td><td>DARK</td><td>e_sig_dark_es</td></tr></table>				LastStable	State	CCState	STOP	STOP	e_sig_stop_es	DARK	DARK	e_sig_dark_es
LastStable	State	CCState										
STOP	STOP	e_sig_stop_es										
DARK	DARK	e_sig_dark_es										
Type:	FUR - Functional Requirement											
Status:	draft	Stability:	stable									
Target Release:	SEA SW GEN 3.0	Priority:	mandatory									
Originator:	Technics (R&D)	Allocation:	SW EC-MUX									
Verification Method:	Test	Verification Step:	module test, subit									
Comments:	OC-LS has probably sent the message as a downgrade aspect for failed command. As it is not possible to forward the command corresponding to SPEED1 aspect before receiving a reply to the previous STOP, it is not necessary to continue waiting for a further reply and thus the CHANGING_TO_SPEED1 state shall be left.											

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Satisfies:	
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EC-MUX-SIG:SSS: REQ:3014	Signal With One Speed Aspect and Stop command - CHANGING_TO_SPEED1 - transition to DARK / STOP														
Given the current element state is CHANGING_TO_SPEED1 And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS within WaitForSpeed ms Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag true to CC And SEA shall report the diagnosis aspect message <DGPMessage> to DGP <table><tr><th>LastStable</th><th>NewState</th><th>CCState</th><th>DGPMessage</th></tr><tr><td>STOP</td><td>DARK</td><td>e_sig_dark_es</td><td>SIG_DARK</td></tr><tr><td>DARK</td><td>STOP</td><td>e_sig_stop_es</td><td>SIG_STOP</td></tr></table>				LastStable	NewState	CCState	DGPMessage	STOP	DARK	e_sig_dark_es	SIG_DARK	DARK	STOP	e_sig_stop_es	SIG_STOP
LastStable	NewState	CCState	DGPMessage												
STOP	DARK	e_sig_dark_es	SIG_DARK												
DARK	STOP	e_sig_stop_es	SIG_STOP												
Type:	FUR - Functional Requirement														
Status:	draft	Stability:	stable												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory												
Originator:	Technics (R&D)	Allocation:	SW EC-MUX												
Verification Method:	Test	Verification Step:	module test, subit												
Comments:	OC-LS has probably sent the message as a downgrade aspect for failed command. As it is not possible to forward the command corresponding to SPEED1 aspect before receiving a reply to the previous STOP, it is not necessary to continue waiting for a further reply and thus the CHANGING_TO_SPEED1 state shall be left. The diagnosis aspect message is sent only if the last stable state is different than the OC-LS reply message.														
Satisfies:															

EC-MUX-SIG:SSS: REQ:3019	Signal With One Speed Aspect and Stop command - CHANGING_TO_SPEED1 - execute STOP command
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Given the current element state is CHANGING_TO_SPEED1

When receiving the command <Command> from CC within WaitForSpeed ms

Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS

And SEA shall set the current element state to CHANGING_TO_STOP

Command
e_set_to_stop_p
e_set_to_stop_2

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	If the STOP command reaches the OC-LS before the SPEED1 command is answered, then it is possible that OC-LS will reply to none of both commands, because the last commanded aspect is still given.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3043	Signal With One Speed Aspect and Stop command - CHANGING_TO_SPEED1 - timeout
Given the current element state is CHANGING_TO_SPEED1 And the last stable element state is <LastStable> When the current state of the element has been CHANGING_TO_SPEED1 for WaitForSpeed ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP And SEA shall set the current element state to <State> And SEA shall report the element state <CCState> with command reply flag true to CC	

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LastStable	State	CCState
STOP	STOP	e_sig_stop_es
DARK	DARK	e_sig_dark_es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Regularly the state CHANGING_TO_SPEED1 will be left after WaitForSpeed ms. No safety reaction OC Disconnection is performed as this is not a safety-critical command. Instead, SEA shall generate a diagnosis and try to fallback to original STOP/DARK.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3015	Signal With One Speed Aspect and Stop command - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state SPEED1									
Given the current element state is CHANGING_TO_STOP And the last stable element state is SPEED1 When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag true to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>NewState</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>STOP</td><td>e_sig_stop_ es</td><td>SIG_STOP</td></tr><tr><td>DARK</td><td>e_sig_dark_ es</td><td>SIG_DARK</td></tr></table>		NewState	CCState	DGPMMessage	STOP	e_sig_stop_ es	SIG_STOP	DARK	e_sig_dark_ es	SIG_DARK
NewState	CCState	DGPMMessage								
STOP	e_sig_stop_ es	SIG_STOP								
DARK	e_sig_dark_ es	SIG_DARK								
Type:	FUR - Functional Requirement									

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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Successful command execution - commanded aspect / downgrade aspect reached.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3047	Signal With One Speed Aspect and Stop command - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state DARK / STOP														
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag true to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>LastStable</th><th>NewState</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>DARK</td><td>STOP</td><td>e_sig_stop_es</td><td>SIG_STOP</td></tr><tr><td>STOP</td><td>DARK</td><td>e_sig_dark_es</td><td>SIG_DARK</td></tr></table>				LastStable	NewState	CCState	DGPMMessage	DARK	STOP	e_sig_stop_es	SIG_STOP	STOP	DARK	e_sig_dark_es	SIG_DARK
LastStable	NewState	CCState	DGPMMessage												
DARK	STOP	e_sig_stop_es	SIG_STOP												
STOP	DARK	e_sig_dark_es	SIG_DARK												
Type:	FUR - Functional Requirement														
Status:	draft	Stability:	stable												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory												
Originator:	Technics (R&D)	Allocation:	SW EC-MUX												
Verification Method:	Test	Verification Step:	module test, subit												
	Successful command execution - commanded aspect / downgrade aspect reached. OC-LS state can be DARK in precondition when a STOP command is														

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Comments:	executed. OC-LS state can be STOP or DARK in the precondition when a STOP command has been executed after a SPEED command without a command reply.
Satisfies:	

EC-MUX-SIG:SSS: REQ:3044	Signal With One Speed Aspect and Stop command - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state STOP / DARK											
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to <State> And SEA shall report the element state <CCState> with command reply flag true to CC <table><tr><th>LastStable</th><th>State</th><th>CCState</th></tr><tr><td>STOP</td><td>STOP</td><td>e_sig_stop_es</td></tr><tr><td>DARK</td><td>DARK</td><td>e_sig_dark_es</td></tr></table>				LastStable	State	CCState	STOP	STOP	e_sig_stop_es	DARK	DARK	e_sig_dark_es
LastStable	State	CCState										
STOP	STOP	e_sig_stop_es										
DARK	DARK	e_sig_dark_es										
Type:	FUR - Functional Requirement											
Status:	draft	Stability:	stable									
Target Release:	SEA SW GEN 3.0	Priority:	mandatory									
Originator:	Technics (R&D)	Allocation:	SW EC-MUX									
Verification Method:	Test	Verification Step:	module test, subit									
Comments:	Successful command execution - commanded aspect / downgrade aspect reached. OC-LS state can be DARK in precondition when a STOP command is executed. OC-LS state can be STOP or DARK in the precondition when a STOP command has been executed after a SPEED command without a command reply.											
Satisfies:												

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EC-MUX-SIG:SSS: REQ:3016	Signal With One Speed Aspect and Stop command - reject commands in CHANGING_TO_STOP						
Given the current element state is CHANGING_TO_STOP When receiving the command <Command> from CC within WaitForSafeState ms Then SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_set_to_stop_2</td></tr><tr><td>e_set_to_stop_p</td></tr><tr><td>e_clear_sig_speed1</td></tr></table>				Command	e_set_to_stop_2	e_set_to_stop_p	e_clear_sig_speed1
Command							
e_set_to_stop_2							
e_set_to_stop_p							
e_clear_sig_speed1							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:	CC shall not send a SPEED command if there is a STOP command not answered yet. CC shall not repeat a STOP command. The timer shall not to be re-started due to the rejected command.						
Satisfies:							

EC-MUX-SIG:SSS: REQ:3017	Signal With One Speed Aspect and Stop command - CHANGING_TO_STOP - ignore SPEED1
Given the current element state is CHANGING_TO_STOP When receiving the LS Indicated Signal Aspect message with Code corresponding to SPEED1 from OC-LS within WaitForSafeState ms Then SEA shall do nothing	

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	It is possible that OC-LS has sent the message corresponding to SPEED1 as reply to the previous SPEED1 command not answered yet. Waiting for the reply to the STOP command is continued by keeping the current state CHANGING_TO_STOP. After timer expiry, if the expected STOP reply is not received, the element state is changed to STOP / DARK or a Safety Reaction OC Disconnection is performed, depending on the last received OC-LS state.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3045	Signal With One Speed Aspect and Stop command - CHANGING_TO_STOP - timeout when OC-LS SPEED1 - OC Disconnection		
Given the current element state is CHANGING_TO_STOP And the last stable element state is SPEED1 When the current state of the element has been CHANGING_TO_STOP for WaitForSafeState ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit

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Comments:	Regularly the state CHANGING_TO_STOP will be left after WaitForSafeState ms. Safety reaction OC Disconnection is applicable only if the OC-LS cannot reach the STOP or DARK aspect and stays in SPEED1. See also EC-MUX-SIG:SSS:REQ:3046.
Satisfies:	

EC-MUX-SIG:SSS: REQ:3046	Signal With One Speed Aspect and Stop command - CHANGING_TO_STOP - timeout when OC-LS DARK / STOP - reach DARK / STOP								
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable> When the current state of the element has been CHANGING_TO_STOP for WaitForSafeState ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP And SEA shall set the current element state to <LastStable> And SEA shall report the element state <CCState> with command reply flag true to CC <table><tr><th>LastStable</th><th>CCState</th></tr><tr><td>STOP</td><td>e_sig_stop_es</td></tr><tr><td>DARK</td><td>e_sig_dark_es</td></tr></table>				LastStable	CCState	STOP	e_sig_stop_es	DARK	e_sig_dark_es
LastStable	CCState								
STOP	e_sig_stop_es								
DARK	e_sig_dark_es								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:	Regularly the state CHANGING_TO_STOP will be left after WaitForSafeState ms. OC-LS state can be STOP or DARK at this point only when a STOP command has been executed after a SPEED command, and no reply has been received for either command. This is an allowed scenario so no safety reaction OC Disconnection is performed.								

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Satisfies:	
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EC-MUX-SIG:SSS: REQ:3020	Signal With One Speed Aspect and Stop command - Disconnection on spontaneous aspect message SPEED1 in STOP / DARK					
Given the current element state is <State> When receiving the LS Indicated Signal Aspect message with Code corresponding to SPEED1 from OC-LS Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: Spontaneous SPEED1 reported for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP <table><tr><td>State</td></tr><tr><td>STOP</td></tr><tr><td>DARK</td></tr></table>				State	STOP	DARK
State						
STOP						
DARK						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

EC-MUX-SIG:SSS: REQ:3042	Signal With One Speed Aspect and Stop command - Ignore repeated messages
Given the current element state is <State> When receiving the LS Indicated Signal Aspect message with Code corresponding to <SeaSigAspect> from OC-LS	

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Then SEA shall do nothing

State	SeaSigAspect
STOP	STOP
SPEED1	SPEED1
DARK	DARK

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS:
REQ:3053

Signal With One Speed Aspect and Stop command - Receive unexpected DARK / SPEED commands from CC

When receiving the command <Command> from CC

Then SEA shall execute EC-MUX safety reaction Deactivation

Command
e_set_to_dark_2
e_set_to_dark_p
e_clear_sig_speed2
e_clear_sig_speed3

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX

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Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

3.1.1.4 Connection - Signal With One Speed Aspect and Stop command

In order to control and supervise a Signal With One Speed Aspect and Stop command

As a SEA

I want to be able to supervise connections and when changes occur then

- on CC connection establishment report the current state to CC
- on CC connection loss execute the safe command STOP if the last state is DARK
- on CC connection loss execute the safe command STOP if the last command forwarded to OC-LS was SPEED1 and it was not answered by STOP / DARK
- on OC-LS connection loss take on the default state UNDEFINED
- on OC-LS connection establishment transition to DARK / STOP state as the OC-LS element state
- report the connection state changes to CC and DGP

For requirement language see Chapter 2.8 "Requirement Language".

CHANGING_TO_SPEED1 and CHANGING_TO_STOP element states are not considered in the mapping between element state and CC State because when reaching these states, nothing is reported to CC.

WaitForSpeed and WaitForSafeState are configurable timers for receiving a command reply from OC-LS. WaitForSpeed is used when waiting for a reply to a SPEED command (e_clear_sig_speed1), its default value is 2000ms.

WaitForSafeState is used when waiting for a reply to a STOP command (e_set_to_stop_2, e_set_to_stop_p), its default value is 1000ms.

Given a Signal With One Speed Aspect and Stop command

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a STOP command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_stop_es
SPEED1	e_sig_speed1_es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

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And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12
SIG_STOP	13
SIG_SPEED_1	16

EC-MUX-SIG:SSS: REQ:3021	Signal With One Speed Aspect and Stop command - [Re-]Connecting to CC - OC-LS CONNECTED - General inquiry in stable states								
Given the current element state is <State> And CC connection state is NOT_CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When establishing the connection to CC Then SEA shall report the element state <CCState> with command reply flag false to CC <table><tr><th>State</th><th>CCState</th></tr><tr><td>STOP</td><td>e_sig_stop_es</td></tr><tr><td>DARK</td><td>e_sig_dark_es</td></tr></table>				State	CCState	STOP	e_sig_stop_es	DARK	e_sig_dark_es
State	CCState								
STOP	e_sig_stop_es								
DARK	e_sig_dark_es								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:	[Re-]Connecting in SPEED1 state is impossible.								
Satisfies:									

EC-MUX-SIG:SSS:	Signal With One Speed Aspect and Stop command - [Re-]Connecting to
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REQ:3022	CC - OC-LS CONNECTED - General inquiry in CHANGING_TO_STOP state - no message after CC disconnect										
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable> And CC connection state is NOT_CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When establishing the connection to CC within WaitForSafeState ms Then SEA shall report the element state <CCState> with command reply flag false to CC <table><tr><th>LastStable</th><th>CCState</th></tr><tr><td>STOP</td><td>e_sig_stop_es</td></tr><tr><td>DARK</td><td>e_sig_dark_es</td></tr><tr><td>SPEED1</td><td>e_sig_speed1_es</td></tr></table>				LastStable	CCState	STOP	e_sig_stop_es	DARK	e_sig_dark_es	SPEED1	e_sig_speed1_es
LastStable	CCState										
STOP	e_sig_stop_es										
DARK	e_sig_dark_es										
SPEED1	e_sig_speed1_es										
Type:	FUR - Functional Requirement										
Status:	draft	Stability:	stable								
Target Release:	SEA SW GEN 3.0	Priority:	mandatory								
Originator:	Technics (R&D)	Allocation:	SW EC-MUX								
Verification Method:	Test	Verification Step:	module test, subit								
Comments:	Re-connecting to CC in CHANGING_TO_STOP state is impossible in real operation, as re-connecting will regularly take much longer time than the configured WaitForSafeState. Due to the execution of the safe command [Re-]Connecting in CHANGING_TO_STOP state is possible only with last forwarded command STOP, but depending on the previous events the last state reported to CC can be different. Following initial states are covered with this requirement: - CHANGING_TO_STOP with last stable element state SPEED1 - CHANGING_TO_STOP with last stable element state DARK/STOP - CHANGING_TO_STOP with last stable element state DARK/STOP and SPEED1 received from OC-LS in CHANGING_TO_STOP										
Satisfies:											

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EC-MUX-SIG:SSS: REQ:3024	Signal With One Speed Aspect and Stop command - [Re-]Connecting to CC - OC-LS NOT_CONNECTED		
Given the current element state is UNDEFINED And CC connection state is NOT_CONNECTED And EC-MUX - OC-LS connection state is NOT_CONNECTED When establishing the connection to CC Then SEA shall report the element state e_undefined_es with command reply flag false to CC			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3025	Signal With One Speed Aspect and Stop command - Connection loss to CC - execute safe command in SPEED1		
Given the current element state is SPEED1 And CC connection state is CONNECTED When losing the connection to CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS And SEA shall set the current element state to CHANGING_TO_STOP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			

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Satisfies:	
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EC-MUX-SIG:SSS: REQ:3035	Signal With One Speed Aspect and Stop command - Connection loss to CC - OC-LS NOT_CONNECTED		
Given the current element state is UNDEFINED And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is NOT_CONNECTED When losing the connection to CC Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3027	Signal With One Speed Aspect and Stop command - Connection loss to CC - do nothing in STOP		
Given the current element state is STOP And CC connection state is CONNECTED When losing the connection to CC Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit

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Comments:	
Satisfies:	

EC-MUX-SIG:SSS: REQ:3029	Signal With One Speed Aspect and Stop command - Connection loss to CC - execute safe command in DARK		
Given the current element state is DARK And CC connection state is CONNECTED When losing the connection to CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS And SEA shall set the current element state to CHANGING_TO_STOP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3026	Signal With One Speed Aspect and Stop command - Connection loss to CC - CHANGING_TO_SPEED1		
Given the current element state is CHANGING_TO_SPEED1 And CC connection state is CONNECTED When losing the connection to CC within WaitForSpeed ms Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS And SEA shall set the current element state to CHANGING_TO_STOP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3028	Signal With One Speed Aspect and Stop command - Connection loss to CC - CHANGING_TO_STOP - do nothing		
Given the current element state is CHANGING_TO_STOP And CC connection state is CONNECTED When losing the connection to CC within WaitForSafeState ms Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3001	Signal With One Speed Aspect and Stop command - [Re-]Connecting between EC-MUX and OC-LS in state STOP
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED And OC-LS state is STOP When establishing the connection between EC-MUX and OC-LS	

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Then SEA shall set the current element state to STOP And SEA shall report the element state e_sig_stop_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_STOP to DGP And SEA shall report the connection_ok diagnosis device fault repair message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3002	Signal With One Speed Aspect and Stop command - [Re-]Connecting between EC-MUX and OC-LS in state DARK		
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED And OC-LS state is DARK When establishing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to DARK And SEA shall report the element state e_sig_dark_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_DARK to DGP And SEA shall report the connection_ok diagnosis device fault repair message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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EC-MUX-SIG:SSS: REQ:3052	Signal With One Speed Aspect and Stop command - [Re-]Connecting between EC-MUX and OC-LS in state SPEED1		
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED And OC-LS state is SPEED1 When establishing the connection between EC-MUX and OC-LS Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: OC-LS connect with SPEED1 for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3030	Signal With One Speed Aspect and Stop command - Connection loss between EC-MUX and OC-LS in regular states			
Given the current element state is <State> And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP <table><tr><td>State</td></tr><tr><td>STOP</td></tr><tr><td>SPEED1</td></tr></table>		State	STOP	SPEED1
State				
STOP				
SPEED1				
Type:	FUR - Functional Requirement			

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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3031	Signal With One Speed Aspect and Stop command - Connection loss between EC-MUX and OC-LS in DARK state		
Given the current element state is DARK And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3048	Signal With One Speed Aspect and Stop command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_SPEED1
Given the current element state is CHANGING_TO_SPEED1 And EC-MUX - OC-LS connection state is CONNECTED	

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When losing the connection between EC-MUX and OC-LS within WaitForSpeed ms

Then SEA shall set the current element state to UNDEFINED

And SEA shall report the element state e_undefined_es with command reply flag true to CC

And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP

And SEA shall report the no_connection diagnosis device fault message to DGP

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3032	Signal With One Speed Aspect and Stop command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_STOP
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Given the current element state is CHANGING_TO_STOP

And EC-MUX - OC-LS connection state is CONNECTED

When losing the connection between EC-MUX and OC-LS within WaitForSafeState ms

Then SEA shall set the current element state to UNDEFINED

And SEA shall report the element state e_undefined_es with command reply flag true to CC

And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP

And SEA shall report the no_connection diagnosis device fault message to DGP

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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3.1.1.5 Diagnosis - Signal With One Speed Aspect and Stop command

In order to provide the aspect diagnosis for a Signal With One Speed Aspect and Stop command
As a SEA

I want to be able to receive an aspect diagnosis query from DGP and answer it by the DGP diagnosis values corresponding to the last diagnosis aspect message reported to DGP for the queried Signal

For requirement language see Chapter 2.8 "Requirement Language".

Given a Signal With One Speed Aspect and Stop command

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a STOP command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_stop_es
SPEED1	e_sig_speed1_es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12
SIG_STOP	13
SIG_SPEED_1	16

EC-MUX-SIG:SSS: REQ:3050	Signal With One Speed Aspect and Stop command - aspect query - last stable state
Given the last stable element state is <LastStable> When receiving an aspect query from DGP Then SEA shall report the diagnosis aspect message <DGPMMessage> to DGP	

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<table><tr><th>LastStable</th><th>DGPMessage</th></tr><tr><td>DARK</td><td>SIG_ DARK</td></tr><tr><td>STOP</td><td>SIG_ STOP</td></tr><tr><td>SPEED1</td><td>SIG_ SPEED_ 1</td></tr></table>				LastStable	DGPMessage	DARK	SIG_ DARK	STOP	SIG_ STOP	SPEED1	SIG_ SPEED_ 1
LastStable	DGPMessage										
DARK	SIG_ DARK										
STOP	SIG_ STOP										
SPEED1	SIG_ SPEED_ 1										
Type:		FUR - Functional Requirement									
Status:		draft	Stability: stable								
Target Release:		SEA SW GEN 3.0	Priority: mandatory								
Originator:		Technics (R&D)	Allocation: SW EC-MUX								
Verification Method:		Test	Verification Step: module test, subit								
Comments:											
Satisfies:											

EC-MUX-SIG:SSS: REQ:3049	Signal With One Speed Aspect and Stop command - aspect query in UNDEFINED - OC-LS NOT_CONNECTED		
Given EC-MUX - OC-LS connection state is NOT_CONNECTED And the current element state is UNDEFINED When receiving an aspect query from DGP Then SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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Used Cmd LS Indicate Signal Aspect and Msg LS Indicated Signal Aspect shall be addressed to the element identified by the Shunting Signal Combined operational identifier. CC command and state mappings to SCI-LS telegrams are defined in following tables.

CC Command	SEA Signal with one speed aspect	Command byte / parameter	Parameter value	OBB aspect
e_set_to_dark_2 OR e_set_to_dark_p	Stop	43 = code for basic aspect types	Stop (1) = 0x01	Halt
e_clear_sig_speed1	Speed1	43 = code for basic aspect types	Shunting allowed = 0x02	Verschubverbot aufgehoben

Table 18: Shunting Signal Combined on Main Signal Mast - Mapping CC command (CC to EC-MUX) to SCI-LS Cmd Indicate Signal Aspect (EC-MUX to OC-LS)

CC Command	SEA Signal with one speed aspect	Command byte / parameter	Parameter value	OBB aspect
e_set_to_dark_2 OR e_set_to_dark_p	Stop	43 = code for basic aspect types	Stop (1) = 0x11	Fahrverbot
e_clear_sig_speed1	Speed1	43 = code for basic aspect types	Shunting allowed = 0x19	Verschubverbot aufgehoben

Table 19: Shunting Signal Combined on Protection Signal Mast - Mapping CC command (CC to EC-MUX) to SCI-LS Cmd Indicate Signal Aspect (EC-MUX to OC-LS)

The SCI-LS Cmd Indicate Signal Aspect irrelevant and relevant parameters for the Shunting Signal Combined are listed in the table *Eulynx Signal Mast configuration*.

Message byte / parameter	Parameter value	SEA Signal with one speed aspect Aspect	OBB aspect	CC State
43 = code for basic aspect types	Shunting allowed = 0x02	Speed 1	Verschubverbot aufgehoben	e_sig_speed1_es
43 = code for basic aspect types	Intended Dark = 0xFF	Dark	Dark	e_sig_dark_es
43 = code for basic aspect types	Halt = 0x01	Stop	Fahrverbot	e_sig_dark_es

Table 20: Shunting Signal Combined on Main Signal Mast - Mapping SCI-LS Msg Indicated Signal Aspect (OC-LS to EC-MUX) to gen_elem_stat (EC-MUX to CC)

Message byte / parameter	Parameter value	SEA Signal with one speed aspect Aspect	OBB aspect	CC State
43 = code for basic	Shunting		Verschubverbot	

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aspect types	allowed = 0x19	Speed 1	aufgehoben	e_sig_speed1_es
43 = code for basic aspect types	Intended Dark = 0xFF	Dark	Dark	e_sig_dark_es
43 = code for basic aspect types	Halt = 0x11	Stop	Fahrverbot	e_sig_dark_es

Table 21: Shunting Signal Combined on Protection Signal Mast - Mapping SCI-LS Msg Indicated Signal Aspect (OC-LS to EC-MUX) to gen_elem_stat (EC-MUX to CC)

For default values of the remaining parameters (byte 44 - 60) of the Msg Indicated Signal Aspect see table for default values of the corresponding command parameters.

3.1.2.2 Operational - Shunting Signal Combined

In order to control and supervise a Shunting Signal Combined

As a SEA

I want to be able to receive commands from CC, translate them into SCI-LS commands and forward them to OC-LS and to receive SCI-LS state messages from OC-LS, translate them into signal state changes and report them to CC.

For requirement language see Chapter 2.8 "Requirement Language".

CHANGING_TO_SPEED1 and CHANGING_TO_STOP element states are not considered in the mapping between element state and CC State because when reaching these states, nothing is reported to CC.

WaitForSpeed and WaitForSafeState are configurable timers for receiving a command reply from OC-LS. WaitForSpeed is used when waiting for a reply to a SPEED command (e_clear_sig_speed1), its default value is 2000ms.

WaitForSafeState is used when waiting for a reply to a STOP command (e_set_to_dark_2, e_set_to_dark_p), its default value is 1000ms.

Given a Shunting Signal Combined

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a STOP command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_dark_es
SPEED1	e_sig_speed1_es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

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SEA diagnosis aspect	DGP diagnosis value
SIG_ UNDEFINED	0
SIG_ DARK	12
SIG_ SPEED_ 1	16

EC-MUX-SIG:SSS: REQ:3800	Shunting Signal Combined - Initial state		
When initializing the element			
Then SEA shall set the current element state to UNDEFINED			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3803	Shunting Signal Combined - reject commands in UNDEFINED state			
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED When receiving the command <Command> from CC Then SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_set_to_dark_p</td></tr><tr><td>e_set_to_dark_2</td></tr></table>		Command	e_set_to_dark_p	e_set_to_dark_2
Command				
e_set_to_dark_p				
e_set_to_dark_2				

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<table><tr><td>Command</td></tr><tr><td>e_clear_sig_speed1</td></tr></table>				Command	e_clear_sig_speed1
Command					
e_clear_sig_speed1					
Type:	FUR - Functional Requirement				
Status:	draft	Stability:	stable		
Target Release:	SEA SW GEN 3.0	Priority:	mandatory		
Originator:	Technics (R&D)	Allocation:	SW EC-MUX		
Verification Method:	Test	Verification Step:	module test, subit		
Comments:					
Satisfies:					

EC-MUX-SIG:SSS: REQ:3804	Shunting Signal Combined - execute e_clear_sig_speed1 command in STOP / DARK					
Given the current element state is <State> When receiving the command e_clear_sig_speed1 from CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to SPEED1 to OC-LS And SEA shall set the current element state to CHANGING_TO_SPEED1 <table><tr><td>State</td></tr><tr><td>STOP</td></tr><tr><td>DARK</td></tr></table>				State	STOP	DARK
State						
STOP						
DARK						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						

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Satisfies:	
------------	--

EC-MUX-SIG:SSS: REQ:3805	Shunting Signal Combined - reject STOP commands in STOP					
Given the current element state is STOP When receiving the command <Command> from CC Then SEA shall report the element state e_sig_dark_es with command reply flag true to CC And SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_set_to_dark_2</td></tr><tr><td>e_set_to_dark_p</td></tr></table>				Command	e_set_to_dark_2	e_set_to_dark_p
Command						
e_set_to_dark_2						
e_set_to_dark_p						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:	CC shall not send the STOP commands if the stop aspect is already given.					
Satisfies:						

EC-MUX-SIG:SSS: REQ:3811	Shunting Signal Combined - execute STOP commands in DARK
Given the current element state is DARK When receiving the command <Command> from CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS And SEA shall set the current element state to CHANGING_TO_STOP	

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<table><tr><th>Command</th></tr><tr><td>e_set_to_dark_2</td></tr><tr><td>e_set_to_dark_p</td></tr></table>				Command	e_set_to_dark_2	e_set_to_dark_p
Command						
e_set_to_dark_2						
e_set_to_dark_p						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

EC-MUX-SIG:SSS: REQ:3806	Shunting Signal Combined - spontaneous transition between STOP / DARK								
Given the current element state is <State> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS Then SEA shall set the current element state to <NewState> <table><tr><th>State</th><th>NewState</th></tr><tr><td>STOP</td><td>DARK</td></tr><tr><td>DARK</td><td>STOP</td></tr></table>				State	NewState	STOP	DARK	DARK	STOP
State	NewState								
STOP	DARK								
DARK	STOP								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						

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Comments:	
Satisfies:	

EC-MUX-SIG:SSS: REQ:3807	Shunting Signal Combined - execute STOP command in SPEED1					
Given the current element state is SPEED1 When receiving the command <Command> from CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS And SEA shall set the current element state to CHANGING_TO_STOP <table><tr><th>Command</th></tr><tr><td>e_set_to_dark_p</td></tr><tr><td>e_set_to_dark_2</td></tr></table>				Command	e_set_to_dark_p	e_set_to_dark_2
Command						
e_set_to_dark_p						
e_set_to_dark_2						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

EC-MUX-SIG:SSS: REQ:3808	Shunting Signal Combined - reject e_clear_sig_speed1 command in SPEED1
Given the current element state is SPEED1 When receiving the command e_clear_sig_speed1 from CC	

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Then SEA shall report the element state e_sig_speed1_es with command reply flag true to CC And SEA shall report the rejected command diagnosis to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	CC shall not send the command e_clear_sig_speed1, if the speed1 aspect is already given.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3809	Shunting Signal Combined - spontaneous transition from SPEED1 to STOP / DARK					
Given the current element state is SPEED1 When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS Then SEA shall set the current element state to <State> And SEA shall report the element state e_sig_dark_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_DARK to DGP <table><tr><td>State</td></tr><tr><td>STOP</td></tr><tr><td>DARK</td></tr></table>				State	STOP	DARK
State						
STOP						
DARK						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

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EC-MUX-SIG:SSS: REQ:3810	Shunting Signal Combined - transition from CHANGING_TO_SPEED1 to SPEED1		
Given the current element state is CHANGING_TO_SPEED1			
When receiving the LS Indicated Signal Aspect message with Code corresponding to SPEED1 from OC-LS within WaitForSpeed ms			
Then SEA shall set the current element state to SPEED1			
And SEA shall report the element state e_sig_speed1_es with command reply flag true to CC			
And SEA shall report the diagnosis aspect message SIG_SPEED_1 to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Successful command execution - commanded aspect reached.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3812	Shunting Signal Combined - reject e_clear_sig_speed1 command in CHANGING_TO_SPEED1		
Given the current element state is CHANGING_TO_SPEED1 When receiving the command e_clear_sig_speed1 from CC within WaitForSpeed ms Then SEA shall report the rejected command diagnosis to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit

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Comments:	CC shall not repeat the command e_clear_sig_speed1. The WaitForSpeed timer shall not be re-started due to the rejected command.
Satisfies:	

EC-MUX-SIG:SSS: REQ:3813	Shunting Signal Combined - CHANGING_TO_SPEED1 - return to STOP / DARK								
Given the current element state is CHANGING_TO_SPEED1 And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS within WaitForSpeed ms Then SEA shall set the current element state to <State> And SEA shall report the element state e_sig_dark_es with command reply flag true to CC <table><tr><th>LastStable</th><th>State</th></tr><tr><td>STOP</td><td>STOP</td></tr><tr><td>DARK</td><td>DARK</td></tr></table>				LastStable	State	STOP	STOP	DARK	DARK
LastStable	State								
STOP	STOP								
DARK	DARK								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:	OC-LS has probably sent the message as a downgrade aspect for failed command. As it is not possible to forward the command corresponding to SPEED1 aspect before receiving a reply to the previous STOP, it is not necessary to continue waiting for a further reply and thus the CHANGING_TO_SPEED1 state shall be left.								
Satisfies:									

EC-MUX-SIG:SSS:	Shunting Signal Combined - CHANGING_TO_SPEED1 - transition to
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REQ:3814	DARK / STOP								
Given the current element state is CHANGING_TO_SPEED1 And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS within WaitForSpeed ms Then SEA shall set the current element state to <NewState> And SEA shall report the element state e_sig_dark_es with command reply flag true to CC <table><tr><th>LastStable</th><th>NewState</th></tr><tr><td>STOP</td><td>DARK</td></tr><tr><td>DARK</td><td>STOP</td></tr></table>				LastStable	NewState	STOP	DARK	DARK	STOP
LastStable	NewState								
STOP	DARK								
DARK	STOP								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:	OC-LS has probably sent the message as a downgrade aspect for failed command. As it is not possible to forward the command corresponding to SPEED1 aspect before receiving a reply to the previous STOP, it is not necessary to continue waiting for a further reply and thus the CHANGING_TO_SPEED1 state shall be left. The diagnosis aspect message is sent only if the last stable state is different than the OC-LS reply message.								
Satisfies:									

EC-MUX-SIG:SSS: REQ:3819	Shunting Signal Combined - CHANGING_TO_SPEED1 - execute STOP command
Given the current element state is CHANGING_TO_SPEED1 When receiving the command <Command> from CC within WaitForSpeed ms Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS And SEA shall set the current element state to CHANGING_TO_STOP	

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<table><tr><th>Command</th></tr><tr><td>e_set_to_dark_p</td></tr><tr><td>e_set_to_dark_2</td></tr></table>				Command	e_set_to_dark_p	e_set_to_dark_2
Command						
e_set_to_dark_p						
e_set_to_dark_2						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:	If the STOP command reaches the OC-LS before the SPEED1 command is answered, then it is possible that OC-LS will reply to none of both commands, because the last commanded aspect is still given.					
Satisfies:						

EC-MUX-SIG:SSS: REQ:3843	Shunting Signal Combined - CHANGING_TO_SPEED1 - timeout								
Given the current element state is CHANGING_TO_SPEED1 And the last stable element state is <LastStable> When the current state of the element has been CHANGING_TO_SPEED1 for WaitForSpeed ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP And SEA shall set the current element state to <State> And SEA shall report the element state e_sig_dark_es with command reply flag true to CC <table><tr><th>LastStable</th><th>State</th></tr><tr><td>STOP</td><td>STOP</td></tr><tr><td>DARK</td><td>DARK</td></tr></table>				LastStable	State	STOP	STOP	DARK	DARK
LastStable	State								
STOP	STOP								
DARK	DARK								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						

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Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Regularly the state CHANGING_TO_SPEED1 will be left after WaitForSpeed ms. No safety reaction OC Disconnection is performed as this is not a safety-critical command. Instead, SEA shall generate a diagnosis and try to fallback to original STOP/DARK.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3815	Shunting Signal Combined - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state SPEED1					
Given the current element state is CHANGING_TO_STOP And the last stable element state is SPEED1 When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to <NewState> And SEA shall report the element state e_sig_dark_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_DARK to DGP <table><tr><td>NewState</td></tr><tr><td>STOP</td></tr><tr><td>DARK</td></tr></table>				NewState	STOP	DARK
NewState						
STOP						
DARK						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:	Successful command execution - commanded aspect / downgrade aspect reached.					
Satisfies:						

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EC-MUX-SIG:SSS: REQ:3847	Shunting Signal Combined - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state DARK / STOP								
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to <NewState> And SEA shall report the element state e_sig_dark_es with command reply flag true to CC <table><tr><th>LastStable</th><th>NewState</th></tr><tr><td>DARK</td><td>STOP</td></tr><tr><td>STOP</td><td>DARK</td></tr></table>				LastStable	NewState	DARK	STOP	STOP	DARK
LastStable	NewState								
DARK	STOP								
STOP	DARK								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:	Successful command execution - commanded aspect / downgrade aspect reached. OC-LS state can be DARK in precondition when a STOP command is executed. OC-LS state can be STOP or DARK in the precondition when a STOP command has been executed after a SPEED command without a command reply.								
Satisfies:									

EC-MUX-SIG:SSS: REQ:3844	Shunting Signal Combined - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state STOP / DARK
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS within WaitForSafeState ms	

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Then SEA shall set the current element state to <State>

And SEA shall report the element state e_sig_dark_es with command reply flag true to CC

LastStable	State
STOP	STOP
DARK	DARK

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Successful command execution - commanded aspect / downgrade aspect reached. OC-LS state can be DARK in precondition when a STOP command is executed. OC-LS state can be STOP or DARK in the precondition when a STOP command has been executed after a SPEED command without a command reply.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3816	Shunting Signal Combined - reject commands in CHANGING_TO_STOP
-------------------------------------	---

Given the current element state is CHANGING_TO_STOP

When receiving the command <Command> from CC within WaitForSafeState ms

Then SEA shall report the rejected command diagnosis to DGP

Command
e_set_to_dark_2
e_set_to_dark_p
e_clear_sig_speed1

Type:	FUR - Functional Requirement
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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	CC shall not send a SPEED command if there is a STOP command not answered yet. CC shall not repeat a STOP command. The timer shall not to be re-started due to the rejected command.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3817	Shunting Signal Combined - CHANGING_TO_STOP - ignore SPEED1		
Given the current element state is CHANGING_TO_STOP			
When receiving the LS Indicated Signal Aspect message with Code corresponding to SPEED1 from OC-LS within WaitForSafeState ms			
Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	It is possible that OC-LS has sent the message corresponding to SPEED1 as reply to the previous SPEED1 command not answered yet. Waiting for the reply to the STOP command is continued by keeping the current state CHANGING_TO_STOP. After timer expiry, if the expected STOP reply is not received, the element state is changed to STOP / DARK or a Safety Reaction OC Disconnection is performed, depending on the last received OC-LS state.		
Satisfies:			

EC-MUX-SIG:SSS:	Shunting Signal Combined - CHANGING_TO_STOP - timeout when OC-

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REQ:3845	LS SPEED1 - OC Disconnection		
Given the current element state is CHANGING_TO_STOP And the last stable element state is SPEED1 When the current state of the element has been CHANGING_TO_STOP for WaitForSafeState ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Regularly the state CHANGING_TO_STOP will be left after WaitForSafeState ms. Safety reaction OC Disconnection is applicable only if the OC-LS cannot reach the STOP or DARK aspect and stays in SPEED1. See also EC-MUX-SIG:SSS:REQ:3846.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3846	Shunting Signal Combined - CHANGING_TO_STOP - timeout when OC-LS DARK / STOP - reach DARK / STOP
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable> When the current state of the element has been CHANGING_TO_STOP for WaitForSafeState ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP And SEA shall set the current element state to <LastStable> And SEA shall report the element state e_sig_dark_es with command reply flag true to CC	

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LastStable STOP DARK			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Regularly the state CHANGING_TO_STOP will be left after WaitForSafeState ms. OC-LS state can be STOP or DARK at this point only when a STOP command has been executed after a SPEED command, and no reply has been received for either command. This is an allowed scenario so no safety reaction OC Disconnection is performed.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3820	Shunting Signal Combined - Disconnection on spontaneous aspect message SPEED1 in STOP / DARK
Given the current element state is <State> When receiving the LS Indicated Signal Aspect message with Code corresponding to SPEED1 from OC-LS Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: Spontaneous SPEED1 reported for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP State STOP DARK	

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3842	Shunting Signal Combined - Ignore repeated messages										
Given the current element state is <State> When receiving the LS Indicated Signal Aspect message with Code corresponding to <SeaSigAspect> from OC-LS Then SEA shall do nothing <table><tr><th>State</th><th>SeaSigAspect</th></tr><tr><td>STOP</td><td>STOP</td></tr><tr><td>SPEED1</td><td>SPEED1</td></tr><tr><td>DARK</td><td>DARK</td></tr></table>				State	SeaSigAspect	STOP	STOP	SPEED1	SPEED1	DARK	DARK
State	SeaSigAspect										
STOP	STOP										
SPEED1	SPEED1										
DARK	DARK										
Type:	FUR - Functional Requirement										
Status:	draft	Stability:	stable								
Target Release:	SEA SW GEN 3.0	Priority:	mandatory								
Originator:	Technics (R&D)	Allocation:	SW EC-MUX								
Verification Method:	Test	Verification Step:	module test, subit								
Comments:											
Satisfies:											

EC-MUX-SIG:SSS:	Shunting Signal Combined - Receive unexpected DARK / SPEED
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REQ:3853	commands from CC							
When receiving the command <Command> from CC Then SEA shall execute EC-MUX safety reaction Deactivation <table><tr><th>Command</th></tr><tr><td>e_set_to_dark_2</td></tr><tr><td>e_set_to_dark_p</td></tr><tr><td>e_clear_sig_speed2</td></tr><tr><td>e_clear_sig_speed3</td></tr></table>				Command	e_set_to_dark_2	e_set_to_dark_p	e_clear_sig_speed2	e_clear_sig_speed3
Command								
e_set_to_dark_2								
e_set_to_dark_p								
e_clear_sig_speed2								
e_clear_sig_speed3								
Type:	FUR - Functional Requirement							
Status:	draft	Stability:	stable					
Target Release:	SEA SW GEN 3.0	Priority:	mandatory					
Originator:	Technics (R&D)	Allocation:	SW EC-MUX					
Verification Method:	Test	Verification Step:	module test, subit					
Comments:								
Satisfies:								

3.1.2.3 Connection - Shunting Signal Combined

In order to control and supervise a Shunting Signal Combined

As a SEA

I want to be able to supervise connections and when changes occur then

- on CC connection establishment report the current state to CC
- on CC connection loss execute the safe command STOP if the last command forwarded to OC-LS was SPEED1 and it was not answered by STOP / DARK
- on OC-LS connection loss take on the default state UNDEFINED
- on OC-LS connection establishment transition to DARK / STOP state as the OC-LS element state
- report the connection state changes to CC and DGP

For requirement language see Chapter 2.8 "Requirement Language".

CHANGING_TO_SPEED1 and CHANGING_TO_STOP element states are not considered in the mapping between element state and CC State because when reaching these states, nothing is reported to CC.

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WaitForSpeed and WaitForSafeState are configurable timers for receiving a command reply from OC-LS. WaitForSpeed is used when waiting for a reply to a SPEED command (e_clear_sig_speed1), its default value is 2000ms.

WaitForSafeState is used when waiting for a reply to a STOP command (e_set_to_dark_2, e_set_to_dark_p), its default value is 1000ms.

Given a Shunting Signal Combined

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a STOP command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_dark_es
SPEED1	e_sig_speed1_es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12
SIG_SPEED_1	16

EC-MUX-SIG:SSS: REQ:3821	Shunting Signal Combined - [Re-]Connecting to CC - OC-LS CONNECTED - General inquiry in stable states		
Given the current element state is <State> And CC connection state is NOT_CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When establishing the connection to CC Then SEA shall report the element state e_sig_dark_es with command reply flag false to CC <table><tr><th>State</th></tr><tr><td>STOP</td></tr></table>		State	STOP
State			
STOP			

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<table><tr><td>State</td></tr><tr><td>DARK</td></tr></table>				State	DARK
State					
DARK					
Type:	FUR - Functional Requirement				
Status:	draft	Stability:	stable		
Target Release:	SEA SW GEN 3.0	Priority:	mandatory		
Originator:	Technics (R&D)	Allocation:	SW EC-MUX		
Verification Method:	Test	Verification Step:	module test, subit		
Comments:	[Re-]Connecting in SPEED1 state is impossible.				
Satisfies:					

EC-MUX-SIG:SSS: REQ:3822	Shunting Signal Combined - [Re-]Connecting to CC - OC-LS CONNECTED - General inquiry in CHANGING_TO_STOP state - no message after CC disconnect										
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable> And CC connection state is NOT_CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When establishing the connection to CC within WaitForSafeState ms Then SEA shall report the element state <CCState> with command reply flag false to CC <table><tr><th>LastStable</th><th>CCState</th></tr><tr><td>STOP</td><td>e_sig_dark_es</td></tr><tr><td>DARK</td><td>e_sig_dark_es</td></tr><tr><td>SPEED1</td><td>e_sig_speed1_es</td></tr></table>				LastStable	CCState	STOP	e_sig_dark_es	DARK	e_sig_dark_es	SPEED1	e_sig_speed1_es
LastStable	CCState										
STOP	e_sig_dark_es										
DARK	e_sig_dark_es										
SPEED1	e_sig_speed1_es										
Type:	FUR - Functional Requirement										
Status:	draft	Stability:	stable								
Target Release:	SEA SW GEN 3.0	Priority:	mandatory								
Originator:	Technics (R&D)	Allocation:	SW EC-MUX								
Verification Method:	Test	Verification Step:	module test, subit								

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Comments:	Re-connecting to CC in CHANGING_TO_STOP state is impossible in real operation, as re-connecting will regularly take much longer time than the configured WaitForSafeState. Due to the execution of the safe command [Re-]Connecting in CHANGING_TO_STOP state is possible only with last forwarded command STOP, but depending on the previous events the last state reported to CC can be different. Following initial states are covered with this requirement: - CHANGING_TO_STOP with last stable element state SPEED1 - CHANGING_TO_STOP with last stable element state DARK/STOP - CHANGING_TO_STOP with last stable element state DARK/STOP and SPEED1 received from OC-LS in CHANGING_TO_STOP
Satisfies:	

EC-MUX-SIG:SSS: REQ:3824	Shunting Signal Combined - [Re-]Connecting to CC - OC-LS NOT_CONNECTED		
Given the current element state is UNDEFINED And CC connection state is NOT_CONNECTED And EC-MUX - OC-LS connection state is NOT_CONNECTED When establishing the connection to CC Then SEA shall report the element state e_undefined_es with command reply flag false to CC			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3825	Shunting Signal Combined - Connection loss to CC - execute safe command in SPEED1
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Given the current element state is SPEED1
And CC connection state is CONNECTED

When losing the connection to CC

Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS

And SEA shall set the current element state to CHANGING_TO_STOP

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3835	Shunting Signal Combined - Connection loss to CC - OC-LS NOT_CONNECTED
-------------------------------------	---

Given the current element state is UNDEFINED
And CC connection state is CONNECTED
And EC-MUX - OC-LS connection state is NOT_CONNECTED

When losing the connection to CC

Then SEA shall do nothing

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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EC-MUX-SIG:SSS: REQ:3827	Shunting Signal Combined - Connection loss to CC - do nothing in STOP / DARK					
Given the current element state is <State> And CC connection state is CONNECTED When losing the connection to CC Then SEA shall do nothing <table><tr><td>State</td></tr><tr><td>STOP</td></tr><tr><td>DARK</td></tr></table>				State	STOP	DARK
State						
STOP						
DARK						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

EC-MUX-SIG:SSS: REQ:3826	Shunting Signal Combined - Connection loss to CC - CHANGING_TO_SPEED1		
Given the current element state is CHANGING_TO_SPEED1 And CC connection state is CONNECTED When losing the connection to CC within WaitForSpeed ms Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS And SEA shall set the current element state to CHANGING_TO_STOP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3828	Shunting Signal Combined - Connection loss to CC - CHANGING_TO_STOP - do nothing		
Given the current element state is CHANGING_TO_STOP And CC connection state is CONNECTED When losing the connection to CC within WaitForSafeState ms Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3801	Shunting Signal Combined - [Re-]Connecting between EC-MUX and OC-LS in state STOP
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED And OC-LS state is STOP When establishing the connection between EC-MUX and OC-LS	

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Then SEA shall set the current element state to STOP And SEA shall report the element state e_sig_dark_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_DARK to DGP And SEA shall report the connection_ok diagnosis device fault repair message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3802	Shunting Signal Combined - [Re-]Connecting between EC-MUX and OC-LS in state DARK		
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED And OC-LS state is DARK When establishing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to DARK And SEA shall report the element state e_sig_dark_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_DARK to DGP And SEA shall report the connection_ok diagnosis device fault repair message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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EC-MUX-SIG:SSS: REQ:3852	Shunting Signal Combined - [Re-]Connecting between EC-MUX and OC-LS in state SPEED1		
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED And OC-LS state is SPEED1 When establishing the connection between EC-MUX and OC-LS Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: OC-LS connect with SPEED1 for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3830	Shunting Signal Combined - Connection loss between EC-MUX and OC-LS in regular states			
Given the current element state is <State> And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP <table><tr><td>State</td></tr><tr><td>STOP</td></tr><tr><td>SPEED1</td></tr></table>		State	STOP	SPEED1
State				
STOP				
SPEED1				
Type:	FUR - Functional Requirement			

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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3831	Shunting Signal Combined - Connection loss between EC-MUX and OC-LS in DARK state		
Given the current element state is DARK And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3848	Shunting Signal Combined - Connection loss between EC-MUX and OC-LS in CHANGING_TO_SPEED1
Given the current element state is CHANGING_TO_SPEED1 And EC-MUX - OC-LS connection state is CONNECTED	

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When losing the connection between EC-MUX and OC-LS within WaitForSpeed ms Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3832	Shunting Signal Combined - Connection loss between EC-MUX and OC-LS in CHANGING_TO_STOP		
Given the current element state is CHANGING_TO_STOP And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS within WaitForSafeState ms Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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3.1.2.4 Diagnosis - Shunting Signal Combined

In order to provide the aspect diagnosis for a Shunting Signal Combined

As a SEA

I want to be able to receive an aspect diagnosis query from DGP and answer it by the DGP diagnosis values corresponding to the last diagnosis aspect message reported to DGP for the queried Signal

For requirement language see Chapter 2.8 "Requirement Language".

Given a Shunting Signal Combined

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a STOP command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_dark_es
SPEED1	e_sig_speed1_es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12
SIG_SPEED_1	16

EC-MUX-SIG:SSS: REQ:3850	Shunting Signal Combined - aspect query - last stable state
Given the last stable element state is <LastStable> When receiving an aspect query from DGP Then SEA shall report the diagnosis aspect message <DGPMMessage> to DGP	

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<table><tr><th>LastStable</th><th>DGPMessage</th></tr><tr><td>DARK</td><td>SIG_ DARK</td></tr><tr><td>STOP</td><td>SIG_ DARK</td></tr><tr><td>SPEED1</td><td>SIG_ SPEED_1</td></tr></table>				LastStable	DGPMessage	DARK	SIG_ DARK	STOP	SIG_ DARK	SPEED1	SIG_ SPEED_1
LastStable	DGPMessage										
DARK	SIG_ DARK										
STOP	SIG_ DARK										
SPEED1	SIG_ SPEED_1										
Type:		FUR - Functional Requirement									
Status:		draft	Stability: stable								
Target Release:		SEA SW GEN 3.0	Priority: mandatory								
Originator:		Technics (R&D)	Allocation: SW EC-MUX								
Verification Method:		Test	Verification Step: module test, subit								
Comments:											
Satisfies:											

EC-MUX-SIG:SSS: REQ:3849	Shunting Signal Combined - aspect query in UNDEFINED - OC-LS NOT_CONNECTED		
Given EC-MUX - OC-LS connection state is NOT_CONNECTED And the current element state is UNDEFINED When receiving an aspect query from DGP Then SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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3.1.3 Signal with Two Speed Aspects and Stop command

SEA Signal with two speed aspects and STOP command can reach following regular signal aspects via CC command:

- Speed1
- Speed2
- Stop

Following signal aspect can be reached only due to the lamp fault:

- Dark

This generic implementation can be configured to respond to specific CC commands and OC-LS messages.

The state machine of the SEA Eulynx Signal with two speed aspects and STOP command is shown in the following figure:

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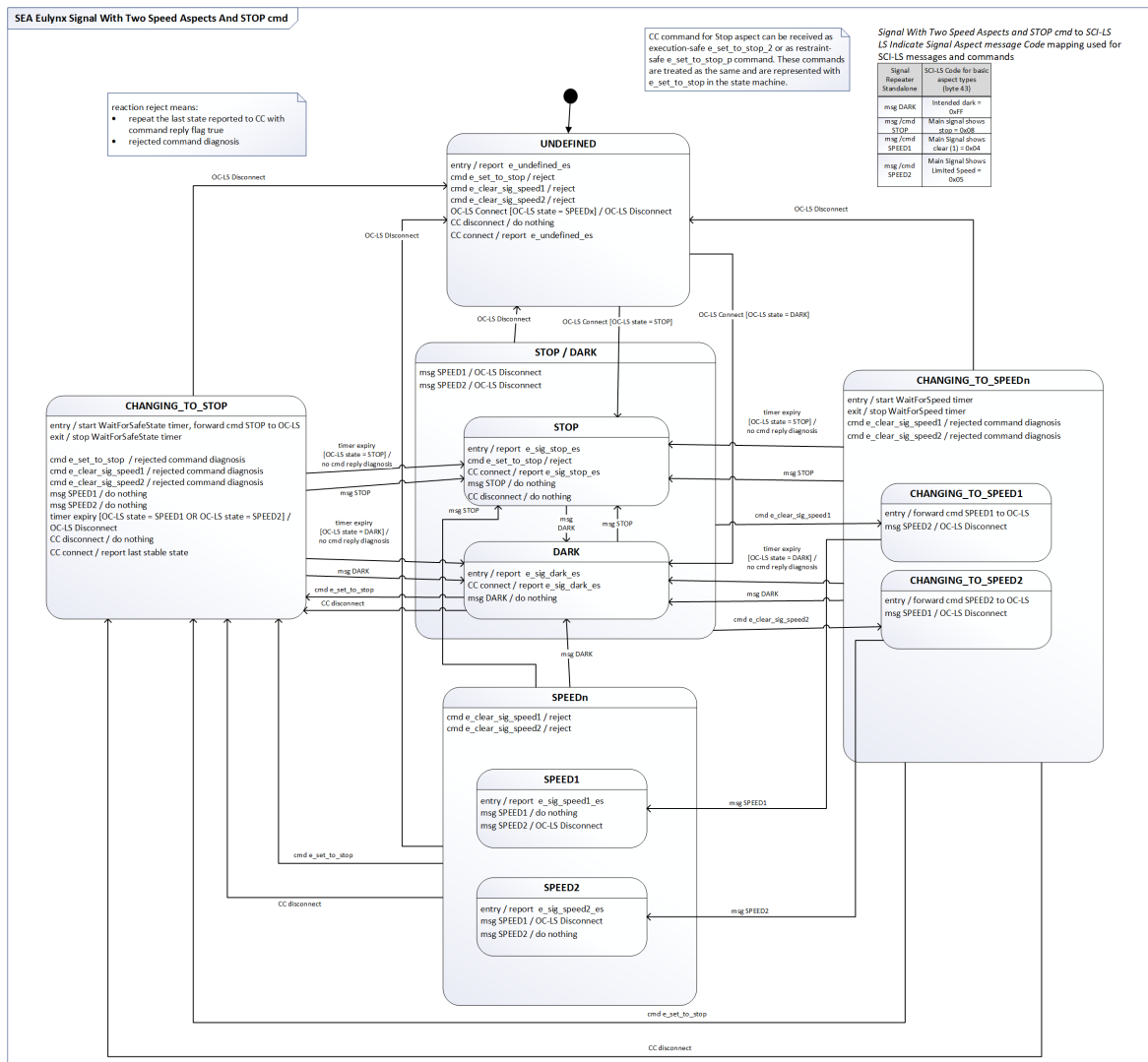


Figure 11: SEA Eulynx Signal with two speed aspects and STOP cmd - State Machine

Note To keep the STM applicable for several Signal types, names based on SEA Signal Aspects are used for SCI-LS messages and commands. A table with mappings to the SCI-LS codes has been provided for each Signal Type.

Following internal element states are defined for the SEA Eulynx Signal with two speed aspects and STOP cmd:

Element State	Functionality	CC State	Faulty
UNDEFINED	no aspect information given, initial / default state	e_undefined_es	yes
	indication for STOP aspect received;		

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STOP	as spontaneous msg or after cmd execution	e_sig_stop_es	??
SPEED1	signal commanded to SPEED1, corresponding aspect indication received already	e_sig_speed1_es	??
SPEED2	signal commanded to SPEED2, corresponding aspect indication received already	e_sig_speed2_es	??
DARK	indication for DARK aspect received; caused by lamp fault	e_sig_dark_es	yes
CHANGING_TO_SPEED1	command for SPEED1 aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history
CHANGING_TO_SPEED2	command for SPEED2 aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history
CHANGING_TO_STOP	command for STOP aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history

Table 22: SEA Eulynx Signal with two speed aspects and STOP cmd - Internal Element States
Behavior in case of connection loss

When losing the connection to CC, SEA shall execute the safe command, i.e. command the signal to the STOP aspect if the last command forwarded to OC-LS was a different one.

3.1.3.1 Signal Repeater Standalone

The Signal Repeater Standalone in SEA is a configuration of the SEA Eulynx Signal with two speed aspects and Stop command.

Used Cmd LS Indicate Signal Aspect and Msg LS Indicated Signal Aspect shall be addressed to the element identified by the Signal Repeater Standalone operational identifier. CC command and state mappings to SCI-LS telegrams are defined in following tables.

CC Command	SEA Signal with two speed aspects	Command byte / parameter	Parameter value	OBB aspect
e_set_to_stop_2 OR e_set_to_stop_p	Stop	43 = code for basic aspect types	Main Signal shows Stop = 0x08	Hauptsignal zeigt Halt
		43 = code for		

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e_clear_sig_speed1	Speed1	basic aspect types	Main Signal shows Clear (1) = 0x04	Hauptsignal zeigt Frei
e_clear_sig_speed2	Speed2	43 = code for basic aspect types	Main Signal shows Limited Speed = 0x05	Hauptsignal zeigt Frei beschr.

Table 23: Signal Repeater Standalone - Mapping CC command (CC to EC-MUX) to SCI-LS Cmd Indicate Signal Aspect (EC-MUX to OC-LS)

(*) TODO: Details for dependencies between different signal types at the same signal mast to be defined in a later project phase.

The SCI-LS Cmd Indicate Signal Aspect irrelevant and relevant parameters for the Signal Repeater Standalone are listed in the table *Eulynx Signal Mast configuration*.

Message byte / parameter	Parameter value	SEA Signal with two speed aspects	OBB aspect	CC State
43 = code for basic aspect types	Main Signal shows Stop = 0x08	Stop	Hauptsignal zeigt Halt	e_sig_stop_es
43 = code for basic aspect types	Main Signal shows Clear (1) = 0x04	Speed 1	Hauptsignal zeigt Frei	e_sig_speed1_es
43 = code for basic aspect types	Main Signal shows Limited Speed = 0x05	Speed 2	Hauptsignal zeigt Frei beschr.	e_sig_speed2_es
43 = code for basic aspect types	Intended dark = 0xFF	Dark	(*)	e_sig_dark_es

Table 24: Signal Repeater Standalone - Mapping SCI-LS Msg Indicated Signal Aspect (OC-LS to EC-MUX) to gen_elem_stat (EC-MUX to CC)

Remark: (*) The aspect defined in Signal aspect table [9] as "Intended dark" cannot be commanded, it can be reached only due to a lamp fault.

For default values of the remaining parameters (byte 44 - 60) of the Msg Indicated Signal Aspect see table for default values of the corresponding command parameters.

3.1.3.2 Operational - Signal With Two Speed Aspects and Stop command

In order to control and supervise a Signal With Two Speed Aspects and Stop command

As a SEA

I want to be able to receive commands from CC, translate them into SCI-LS commands and forward them to OC-LS and to receive SCI-LS state messages from OC-LS, translate them into signal state changes and report them to CC.

For requirement language see Chapter 2.8 "Requirement Language".

CHANGING_TO_SPEEDn and CHANGING_TO_STOP element states are not considered in the mapping between element state and CC State because when reaching these states, nothing is reported

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to CC.

WaitForSpeed and WaitForSafeState are configurable timers for receiving a command reply from OC-LS. WaitForSpeed is used when waiting for a reply to a SPEED command (e_clear_sig_speed1, e_clear_sig_speed2), its default value is 2000ms.

WaitForSafeState is used when waiting for a reply to a STOP command (e_set_to_stop_2, e_set_to_stop_p), its default value is 1000ms.

Given a Signal With Two Speed Aspects and Stop command

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a STOP command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_stop_es
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12
SIG_STOP	13
SIG_SPEED_1	16
SIG_SPEED_2	17

EC-MUX-SIG:SSS: REQ:3403	Signal With Two Speed Aspects and Stop command - reject commands in UNDEFINED state
Given EC-MUX - OC-LS connection state is NOT_CONNECTED And the current element state is UNDEFINED When receiving the command <Command> from CC	

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Then SEA shall report the element state e_undefined_es with command reply flag true to CC
And SEA shall report the rejected command diagnosis to DGP

Command
e_set_to_stop_p
e_set_to_stop_2
e_clear_sig_speed1
e_clear_sig_speed2

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3404	Signal With Two Speed Aspects and Stop command - execute SPEED command in STOP / DARK
-------------------------------------	--

Given the current element state is <State>

When receiving the command <Command> from CC

Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to <NewState> to OC-LS

And SEA shall set the current element state to <ChangingToSpeedState>

State	Command	NewState	ChangingToSpeedState
STOP	e_clear_sig_speed1	SPEED1	CHANGING_TO_SPEED1
STOP	e_clear_sig_speed2	SPEED2	CHANGING_TO_SPEED2
DARK	e_clear_sig_speed1	SPEED1	CHANGING_TO_SPEED1

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State	Command	NewState	ChangingToSpeedState
DARK	e_clear_sig_speed2	SPEED2	CHANGING_TO_SPEED2
Type: FUR - Functional Requirement			
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3405	Signal With Two Speed Aspects and Stop command - reject STOP commands in STOP					
Given the current element state is STOP When receiving the command <Command> from CC Then SEA shall report the element state e_sig_stop_es with command reply flag true to CC And SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_set_to_stop_2</td></tr><tr><td>e_set_to_stop_p</td></tr></table>				Command	e_set_to_stop_2	e_set_to_stop_p
Command						
e_set_to_stop_2						
e_set_to_stop_p						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:	CC shall not send the STOP command, if the stop aspect is already given.					

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Satisfies:	
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EC-MUX-SIG:SSS: REQ:3439	Signal With Two Speed Aspects and Stop command - execute STOP commands in DARK					
Given the current element state is DARK When receiving the command <Command> from CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS And SEA shall set the current element state to CHANGING_TO_STOP <table><tr><th>Command</th></tr><tr><td>e_set_to_stop_2</td></tr><tr><td>e_set_to_stop_p</td></tr></table>				Command	e_set_to_stop_2	e_set_to_stop_p
Command						
e_set_to_stop_2						
e_set_to_stop_p						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

EC-MUX-SIG:SSS: REQ:3406	Signal With Two Speed Aspects and Stop command - spontaneous transition between STOP / DARK
Given the current element state is <State> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS	

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Then SEA shall set the current element state to <NewState>

And SEA shall report the element state <CCState> with command reply flag false to CC

And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP

State	NewState	CCState	DGPMMessage
STOP	DARK	e_sig_dark_es	SIG_DARK
DARK	STOP	e_sig_stop_es	SIG_STOP

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3407	Signal With Two Speed Aspects and Stop command - execute STOP commands in SPEEDx
-------------------------------------	---

Given the current element state is <State>

When receiving the command <Command> from CC

Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS

And SEA shall set the current element state to CHANGING_TO_STOP

State	Command
SPEED1	e_set_to_stop_p
SPEED2	e_set_to_stop_p
SPEED1	e_set_to_stop_2

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State	Command
SPEED2	e_set_to_stop_2

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3408	Signal With Two Speed Aspects and Stop command - reject SPEED command in SPEEDx																	
Given the current element state is <State> When receiving the command <Command> from CC Then SEA shall report the element state <CCState> with command reply flag true to CC And SEA shall report the rejected command diagnosis to DGP <table><tr><th>State</th><th>Command</th><th>CCState</th></tr><tr><td>SPEED1</td><td>e_clear_sig_speed1</td><td>e_sig_speed1_es</td></tr><tr><td>SPEED1</td><td>e_clear_sig_speed2</td><td>e_sig_speed1_es</td></tr><tr><td>SPEED2</td><td>e_clear_sig_speed1</td><td>e_sig_speed2_es</td></tr><tr><td>SPEED2</td><td>e_clear_sig_speed2</td><td>e_sig_speed2_es</td></tr></table>				State	Command	CCState	SPEED1	e_clear_sig_speed1	e_sig_speed1_es	SPEED1	e_clear_sig_speed2	e_sig_speed1_es	SPEED2	e_clear_sig_speed1	e_sig_speed2_es	SPEED2	e_clear_sig_speed2	e_sig_speed2_es
State	Command	CCState																
SPEED1	e_clear_sig_speed1	e_sig_speed1_es																
SPEED1	e_clear_sig_speed2	e_sig_speed1_es																
SPEED2	e_clear_sig_speed1	e_sig_speed2_es																
SPEED2	e_clear_sig_speed2	e_sig_speed2_es																
Type:	FUR - Functional Requirement																	
Status:	draft	Stability:	stable															
Target Release:	SEA SW GEN 3.0	Priority:	mandatory															
Originator:	Technics (R&D)	Allocation:	SW EC-MUX															

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Verification Method:	Test	Verification Step:	module test, subit
Comments:	CC shall not send the command SPEED, if the corresponding aspect is already given. CC shall not send any other SPEED command (downgrade / upgrade the speed) without STOP command between.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3409	Signal With Two Speed Aspects and Stop command - spontaneous transition from SPEEDx to STOP / DARK																						
Given the current element state is <State> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag false to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>State</th><th>NewState</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>SPEED1</td><td>STOP</td><td>e_sig_stop_es</td><td>SIG_STOP</td></tr><tr><td>SPEED1</td><td>DARK</td><td>e_sig_dark_es</td><td>SIG_DARK</td></tr><tr><td>SPEED2</td><td>STOP</td><td>e_sig_stop_es</td><td>SIG_STOP</td></tr><tr><td>SPEED2</td><td>DARK</td><td>e_sig_dark_es</td><td>SIG_DARK</td></tr></table>				State	NewState	CCState	DGPMMessage	SPEED1	STOP	e_sig_stop_es	SIG_STOP	SPEED1	DARK	e_sig_dark_es	SIG_DARK	SPEED2	STOP	e_sig_stop_es	SIG_STOP	SPEED2	DARK	e_sig_dark_es	SIG_DARK
State	NewState	CCState	DGPMMessage																				
SPEED1	STOP	e_sig_stop_es	SIG_STOP																				
SPEED1	DARK	e_sig_dark_es	SIG_DARK																				
SPEED2	STOP	e_sig_stop_es	SIG_STOP																				
SPEED2	DARK	e_sig_dark_es	SIG_DARK																				
Type:	FUR - Functional Requirement																						
Status:	draft	Stability:	stable																				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory																				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX																				
Verification Method:	Test	Verification Step:	module test, subit																				
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Satisfies:																							

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EC-MUX-SIG:SSS: REQ:3410	Signal With Two Speed Aspects and Stop command - CHANGING_TO_SPEEDn - transition from CHANGING_TO_SPEEDx to SPEEDx														
Given the current element state is <ChangingToSpeedState> When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS within WaitForSpeed ms Then SEA shall set the current element state to <State> And SEA shall report the element state <CCState> with command reply flag true to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>ChangingToSpeedState</th><th>State</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>CHANGING_TO_SPEED1</td><td>SPEED1</td><td>e_sig_speed1_es</td><td>SIG_SPEED_1</td></tr><tr><td>CHANGING_TO_SPEED2</td><td>SPEED2</td><td>e_sig_speed2_es</td><td>SIG_SPEED_2</td></tr></table>				ChangingToSpeedState	State	CCState	DGPMMessage	CHANGING_TO_SPEED1	SPEED1	e_sig_speed1_es	SIG_SPEED_1	CHANGING_TO_SPEED2	SPEED2	e_sig_speed2_es	SIG_SPEED_2
ChangingToSpeedState	State	CCState	DGPMMessage												
CHANGING_TO_SPEED1	SPEED1	e_sig_speed1_es	SIG_SPEED_1												
CHANGING_TO_SPEED2	SPEED2	e_sig_speed2_es	SIG_SPEED_2												
Type:	FUR - Functional Requirement														
Status:	draft	Stability:	stable												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory												
Originator:	Technics (R&D)	Allocation:	SW EC-MUX												
Verification Method:	Test	Verification Step:	module test, subit												
Comments:	Successful SPEED command execution - commanded aspect reached.														
Satisfies:															

EC-MUX-SIG:SSS: REQ:3411	Signal With Two Speed Aspects and Stop command - CHANGING_TO_SPEEDn - reject repeated SPEED command in CHANGING_TO_SPEEDn
Given the current element state is CHANGING_TO_SPEEDn When receiving the command <Command> from CC within WaitForSpeed ms Then SEA shall report the rejected command diagnosis to DGP	

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<table border="1"> <tr> <th colspan="4">Command</th></tr> <tr> <td colspan="4">e_clear_sig_speed1</td></tr> <tr> <td colspan="4">e_clear_sig_speed2</td></tr> </table>				Command				e_clear_sig_speed1				e_clear_sig_speed2			
Command															
e_clear_sig_speed1															
e_clear_sig_speed2															
Type:	FUR - Functional Requirement														
Status:	draft	Stability:	stable												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory												
Originator:	Technics (R&D)	Allocation:	SW EC-MUX												
Verification Method:	Test	Verification Step:	module test, subit												
Comments:	CC shall not repeat the command SPEED. CC shall not switch to another SPEED aspect without STOP command between them. The WaitForSpeed timer shall not to be re-started due to the rejected command.														
Satisfies:															

EC-MUX-SIG:SSS: REQ:3412	Signal With Two Speed Aspects and Stop command - CHANGING_TO_SPEEDn - return to STOP / DARK									
Given the current element state is CHANGING_TO_SPEEDn And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS within WaitForSpeed ms Then SEA shall set the current element state to <State> And SEA shall report the element state <CCState> with command reply flag true to CC <table><tr><th>LastStable</th><th>State</th><th>CCState</th></tr><tr><td>STOP</td><td>STOP</td><td>e_sig_stop_es</td></tr><tr><td>DARK</td><td>DARK</td><td>e_sig_dark_es</td></tr></table>		LastStable	State	CCState	STOP	STOP	e_sig_stop_es	DARK	DARK	e_sig_dark_es
LastStable	State	CCState								
STOP	STOP	e_sig_stop_es								
DARK	DARK	e_sig_dark_es								
Type:	FUR - Functional Requirement									

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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	OC-LS has probably sent the message as a downgrade aspect for failed clear command. As it is not possible to forward the command corresponding to SPEEDx aspect before receiving a reply to the previous STOP, it is not necessary to continue waiting for a further reply and thus the CHANGING_TO_SPEEDx state shall be left.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3413	Signal With Two Speed Aspects and Stop command - CHANGING_TO_SPEEDn - transition to STOP / DARK														
Given the current element state is CHANGING_TO_SPEEDn And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS within WaitForSpeed ms Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag true to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>LastStable</th><th>NewState</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>STOP</td><td>DARK</td><td>e_sig_dark_es</td><td>SIG_DARK</td></tr><tr><td>DARK</td><td>STOP</td><td>e_sig_stop_es</td><td>SIG_STOP</td></tr></table>				LastStable	NewState	CCState	DGPMMessage	STOP	DARK	e_sig_dark_es	SIG_DARK	DARK	STOP	e_sig_stop_es	SIG_STOP
LastStable	NewState	CCState	DGPMMessage												
STOP	DARK	e_sig_dark_es	SIG_DARK												
DARK	STOP	e_sig_stop_es	SIG_STOP												
Type:		FUR - Functional Requirement													
Status:	draft	Stability:	stable												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory												
Originator:	Technics (R&D)	Allocation:	SW EC-MUX												
Verification Method:	Test	Verification Step:	module test, subit												

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Comments:	OC-LS has probably sent the message as a downgrade aspect for failed command. As it is not possible to forward the command corresponding to SPEEDx aspect before receiving a reply to the previous STOP, it is not necessary to continue waiting for a further reply and thus the CHANGING_TO_SPEEDx state shall be left. The diagnosis aspect message is sent only if the last stable state is different than the OC-LS reply message.
Satisfies:	

EC-MUX-SIG:SSS: REQ:3414	Signal With Two Speed Aspects and Stop command - CHANGING_TO_SPEEDn - timeout								
Given the current element state is CHANGING_TO_SPEEDn And the last stable element state is <LastStable> When the current state of the element has been CHANGING_TO_SPEEDn for WaitForSpeed ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x". to DGP And SEA shall set the current element state to <LastStable> And SEA shall report the element state <CCState> with command reply flag true to CC <table><tr><th>LastStable</th><th>CCState</th></tr><tr><td>STOP</td><td>e_sig_dark_es</td></tr><tr><td>DARK</td><td>e_sig_stop_es</td></tr></table>				LastStable	CCState	STOP	e_sig_dark_es	DARK	e_sig_stop_es
LastStable	CCState								
STOP	e_sig_dark_es								
DARK	e_sig_stop_es								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:	Regularly the state CHANGING_TO_SPEEDn will be left after WaitForSpeed ms. No safety reaction OC Disconnection is performed as this is not a safety-critical command. Instead, SEA shall generate a diagnosis and try to fallback to original STOP/DARK.								

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Satisfies:	
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EC-MUX-SIG:SSS: REQ:3415	Signal With Two Speed Aspects and Stop command - CHANGING_TO_SPEEDn - execute STOP command					
Given the current element state is CHANGING_TO_SPEEDn When receiving the command <Command> from CC within WaitForSpeed ms Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS And SEA shall set the current element state to CHANGING_TO_STOP <table><tr><th>Command</th></tr><tr><td>e_set_to_stop_p</td></tr><tr><td>e_set_to_stop_2</td></tr></table>				Command	e_set_to_stop_p	e_set_to_stop_2
Command						
e_set_to_stop_p						
e_set_to_stop_2						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:	If the STOP command reaches the OC-LS before the SPEEDx command is answered, then it is possible that OC-LS will reply to none of both commands, because the last commanded aspect is still given.					
Satisfies:						

EC-MUX-SIG:SSS: REQ:3416	Signal With Two Speed Aspects and Stop command - CHANGING_TO_STOP - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state SPEEDx
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable>	

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When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS within WaitForSafeState ms

Then SEA shall set the current element state to <NewState>

And SEA shall report the element state <CCState> with command reply flag true to CC

And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP

LastStable	NewState	CCState	DGPMMessage
SPEED1	STOP	e_sig_stop_es	SIG_STOP
SPEED1	DARK	e_sig_dark_es	SIG_DARK
SPEED2	STOP	e_sig_stop_es	SIG_STOP
SPEED2	DARK	e_sig_dark_es	SIG_DARK

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Successful stop command execution - commanded aspect / downgrade aspect reached.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3417	Signal With Two Speed Aspects and Stop command - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state STOP / DARK
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to <State> And SEA shall report the element state <CCState> with command reply flag true to CC	

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LastStable	State	CCState
STOP	STOP	e_sig_stop_es
DARK	DARK	e_sig_dark_es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	After receiving the STOP / DARK as reply to the STOP command, the state will be reported to the CC with command reply flag true.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3418	Signal With Two Speed Aspects and Stop command - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state DARK / STOP														
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag true to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>LastStable</th><th>NewState</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>DARK</td><td>STOP</td><td>e_sig_stop_ es</td><td>SIG_STOP</td></tr><tr><td>STOP</td><td>DARK</td><td>e_sig_dark_ es</td><td>SIG_DARK</td></tr></table>				LastStable	NewState	CCState	DGPMMessage	DARK	STOP	e_sig_stop_ es	SIG_STOP	STOP	DARK	e_sig_dark_ es	SIG_DARK
LastStable	NewState	CCState	DGPMMessage												
DARK	STOP	e_sig_stop_ es	SIG_STOP												
STOP	DARK	e_sig_dark_ es	SIG_DARK												
Type:	FUR - Functional Requirement														
Status:	draft	Stability:	stable												

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	After receiving the STOP / DARK as reply to the STOP command, the state will be reported to the CC with command reply flag true; aspect diagnosis is sent also as the state is changed.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3419	Signal With Two Speed Aspects and Stop command - CHANGING_TO_STOP - reject commands in CHANGING_TO_STOP							
Given the current element state is CHANGING_TO_STOP When receiving the command <Command> from CC within WaitForSafeState ms Then SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_set_to_stop_2</td></tr><tr><td>e_set_to_stop_p</td></tr><tr><td>e_clear_sig_speed1</td></tr><tr><td>e_clear_sig_speed2</td></tr></table>				Command	e_set_to_stop_2	e_set_to_stop_p	e_clear_sig_speed1	e_clear_sig_speed2
Command								
e_set_to_stop_2								
e_set_to_stop_p								
e_clear_sig_speed1								
e_clear_sig_speed2								
Type:	FUR - Functional Requirement							
Status:	draft	Stability:	stable					
Target Release:	SEA SW GEN 3.0	Priority:	mandatory					
Originator:	Technics (R&D)	Allocation:	SW EC-MUX					
Verification Method:	Test	Verification Step:	module test, subit					
Comments:	CC shall not send a clear command if there is STOP command not answered yet. CC shall not repeat the command STOP. The WaitForSafeState timer shall not to be re-started due to the rejected command.							

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Satisfies:	
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EC-MUX-SIG:SSS: REQ:3420	Signal With Two Speed Aspects and Stop command - CHANGING_TO_STOP - timeout when OC-LS SPEEDx - OC Disconnection					
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable> When the current state of the element has been CHANGING_TO_STOP for WaitForSafeState ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP <table><tr><td>LastStable</td></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr></table>				LastStable	SPEED1	SPEED2
LastStable						
SPEED1						
SPEED2						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:	Regularly the state CHANGING_TO_STOP will be left after WaitForSafeState ms. Safety reaction OC Disconnection is applicable only if the OC-LS cannot reach the STOP or DARK aspect and stays in SPEEDx. See also EC-MUX-SIG:SSS:REQ:3421.					
Satisfies:						

EC-MUX-SIG:SSS: REQ:3421	Signal With Two Speed Aspects and Stop command - CHANGING_TO_STOP - timeout when OC-LS DARK / STOP - reach DARK / STOP
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Given the current element state is CHANGING_TO_STOP
And the last stable element state is <LastStable>

When the current state of the element has been CHANGING_TO_STOP for WaitForSafeState ms

Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP
And SEA shall set the current element state to <LastStable> after WaitForSafeState ms
And SEA shall report the element state <CCState> with command reply flag true to CC

LastStable	CCState
STOP	e_sig_stop_es
DARK	e_sig_dark_es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Regularly the state CHANGING_TO_STOP will be left after WaitForSafeState ms. OC-LS state can be STOP or DARK at this point only when a STOP command has been executed after a SPEED command, and no reply has been received for either command. This is an allowed scenario so no safety reaction OC Disconnection is performed.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3422	Signal With Two Speed Aspects and Stop command - CHANGING_TO_STOP - ignore SPEED message
Given the current element state is CHANGING_TO_STOP When receiving the LS Indicated Signal Aspect message with Code corresponding to <SpeedAspect> from OC-LS within WaitForSafeState ms Then SEA shall do nothing	

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<table><tr><th>SpeedAspect</th></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr></table>				SpeedAspect	SPEED1	SPEED2
SpeedAspect						
SPEED1						
SPEED2						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:	It is possible that OC-LS has sent the message corresponding to SPEED as reply to the previous SPEED command not answered yet. Waiting for the reply to the STOP command is continued by keeping the current state CHANGING_TO_STOP. After timer expiry, if the expected STOP reply is not received, the element state is changed to STOP / DARK or a Safety Reaction OC Disconnection is performed, depending on the last received OC-LS state.					
Satisfies:						

EC-MUX-SIG:SSS: REQ:3423	Signal With Two Speed Aspects and Stop command - Disconnection on spontaneous SPEED message in STOP / SPEED								
Given the current element state is <State> When receiving the LS Indicated Signal Aspect message with Code corresponding to <SpeedAspect> from OC-LS Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: Spontaneous <SpeedAspect> reported for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP <table border="1"> <tr> <th>State</th><th>SpeedAspect</th></tr> <tr> <td>STOP</td><td>SPEED1</td></tr> <tr> <td>STOP</td><td>SPEED2</td></tr> <tr> <td>SPEED1</td><td>SPEED2</td></tr> </table>		State	SpeedAspect	STOP	SPEED1	STOP	SPEED2	SPEED1	SPEED2
State	SpeedAspect								
STOP	SPEED1								
STOP	SPEED2								
SPEED1	SPEED2								

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<table><tr><th>State</th><th>SpeedAspect</th></tr><tr><td>SPEED2</td><td>SPEED1</td></tr></table>		State	SpeedAspect	SPEED2	SPEED1
State	SpeedAspect				
SPEED2	SPEED1				
Type:	FUR - Functional Requirement				
Status:	draft	Stability:	stable		
Target Release:	SEA SW GEN 3.0	Priority:	mandatory		
Originator:	Technics (R&D)	Allocation:	SW EC-MUX		
Verification Method:	Test	Verification Step:	module test, subit		
Comments:					
Satisfies:					

EC-MUX-SIG:SSS: REQ:3424	Signal With Two Speed Aspects and Stop command - CHANGING_TO_SPEEDx - Disconnection on unexpected command reply											
Given the current element state is <ChangingToSpeedState> When receiving the LS Indicated Signal Aspect message with Code corresponding to <SpeedMsg> from OC-LS within WaitForSpeed ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: Cmd <SpeedCmd> replied by Msg <SpeedMsg> for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP <table><tr><th>ChangingToSpeedState</th><th>SpeedMsg</th><th>SpeedCmd</th></tr><tr><td>CHANGING_TO_SPEED1</td><td>SPEED2</td><td>SPEED1</td></tr><tr><td>CHANGING_TO_SPEED2</td><td>SPEED1</td><td>SPEED2</td></tr></table>				ChangingToSpeedState	SpeedMsg	SpeedCmd	CHANGING_TO_SPEED1	SPEED2	SPEED1	CHANGING_TO_SPEED2	SPEED1	SPEED2
ChangingToSpeedState	SpeedMsg	SpeedCmd										
CHANGING_TO_SPEED1	SPEED2	SPEED1										
CHANGING_TO_SPEED2	SPEED1	SPEED2										
Type:	FUR - Functional Requirement											
Status:	draft	Stability:	stable									
Target Release:	SEA SW GEN 3.0	Priority:	mandatory									

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Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3425	Signal With Two Speed Aspects and Stop command - Ignore repeated messages												
Given the current element state is <State> When receiving the LS Indicated Signal Aspect message with Code corresponding to <SpeedAspect> from OC-LS Then SEA shall do nothing <table><tr><th>State</th><th>SpeedAspect</th></tr><tr><td>STOP</td><td>STOP</td></tr><tr><td>SPEED1</td><td>SPEED1</td></tr><tr><td>SPEED2</td><td>SPEED2</td></tr><tr><td>DARK</td><td>DARK</td></tr></table>				State	SpeedAspect	STOP	STOP	SPEED1	SPEED1	SPEED2	SPEED2	DARK	DARK
State	SpeedAspect												
STOP	STOP												
SPEED1	SPEED1												
SPEED2	SPEED2												
DARK	DARK												
Type:	FUR - Functional Requirement												
Status:	draft	Stability:	stable										
Target Release:	SEA SW GEN 3.0	Priority:	mandatory										
Originator:	Technics (R&D)	Allocation:	SW EC-MUX										
Verification Method:	Test	Verification Step:	module test, subit										
Comments:													
Satisfies:													

EC-MUX-SIG:SSS: REQ:3426	Signal With Two Speed Aspects and Stop command - Receive unexpected DARK / SPEED commands from CC
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When receiving the command <Command> from CC

Then SEA shall execute EC-MUX safety reaction Deactivation

Command			
e_set_to_dark_p			
e_set_to_dark_2			
e_clear_sig_speed3			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

3.1.3.3 Operational - Signal With Two Speed Aspects and Stop command - DARK state

In order to control and supervise a Signal With Two Speed Aspects and Stop command

As a SEA

I want to be able to receive commands from CC, translate them into SCI-LS commands and forward them to OC-LS and to receive SCI-LS state messages from OC-LS, translate them into signal state changes and report them to CC.

For requirement language see Chapter 2.8 "Requirement Language".

Given a Signal With Two Speed Aspects and Stop command

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_stop_es
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es

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State	CC State
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12
SIG_STOP	13
SIG_SPEED_1	16
SIG_SPEED_2	17

EC-MUX-SIG:SSS: REQ:3400	Signal With Two Speed Aspects and Stop command - Initial state		
When initializing the element			
Then SEA shall set the current element state to UNDEFINED			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3450	Signal With Two Speed Aspects and Stop command - Disconnection on spontaneous SPEED message in DARK
Given the current element state is DARK	

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When receiving the LS Indicated Signal Aspect message with Code corresponding to <SpeedAspect> from OC-LS

Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: Spontaneous <SpeedAspect> reported for Signal x." to DGP

And SEA shall execute EC-MUX safety reaction OC Disconnection

And SEA shall set the current element state to UNDEFINED

And SEA shall report the element state e_undefined_es with command reply flag false to CC

And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP

And SEA shall report the no_connection diagnosis device fault message to DGP

SpeedAspect

SPEED1

SPEED2

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

3.1.3.4 Connection - Signal With Two Speed Aspects and Stop command

In order to control and supervise a Signal With Two Speed Aspects and Stop command

As a SEA

I want to be able to supervise connections and when changes occur then

- on CC connection establishment report the current state to CC
- on CC connection loss execute the safe command STOP if the last state is DARK
- on CC connection loss execute the safe command STOP if the last command forwarded to OC-LS was SPEED1 / SPEED2 and it was not answered by STOP / DARK
- on OC-LS connection loss take on the default state UNDEFINED
- on OC-LS connection establishment transition to DARK / STOP state as the OC-LS element state
- report the connection state changes to CC and DGP

For requirement language see Chapter 2.8 "Requirement Language".

CHANGING_TO_SPEEDn and CHANGING_TO_STOP element states are not considered in the

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mapping between element state and CC State because when reaching these states, nothing is reported to CC.

WaitForSpeed and WaitForSafeState are configurable timers for receiving a command reply from OC-LS. WaitForSpeed is used when waiting for a reply to a SPEED command (e_clear_sig_speed1, e_clear_sig_speed2), its default value is 2000ms.

WaitForSafeState is used when waiting for a reply to a STOP command (e_set_to_stop_2, e_set_to_stop_p), its default value is 1000ms.

Given a Signal With Two Speed Aspects and Stop command

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a STOP command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_stop_es
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12
SIG_STOP	13
SIG_SPEED_1	16
SIG_SPEED_2	17

EC-MUX-SIG:SSS: REQ:3427	Signal With Two Speed Aspects and Stop command - [Re-]Connecting to CC - OC-LS CONNECTED - General inquiry in stable states
Given the current element state is <State> And CC connection state is NOT_CONNECTED And EC-MUX - OC-LS connection state is CONNECTED	

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When establishing the connection to CC

Then SEA shall report the element state <CCState> with command reply flag false to CC

State	CCState
STOP	e_sig_stop_es
DARK	e_sig_dark_es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	[Re-]Connecting in SPEEDx state is impossible.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3428	Signal With Two Speed Aspects and Stop command - [Re-]Connecting to CC - OC-LS CONNECTED - General inquiry in CHANGING_TO_STOP - no message after CC disconnect
-------------------------------------	--

Given the current element state is CHANGING_TO_STOP

And the last stable element state is <LastStable>

And CC connection state is NOT_CONNECTED

And EC-MUX - OC-LS connection state is CONNECTED

When establishing the connection to CC within WaitForSafeState ms

Then SEA shall report the element state <CCState> with command reply flag false to CC

LastStable	CCState
STOP	e_sig_stop_es
DARK	e_sig_dark_es
SPEED1	e_sig_speed1_es
	e_sig_speed2_

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LastStable		CCState	
SPEED2		es	
Type:		FUR - Functional Requirement	
Status:		draft	Stability: stable
Target Release:		SEA SW GEN 3.0	Priority: mandatory
Originator:		Technics (R&D)	Allocation: SW EC-MUX
Verification Method:		Test	Verification Step: module test, subit
Comments:		Re-connecting to CC in CHANGING_TO_STOP state is impossible in real operation, as re-connecting will regularly take much longer time than the configured WaitForSafeState. Due to the execution of the safe command [Re-]Connecting in CHANGING_TO_STOP state is possible only with last forwarded command STOP, but depending on the previous events the last state reported to CC can be different. Following initial states are covered with this requirement: - CHANGING_TO_STOP with last stable element state SPEED1/SPEED2 - CHANGING_TO_STOP with last stable element state DARK/STOP - CHANGING_TO_STOP with last stable element state DARK/STOP and SPEED1/SPEED2 received from OC-LS in CHANGING_TO_STOP	
Satisfies:			

EC-MUX-SIG:SSS: REQ:3429	Signal With Two Speed Aspects and Stop command - [Re-]Connecting to CC - OC-LS NOT_CONNECTED - General inquiry		
Given the current element state is UNDEFINED And CC connection state is NOT_CONNECTED And EC-MUX - OC-LS connection state is NOT_CONNECTED When establishing the connection to CC Then SEA shall report the element state e_undefined_es with command reply flag false to CC			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX

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Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3430	Signal With Two Speed Aspects and Stop command - Connection loss to CC - execute safe command in SPEEDx					
Given the current element state is <State> And CC connection state is CONNECTED When losing the connection to CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS And SEA shall set the current element state to CHANGING_TO_STOP <table><tr><td>State</td></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr></table>				State	SPEED1	SPEED2
State						
SPEED1						
SPEED2						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

EC-MUX-SIG:SSS: REQ:3431	Signal With Two Speed Aspects and Stop command - Connection loss to CC - do nothing in STOP / UNDEFINED
Given the current element state is <State> And CC connection state is CONNECTED	

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When losing the connection to CC

Then SEA shall do nothing

State
STOP
UNDEFINED

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3451	Signal With Two Speed Aspects and Stop command - Connection loss to CC - do nothing in DARK
-------------------------------------	--

Given the current element state is DARK

And CC connection state is CONNECTED

When losing the connection to CC

Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS

And SEA shall set the current element state to CHANGING_TO_STOP

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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EC-MUX-SIG:SSS: REQ:3432	Signal With Two Speed Aspects and Stop command - Connection loss to CC - CHANGING_TO_SPEEDn		
Given the current element state is CHANGING_TO_SPEEDn And CC connection state is CONNECTED When losing the connection to CC within WaitForSpeed ms Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS And SEA shall set the current element state to CHANGING_TO_STOP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3433	Signal With Two Speed Aspects and Stop command - Connection loss to CC - CHANGING_TO_STOP - do nothing		
Given the current element state is CHANGING_TO_STOP And CC connection state is CONNECTED When losing the connection to CC within WaitForSafeState ms Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			

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Satisfies:	
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EC-MUX-SIG:SSS: REQ:3401	Signal With Two Speed Aspects and Stop command - [Re-]Connecting between EC-MUX and OC-LS in state STOP		
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED And OC-LS state is STOP When establishing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to STOP And SEA shall report the element state e_sig_stop_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_STOP to DGP And SEA shall report the connection_ok diagnosis device fault repair message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3402	Signal With Two Speed Aspects and Stop command - [Re-]Connecting between EC-MUX and OC-LS in state DARK
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED And OC-LS state is DARK When establishing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to DARK And SEA shall report the element state e_sig_dark_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_DARK to DGP And SEA shall report the connection_ok diagnosis device fault repair message to DGP	
Type:	FUR - Functional Requirement

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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3434	Signal With Two Speed Aspects and Stop command - [Re-]Connecting between EC-MUX and OC-LS in state SPEEDx					
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED And OC-LS state is <SpeedAspect> When establishing the connection between EC-MUX and OC-LS Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: OC-LS connect with <SpeedAspect> for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection <table><tr><td>SpeedAspect</td></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr></table>				SpeedAspect	SPEED1	SPEED2
SpeedAspect						
SPEED1						
SPEED2						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

EC-MUX-SIG:SSS:	Signal With Two Speed Aspects and Stop command - Connection loss
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REQ:3435	between EC-MUX and OC-LS in regular states						
Given the current element state is <State> And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP <table><tr><td>State</td></tr><tr><td>STOP</td></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr></table>				State	STOP	SPEED1	SPEED2
State							
STOP							
SPEED1							
SPEED2							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:							
Satisfies:							

EC-MUX-SIG:SSS: REQ:3436	Signal With Two Speed Aspects and Stop command - Connection loss between EC-MUX and OC-LS in DARK state
Given the current element state is DARK And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP	

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3437	Signal With Two Speed Aspects and Stop command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_SPEEDn		
Given the current element state is CHANGING_TO_SPEEDn And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS within WaitForSpeed ms Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3438	Signal With Two Speed Aspects and Stop command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_STOP		
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Given the current element state is CHANGING_TO_STOP
And CC connection state is CONNECTED
And EC-MUX - OC-LS connection state is CONNECTED

When losing the connection between EC-MUX and OC-LS within WaitForSafeState ms

Then SEA shall set the current element state to UNDEFINED
And SEA shall report the element state e_undefined_es with command reply flag true to CC
And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP
And SEA shall report the no_connection diagnosis device fault message to DGP

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

3.1.3.5 Diagnosis - Signal With Two Speed Aspects and Stop command

In order to control and supervise a Signal With Two Speed Aspects and Stop command

As a SEA

I want to be able to receive an aspect diagnosis query from DGP and answer it by the DGP diagnosis values corresponding to the last diagnosis aspect message reported to DGP for the queried Signal

For requirement language see Chapter 2.8 "Requirement Language".

Given a Signal With Two Speed Aspects and Stop command

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a STOP command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_stop_es
SPEED1	e_sig_speed1_es
	e_sig_speed2_

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State	CC State
SPEED2	es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12
SIG_STOP	13
SIG_SPEED_1	16
SIG_SPEED_2	17

EC-MUX-SIG:SSS: REQ:3440	Signal With Two Speed Aspects and Stop command - aspect query - last stable state												
Given the last stable element state is <LastStable> When receiving an aspect query from DGP Then SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>LastStable</th><th>DGPMMessage</th></tr><tr><td>DARK</td><td>SIG_ DARK</td></tr><tr><td>STOP</td><td>SIG_ STOP</td></tr><tr><td>SPEED1</td><td>SIG_ SPEED_1</td></tr><tr><td>SPEED2</td><td>SIG_ SPEED_2</td></tr></table>				LastStable	DGPMMessage	DARK	SIG_ DARK	STOP	SIG_ STOP	SPEED1	SIG_ SPEED_1	SPEED2	SIG_ SPEED_2
LastStable	DGPMMessage												
DARK	SIG_ DARK												
STOP	SIG_ STOP												
SPEED1	SIG_ SPEED_1												
SPEED2	SIG_ SPEED_2												
Type:	FUR - Functional Requirement												
Status:	draft	Stability:	stable										
Target Release:	SEA SW GEN 3.0	Priority:	mandatory										
Originator:	Technics (R&D)	Allocation:	SW EC-MUX										
Verification Method:	Test	Verification Step:	module test, subit										

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Comments:	
Satisfies:	

EC-MUX-SIG:SSS: REQ:3442	Signal With Two Speed Aspects and Stop command - aspect query in UNDEFINED - OC-LS NOT_CONNECTED		
Given EC-MUX - OC-LS connection state is NOT_CONNECTED And the current element state is UNDEFINED			
When receiving an aspect query from DGP			
Then SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

3.1.4 Signal with Three Speed Aspects and Stop command

SEA Signal with three speed aspects and STOP command can reach following regular signal aspects via CC command:

- Speed1
- Speed2
- Speed3
- Stop

Following signal aspect can be reached only due to the lamp fault:

- Dark

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This generic implementation can be configured to respond to specific CC commands and OC-LS messages.

The state machine of the SEA Eulynx Signal with three speed aspects and STOP command is shown in the following figure:

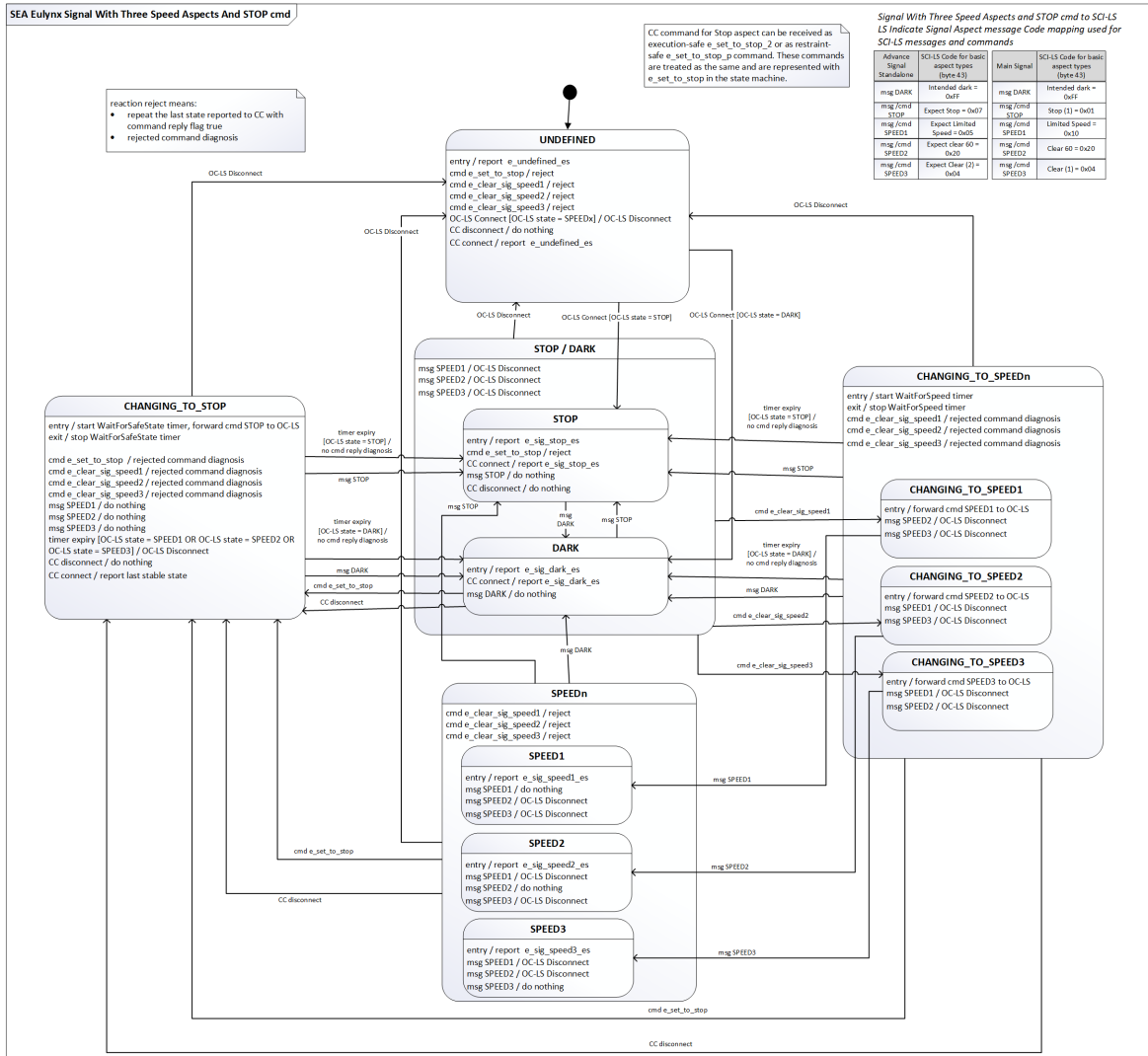


Figure 12: SEA Eulynx Signal with three speed aspects and STOP cmd - State Machine

Note To keep the STM applicable for several Signal types, names based on SEA Signal Aspects are used for SCI-LS messages and commands. A table with mappings to the SCI-LS codes has been provided for each Signal Type.

Following internal element states are defined for the SEA Eulynx Signal with three speed aspects and STOP cmd:

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Element State	Functionality	CC State	Faulty
UNDEFINED	no aspect information given, initial / default state	e_undefined_es	yes
STOP	indication for STOP aspect received; as spontaneous msg or after cmd execution	e_sig_stop_es	??
SPEED1	signal commanded to SPEED1, corresponding aspect indication received already	e_sig_speed1_es	??
SPEED2	signal commanded to SPEED2, corresponding aspect indication received already	e_sig_speed2_es	??
SPEED3	signal commanded to SPEED3, corresponding aspect indication received already	e_sig_speed3_es	??
DARK	indication for DARK aspect received; caused by lamp fault	e_sig_dark_es	yes
CHANGING_TO_SPEED1	command for SPEED1 aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history
CHANGING_TO_SPEED2	command for SPEED2 aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history
CHANGING_TO_SPEED3	command for SPEED3 aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history
CHANGING_TO_STOP	command for STOP aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history

Table 25: SEA Eulynx Signal with three speed aspects and STOP cmd - Internal Element States
Behavior in case of connection loss

When losing the connection to CC, SEA shall execute the safe command, i.e. command the signal to the STOP aspect if the last command forwarded to OC-LS was a different one.

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3.1.4.1 Advance Signal Standalone

The Advance Signal Standalone in SEA is a configuration of the SEA Eulynx Signal with three speed aspects and Stop command.

Used Cmd LS Indicate Signal Aspect and Msg LS Indicated Signal Aspect shall be addressed to the element identified by the Advance Signal Standalone operational identifier. CC command and state mappings to SCI-LS telegrams are defined in following tables.

CC Command	SEA Signal with three speed aspects	Command byte / parameter	Parameter value	OBB aspect
e_set_to_stop_2 OR e_set_to_stop_p	Stop	43 = code for basic aspect types	Expect stop = 0x07	Vorsicht
e_clear_sig_speed1	Speed1	43 = code for basic aspect types	Expect limited speed = 0x05	Hauptsignal Frei 40
e_clear_sig_speed2	Speed2	43 = code for basic aspect types	Expect clear 60 = 0x20	Hauptsignal Frei 60
e_clear_sig_speed3	Speed3	43 = code for basic aspect types	Expect clear (2) = 0x04 (*)	Hauptsignal Frei

Table 26: Advance Signal Standalone - Mapping CC command (CC to EC-MUX) to SCI-LS Cmd Indicate Signal Aspect (EC-MUX to OC-LS)

(*) TODO: Details for dependencies between different signal types at the same signal mast to be defined in a later project phase.

The SCI-LS Cmd Indicate Signal Aspect irrelevant and relevant parameters for the Advance Signal Standalone are listed in the table *Eulynx Signal Mast configuration*.

Message byte / parameter	Parameter value	SEA Signal with three speed aspects	OBB aspect	CC State
43 = code for basic aspect types	Expect stop = 0x07	Stop	Vorsicht	e_sig_stop_es
43 = code for basic aspect types	Expect limited speed = 0x05	Speed 1	Hauptsignal Frei 40	e_sig_speed1_es
43 = code for basic aspect types	Expect clear 60 = 0x20	Speed 2	Hauptsignal Frei 60	e_sig_speed2_es
43 = code for basic aspect types	Expect clear (2) = 0x04	Speed 3	Hauptsignal Frei	e_sig_speed3_es
43 = code for basic aspect types	Intended dark = 0xFF	Dark	(*)	e_sig_dark_es

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Table 27: Advance Signal Standalone - Mapping SCI-LS Msg Indicated Signal Aspect (OC-LS to EC-MUX) to gen_elem_stat (EC-MUX to CC)

Remark: (*) The aspect defined in Signal aspect table [9] as “Intended dark” cannot be commanded, it can be reached only due to a lamp fault.

For default values of the remaining parameters (byte 44 - 60) of the Msg Indicated Signal Aspect see table for default values of the corresponding command parameters.

3.1.4.2 Main Signal

The Main Signal in SEA is a configuration of the SEA Eulynx Signal with three speed aspects and Stop command.

Used Cmd LS Indicate Signal Aspect and Msg LS Indicated Signal Aspect shall be addressed to the element identified by the Main Signal operational identifier. CC command and state mappings to SCI-LS telegrams are defined in following tables.

CC Command	SEA Signal with three speed aspects	Command byte / parameter	Parameter value	OBB aspect
e_set_to_stop_2 OR e_set_to_stop_p	Stop	43 = code for basic aspect types	Stop (1) = 0x01	Halt
e_clear_sig_speed1	Speed1	43 = code for basic aspect types	Limited speed = 0x10	Frei 40
e_clear_sig_speed2	Speed2	43 = code for basic aspect types	Clear 60 = 0x20	Frei 60
e_clear_sig_speed3	Speed3	43 = code for basic aspect types	Clear (1) = 0x04 (*)	Frei

Table 28: Main Signal - Mapping CC command (CC to EC-MUX) to SCI-LS Cmd Indicate Signal Aspect (EC-MUX to OC-LS)

(*) TODO: Details for dependencies between different signal types at the same signal mast to be defined in a later project phase.

The SCI-LS Cmd Indicate Signal Aspect irrelevant and relevant parameters for the Main Signal are listed in the table *Eulynx Signal Mast configuration*.

Message byte / parameter	Parameter value	SEA Signal with three speed aspects	OBB aspect	CC State
43 = code for basic aspect types	Stop (1) = 0x01	Stop	Halt	e_sig_stop_es
43 = code for basic aspect types	Limited speed = 0x10	Speed 1	Frei 40	e_sig_speed1_es

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43 = code for basic aspect types	Clear 60 = 0x20	Speed 2	Frei 60	e_sig_speed2_es
43 = code for basic aspect types	Clear (1) = 0x04	Speed 3	Frei	e_sig_speed3_es
43 = code for basic aspect types	Intended dark = 0xFF	Dark	(*)	e_sig_dark_es

Table 29: Main Signal - Mapping SCI-LS Msg Indicated Signal Aspect (OC-LS to EC-MUX) to gen_elem_stat (EC-MUX to CC)

Remark: (*) The aspect defined in Signal aspect table [9] as "Intended dark" cannot be commanded, it can be reached only due to a lamp fault.

The aspect defined in Signal aspect table [9] as "Shunting Allowed" (0x02) is received in the byte 43 but is handled by the Shunting Signal, not the Main Signal.

For default values of the remaining parameters (byte 44 - 60) of the Msg Indicated Signal Aspect see table for default values of the corresponding command parameters.

3.1.4.3 Operational - Signal With Three Speed Aspects and Stop command

In order to control and supervise a Signal With Three Speed Aspects and Stop command
As a SEA

I want to be able to receive commands from CC, translate them into SCI-LS commands and forward them to OC-LS and to receive SCI-LS state messages from OC-LS, translate them into signal state changes and report them to CC.

For requirement language see Chapter 2.8 "Requirement Language".

CHANGING_TO_SPEEDn and CHANGING_TO_STOP element states are not considered in the mapping between element state and CC State because when reaching these states, nothing is reported to CC.

WaitForSpeed and WaitForSafeState are configurable timers for receiving a command reply from OC-LS. WaitForSpeed is used when waiting for a reply to a SPEED command (e_clear_sig_speed1, e_clear_sig_speed2, e_clear_sig_speed3), its default value is 2000ms. WaitForSafeState is used when waiting for a reply to a STOP command (e_set_to_stop_2, e_set_to_stop_p), its default value is 1000ms.

Given a Signal With Three Speed Aspects and Stop command

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a STOP command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_stop_es

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State	CC State
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es
SPEED3	e_sig_speed3_es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12
SIG_STOP	13
SIG_SPEED_1	16
SIG_SPEED_2	17
SIG_SPEED_3	18

EC-MUX-SIG:SSS: REQ:3092	Signal With Three Speed Aspects and Stop command - Initial state		
When initializing the element			
Then SEA shall set the current element state to UNDEFINED			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test
Comments:			
Satisfies:			

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EC-MUX-SIG:SSS: REQ:3095	Signal With Three Speed Aspects and Stop command - reject commands in UNDEFINED state								
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED When receiving the command <Command> from CC Then SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_set_to_stop_p</td></tr><tr><td>e_set_to_stop_2</td></tr><tr><td>e_clear_sig_speed1</td></tr><tr><td>e_clear_sig_speed2</td></tr><tr><td>e_clear_sig_speed3</td></tr></table>				Command	e_set_to_stop_p	e_set_to_stop_2	e_clear_sig_speed1	e_clear_sig_speed2	e_clear_sig_speed3
Command									
e_set_to_stop_p									
e_set_to_stop_2									
e_clear_sig_speed1									
e_clear_sig_speed2									
e_clear_sig_speed3									
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:									
Satisfies:									

EC-MUX-SIG:SSS: REQ:3096	Signal With Three Speed Aspects and Stop command - execute SPEED command in STOP / DARK
Given the current element state is <State> When receiving the command <Command> from CC	

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Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to <NewState> to OC-LS

And SEA shall set the current element state to <ChangingToSpeedState>

State	Command	NewState	ChangingToSpeedState
STOP	e_clear_sig_speed1	SPEED1	CHANGING_TO_SPEED1
STOP	e_clear_sig_speed2	SPEED2	CHANGING_TO_SPEED2
STOP	e_clear_sig_speed3	SPEED3	CHANGING_TO_SPEED3
DARK	e_clear_sig_speed1	SPEED1	CHANGING_TO_SPEED1
DARK	e_clear_sig_speed2	SPEED2	CHANGING_TO_SPEED2
DARK	e_clear_sig_speed3	SPEED3	CHANGING_TO_SPEED3

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3097	Signal With Three Speed Aspects and Stop command - reject STOP commands in STOP
Given the current element state is STOP When receiving the command <Command> from CC Then SEA shall report the element state e_sig_stop_es with command reply flag true to CC And SEA shall report the rejected command diagnosis to DGP	

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<table><tr><th colspan="2">Command</th></tr><tr><td>e_set_to_stop_2</td><td></td></tr><tr><td>e_set_to_stop_p</td><td></td></tr></table>				Command		e_set_to_stop_2		e_set_to_stop_p	
Command									
e_set_to_stop_2									
e_set_to_stop_p									
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:	CC shall not send the STOP command, if the stop aspect is already given.								
Satisfies:									

EC-MUX-SIG:SSS: REQ:3018	Signal With Three Speed Aspects and Stop command - execute STOP commands in DARK					
Given the current element state is DARK When receiving the command <Command> from CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS And SEA shall set the current element state to CHANGING_TO_STOP <table><tr><th>Command</th></tr><tr><td>e_set_to_stop_2</td></tr><tr><td>e_set_to_stop_p</td></tr></table>				Command	e_set_to_stop_2	e_set_to_stop_p
Command						
e_set_to_stop_2						
e_set_to_stop_p						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			

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Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3098	Signal With Three Speed Aspects and Stop command - spontaneous transition between STOP / DARK														
Given the current element state is <State> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag false to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>State</th><th>NewState</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>STOP</td><td>DARK</td><td>e_sig_dark_es</td><td>SIG_DARK</td></tr><tr><td>DARK</td><td>STOP</td><td>e_sig_stop_es</td><td>SIG_STOP</td></tr></table>				State	NewState	CCState	DGPMMessage	STOP	DARK	e_sig_dark_es	SIG_DARK	DARK	STOP	e_sig_stop_es	SIG_STOP
State	NewState	CCState	DGPMMessage												
STOP	DARK	e_sig_dark_es	SIG_DARK												
DARK	STOP	e_sig_stop_es	SIG_STOP												
Type:	FUR - Functional Requirement														
Status:	draft	Stability:	stable												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory												
Originator:	Technics (R&D)	Allocation:	SW EC-MUX												
Verification Method:	Test	Verification Step:	module test, subit												
Comments:															
Satisfies:															

EC-MUX-SIG:SSS: REQ:3099	Signal With Three Speed Aspects and Stop command - execute STOP commands in SPEEDx
Given the current element state is <State>	

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When receiving the command <Command> from CC

Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS

And SEA shall set the current element state to CHANGING_TO_STOP

State	Command
SPEED1	e_set_to_stop_p
SPEED2	e_set_to_stop_p
SPEED3	e_set_to_stop_p
SPEED1	e_set_to_stop_2
SPEED2	e_set_to_stop_2
SPEED3	e_set_to_stop_2

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3100	Signal With Three Speed Aspects and Stop command - reject SPEED command in SPEEDx
Given the current element state is <State> When receiving the command <Command> from CC Then SEA shall report the element state <CCState> with command reply flag true to CC And SEA shall report the rejected command diagnosis to DGP	

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State	Command	CCState
SPEED1	e_clear_sig_speed1	e_sig_speed1_es
SPEED1	e_clear_sig_speed2	e_sig_speed1_es
SPEED1	e_clear_sig_speed3	e_sig_speed1_es
SPEED2	e_clear_sig_speed1	e_sig_speed2_es
SPEED2	e_clear_sig_speed2	e_sig_speed2_es
SPEED2	e_clear_sig_speed3	e_sig_speed2_es
SPEED3	e_clear_sig_speed1	e_sig_speed3_es
SPEED3	e_clear_sig_speed2	e_sig_speed3_es
SPEED3	e_clear_sig_speed3	e_sig_speed3_es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	CC shall not send the command SPEED, if the corresponding aspect is already given. CC shall not send any other SPEED command (downgrade / upgrade the speed) without STOP command between.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3101	Signal With Three Speed Aspects and Stop command - spontaneous transition from SPEEDx to STOP / DARK
Given the current element state is <State>	

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When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS

Then SEA shall set the current element state to <NewState>

And SEA shall report the element state <CCState> with command reply flag false to CC

And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP

State	NewState	CCState	DGPMMessage
SPEED1	STOP	e_sig_stop_es	SIG_STOP
SPEED1	DARK	e_sig_dark_es	SIG_DARK
SPEED2	STOP	e_sig_stop_es	SIG_STOP
SPEED2	DARK	e_sig_dark_es	SIG_DARK
SPEED3	STOP	e_sig_stop_es	SIG_STOP
SPEED3	DARK	e_sig_dark_es	SIG_DARK

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3102	Signal With Three Speed Aspects and Stop command - CHANGING_TO_SPEEDn - transition from CHANGING_TO_SPEEDx to SPEEDx
Given the current element state is <ChangingToSpeedState> When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS within WaitForSpeed ms	

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Then SEA shall set the current element state to <State>

And SEA shall report the element state <CCState> with command reply flag true to CC

And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP

ChangingToSpeedState	State	CCState	DGPMMessage
CHANGING_TO_SPEED1	SPEED1	e_sig_speed1_es	SIG_SPEED_1
CHANGING_TO_SPEED2	SPEED2	e_sig_speed2_es	SIG_SPEED_2
CHANGING_TO_SPEED3	SPEED3	e_sig_speed3_es	SIG_SPEED_3

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Successful SPEED command execution - commanded aspect reached.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3103	Signal With Three Speed Aspects and Stop command - CHANGING_TO_SPEEDn - reject repeated SPEED command in CHANGING_TO_SPEEDn
-------------------------------------	--

Given the current element state is CHANGING_TO_SPEEDn

When receiving the command <Command> from CC within WaitForSpeed ms

Then SEA shall report the rejected command diagnosis to DGP

Command
e_clear_sig_speed1
e_clear_sig_speed2
e_clear_sig_speed3

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	CC shall not repeat the command SPEED. CC shall not switch to another SPEED aspect without STOP command between them. The WaitForSpeed timer shall not to be re-started due to the rejected command.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3104	Signal With Three Speed Aspects and Stop command - CHANGING_TO_SPEEDn - return to STOP / DARK											
Given the current element state is CHANGING_TO_SPEEDn And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS within WaitForSpeed ms Then SEA shall set the current element state to <State> And SEA shall report the element state <CCState> with command reply flag true to CC <table><tr><th>LastStable</th><th>State</th><th>CCState</th></tr><tr><td>STOP</td><td>STOP</td><td>e_sig_stop_es</td></tr><tr><td>DARK</td><td>DARK</td><td>e_sig_dark_es</td></tr></table>				LastStable	State	CCState	STOP	STOP	e_sig_stop_es	DARK	DARK	e_sig_dark_es
LastStable	State	CCState										
STOP	STOP	e_sig_stop_es										
DARK	DARK	e_sig_dark_es										
Type:		FUR - Functional Requirement										
Status:	draft	Stability:	stable									
Target Release:	SEA SW GEN 3.0	Priority:	mandatory									
Originator:	Technics (R&D)	Allocation:	SW EC-MUX									
Verification Method:	Test	Verification Step:	module test, subit									

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Comments:	OC-LS has probably sent the message as a downgrade aspect for failed clear command. As it is not possible to forward the command corresponding to SPEEDx aspect before receiving a reply to the previous STOP, it is not necessary to continue waiting for a further reply and thus the CHANGING_TO_SPEEDx state shall be left.
Satisfies:	

EC-MUX-SIG:SSS: REQ:3105	Signal With Three Speed Aspects and Stop command - CHANGING_TO_SPEEDn - transition to STOP / DARK														
Given the current element state is CHANGING_TO_SPEEDn And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS within WaitForSpeed ms Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag true to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>LastStable</th><th>NewState</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>STOP</td><td>DARK</td><td>e_sig_dark_es</td><td>SIG_DARK</td></tr><tr><td>DARK</td><td>STOP</td><td>e_sig_stop_es</td><td>SIG_STOP</td></tr></table>				LastStable	NewState	CCState	DGPMMessage	STOP	DARK	e_sig_dark_es	SIG_DARK	DARK	STOP	e_sig_stop_es	SIG_STOP
LastStable	NewState	CCState	DGPMMessage												
STOP	DARK	e_sig_dark_es	SIG_DARK												
DARK	STOP	e_sig_stop_es	SIG_STOP												
Type:	FUR - Functional Requirement														
Status:	draft	Stability:	stable												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory												
Originator:	Technics (R&D)	Allocation:	SW EC-MUX												
Verification Method:	Test	Verification Step:	module test, subit												
Comments:	OC-LS has probably sent the message as a downgrade aspect for failed command. As it is not possible to forward the command corresponding to SPEED1 aspect before receiving a reply to the previous STOP, it is not necessary to continue waiting for a further reply and thus the CHANGING_TO_SPEED1 state shall be left. The diagnosis aspect message is sent only if the last stable state is different than the OC-LS reply message.														

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Satisfies:	
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EC-MUX-SIG:SSS: REQ:3140	Signal With Three Speed Aspects and Stop command - CHANGING_TO_SPEEDn - timeout								
Given the current element state is CHANGING_TO_SPEEDn And the last stable element state is <LastStable> When the current state of the element has been CHANGING_TO_SPEEDn for WaitForSpeed ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x". to DGP And SEA shall set the current element state to <LastStable> And SEA shall report the element state <CCState> with command reply flag true to CC <table><tr><th>LastStable</th><th>CCState</th></tr><tr><td>STOP</td><td>e_sig_dark_es</td></tr><tr><td>DARK</td><td>e_sig_stop_es</td></tr></table>				LastStable	CCState	STOP	e_sig_dark_es	DARK	e_sig_stop_es
LastStable	CCState								
STOP	e_sig_dark_es								
DARK	e_sig_stop_es								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:	Regularly the state CHANGING_TO_SPEEDn will be left after WaitForSpeed ms. No safety reaction OC Disconnection is performed as this is not a safety-critical command. Instead, SEA shall generate a diagnosis and try to fallback to original STOP/DARK.								
Satisfies:									

EC-MUX-SIG:SSS: REQ:3143	Signal With Three Speed Aspects and Stop command - CHANGING_TO_SPEEDn - execute STOP command
Given the current element state is CHANGING_TO_SPEEDn	

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When receiving the command <Command> from CC within WaitForSpeed ms

Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS

And SEA shall set the current element state to CHANGING_TO_STOP

Command
e_set_to_stop_p
e_set_to_stop_2

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	If the STOP command reaches the OC-LS before the SPEEDx command is answered, then it is possible that OC-LS will reply to none of both commands, because the last commanded aspect is still given.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3107	Signal With Three Speed Aspects and Stop command - CHANGING_TO_STOP - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state SPEEDx										
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag true to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>LastStable</th><th>NewState</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>SPEED1</td><td>STOP</td><td>e_sig_stop_ es</td><td>SIG_STOP</td></tr></table>				LastStable	NewState	CCState	DGPMMessage	SPEED1	STOP	e_sig_stop_ es	SIG_STOP
LastStable	NewState	CCState	DGPMMessage								
SPEED1	STOP	e_sig_stop_ es	SIG_STOP								

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LastStable	NewState	CCState	DGPMessage
SPEED1	DARK	e_sig_dark_es	SIG_DARK
SPEED2	STOP	e_sig_stop_es	SIG_STOP
SPEED2	DARK	e_sig_dark_es	SIG_DARK
SPEED3	STOP	e_sig_stop_es	SIG_STOP
SPEED3	DARK	e_sig_dark_es	SIG_DARK

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Successful stop command execution - commanded aspect / downgrade aspect reached.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3111	Signal With Three Speed Aspects and Stop command - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state STOP / DARK									
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to <State> And SEA shall report the element state <CCState> with command reply flag true to CC <table><tr><th>LastStable</th><th>State</th><th>CCState</th></tr><tr><td>STOP</td><td>STOP</td><td>e_sig_stop_ es</td></tr><tr><td></td><td></td><td>e_sig_dark_ es</td></tr></table>		LastStable	State	CCState	STOP	STOP	e_sig_stop_ es			e_sig_dark_ es
LastStable	State	CCState								
STOP	STOP	e_sig_stop_ es								
		e_sig_dark_ es								

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LastStable	State	CCState
DARK	DARK	es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	After receiving the STOP / DARK as reply to the STOP command, the state will be reported to the CC with command reply flag true.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3112	Signal With Three Speed Aspects and Stop command - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state DARK / STOP														
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag true to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>LastStable</th><th>NewState</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>DARK</td><td>STOP</td><td>e_sig_stop_ es</td><td>SIG_STOP</td></tr><tr><td>STOP</td><td>DARK</td><td>e_sig_dark_ es</td><td>SIG_DARK</td></tr></table>				LastStable	NewState	CCState	DGPMMessage	DARK	STOP	e_sig_stop_ es	SIG_STOP	STOP	DARK	e_sig_dark_ es	SIG_DARK
LastStable	NewState	CCState	DGPMMessage												
DARK	STOP	e_sig_stop_ es	SIG_STOP												
STOP	DARK	e_sig_dark_ es	SIG_DARK												
Type:	FUR - Functional Requirement														
Status:	draft	Stability:	stable												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory												
Originator:	Technics (R&D)	Allocation:	SW EC-MUX												
Verification Method:	Test	Verification Step:	module test, subit												

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Comments:	After receiving the STOP / DARK as reply to the STOP command, the state will be reported to the CC with command reply flag true; aspect diagnosis is sent also as the state is changed.
Satisfies:	

EC-MUX-SIG:SSS: REQ:3108	Signal With Three Speed Aspects and Stop command - CHANGING_TO_STOP - reject commands in CHANGING_TO_STOP								
Given the current element state is CHANGING_TO_STOP When receiving the command <Command> from CC within WaitForSafeState ms Then SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_set_to_stop_2</td></tr><tr><td>e_set_to_stop_p</td></tr><tr><td>e_clear_sig_speed1</td></tr><tr><td>e_clear_sig_speed2</td></tr><tr><td>e_clear_sig_speed3</td></tr></table>				Command	e_set_to_stop_2	e_set_to_stop_p	e_clear_sig_speed1	e_clear_sig_speed2	e_clear_sig_speed3
Command									
e_set_to_stop_2									
e_set_to_stop_p									
e_clear_sig_speed1									
e_clear_sig_speed2									
e_clear_sig_speed3									
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:	CC shall not send a clear command if there is STOP command not answered yet. CC shall not repeat the command STOP. The WaitForSafeState timer shall not to be re-started due to the rejected command.								
Satisfies:									

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EC-MUX-SIG:SSS: REQ:3141	Signal With Three Speed Aspects and Stop command - CHANGING_TO_STOP - timeout when OC-LS SPEEDx - OC Disconnection						
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable> When the current state of the element has been CHANGING_TO_STOP for WaitForSafeState ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP <table><tr><td>LastStable</td></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr><tr><td>SPEED3</td></tr></table>				LastStable	SPEED1	SPEED2	SPEED3
LastStable							
SPEED1							
SPEED2							
SPEED3							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:	Regularly the state CHANGING_TO_STOP will be left after WaitForSafeState ms. Safety reaction OC Disconnection is applicable only if the OC-LS cannot reach the STOP or DARK aspect and stays in SPEEDx. See also EC-MUX-SIG:SSS:REQ:3142.						
Satisfies:							

EC-MUX-SIG:SSS: REQ:3142	Signal With Three Speed Aspects and Stop command - CHANGING_TO_STOP - timeout when OC-LS DARK / STOP - reach DARK / STOP
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Given the current element state is CHANGING_TO_STOP
And the last stable element state is <LastStable>

When the current state of the element has been CHANGING_TO_STOP for WaitForSafeState ms

Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP

And SEA shall set the current element state to <LastStable> after WaitForSafeState ms

And SEA shall report the element state <CCState> with command reply flag true to CC

LastStable	CCState
STOP	e_sig_stop_ es
DARK	e_sig_dark_ es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Regularly the state CHANGING_TO_STOP will be left after WaitForSafeState ms. OC-LS state can be STOP or DARK at this point only when a STOP command has been executed after a SPEED command, and no reply has been received for either command. This is an allowed scenario so no safety reaction OC Disconnection is performed.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3110	Signal With Three Speed Aspects and Stop command - CHANGING_TO_STOP - ignore SPEED message
Given the current element state is CHANGING_TO_STOP When receiving the LS Indicated Signal Aspect message with Code corresponding to <SpeedAspect> from OC-LS within WaitForSafeState ms Then SEA shall do nothing	

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<table><tr><th>SpeedAspect</th></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr><tr><td>SPEED3</td></tr></table>				SpeedAspect	SPEED1	SPEED2	SPEED3
SpeedAspect							
SPEED1							
SPEED2							
SPEED3							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:	It is possible that OC-LS has sent the message corresponding to SPEED as reply to the previous SPEED command not answered yet. Waiting for the reply to the STOP command is continued by keeping the current state CHANGING_TO_STOP. After timer expiry, if the expected STOP reply is not received, the element state is changed to STOP / DARK or a Safety Reaction OC Disconnection is performed, depending on the last received OC-LS state.						
Satisfies:							

EC-MUX-SIG:SSS: REQ:3114	Signal With Three Speed Aspects and Stop command - Disconnection on spontaneous SPEED messages in STOP / DARK / SPEED						
Given the current element state is <State> When receiving the LS Indicated Signal Aspect message with Code corresponding to <SpeedAspect> from OC-LS Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: Spontaneous <SpeedAspect> reported for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP <table border="1"> <tr> <th>State</th><th>SpeedAspect</th></tr> <tr> <td>STOP</td><td>SPEED1</td></tr> <tr> <td>STOP</td><td>SPEED2</td></tr> </table>		State	SpeedAspect	STOP	SPEED1	STOP	SPEED2
State	SpeedAspect						
STOP	SPEED1						
STOP	SPEED2						

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State	SpeedAspect
STOP	SPEED3
DARK	SPEED1
DARK	SPEED2
DARK	SPEED3
SPEED1	SPEED2
SPEED1	SPEED3
SPEED2	SPEED1
SPEED2	SPEED3
SPEED3	SPEED1
SPEED3	SPEED2

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3106	Signal With Three Speed Aspects and Stop command - CHANGING_TO_SPEEDx - Disconnection on unexpected command reply
Given the current element state is <ChangingToSpeedState> When receiving the LS Indicated Signal Aspect message with Code corresponding to <SpeedMsg> from OC-LS within WaitForSpeed ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: Cmd <SpeedCmd> replied by Msg <SpeedMsg> for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection	

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And SEA shall set the current element state to UNDEFINED

And SEA shall report the element state e_undefined_es with command reply flag true to CC

And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP

And SEA shall report the no_connection diagnosis device fault message to DGP

ChangingToSpeedState	SpeedMsg	SpeedCmd
CHANGING_TO_SPEED1	SPEED2	SPEED1
CHANGING_TO_SPEED1	SPEED3	SPEED1
CHANGING_TO_SPEED2	SPEED1	SPEED2
CHANGING_TO_SPEED2	SPEED3	SPEED2
CHANGING_TO_SPEED3	SPEED1	SPEED3
CHANGING_TO_SPEED3	SPEED2	SPEED3

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3115	Signal With Three Speed Aspects and Stop command - Ignore repeated messages
Given the current element state is <State> When receiving the LS Indicated Signal Aspect message with Code corresponding to <SpeedAspect> from OC-LS Then SEA shall do nothing	

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<table><tr><th>State</th><th>SpeedAspect</th></tr><tr><td>STOP</td><td>STOP</td></tr><tr><td>SPEED1</td><td>SPEED1</td></tr><tr><td>SPEED2</td><td>SPEED2</td></tr><tr><td>SPEED3</td><td>SPEED3</td></tr><tr><td>DARK</td><td>DARK</td></tr></table>				State	SpeedAspect	STOP	STOP	SPEED1	SPEED1	SPEED2	SPEED2	SPEED3	SPEED3	DARK	DARK
State	SpeedAspect														
STOP	STOP														
SPEED1	SPEED1														
SPEED2	SPEED2														
SPEED3	SPEED3														
DARK	DARK														
Type:		FUR - Functional Requirement													
Status:		draft	Stability: stable												
Target Release:		SEA SW GEN 3.0	Priority: mandatory												
Originator:		Technics (R&D)	Allocation: SW EC-MUX												
Verification Method:		Test	Verification Step: module test, subit												
Comments:															
Satisfies:															

EC-MUX-SIG:SSS: REQ:3146	Signal With Three Speed Aspects and Stop command - Receive unexpected DARK commands from CC					
When receiving the command <Command> from CC Then SEA shall execute EC-MUX safety reaction Deactivation <table><tr><th>Command</th></tr><tr><td>e_set_to_dark_2</td></tr><tr><td>e_set_to_dark_p</td></tr></table>				Command	e_set_to_dark_2	e_set_to_dark_p
Command						
e_set_to_dark_2						
e_set_to_dark_p						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			

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Comments:	
Satisfies:	

3.1.4.4 Connection - Signal With Three Speed Aspects and Stop command

In order to control and supervise a Signal With Three Speed Aspects and Stop command

As a SEA

I want to be able to supervise connections and when changes occur then

- on CC connection establishment report the current state to CC
- on CC connection loss execute the safe command STOP if the last state is DARK
- on CC connection loss execute the safe command STOP if the last command forwarded to OC-LS was SPEED1 / SPEED2 / SPEED3 and it was not answered by STOP / DARK
- on OC-LS connection loss take on the default state UNDEFINED
- on OC-LS connection establishment transition to DARK / STOP state as the OC-LS element state
- report the connection state changes to CC and DGP

For requirement language see Chapter 2.8 "Requirement Language".

CHANGING_TO_SPEEDn and CHANGING_TO_STOP element states are not considered in the mapping between element state and CC State because when reaching these states, nothing is reported to CC.

WaitForSpeed and WaitForSafeState are configurable timers for receiving a command reply from OC-LS.

WaitForSpeed is used when waiting for a reply to a SPEED command (e_clear_sig_speed1, e_clear_sig_speed2, e_clear_sig_speed3), its default value is 2000ms.

WaitForSafeState is used when waiting for a reply to a STOP command (e_set_to_stop_2, e_set_to_stop_p), its default value is 1000ms.

Given a Signal With Three Speed Aspects and Stop command

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a STOP command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_stop_es
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es
SPEED3	e_sig_speed3_es

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State	CC State
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12
SIG_STOP	13
SIG_SPEED_1	16
SIG_SPEED_2	17
SIG_SPEED_3	18

EC-MUX-SIG:SSS: REQ:3116	Signal With Three Speed Aspects and Stop command - [Re-]Connecting to CC - OC-LS CONNECTED - General inquiry in stable states								
Given the current element state is <State> And CC connection state is NOT_CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When establishing the connection to CC Then SEA shall report the element state <CCState> with command reply flag false to CC <table><tr><th>State</th><th>CCState</th></tr><tr><td>STOP</td><td>e_sig_stop_es</td></tr><tr><td>DARK</td><td>e_sig_dark_es</td></tr></table>				State	CCState	STOP	e_sig_stop_es	DARK	e_sig_dark_es
State	CCState								
STOP	e_sig_stop_es								
DARK	e_sig_dark_es								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						

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Comments:	[Re-]Connecting in SPEEDx state is impossible.
Satisfies:	

EC-MUX-SIG:SSS: REQ:3117	Signal With Three Speed Aspects and Stop command - [Re-]Connecting to CC - OC-LS CONNECTED - General inquiry in CHANGING_TO_STOP - no message after CC disconnect
-------------------------------------	--

Given the current element state is CHANGING_TO_STOP
And the last stable element state is <LastStable>
And CC connection state is NOT_CONNECTED
And EC-MUX - OC-LS connection state is CONNECTED

When establishing the connection to CC within WaitForSafeState ms

Then SEA shall report the element state <CCState> with command reply flag false to CC

LastStable	CCState
STOP	e_sig_stop_es
DARK	e_sig_dark_es
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es
SPEED3	e_sig_speed3_es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
	Re-connecting to CC in CHANGING_TO_STOP state is impossible in real operation, as re-connecting will regularly take much longer time than the configured WaitForSafeState. Due to the execution of the safe command [Re-]Connecting in CHANGING_TO_STOP state is possible only with last forwarded command STOP, but depending on the previous events the last state reported to CC		

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Comments:	can be different. Following initial states are covered with this requirement: - CHANGING_TO_STOP with last stable element state SPEED1/SPEED2 /SPEED3 - CHANGING_TO_STOP with last stable element state DARK/STOP - CHANGING_TO_STOP with last stable element state DARK/STOP and SPEED1/SPEED2/SPEED3 received from OC-LS in CHANGING_TO_STOP
Satisfies:	

EC-MUX-SIG:SSS: REQ:3139	Signal With Three Speed Aspects and Stop command - [Re-]Connecting to CC - OC-LS NOT_CONNECTED		
Given the current element state is UNDEFINED And CC connection state is NOT_CONNECTED And EC-MUX - OC-LS connection state is NOT_CONNECTED When establishing the connection to CC Then SEA shall report the element state e_undefined_es with command reply flag false to CC			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3120	Signal With Three Speed Aspects and Stop command - Connection loss to CC - execute safe command in SPEEDx
Given the current element state is <State> And CC connection state is CONNECTED When losing the connection to CC	

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Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS

And SEA shall set the current element state to CHANGING_TO_STOP

State
SPEED1
SPEED2
SPEED3

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3118	Signal With Three Speed Aspects and Stop command - Connection loss to CC - do nothing in STOP / UNDEFINED
-------------------------------------	--

Given the current element state is <State>

And CC connection state is CONNECTED

When losing the connection to CC

Then SEA shall do nothing

State
STOP
UNDEFINED

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX

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Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3150	Signal With Three Speed Aspects and Stop command - Connection loss to CC - execute safe command in DARK		
Given the current element state is DARK And CC connection state is CONNECTED When losing the connection to CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS And SEA shall set the current element state to CHANGING_TO_STOP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3123	Signal With Three Speed Aspects and Stop command - Connection loss to CC - CHANGING_TO_SPEEDn
Given the current element state is CHANGING_TO_SPEEDn And CC connection state is CONNECTED When losing the connection to CC within WaitForSpeed ms Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS And SEA shall set the current element state to CHANGING_TO_STOP	
Type:	FUR - Functional Requirement

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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3122	Signal With Three Speed Aspects and Stop command - Connection loss to CC - CHANGING_TO_STOP - do nothing		
Given the current element state is CHANGING_TO_STOP And CC connection state is CONNECTED When losing the connection to CC within WaitForSafeState ms Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3093	Signal With Three Speed Aspects and Stop command - [Re-]Connecting between EC-MUX and OC-LS in state STOP
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED And OC-LS state is STOP When establishing the connection between EC-MUX and OC-LS	

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Then SEA shall set the current element state to STOP And SEA shall report the element state e_sig_stop_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_STOP to DGP And SEA shall report the connection_ok diagnosis device fault repair message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3094	Signal With Three Speed Aspects and Stop command - [Re-]Connecting between EC-MUX and OC-LS in state DARK		
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED And OC-LS state is DARK When establishing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to DARK And SEA shall report the element state e_sig_dark_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_DARK to DGP And SEA shall report the connection_ok diagnosis device fault repair message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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EC-MUX-SIG:SSS: REQ:3147	Signal With Three Speed Aspects and Stop command - [Re-]Connecting between EC-MUX and OC-LS in state SPEEDx						
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED And OC-LS state is <SpeedAspect> When establishing the connection between EC-MUX and OC-LS Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: OC-LS connect with <SpeedAspect> for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection <table><tr><td>SpeedAspect</td></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr><tr><td>SPEED3</td></tr></table>				SpeedAspect	SPEED1	SPEED2	SPEED3
SpeedAspect							
SPEED1							
SPEED2							
SPEED3							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:							
Satisfies:							

EC-MUX-SIG:SSS: REQ:3127	Signal With Three Speed Aspects and Stop command - Connection loss between EC-MUX and OC-LS in regular states
Given the current element state is <State> And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS	

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Then SEA shall set the current element state to UNDEFINED
And SEA shall report the element state e_undefined_es with command reply flag false to CC
And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP
And SEA shall report the no_connection diagnosis device fault message to DGP

State
STOP
SPEED1
SPEED2
SPEED3

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3128	Signal With Three Speed Aspects and Stop command - Connection loss between EC-MUX and OC-LS in DARK state		
Given the current element state is DARK And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory

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Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3129	Signal With Three Speed Aspects and Stop command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_SPEEDn		
Given the current element state is CHANGING_TO_SPEEDn And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS within WaitForSpeed ms Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3144	Signal With Three Speed Aspects and Stop command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_STOP
Given the current element state is CHANGING_TO_STOP And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS within WaitForSafeState ms	

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Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

3.1.4.5 Diagnosis - Signal With Three Speed Aspects and Stop command

In order to control and supervise a Signal With Three Speed Aspects and Stop command

As a SEA

I want to be able to receive an aspect diagnosis query from DGP and answer it by the DGP diagnosis values corresponding to the last diagnosis aspect message reported to DGP for the queried Signal

For requirement language see Chapter 2.8 "Requirement Language".

Given a Signal With Three Speed Aspects and Stop command

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a STOP command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_stop_es
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es
SPEED3	e_sig_speed3_es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

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And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12
SIG_STOP	13
SIG_SPEED_1	16
SIG_SPEED_2	17
SIG_SPEED_3	18

EC-MUX-SIG:SSS: REQ:3138	Signal With Three Speed Aspects and Stop command - aspect query - last stable state														
Given the last stable element state is <LastStable> When receiving an aspect query from DGP Then SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>LastStable</th><th>DGPMMessage</th></tr><tr><td>DARK</td><td>SIG_ DARK</td></tr><tr><td>STOP</td><td>SIG_ STOP</td></tr><tr><td>SPEED1</td><td>SIG_ SPEED_ 1</td></tr><tr><td>SPEED2</td><td>SIG_ SPEED_ 2</td></tr><tr><td>SPEED3</td><td>SIG_ SPEED_ 3</td></tr></table>				LastStable	DGPMMessage	DARK	SIG_ DARK	STOP	SIG_ STOP	SPEED1	SIG_ SPEED_ 1	SPEED2	SIG_ SPEED_ 2	SPEED3	SIG_ SPEED_ 3
LastStable	DGPMMessage														
DARK	SIG_ DARK														
STOP	SIG_ STOP														
SPEED1	SIG_ SPEED_ 1														
SPEED2	SIG_ SPEED_ 2														
SPEED3	SIG_ SPEED_ 3														
Type:	FUR - Functional Requirement														
Status:	draft	Stability:	stable												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory												
Originator:	Technics (R&D)	Allocation:	SW EC-MUX												
Verification Method:	Test	Verification Step:	module test, subit												
Comments:															

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Satisfies:	
------------	--

EC-MUX-SIG:SSS: REQ:3145	Signal With Three Speed Aspects and Stop command - aspect query in UNDEFINED - OC-LS NOT_CONNECTED		
Given EC-MUX - OC-LS connection state is NOT_CONNECTED And the current element state is UNDEFINED When receiving an aspect query from DGP Then SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

3.1.5 Signal with One Speed Aspect and Dark command

SEA Eulynx Signal with one speed aspect and Dark command can reach following regular signal aspects via CC command:

- Speed1
- Dark

This generic implementation can be configured to respond to specific CC commands and OC-LS messages.

The state machine of the SEA Eulynx Signal with one speed aspect and DARK command is shown in the following figure:

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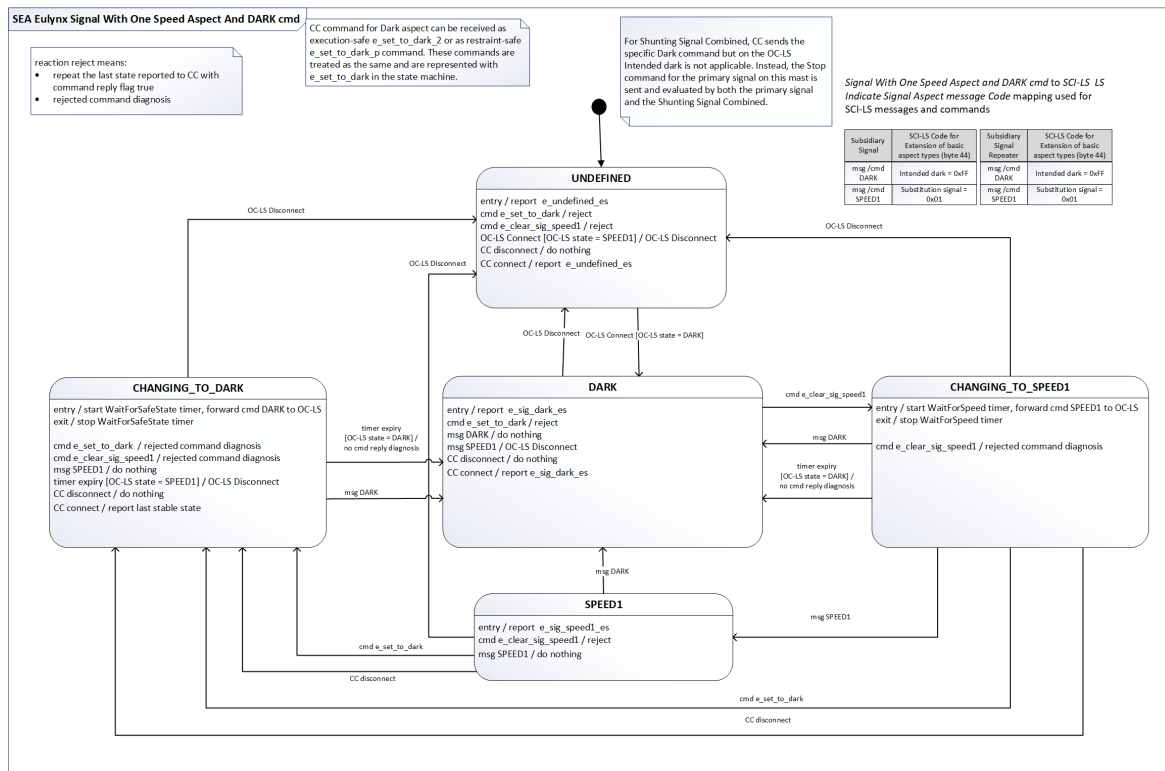


Figure 13: SEA Eulynx Signal with one speed aspect and DARK cmd - State Machine

Note To keep the STM applicable for several Signal types, names based on SEA Signal Aspects are used for SCI-LS messages and commands. A table with mappings to the SCI-LS codes has been provided for each Signal Type.

Following internal element states are defined for the SEA Eulynx Signal with one speed aspect and DARK cmd:

Element State	Functionality	CC State	Faulty
UNDEFINED	no aspect information given, initial / default state	e_undefined_es	yes
SPEED1	signal commanded to SPEED1, corresponding aspect indication received already	e_sig_speed1_es	??
DARK	indication for DARK aspect received; caused by lamp fault or after cmd execution	e_sig_dark_es	yes
CHANGING_TO_SPEED1	command for SPEED1 aspect forwarded to OC-LS, waiting for the reply;	none on entry	no information, depending on

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	timed state		history
CHANGING_TO_DARK	command for DARK aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history

Table 30: SEA Eulynx Signal with one speed aspect and DARK cmd - Internal Element States
Behavior in case of connection loss

When losing the connection to CC, SEA shall execute the safe command, i.e. command the signal to the DARK aspect if the last command forwarded to OC-LS was a different one.

3.1.5.1 Subsidiary / Caution Signal

The Subsidiary Signal in SEA is a configuration of the SEA Eulynx Signal with one speed aspect and Dark command.

Used Cmd LS Indicate Signal Aspect and Msg LS Indicated Signal Aspect shall be addressed to the element identified by the Subsidiary Signal operational identifier. CC command and state mappings to SCI-LS telegrams are defined in following tables.

CC Command	SEA Signal with one speed aspect	Command byte / parameter	Parameter value	OBB aspect
e_set_to_dark_2 OR e_set_to_dark_p	Dark	44 = code for extension of basic aspect types	Intended Dark = 0xFF	Dark
e_clear_sig_speed1	Speed1	44 = code for extension of basic aspect types	Substitution Signal = 0x01	Ersatzsignal

Table 31: Subsidiary Signal - Mapping CC command (CC to EC-MUX) to SCI-LS Cmd Indicate Signal Aspect (EC-MUX to OC-LS)

The SCI-LS Cmd Indicate Signal Aspect irrelevant and relevant parameters for the Subsidiary Signal are listed in the table *Eulynx Signal Mast configuration*.

Message byte / parameter	Parameter value	SEA Signal with one speed aspect Aspect	OBB aspect	CC State
44 = code for extension of basic aspect types	Substitution Signal = 0x01	Speed 1	Ersatzsignal	e_sig_speed1_es
44 = code for extension of basic aspect types	Intended dark = 0xFF	Dark	Dark	e_sig_dark_es

Table 32: Subsidiary Signal - Mapping SCI-LS Msg Indicated Signal Aspect (OC-LS to EC-MUX) to gen_elem_stat (EC-MUX to CC)

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For default values of the remaining parameters (byte 43, 45 - 60) of the Msg Indicated Signal Aspect see table for default values of the corresponding command parameters.

3.1.5.2 Subsidiary Signal Repeater

The Subsidiary Signal Repeater in SEA is a configuration of the SEA Eulynx Signal with one speed aspect and Dark command.

Used Cmd LS Indicate Signal Aspect and Msg LS Indicated Signal Aspect shall be addressed to the element identified by the Subsidiary Signal operational identifier. CC command and state mappings to SCI-LS telegrams are defined in following tables.

CC Command	SEA Signal with one speed aspect	Command byte / parameter	Parameter value	OBB aspect
e_set_to_dark_2 OR e_set_to_dark_p	Dark	44 = code for extension of basic aspect types	Intended Dark = 0xFF	Dark
e_clear_sig_speed1	Speed1	44 = code for extension of basic aspect types	Substitution Signal = 0x01	Ersatzsignal

Table 33: Subsidiary Signal Repeater - Mapping CC command (CC to EC-MUX) to SCI-LS Cmd Indicate Signal Aspect (EC-MUX to OC-LS)

The SCI-LS Cmd Indicate Signal Aspect irrelevant and relevant parameters for the Subsidiary Signal Repeater are listed in the table *Eulynx Signal Mast configuration*.

Message byte / parameter	Parameter value	SEA Signal with one speed aspect Aspect	OBB aspect	CC State
44 = code for extension of basic aspect types	Substitution Signal = 0x01	Speed 1	Ersatzsignal	e_sig_speed1_es
44 = code for extension of basic aspect types	Intended dark = 0xFF	Dark	Dark	e_sig_dark_es

Table 34: Subsidiary Signal Repeater - Mapping SCI-LS Msg Indicated Signal Aspect (OC-LS to EC-MUX) to gen_elem_stat (EC-MUX to CC)

For default values of the remaining parameters (byte 43, 45 - 60) of the Msg Indicated Signal Aspect see table for default values of the corresponding command parameters.

3.1.5.3 Operational - Signal With One Speed Aspect and Dark command

In order to control and supervise a Signal With One Speed Aspect and Dark command
As a SEA

I want to be able to receive commands from CC, translate them into SCI-LS commands and forward them to OC-LS and to receive SCI-LS state messages from OC-LS, translate them into signal state changes

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and report them to CC.

For requirement language see Chapter 2.8 "Requirement Language".

CHANGING_TO_SPEED1 and CHANGING_TO_DARK element states are not considered in the mapping between element state and CC State because when reaching these states, nothing is reported to CC.

WaitForSpeed and WaitForSafeState are configurable timers for receiving a command reply from OC-LS. WaitForSpeed is used when waiting for a reply to a SPEED command (e_clear_sig_speed1), its default value is 2000ms.

WaitForSafeState is used when waiting for a reply to a DARK command (e_set_to_dark_2, e_set_to_dark_p), its default value is 1000ms.

Given a Signal With One Speed Aspect and Dark command

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a DARK command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
SPEED1	e_sig_speed1_es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12
SIG_SPEED_1	16

EC-MUX-SIG:SSS: REQ:3200	Signal With One Speed Aspect and Dark command - Initial state		
When initializing the element			
Then SEA shall set the current element state to UNDEFINED			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3202	Signal With One Speed Aspect and Dark command - reject commands in UNDEFINED state						
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED When receiving the command <Command> from CC Then SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_set_to_dark_p</td></tr><tr><td>e_set_to_dark_2</td></tr><tr><td>e_clear_sig_speed1</td></tr></table>				Command	e_set_to_dark_p	e_set_to_dark_2	e_clear_sig_speed1
Command							
e_set_to_dark_p							
e_set_to_dark_2							
e_clear_sig_speed1							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:							
Satisfies:							

EC-MUX-SIG:SSS: REQ:3203	Signal With One Speed Aspect and Dark command - execute e_clear_sig_speed1 command in DARK
-----------------------------	--

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Given the current element state is DARK

When receiving the command e_clear_sig_speed1 from CC

Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to SPEED1 to OC-LS

And SEA shall set the current element state to CHANGING_TO_SPEED1

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3204	Signal With One Speed Aspect and Dark command - reject DARK commands in DARK
-------------------------------------	---

Given the current element state is DARK

When receiving the command <Command> from CC

Then SEA shall report the element state e_sig_dark_es with command reply flag true to CC

And SEA shall report the rejected command diagnosis to DGP

Command
e_set_to_dark_2
e_set_to_dark_p

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit

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Comments:	CC shall not send the DARK commands if the dark aspect is already given.
Satisfies:	

EC-MUX-SIG:SSS: REQ:3205	Signal With One Speed Aspect and Dark command - execute DARK command in SPEED1					
Given the current element state is SPEED1 When receiving the command <Command> from CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to CHANGING_TO_DARK <table><tr><th>Command</th></tr><tr><td>e_set_to_dark_2</td></tr><tr><td>e_set_to_dark_p</td></tr></table>				Command	e_set_to_dark_2	e_set_to_dark_p
Command						
e_set_to_dark_2						
e_set_to_dark_p						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

EC-MUX-SIG:SSS: REQ:3206	Signal With One Speed Aspect and Dark command - reject e_clear_sig_speed1 command in SPEED1
Given the current element state is SPEED1	
When receiving the command e_clear_sig_speed1 from CC	

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Then SEA shall report the element state e_sig_speed1_es with command reply flag true to CC And SEA shall report the rejected command diagnosis to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	CC shall not send the command e_clear_sig_speed1, if the speed1 aspect is already given.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3207	Signal With One Speed Aspect and Dark command - spontaneous transition from SPEED1 to DARK		
Given the current element state is SPEED1			
When receiving the LS Indicated Signal Aspect message with Code corresponding to DARK from OC-LS			
Then SEA shall set the current element state to DARK			
And SEA shall report the element state e_sig_dark_es with command reply flag false to CC			
And SEA shall report the diagnosis aspect message SIG_DARK to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3208	Signal With One Speed Aspect and Dark command - transition from CHANGING_TO_SPEED1 to SPEED1		
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Given the current element state is CHANGING_TO_SPEED1

When receiving the LS Indicated Signal Aspect message with Code corresponding to SPEED1 from OC-LS within WaitForSpeed ms

Then SEA shall set the current element state to SPEED1

And SEA shall report the element state e_sig_speed1_es with command reply flag true to CC

And SEA shall report the diagnosis aspect message SIG_SPEED_1 to DGP

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Successful command execution - commanded aspect reached.		
Satisfies:			

EC-MUX-SIG:SSS:
REQ:3209 **Signal With One Speed Aspect and Dark command - reject e_clear_sig_speed1 command in CHANGING_TO_SPEED1**

Given the current element state is CHANGING_TO_SPEED1

When receiving the command e_clear_sig_speed1 from CC within WaitForSpeed ms

Then SEA shall report the rejected command diagnosis to DGP

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	CC shall not repeat the command e_clear_sig_speed1. The WaitForSpeed timer shall not be re-started due to the rejected command.		
Satisfies:			

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EC-MUX-SIG:SSS: REQ:3210	Signal With One Speed Aspect and Dark command - CHANGING_TO_SPEED1 - return to DARK		
Given the current element state is CHANGING_TO_SPEED1			
When receiving the LS Indicated Signal Aspect message with Code corresponding to DARK from OC-LS within WaitForSpeed ms			
Then SEA shall set the current element state to DARK			
And SEA shall report the element state e_sig_dark_es with command reply flag true to CC			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	OC-LS has probably sent the message as a downgrade aspect for failed command. As it is not possible to forward the command corresponding to SPEED1 aspect before receiving a reply to the previous DARK, it is not necessary to continue waiting for a further reply and thus the CHANGING_TO_SPEED1 state shall be left.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3211	Signal With One Speed Aspect and Dark command - CHANGING_TO_SPEED1 - execute DARK command		
Given the current element state is CHANGING_TO_SPEED1 When receiving the command <Command> from CC within WaitForSpeed ms Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to CHANGING_TO_DARK			
<table><tr><th>Command</th></tr><tr><td>e_set_to_dark_2</td></tr></table>		Command	e_set_to_dark_2
Command			
e_set_to_dark_2			

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<table border="1"> <tr> <td>Command</td> <td></td> <td></td> <td></td> </tr> <tr> <td>e_set_to_dark_p</td> <td></td> <td></td> <td></td> </tr> </table>				Command				e_set_to_dark_p			
Command											
e_set_to_dark_p											
Type:	FUR - Functional Requirement										
Status:	draft	Stability:	stable								
Target Release:	SEA SW GEN 3.0	Priority:	mandatory								
Originator:	Technics (R&D)	Allocation:	SW EC-MUX								
Verification Method:	Test	Verification Step:	module test, subit								
Comments:	If the DARK command reaches the OC-LS before the SPEED1 command is answered, then it is possible that OC-LS will reply to none of both commands, because the last commanded aspect is still given.										
Satisfies:											

EC-MUX-SIG:SSS: REQ:3212	Signal With One Speed Aspect and Dark command - CHANGING_TO_SPEED1 - timeout		
Given the current element state is CHANGING_TO_SPEED1 When the current state of the element has been CHANGING_TO_SPEED1 for WaitForSpeed ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP And SEA shall set the current element state to DARK And SEA shall report the element state e_sig_dark_es with command reply flag true to CC			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Regularly the state CHANGING_TO_SPEED1 will be left after WaitForSpeed ms. No safety reaction OC Disconnection is performed as this is not a safety-critical command. Instead, SEA shall generate a diagnosis and try to fallback to original DARK.		
Satisfies:			

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EC-MUX-SIG:SSS: REQ:3213	Signal With One Speed Aspect and Dark command - transition from CHANGING_TO_DARK to DARK - from OC-LS state SPEED1		
Given the current element state is CHANGING_TO_DARK And the last stable element state is SPEED1 When receiving the LS Indicated Signal Aspect message with Code corresponding to DARK from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to DARK And SEA shall report the element state e_sig_dark_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_DARK to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Successful command execution - commanded aspect / downgrade aspect reached.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3214	Signal With One Speed Aspect and Dark command - transition from CHANGING_TO_DARK to DARK - from OC-LS state DARK		
Given the current element state is CHANGING_TO_DARK And the last stable element state is DARK When receiving the LS Indicated Signal Aspect message with Code corresponding to DARK from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to DARK And SEA shall report the element state e_sig_dark_es with command reply flag true to CC			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX

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Verification Method:	Test	Verification Step:	module test, subit
Comments:	Successful command execution - commanded aspect / downgrade aspect reached. OC-LS state can be DARK in the precondition when a DARK command has been executed after a SPEED command without a command reply.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3215	Signal With One Speed Aspect and Dark command - reject commands in CHANGING_TO_DARK						
Given the current element state is CHANGING_TO_DARK When receiving the command <Command> from CC within WaitForSafeState ms Then SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_set_to_dark_2</td></tr><tr><td>e_set_to_dark_p</td></tr><tr><td>e_clear_sig_speed1</td></tr></table>				Command	e_set_to_dark_2	e_set_to_dark_p	e_clear_sig_speed1
Command							
e_set_to_dark_2							
e_set_to_dark_p							
e_clear_sig_speed1							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:	CC shall not send a SPEED command if there is a DARK command not answered yet. CC shall not repeat a DARK command. The timer shall not to be re-started due to the rejected command.						
Satisfies:							

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EC-MUX-SIG:SSS: REQ:3216	Signal With One Speed Aspect and Dark command - CHANGING_TO_DARK - ignore SPEED1		
Given the current element state is CHANGING_TO_DARK			
When receiving the LS Indicated Signal Aspect message with Code corresponding to SPEED1 from OC-LS within WaitForSafeState ms			
Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	It is possible that OC-LS has sent the message corresponding to SPEED1 as reply to the previous SPEED1 command not answered yet. Waiting for the reply to the DARK command is continued by keeping the current state CHANGING_TO_DARK. After timer expiry, if the expected DARK reply is not received, the element state is changed to DARK or a Safety Reaction OC Disconnection is performed, depending on the last received OC-LS state.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3217	Signal With One Speed Aspect and Dark command - CHANGING_TO_DARK - timeout when OC-LS SPEED1 - OC Disconnection
Given the current element state is CHANGING_TO_DARK And the last stable element state is SPEED1 When the current state of the element has been CHANGING_TO_DARK for WaitForSafeState ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP	
Type:	FUR - Functional Requirement

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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Regularly the state CHANGING_TO_DARK will be left after WaitForSafeState ms. Safety reaction OC Disconnection is applicable only if the OC-LS cannot reach the DARK aspect and stays in SPEED1. See also EC-MUX-SIG:SSS:REQ:3218.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3218	Signal With One Speed Aspect and Dark command - CHANGING_TO_DARK - timeout when OC-LS DARK - reach DARK		
Given the current element state is CHANGING_TO_DARK And the last stable element state is DARK When the current state of the element has been CHANGING_TO_DARK for WaitForSafeState ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP And SEA shall set the current element state to DARK And SEA shall report the element state e_sig_dark_es with command reply flag true to CC			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Regularly the state CHANGING_TO_DARK will be left after WaitForSafeState ms. OC-LS state can be DARK at this point only when a DARK command has been executed after a SPEED command, and no reply has been received for either command. This is an allowed scenario so no safety reaction OC Disconnection is performed.		
Satisfies:			

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EC-MUX-SIG:SSS: REQ:3219	Signal With One Speed Aspect and Dark command - Disconnection on spontaneous aspect message SPEED1 in DARK		
Given the current element state is DARK When receiving the LS Indicated Signal Aspect message with Code corresponding to SPEED1 from OC-LS Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: Spontaneous SPEED1 reported for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3221	Signal With One Speed Aspect and Dark command - Ignore repeated messages						
Given the current element state is <State> When receiving the LS Indicated Signal Aspect message with Code corresponding to <SeaSigAspect> from OC-LS Then SEA shall do nothing <table><tr><th>State</th><th>SeaSigAspect</th></tr><tr><td>SPEED1</td><td>SPEED1</td></tr><tr><td>DARK</td><td>DARK</td></tr></table>		State	SeaSigAspect	SPEED1	SPEED1	DARK	DARK
State	SeaSigAspect						
SPEED1	SPEED1						
DARK	DARK						
Type:	FUR - Functional Requirement						

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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3238	Signal With One Speed Aspect and Dark command - Receive unexpected STOP / SPEED commands from CC							
When receiving the command <Command> from CC Then SEA shall execute EC-MUX safety reaction Deactivation <table><tr><th>Command</th></tr><tr><td>e_set_to_stop_p</td></tr><tr><td>e_set_to_stop_2</td></tr><tr><td>e_clear_sig_speed2</td></tr><tr><td>e_clear_sig_speed3</td></tr></table>				Command	e_set_to_stop_p	e_set_to_stop_2	e_clear_sig_speed2	e_clear_sig_speed3
Command								
e_set_to_stop_p								
e_set_to_stop_2								
e_clear_sig_speed2								
e_clear_sig_speed3								
Type:	FUR - Functional Requirement							
Status:	draft	Stability:	stable					
Target Release:	SEA SW GEN 3.0	Priority:	mandatory					
Originator:	Technics (R&D)	Allocation:	SW EC-MUX					
Verification Method:	Test	Verification Step:	module test, subit					
Comments:								
Satisfies:								

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3.1.5.4 Connection - Signal With One Speed Aspect and Dark command

In order to control and supervise a Signal With One Speed Aspect and Dark command
 As a SEA

I want to be able to supervise connections and when changes occur then

- on CC connection establishment report the current state to CC
- on CC connection loss execute the safe command DARK if the last command forwarded to OC-LS was SPEED1 and it was not answered by DARK
- on OC-LS connection loss take on the default state UNDEFINED
- on OC-LS connection establishment transition to DARK state as the OC-LS element state
- report the connection state changes to CC and DGP

For requirement language see Chapter 2.8 "Requirement Language".

CHANGING_TO_SPEED1 and CHANGING_TO_DARK element states are not considered in the mapping between element state and CC State because when reaching these states, nothing is reported to CC.

WaitForSpeed and WaitForSafeState are configurable timers for receiving a command reply from OC-LS. WaitForSpeed is used when waiting for a reply to a SPEED command (e_clear_sig_speed1), its default value is 2000ms.

WaitForSafeState is used when waiting for a reply to a DARK command (e_set_to_dark_2, e_set_to_dark_p), its default value is 1000ms.

Given a Signal With One Speed Aspect and Dark command

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a DARK command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
SPEED1	e_sig_speed1_es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12
SIG_SPEED_1	16

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EC-MUX-SIG:SSS: REQ:3222	Signal With One Speed Aspect and Dark command - [Re-]Connecting to CC - OC-LS CONNECTED - General inquiry in stable states		
Given the current element state is DARK And CC connection state is NOT_CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When establishing the connection to CC Then SEA shall report the element state e_sig_dark_es with command reply flag false to CC			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	[Re-]Connecting in SPEED1 state is impossible.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3223	Signal With One Speed Aspect and Dark command - [Re-]Connecting to CC - OC-LS CONNECTED - General inquiry in CHANGING_TO_DARK state - no message after CC disconnect						
Given the current element state is CHANGING_TO_DARK And the last stable element state is <LastStable> And CC connection state is NOT_CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When establishing the connection to CC within WaitForSafeState ms Then SEA shall report the element state <CCState> with command reply flag false to CC <table><tr><th>LastStable</th><th>CCState</th></tr><tr><td>DARK</td><td>e_sig_dark_es</td></tr><tr><td>SPEED1</td><td>e_sig_speed1_es</td></tr></table>		LastStable	CCState	DARK	e_sig_dark_es	SPEED1	e_sig_speed1_es
LastStable	CCState						
DARK	e_sig_dark_es						
SPEED1	e_sig_speed1_es						
Type:	FUR - Functional Requirement						

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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Re-connecting to CC in CHANGING_TO_DARK state is impossible in real operation, as re-connecting will regularly take much longer time than the configured WaitForSafeState. Due to the execution of the safe command [Re-]Connecting in CHANGING_TO_DARK state is possible only with last forwarded command DARK, but depending on the previous events the last state reported to CC can be different. Following initial states are covered with this requirement: - CHANGING_TO_DARK with last stable element state SPEED1 - CHANGING_TO_DARK with last stable element state DARK/STOP - CHANGING_TO_DARK with last stable element state DARK/STOP and SPEED1 received from OC-LS in CHANGING_TO_DARK		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3224	Signal With One Speed Aspect and Dark command - [Re-]Connecting to CC - OC-LS NOT_CONNECTED - General inquiry		
Given the current element state is UNDEFINED And CC connection state is NOT_CONNECTED And EC-MUX - OC-LS connection state is NOT_CONNECTED When establishing the connection to CC Then SEA shall report the element state e_undefined_es with command reply flag false to CC			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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EC-MUX-SIG:SSS: REQ:3225	Signal With One Speed Aspect and Dark command - Connection loss to CC - execute safe command in SPEED1		
Given the current element state is SPEED1 And CC connection state is CONNECTED When losing the connection to CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to CHANGING_TO_DARK			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3226	Signal With One Speed Aspect and Dark command - Connection loss to CC - do nothing in DARK / UNDEFINED					
Given the current element state is <State> And CC connection state is CONNECTED When losing the connection to CC Then SEA shall do nothing <table><tr><th>State</th></tr><tr><td>DARK</td></tr><tr><td>UNDEFINED</td></tr></table>				State	DARK	UNDEFINED
State						
DARK						
UNDEFINED						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3227	Signal With One Speed Aspect and Dark command - Connection loss to CC - CHANGING_TO_SPEED1		
Given the current element state is CHANGING_TO_SPEED1 And CC connection state is CONNECTED When losing the connection to CC within WaitForSpeed ms Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to CHANGING_TO_DARK			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3228	Signal With One Speed Aspect and Dark command - Connection loss to CC - CHANGING_TO_DARK - do nothing
Given the current element state is CHANGING_TO_DARK And CC connection state is CONNECTED When losing the connection to CC within WaitForSafeState ms Then SEA shall do nothing	

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3201	Signal With One Speed Aspect and Dark command - [Re-]Connecting between EC-MUX and OC-LS in state DARK		
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED And OC-LS state is DARK When establishing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to DARK And SEA shall report the element state e_sig_dark_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_DARK to DGP And SEA shall report the connection_ok diagnosis device fault repair message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3220	Signal With One Speed Aspect and Dark command - [Re-]Connecting between EC-MUX and OC-LS in state SPEED1
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Given the current element state is UNDEFINED
And EC-MUX - OC-LS connection state is NOT_CONNECTED
And OC-LS state is SPEED1

When establishing the connection between EC-MUX and OC-LS

Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: OC-LS connect with SPEED1 for Signal x." to DGP
And SEA shall execute EC-MUX safety reaction OC Disconnection

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3230	Signal With One Speed Aspect and Dark command - Connection loss between EC-MUX and OC-LS in SPEED1 state
-------------------------------------	---

Given the current element state is SPEED1
And EC-MUX - OC-LS connection state is CONNECTED

When losing the connection between EC-MUX and OC-LS

Then SEA shall set the current element state to UNDEFINED
And SEA shall report the element state e_undefined_es with command reply flag false to CC
And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP
And SEA shall report the no_connection diagnosis device fault message to DGP

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			

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Satisfies:	
------------	--

EC-MUX-SIG:SSS: REQ:3231	Signal With One Speed Aspect and Dark command - Connection loss between EC-MUX and OC-LS in DARK state		
Given the current element state is DARK And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3232	Signal With One Speed Aspect and Dark command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_SPEED1		
Given the current element state is CHANGING_TO_SPEED1 And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS within WaitForSpeed ms Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3233	Signal With One Speed Aspect and Dark command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_DARK		
Given the current element state is CHANGING_TO_DARK And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS within WaitForSafeState ms Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

3.1.5.5 Diagnosis - Signal With One Speed Aspect and Dark command

In order to provide the aspect diagnosis for a Signal With One Speed Aspect and Dark command
As a SEA

I want to be able to receive an aspect diagnosis query from DGP and answer it by the DGP diagnosis values corresponding to the last diagnosis aspect message reported to DGP for the queried Signal

For requirement language see Chapter 2.8 "Requirement Language".

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Given a Signal With One Speed Aspect and Dark command

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a DARK command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
SPEED1	e_sig_speed1_es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12
SIG_SPEED_1	16

EC-MUX-SIG:SSS: REQ:3235	Signal With One Speed Aspect and Dark command - aspect query - last stable state								
Given the last stable element state is <LastStable> When receiving an aspect query from DGP Then SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>LastStable</th><th>DGPMMessage</th></tr><tr><td>DARK</td><td>SIG_ DARK</td></tr><tr><td>SPEED1</td><td>SIG_ SPEED_ 1</td></tr></table>				LastStable	DGPMMessage	DARK	SIG_ DARK	SPEED1	SIG_ SPEED_ 1
LastStable	DGPMMessage								
DARK	SIG_ DARK								
SPEED1	SIG_ SPEED_ 1								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						

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Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3237	Signal With One Speed Aspect and Dark command - aspect query in UNDEFINED - OC-LS NOT_CONNECTED		
Given EC-MUX - OC-LS connection state is NOT_CONNECTED And the current element state is UNDEFINED When receiving an aspect query from DGP Then SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

3.1.6 Signal with Three Speed Aspects and Dark command

SEA Eulynx Signal with three speed aspects and DARK command can reach following regular signal aspects via CC command:

- Speed1
- Speed2
- Speed3
- Dark

This generic implementation can be configured to respond to specific CC commands and OC-LS messages.

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The state machine of the SEA Eulynx Signal with three speed aspects and DARK command is shown in the following figure:

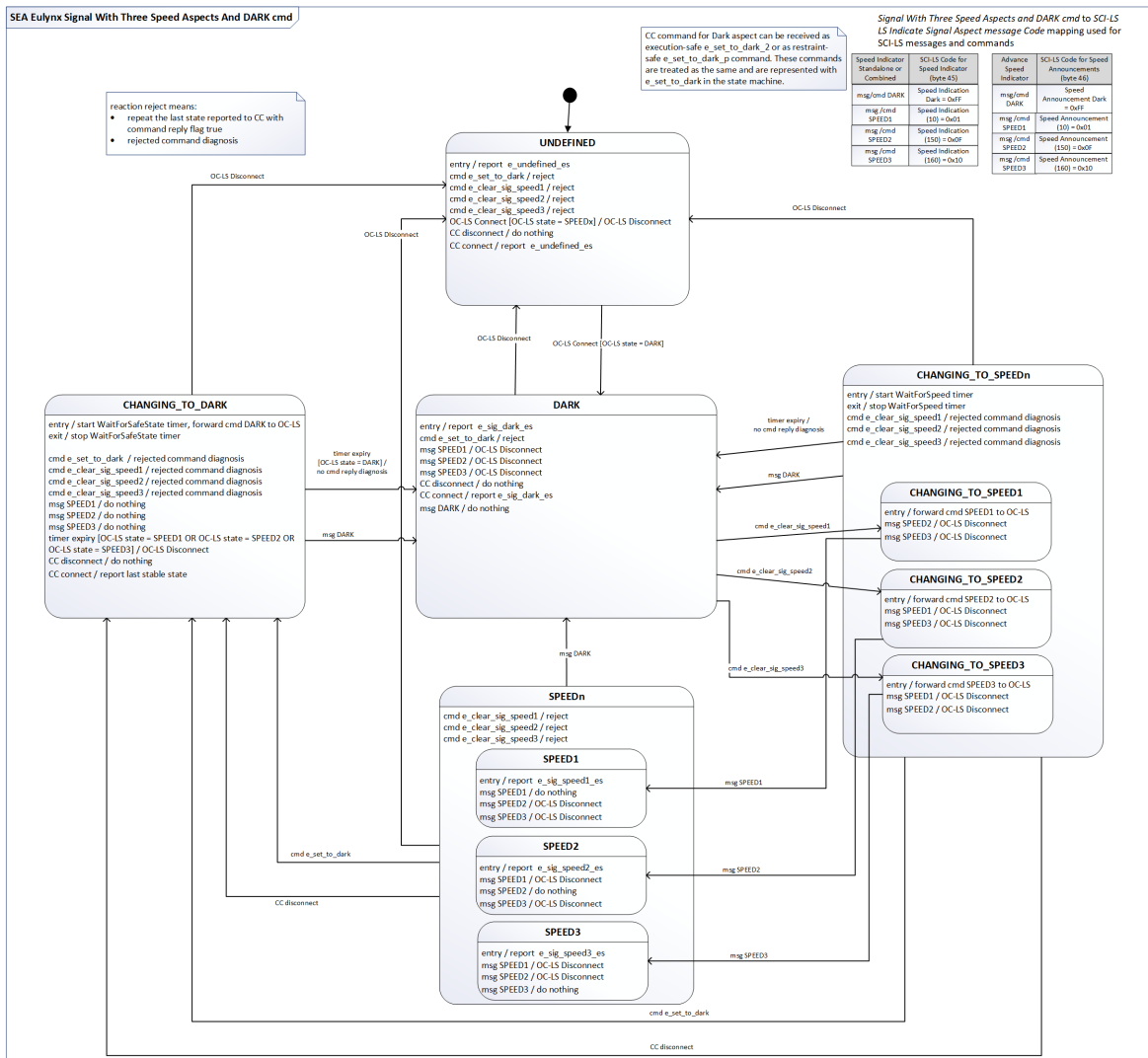


Figure 14: SEA Eulynx Signal with three speed aspects and DARK cmd - State Machine

Note To keep the STM applicable for several Signal types, names based on SEA Signal Aspects are used for SCI-LS messages and commands. A table with mappings to the SCI-LS codes has been provided for each Signal Type.

Following internal element states are defined for the SEA Eulynx Signal with three speed aspects and DARK cmd:

Element State	Functionality	CC State	Faulty
---------------	---------------	----------	--------

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UNDEFINED	no aspect information given, initial / default state	e_undefined_es	yes
SPEED1	signal commanded to SPEED1, corresponding aspect indication received already	e_sig_speed1_es	??
SPEED2	signal commanded to SPEED2, corresponding aspect indication received already	e_sig_speed2_es	??
SPEED3	signal commanded to SPEED3, corresponding aspect indication received already	e_sig_speed3_es	??
DARK	indication for DARK aspect received; caused by lamp fault or after cmd execution	e_sig_dark_es	yes
CHANGING_TO_SPEED1	command for SPEED1 aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history
CHANGING_TO_SPEED2	command for SPEED2 aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history
CHANGING_TO_SPEED3	command for SPEED3 aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history
CHANGING_TO_DARK	command for DARK aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history

Table 35: SEA Eulynx Signal with three speed aspects and DARK cmd - Internal Element States
Behavior in case of connection loss

When losing the connection to CC, SEA shall execute the safe command, i.e. command the signal to the DARK aspect if the last command forwarded to OC-LS was a different one.

3.1.6.1 Speed Indicator

The following information is applicable for Speed Indicator Standalone and Combined.

The Speed Indicator Standalone and Combined in SEA are configurations of the SEA Eulynx Signal with three speed aspects and Dark command.

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Used Cmd LS Indicate Signal Aspect and Msg LS Indicated Signal Aspect shall be addressed to the element identified by the Speed Indicator Standalone and Combined operational identifier. CC command and state mappings to SCI-LS telegrams are defined in following tables.

CC Command	SEA Signal with three speed aspects	Command byte / parameter	Parameter value	OBB aspect
e_set_to_dark_2 OR e_set_to_dark_p	Dark	45 = code for speed indicators	Speed Announcement Dark = 0xFF	dunkel
e_clear_sig_speed1	Speed1	45 = code for speed indicators	Speed Announcement (10) = 0x01	speed 1
e_clear_sig_speed2	Speed2	45 = code for speed indicators	Speed Announcement (150) = 0x0F	speed 2
e_clear_sig_speed3	Speed3	45 = code for speed indicators	Speed Announcement (160) = 0x10	speed 3

Table 36: Speed Indicator Standalone and Combined - Mapping CC command (CC to EC-MUX) to SCI-LS Cmd Indicate Signal Aspect (EC-MUX to OC-LS)

The SCI-LS Cmd Indicate Signal Aspect irrelevant and relevant parameters for the Speed Indicator Standalone and Combined are listed in the table *Eulynx Signal Mast configuration*.

Message byte / parameter	Parameter value	SEA Signal with three speed aspects	OBB aspect	CC State
45 = code for speed indicators	Speed Announcement Dark = 0xFF	Dark	dunkel	e_sig_dark_es
45 = code for speed indicators	Speed Announcement (10) = 0x01	speed 1	speed 1	e_sig_speed1_es
45 = code for speed indicators	Speed Announcement (150) = 0x0F	speed 2	speed 2	e_sig_speed2_es
45 = code for speed indicators	Speed Announcement (160) = 0x10	speed 3	speed 3	e_sig_speed3_es

Table 37: Speed Indicator Standalone and Combined - Mapping SCI-LS Msg Indicated Signal Aspect (OC-LS to EC-MUX) to gen_elem_stat (EC-MUX to CC)

For default values of the remaining parameters (byte 43, 44, 46 - 60) of the Msg Indicated Signal Aspect see table for default values of the corresponding command parameters.

3.1.6.2 Advance Speed Indicator

The Advance Speed Indicator in SEA is a configuration of the SEA Eulynx Signal with three speed aspects and Dark command.

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Used Cmd LS Indicate Signal Aspect and Msg LS Indicated Signal Aspect shall be addressed to the element identified by the Advance Speed Indicator operational identifier. CC command and state mappings to SCI-LS telegrams are defined in following tables.

CC Command	SEA Signal with three speed aspects	Command byte / parameter	Parameter value	OBB aspect
e_set_to_dark_2 OR e_set_to_dark_p	Dark	46 = code for speed indicator announcements	Speed Announcement Dark = 0xFF	dunkel
e_clear_sig_speed1	Speed1	46 = code for speed indicator announcements	Speed Announcement (10) = 0x01	speed 1
e_clear_sig_speed2	Speed2	46 = code for speed indicator announcements	Speed Announcement (150) = 0x0F	speed 2
e_clear_sig_speed3	Speed3	46 = code for speed indicator announcements	Speed Announcement (160) = 0x10	speed 3

Table 38: Advance Speed Indicator - Mapping CC command (CC to EC-MUX) to SCI-LS Cmd Indicate Signal Aspect (EC-MUX to OC-LS)

(*) TODO: Details for dependencies between different signal types at the same signal mast to be defined in a later project phase.

The SCI-LS Cmd Indicate Signal Aspect irrelevant and relevant parameters for the Advance Speed Indicator are listed in the table *Eulynx Signal Mast configuration*.

Message byte / parameter	Parameter value	SEA Signal with three speed aspects	OBB aspect	CC State
46 = code for speed indicator announcements	Speed Announcement Dark = 0xFF	Dark	dunkel	e_sig_dark_es
46 = code for speed indicator announcements	Speed Announcement (10) = 0x01	speed 1	speed 1	e_sig_speed1_es
46 = code for speed indicator announcements	Speed Announcement (150) = 0x0F	speed 2	speed 2	e_sig_speed2_es
46 = code for speed indicator announcements	Speed Announcement (160) = 0x10	speed 3	speed 3	e_sig_speed3_es

Table 39: Advance Speed Indicator - Mapping SCI-LS Msg Indicated Signal Aspect (OC-LS to EC-MUX) to gen_elem_stat (EC-MUX to CC)

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For default values of the remaining parameters (byte 43 - 45, 47 - 60) of the Msg Indicated Signal Aspect see table for default values of the corresponding command parameters.

3.1.6.3 Operational - Signal With Three Speed Aspects and Dark command

In order to control and supervise a Signal With Three Speed Aspects and Dark command
As a SEA

I want to be able to receive commands from CC, translate them into SCI-LS commands and forward them to OC-LS and to receive SCI-LS state messages from OC-LS, translate them into signal state changes and report them to CC.

For requirement language see Chapter 2.8 "Requirement Language".

CHANGING_TO_SPEED_n and CHANGING_TO_DARK element states are not considered in the mapping between element state and CC State because when reaching these states, nothing is reported to CC.

WaitForSpeed and WaitForSafeState are configurable timers for receiving a command reply from OC-LS. WaitForSpeed is used when waiting for a reply to a SPEED command (e_clear_sig_speed1, e_clear_sig_speed2, e_clear_sig_speed3), its default value is 2000ms. WaitForSafeState is used when waiting for a reply to a DARK command (e_set_to_dark_2, e_set_to_dark_p), its default value is 1000ms.

Given a Signal With Three Speed Aspects and Dark command

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a DARK command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es
SPEED3	e_sig_speed3_es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12

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SEA diagnosis aspect	DGP diagnosis value
SIG_SPEED_1	16
SIG_SPEED_2	17
SIG_SPEED_3	18

EC-MUX-SIG:SSS: REQ:3300	Signal With Three Speed Aspects and Dark command - Initial state		
When initializing the element			
Then SEA shall set the current element state to UNDEFINED			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3302	Signal With Three Speed Aspects and Dark command - reject commands in UNDEFINED state			
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED When receiving the command <Command> from CC Then SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_set_to_dark_p</td></tr><tr><td>e_set_to_dark_2</td></tr></table>		Command	e_set_to_dark_p	e_set_to_dark_2
Command				
e_set_to_dark_p				
e_set_to_dark_2				

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<table><tr><th>Command</th></tr><tr><td>e_clear_sig_speed1</td></tr><tr><td>e_clear_sig_speed2</td></tr><tr><td>e_clear_sig_speed3</td></tr></table>				Command	e_clear_sig_speed1	e_clear_sig_speed2	e_clear_sig_speed3
Command							
e_clear_sig_speed1							
e_clear_sig_speed2							
e_clear_sig_speed3							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:							
Satisfies:							

EC-MUX-SIG:SSS: REQ:3303	Signal With Three Speed Aspects and Dark command - execute SPEED command in DARK													
Given the current element state is DARK When receiving the command <Command> from CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to <State> to OC-LS And SEA shall set the current element state to <ChangingToSpeedState> <table><tr><th>Command</th><th>State</th><th>ChangingToSpeedState</th></tr><tr><td>e_clear_sig_speed1</td><td>SPEED1</td><td>CHANGING_TO_SPEED1</td></tr><tr><td>e_clear_sig_speed2</td><td>SPEED2</td><td>CHANGING_TO_SPEED2</td></tr><tr><td>e_clear_sig_speed3</td><td>SPEED3</td><td>CHANGING_TO_SPEED3</td></tr></table>			Command	State	ChangingToSpeedState	e_clear_sig_speed1	SPEED1	CHANGING_TO_SPEED1	e_clear_sig_speed2	SPEED2	CHANGING_TO_SPEED2	e_clear_sig_speed3	SPEED3	CHANGING_TO_SPEED3
Command	State	ChangingToSpeedState												
e_clear_sig_speed1	SPEED1	CHANGING_TO_SPEED1												
e_clear_sig_speed2	SPEED2	CHANGING_TO_SPEED2												
e_clear_sig_speed3	SPEED3	CHANGING_TO_SPEED3												
Type:	FUR - Functional Requirement													

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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3304	Signal With Three Speed Aspects and Dark command - reject DARK commands in DARK					
Given the current element state is DARK When receiving the command <Command> from CC Then SEA shall report the element state e_sig_dark_es with command reply flag true to CC And SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_set_to_dark_2</td></tr><tr><td>e_set_to_dark_p</td></tr></table>				Command	e_set_to_dark_2	e_set_to_dark_p
Command						
e_set_to_dark_2						
e_set_to_dark_p						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:	CC shall not send the DARK command, if the dark aspect is already given.					
Satisfies:						

EC-MUX-SIG:SSS: REQ:3305	Signal With Three Speed Aspects and Dark command - execute DARK commands in SPEEDx
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Given the current element state is <State>

When receiving the command <Command> from CC

Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS

And SEA shall set the current element state to CHANGING_TO_DARK

State	Command
SPEED1	e_set_to_dark_p
SPEED2	e_set_to_dark_p
SPEED3	e_set_to_dark_p
SPEED1	e_set_to_dark_2
SPEED2	e_set_to_dark_2
SPEED3	e_set_to_dark_2

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3306	Signal With Three Speed Aspects and Dark command - reject SPEED command in SPEEDx
Given the current element state is <State> When receiving the command <Command> from CC	

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Then SEA shall report the element state <CCState> with command reply flag true to CC
And SEA shall report the rejected command diagnosis to DGP

State	Command	CCState
SPEED1	e_clear_sig_speed1	e_sig_speed1_es
SPEED1	e_clear_sig_speed2	e_sig_speed1_es
SPEED1	e_clear_sig_speed3	e_sig_speed1_es
SPEED2	e_clear_sig_speed1	e_sig_speed2_es
SPEED2	e_clear_sig_speed2	e_sig_speed2_es
SPEED2	e_clear_sig_speed3	e_sig_speed2_es
SPEED3	e_clear_sig_speed1	e_sig_speed3_es
SPEED3	e_clear_sig_speed2	e_sig_speed3_es
SPEED3	e_clear_sig_speed3	e_sig_speed3_es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	CC shall not send the command SPEED, if the corresponding aspect is already given. CC shall not send any other SPEED command (downgrade / upgrade the speed) without DARK command between.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3307	Signal With Three Speed Aspects and Dark command - spontaneous transition from SPEEDx to DARK
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Given the current element state is <State>

When receiving the LS Indicated Signal Aspect message with Code corresponding to DARK from OC-LS

Then SEA shall set the current element state to DARK

And SEA shall report the element state e_sig_dark_es with command reply flag false to CC

And SEA shall report the diagnosis aspect message SIG_DARK to DGP

State
SPEED1
SPEED2
SPEED3

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3308	Signal With Three Speed Aspects and Dark command - CHANGING_TO_SPEEDn - transition from CHANGING_TO_SPEEDx to SPEEDx
-------------------------------------	---

Given the current element state is <ChangingToSpeedState>

When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS within WaitForSpeed ms

Then SEA shall set the current element state to <State>

And SEA shall report the element state <CCState> with command reply flag true to CC

And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP

ChangingToSpeedState	State	CCState	DGPMMessage
CHANGING_TO_		e_sig_speed1_	SIG_SPEED_

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ChangingToSpeedState	State	CCState	DGPMessage
SPEED1	SPEED1	es	1
CHANGING_TO_SPEED2	SPEED2	e_sig_speed2_es	SIG_SPEED_2
CHANGING_TO_SPEED3	SPEED3	e_sig_speed3_es	SIG_SPEED_3

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Successful SPEED command execution - commanded aspect reached.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3309	Signal With Three Speed Aspects and Dark command - CHANGING_TO_SPEEDn - reject repeated SPEED command in CHANGING_TO_SPEEDn						
Given the current element state is CHANGING_TO_SPEEDn When receiving the command <Command> from CC within WaitForSpeed ms Then SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_clear_sig_speed1</td></tr><tr><td>e_clear_sig_speed2</td></tr><tr><td>e_clear_sig_speed3</td></tr></table>				Command	e_clear_sig_speed1	e_clear_sig_speed2	e_clear_sig_speed3
Command							
e_clear_sig_speed1							
e_clear_sig_speed2							
e_clear_sig_speed3							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				

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Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	CC shall not repeat the command SPEED. CC shall not switch to another SPEED aspect without DARK command between them. The WaitForSpeed timer shall not to be re-started due to the rejected command.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3310	Signal With Three Speed Aspects and Dark command - CHANGING_TO_SPEEDn - return to DARK		
Given the current element state is CHANGING_TO_SPEEDn			
When receiving the LS Indicated Signal Aspect message with Code corresponding to DARK from OC-LS within WaitForSpeed ms			
Then SEA shall set the current element state to DARK			
And SEA shall report the element state e_sig_dark_es with command reply flag true to CC			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	OC-LS has probably sent the message as a downgrade aspect for failed clear command. As it is not possible to forward the command corresponding to SPEEDx aspect before receiving a reply to the previous DARK, it is not necessary to continue waiting for a further reply and thus the CHANGING_TO_SPEEDx state shall be left.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3311	Signal With Three Speed Aspects and Dark command - CHANGING_TO_SPEEDn - timeout		
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Given the current element state is CHANGING_TO_SPEEDn			
When the current state of the element has been CHANGING_TO_SPEEDn for WaitForSpeed ms			
Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x". to DGP			
And SEA shall set the current element state to DARK			
And SEA shall report the element state e_sig_dark_es with command reply flag true to CC			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Regularly the state CHANGING_TO_SPEEDn will be left after WaitForSpeed ms. No safety reaction OC Disconnection is performed as this is not a safety-critical command. Instead, SEA shall generate a diagnosis and try to fallback to original DARK.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3312	Signal With Three Speed Aspects and Dark command - CHANGING_TO_SPEEDn - execute DARK command					
Given the current element state is CHANGING_TO_SPEEDn						
When receiving the command <Command> from CC within WaitForSpeed ms						
Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS						
And SEA shall set the current element state to CHANGING_TO_DARK						
<table><tr><th>Command</th></tr><tr><td>e_set_to_dark_p</td></tr><tr><td>e_set_to_dark_2</td></tr></table>				Command	e_set_to_dark_p	e_set_to_dark_2
Command						
e_set_to_dark_p						
e_set_to_dark_2						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	If the DARK command reaches the OC-LS before the SPEEDx command is answered, then it is possible that OC-LS will reply to none of both commands, because the last commanded aspect is still given.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3313	Signal With Three Speed Aspects and Dark command - CHANGING_TO_DARK - transition from CHANGING_TO_DARK to DARK - from OC-LS state SPEEDx						
Given the current element state is CHANGING_TO_DARK And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to DARK from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to DARK And SEA shall report the element state e_sig_dark_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_DARK to DGP <table><tr><td>LastStable</td></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr><tr><td>SPEED3</td></tr></table>				LastStable	SPEED1	SPEED2	SPEED3
LastStable							
SPEED1							
SPEED2							
SPEED3							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:	Successful DARK command execution - commanded aspect / downgrade aspect reached.						
Satisfies:							

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EC-MUX-SIG:SSS: REQ:3314	Signal With Three Speed Aspects and Dark command - CHANGING_TO_DARK - reject commands in CHANGING_TO_DARK								
Given the current element state is CHANGING_TO_DARK When receiving the command <Command> from CC within WaitForSafeState ms Then SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_set_to_dark_2</td></tr><tr><td>e_set_to_dark_p</td></tr><tr><td>e_clear_sig_speed1</td></tr><tr><td>e_clear_sig_speed2</td></tr><tr><td>e_clear_sig_speed3</td></tr></table>				Command	e_set_to_dark_2	e_set_to_dark_p	e_clear_sig_speed1	e_clear_sig_speed2	e_clear_sig_speed3
Command									
e_set_to_dark_2									
e_set_to_dark_p									
e_clear_sig_speed1									
e_clear_sig_speed2									
e_clear_sig_speed3									
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:	CC shall not send a clear command if there is DARK command not answered yet. CC shall not repeat the command DARK. The WaitForSafeState timer shall not to be re-started due to the rejected command.								
Satisfies:									

EC-MUX-SIG:SSS: REQ:3315	Signal With Three Speed Aspects and Dark command - CHANGING_TO_DARK - timeout when OC-LS SPEEDx - OC Disconnection
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Given the current element state is CHANGING_TO_DARK
And the last stable element state is <LastStable>

When the current state of the element has been CHANGING_TO_DARK for WaitForSafeState ms

Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP

And SEA shall execute EC-MUX safety reaction OC Disconnection

And SEA shall set the current element state to UNDEFINED

And SEA shall report the element state e_undefined_es with command reply flag true to CC

And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP

And SEA shall report the no_connection diagnosis device fault message to DGP

LastStable

SPEED1

SPEED2

SPEED3

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Regularly the state CHANGING_TO_DARK will be left after WaitForSafeState ms. Safety reaction OC Disconnection is applicable only if the OC-LS cannot reach the DARK aspect and stays in SPEEDx. See also EC-MUX-SIG:SSS:REQ:3316.		
Satisfies:			

EC-MUX-SIG:SSS:REQ:3316	Signal With Three Speed Aspects and Dark command - CHANGING_TO_DARK - ignore SPEED message
Given the current element state is CHANGING_TO_DARK When receiving the LS Indicated Signal Aspect message with Code corresponding to <SpeedAspect> from OC-LS within WaitForSafeState ms Then SEA shall do nothing	

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<table><tr><td>SpeedAspect</td></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr><tr><td>SPEED3</td></tr></table>				SpeedAspect	SPEED1	SPEED2	SPEED3
SpeedAspect							
SPEED1							
SPEED2							
SPEED3							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:	It is possible that OC-LS has sent the message corresponding to SPEED as reply to the previous SPEED command not answered yet. Waiting for the reply to the DARK command is continued by keeping the current state CHANGING_TO_DARK. After timer expiry, if the expected DARK reply is not received, the element state is changed to DARK or a Safety Reaction OC Disconnection is performed, depending on the last received OC-LS state.						
Satisfies:							

EC-MUX-SIG:SSS: REQ:3317	Signal With Three Speed Aspects and Dark command - transition from CHANGING_TO_DARK to DARK - from OC-LS state DARK		
Given the current element state is CHANGING_TO_DARK And the last stable element state is DARK When receiving the LS Indicated Signal Aspect message with Code corresponding to DARK from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to DARK And SEA shall report the element state e_sig_dark_es with command reply flag true to CC			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit

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Comments:	After receiving the DARK as reply to the DARK command, the state will be reported to the CC with command reply flag true.
Satisfies:	

EC-MUX-SIG:SSS: REQ:3336	Signal With Three Speed Aspects and Dark command - CHANGING_TO_DARK - timeout when OC-LS DARK - reach DARK		
Given the current element state is CHANGING_TO_DARK And OC-LS state is DARK When the current state of the element has been CHANGING_TO_DARK for WaitForSafeState ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP And SEA shall set the current element state to DARK And SEA shall report the element state e_sig_dark_es with command reply flag true to CC			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Regularly the state CHANGING_TO_DARK will be left after WaitForSafeState ms. OC-LS state can be DARK at this point only when a DARK command has been executed after a SPEED command, and no reply has been received for either command. This is an allowed scenario so no safety reaction OC Disconnection is performed.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3318	Signal With Three Speed Aspects and Dark command - Disconnection on spontaneous SPEED message in DARK / SPEED
Given the current element state is <State> When receiving the LS Indicated Signal Aspect message with Code corresponding to <SpeedAspect> from OC-LS	

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Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: Spontaneous <SpeedAspect> reported for Signal x." to DGP

And SEA shall execute EC-MUX safety reaction OC Disconnection

And SEA shall set the current element state to UNDEFINED

And SEA shall report the element state e_undefined_es with command reply flag false to CC

And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP

And SEA shall report the no_connection diagnosis device fault message to DGP

State	SpeedAspect
DARK	SPEED1
DARK	SPEED2
DARK	SPEED3
SPEED1	SPEED2
SPEED1	SPEED3
SPEED2	SPEED1
SPEED2	SPEED3
SPEED3	SPEED1
SPEED3	SPEED2

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3319	Signal With Three Speed Aspects and Dark command - CHANGING_TO_SPEEDn - Disconnection on unexpected command reply
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Given the current element state is <ChangingToSpeedState>

When receiving the LS Indicated Signal Aspect message with Code corresponding to <SpeedMsg> from OC-LS within WaitForSpeed ms

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Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: Cmd <SpeedCmd> replied by Msg <SpeedMsg> for Signal x." to DGP

And SEA shall execute EC-MUX safety reaction OC Disconnection

And SEA shall set the current element state to UNDEFINED

And SEA shall report the element state e_undefined_es with command reply flag true to CC

And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP

And SEA shall report the no_connection diagnosis device fault message to DGP

ChangingToSpeedState	SpeedMsg	SpeedCmd
CHANGING_TO_SPEED1	SPEED2	SPEED1
CHANGING_TO_SPEED1	SPEED3	SPEED1
CHANGING_TO_SPEED2	SPEED1	SPEED2
CHANGING_TO_SPEED2	SPEED3	SPEED2
CHANGING_TO_SPEED3	SPEED1	SPEED3
CHANGING_TO_SPEED3	SPEED2	SPEED3

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3320	Signal With Three Speed Aspects and Dark command - Ignore repeated messages
Given the current element state is <State> When receiving the LS Indicated Signal Aspect message with Code corresponding to <SpeedAspect> from OC-LS	

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Then SEA shall do nothing

State	SpeedAspect
SPEED1	SPEED1
SPEED2	SPEED2
SPEED3	SPEED3
DARK	DARK

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS:
REQ:3338 **Signal With Three Speed Aspects and Dark command - Receive unexpected STOP commands from CC**

When receiving the command <Command> from CC

Then SEA shall execute EC-MUX safety reaction Deactivation

Command
e_set_to_stop_p
e_set_to_stop_2

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit

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Comments:	
Satisfies:	

3.1.6.4 Connection - Signal With Three Speed Aspects and Dark command

In order to control and supervise a Signal With Three Speed Aspects and Dark command
As a SEA

I want to be able to supervise connections and when changes occur then

- on CC connection establishment report the current state to CC
- on CC connection loss execute the safe command DARK if the last command forwarded to OC-LS was SPEED1 / SPEED2 / SPEED3 and it was not answered by DARK
- on OC-LS connection loss take on the default state UNDEFINED
- on OC-LS connection establishment transition to DARK state as the OC-LS element state
- report the connection state changes to CC and DGP

For requirement language see Chapter 2.8 "Requirement Language".

CHANGING_TO_SPEEDn and CHANGING_TO_DARK element states are not considered in the mapping between element state and CC State because when reaching these states, nothing is reported to CC.

WaitForSpeed and WaitForSafeState are configurable timers for receiving a command reply from OC-LS.

WaitForSpeed is used when waiting for a reply to a SPEED command (e_clear_sig_speed1, e_clear_sig_speed2, e_clear_sig_speed3), its default value is 2000ms.

WaitForSafeState is used when waiting for a reply to a DARK command (e_set_to_dark_2, e_set_to_dark_p), its default value is 1000ms.

Given a Signal With Three Speed Aspects and Dark command

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a DARK command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es
SPEED3	e_sig_speed3_es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

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And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12
SIG_SPEED_1	16
SIG_SPEED_2	17
SIG_SPEED_3	18

EC-MUX-SIG:SSS: REQ:3321	Signal With Three Speed Aspects and Dark command - [Re-]Connecting to CC - OC-LS CONNECTED - General inquiry in stable states		
Given the current element state is DARK And CC connection state is NOT_CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When establishing the connection to CC Then SEA shall report the element state e_sig_dark_es with command reply flag false to CC			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	[Re-]Connecting in SPEEDx state is impossible.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3322	Signal With Three Speed Aspects and Dark command - [Re-]Connecting to CC - OC-LS CONNECTED - General inquiry in CHANGING_TO_DARK - no message after CC disconnect
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Given the current element state is CHANGING_TO_DARK
And the last stable element state is <LastStable>
And CC connection state is NOT_CONNECTED
And EC-MUX - OC-LS connection state is CONNECTED

When establishing the connection to CC within WaitForSafeState ms

Then SEA shall report the element state <CCState> with command reply flag false to CC

LastStable	CCState
DARK	e_sig_dark_es
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es
SPEED3	e_sig_speed3_es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Re-connecting to CC in CHANGING_TO_DARK state is impossible in real operation, as re-connecting will regularly take much longer time than the configured WaitForSafeState. Due to the execution of the safe command [Re-]Connecting in CHANGING_TO_DARK state is possible only with last forwarded command DARK, but depending on the previous events the last state reported to CC can be different. Following initial states are covered with this requirement: - CHANGING_TO_DARK with last stable element state SPEED1/SPEED2 /SPEED3 - CHANGING_TO_DARK with last stable element state DARK - CHANGING_TO_DARK with last stable element state DARK and SPEED1 /SPEED2/SPEED3 received from OC-LS in CHANGING_TO_DARK		
Satisfies:			

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EC-MUX-SIG:SSS: REQ:3323	Signal With Three Speed Aspects and Dark command - [Re-]Connecting to CC - OC-LS NOT_CONNECTED		
Given the current element state is UNDEFINED And CC connection state is NOT_CONNECTED And EC-MUX - OC-LS connection state is NOT_CONNECTED When establishing the connection to CC Then SEA shall report the element state e_undefined_es with command reply flag false to CC			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3324	Signal With Three Speed Aspects and Dark command - Connection loss to CC - execute safe command in SPEEDx						
Given the current element state is <State> And CC connection state is CONNECTED When losing the connection to CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to CHANGING_TO_DARK <table><tr><th>State</th></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr><tr><td>SPEED3</td></tr></table>				State	SPEED1	SPEED2	SPEED3
State							
SPEED1							
SPEED2							
SPEED3							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3325	Signal With Three Speed Aspects and Dark command - Connection loss to CC - do nothing in DARK / UNDEFINED					
Given the current element state is <State> And CC connection state is CONNECTED When losing the connection to CC Then SEA shall do nothing <table><tr><th>State</th></tr><tr><td>DARK</td></tr><tr><td>UNDEFINED</td></tr></table>				State	DARK	UNDEFINED
State						
DARK						
UNDEFINED						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

EC-MUX-SIG:SSS: REQ:3326	Signal With Three Speed Aspects and Dark command - Connection loss to CC - CHANGING_TO_SPEEDn
Given the current element state is CHANGING_TO_SPEEDn And CC connection state is CONNECTED	

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When losing the connection to CC within WaitForSpeed ms			
Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS			
And SEA shall set the current element state to CHANGING_TO_DARK			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3327	Signal With Three Speed Aspects and Dark command - Connection loss to CC - CHANGING_TO_DARK - do nothing		
Given the current element state is CHANGING_TO_DARK And CC connection state is CONNECTED			
When losing the connection to CC within WaitForSafeState ms			
Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3301	Signal With Three Speed Aspects and Dark command - [Re-]Connecting between EC-MUX and OC-LS in state DARK
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Given the current element state is UNDEFINED
And EC-MUX - OC-LS connection state is NOT_CONNECTED
And OC-LS state is DARK

When establishing the connection between EC-MUX and OC-LS

Then SEA shall set the current element state to DARK
And SEA shall report the element state e_sig_dark_es with command reply flag false to CC
And SEA shall report the diagnosis aspect message SIG_DARK to DGP
And SEA shall report the connection_ok diagnosis device fault repair message to DGP

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3339	Signal With Three Speed Aspects and Dark command - [Re-]Connecting between EC-MUX and OC-LS in state SPEEDx
-------------------------------------	--

Given the current element state is UNDEFINED
And EC-MUX - OC-LS connection state is NOT_CONNECTED
And OC-LS state is <SpeedAspect>

When establishing the connection between EC-MUX and OC-LS

Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: OC-LS connect with <SpeedAspect> for Signal x." to DGP
And SEA shall execute EC-MUX safety reaction OC Disconnection

SpeedAspect

SPEED1

SPEED2

SPEED3

Type:	FUR - Functional Requirement
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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3329	Signal With Three Speed Aspects and Dark command - Connection loss between EC-MUX and OC-LS in regular states						
Given the current element state is <State> And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP <table><tr><th>State</th></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr><tr><td>SPEED3</td></tr></table>				State	SPEED1	SPEED2	SPEED3
State							
SPEED1							
SPEED2							
SPEED3							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:							
Satisfies:							

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EC-MUX-SIG:SSS: REQ:3330	Signal With Three Speed Aspects and Dark command - Connection loss between EC-MUX and OC-LS in DARK state		
Given the current element state is DARK And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3331	Signal With Three Speed Aspects and Dark command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_SPEEDn		
Given the current element state is CHANGING_TO_SPEEDn And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS within WaitForSpeed ms Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory

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Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3332	Signal With Three Speed Aspects and Dark command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_DARK		
Given the current element state is CHANGING_TO_DARK And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS within WaitForSafeState ms Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

3.1.6.5 Diagnosis - Signal With Three Speed Aspects and Dark command

In order to provide the aspect diagnosis for a Signal With Three Speed Aspects and Dark command
As a SEA

I want to be able to receive an aspect diagnosis query from DGP and answer it by the DGP diagnosis values corresponding to the last diagnosis aspect message reported to DGP for the queried Signal

For requirement language see Chapter 2.8 "Requirement Language".

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Given a Signal With Three Speed Aspects and Dark command

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a DARK command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es
SPEED3	e_sig_speed3_es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12
SIG_SPEED_1	16
SIG_SPEED_2	17
SIG_SPEED_3	18

EC-MUX-SIG:SSS: REQ:3333	Signal With Three Speed Aspects and Dark command - aspect query - last diagnosis aspect message						
Given the last stable element state is <LastStable> When receiving an aspect query from DGP Then SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table border="1"> <tr> <th>LastStable</th><th>DGPMMessage</th></tr> <tr> <td>DARK</td><td>SIG_DARK</td></tr> <tr> <td></td><td>SIG_SPEED_</td></tr> </table>		LastStable	DGPMMessage	DARK	SIG_DARK		SIG_SPEED_
LastStable	DGPMMessage						
DARK	SIG_DARK						
	SIG_SPEED_						

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LastStable	DGPMessage
SPEED1	1
SPEED2	SIG_ SPEED_ 2
SPEED3	SIG_ SPEED_ 3

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3335	Signal With Three Speed Aspects and Dark command - aspect query in UNDEFINED - OC-LS NOT_CONNECTED		
Given EC-MUX - OC-LS connection state is NOT_CONNECTED And the current element state is UNDEFINED When receiving an aspect query from DGP Then SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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3.1.7 Signal with Three Speed Aspects and Stop and Dark commands

SEA Signal with three speed aspects and STOP and DARK commands can reach following regular signal aspects via CC command:

- Speed1
- Speed2
- Speed3
- Stop
- Dark

This generic implementation can be configured to respond to specific CC commands and OC-LS messages.

The state machine of the SEA Eulynx Signal with three speed aspects and STOP command is shown in the following figure:

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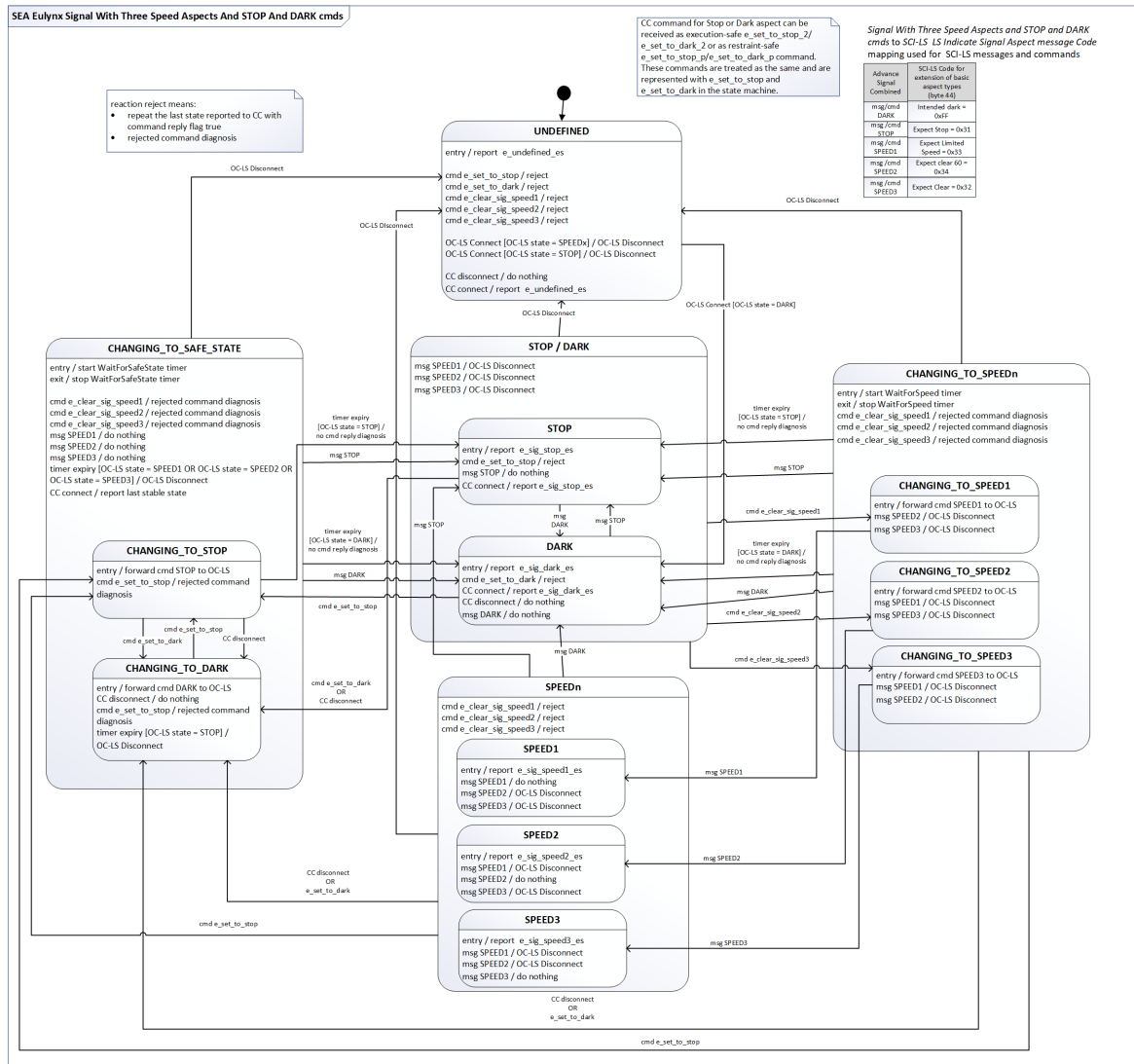


Figure 15: SEA Eulynx Signal with three speed aspects and STOP and DARK cmds - State Machine

Note To keep the STM applicable for several Signal types, names based on SEA Signal Aspects are used for SCI-LS messages and commands. A table with mappings to the SCI-LS codes has been provided for each Signal Type.

Following internal element states are defined for the SEA Eulynx Signal with three speed aspects and STOP and DARK cmds:

Element State	Functionality	CC State	Faulty
UNDEFINED	no aspect information given, initial / default state	e_undefined_es	yes
	indication for STOP aspect		

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STOP	received; as spontaneous msg or after cmd execution	e_sig_stop_es	??
SPEED1	signal commanded to SPEED1, corresponding aspect indication received already	e_sig_speed1_es	??
SPEED2	signal commanded to SPEED2, corresponding aspect indication received already	e_sig_speed2_es	??
SPEED3	signal commanded to SPEED3, corresponding aspect indication received already	e_sig_speed3_es	??
DARK	indication for DARK aspect received; as spontaneous msg or after cmd execution	e_sig_dark_es	yes
CHANGING_TO_SPEED1	command for SPEED1 aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history
CHANGING_TO_SPEED2	command for SPEED2 aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history
CHANGING_TO_SPEED3	command for SPEED3 aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history
CHANGING_TO_STOP	command for STOP aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history
CHANGING_TO_DARK	command for DARK aspect forwarded to OC-LS, waiting for the reply; timed state	none on entry	no information, depending on history

Table 40: SEA Eulynx Signal with three speed aspects and STOP and DARK cmds - Internal Element States

Behavior in case of connection loss

When losing the connection to CC, SEA shall execute the safe command, i.e. command the signal to the STOP aspect if the last command forwarded to OC-LS was a different one.

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3.1.7.1 Advance Signal Combined

The Advance Signal Combined in SEA is a configuration of the SEA Eulynx Signal with three speed aspects and Stop and Dark commands.

Used Cmd LS Indicate Signal Aspect and Msg LS Indicated Signal Aspect shall be addressed to the element identified by the Advance Signal Combined operational identifier. CC command and state mappings to SCI-LS telegrams are defined in following tables.

CC Command	SEA Signal with three speed aspects	Command byte / parameter	Parameter value	OBB aspect
e_set_to_stop_2 OR e_set_to_stop_p	Stop	44 = code for extension of basic aspect types	Expect stop = 0x31	Vorsicht
e_clear_sig_speed1	Speed1	44 = code for extension of basic aspect types	Expect limited speed = 0x33	Hauptsignal Frei 40
e_clear_sig_speed2	Speed2	44 = code for extension of basic aspect types	Expect clear 60 = 0x34	Hauptsignal Frei 60
e_clear_sig_speed3	Speed3	44 = code for extension of basic aspect types	Expect clear = 0x32	Hauptsignal Frei
e_set_to_dark_2 OR e_set_to_dark_p	Dark	44 = code for extension of basic aspect types	Intended Dark = 0xFF	Dark

Table 41: Advance Signal Combined - Mapping CC command (CC to EC-MUX) to SCI-LS Cmd Indicate Signal Aspect (EC-MUX to OC-LS)

The SCI-LS Cmd Indicate Signal Aspect irrelevant and relevant parameters for the Advance Signal Combined are listed in the table *Eulynx Signal Mast configuration*.

Message byte / parameter	Parameter value	SEA Signal with three speed aspects	OBB aspect	CC State
44 = code for extension of basic aspect types	Expect stop = 0x31	Stop	Vorsicht	e_sig_stop_es
44 = code for extension of basic aspect types	Expect limited speed = 0x33	Speed 1	Hauptsignal Frei 40	e_sig_speed1_es
44 = code for extension of basic aspect types	Expect clear 60 = 0x34	Speed 2	Hauptsignal Frei 60	e_sig_speed2_es
44 = code for extension of basic aspect types	Expect clear = 0x32	Speed 3	Hauptsignal Frei	e_sig_speed3_es
44 = code for extension of basic aspect types	Intended dark = 0xFF	Dark	Dark	e_sig_dark_es

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Table 42: Advance Signal Combined - Mapping SCI-LS Msg Indicated Signal Aspect (OC-LS to EC-MUX) to gen_elem_stat (EC-MUX to CC)

For default values of the remaining parameters (byte 43, 45 - 60) of the Msg Indicated Signal Aspect see table for default values of the corresponding command parameters.

3.1.7.2 Operational - Signal With Three Speed Aspects and Stop and Dark commands

In order to control and supervise a Signal With Three Speed Aspects and Stop and Dark commands
As a SEA

I want to be able to receive commands from CC, translate them into SCI-LS commands and forward them to OC-LS and to receive SCI-LS state messages from OC-LS, translate them into signal state changes and report them to CC.

For requirement language see Chapter 2.8 "Requirement Language".

CHANGING_TO_SPEED_n, CHANGING_TO_STOP and CHANGING_TO_DARK element states are not considered in the mapping between element state and CC State because when reaching these states, nothing is reported to CC.

WaitForSpeed and WaitForSafeState are configurable timers for receiving a command reply from OC-LS. WaitForSpeed is used when waiting for a reply to a SPEED command (e_clear_sig_speed1, e_clear_sig_speed2, e_clear_sig_speed3), its default value is 2000ms. WaitForSafeState is used when waiting for a reply to a STOP command (e_set_to_stop_2, e_set_to_stop_p) or to a DARK command (e_set_to_dark_2, e_set_to_dark_p), its default value is 1000ms.

Given a Signal With Three Speed Aspects and Stop and Dark commands

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a STOP command or a DARK command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_stop_es
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es
SPEED3	e_sig_speed3_es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

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SEA diagnosis aspect	DGP diagnosis value
SIG_ UNDEFINED	0
SIG_ DARK	12
SIG_ STOP	13
SIG_ SPEED_ 1	16
SIG_ SPEED_ 2	17
SIG_ SPEED_ 3	18

EC-MUX-SIG:SSS: REQ:3500	Signal With Three Speed Aspects and Stop and Dark commands - Initial state		
When initializing the element			
Then SEA shall set the current element state to UNDEFINED			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3503	Signal With Three Speed Aspects and Stop and Dark commands - reject commands in UNDEFINED state
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED When receiving the command <Command> from CC Then SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the rejected command diagnosis to DGP	

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<table border="1"> <thead> <tr> <th colspan="4">Command</th></tr> </thead> <tbody> <tr><td colspan="4">e_set_to_stop_p</td></tr> <tr><td colspan="4">e_set_to_stop_2</td></tr> <tr><td colspan="4">e_set_to_dark_p</td></tr> <tr><td colspan="4">e_set_to_dark_2</td></tr> <tr><td colspan="4">e_clear_sig_speed1</td></tr> <tr><td colspan="4">e_clear_sig_speed2</td></tr> <tr><td colspan="4">e_clear_sig_speed3</td></tr> </tbody> </table>				Command				e_set_to_stop_p				e_set_to_stop_2				e_set_to_dark_p				e_set_to_dark_2				e_clear_sig_speed1				e_clear_sig_speed2				e_clear_sig_speed3			
Command																																			
e_set_to_stop_p																																			
e_set_to_stop_2																																			
e_set_to_dark_p																																			
e_set_to_dark_2																																			
e_clear_sig_speed1																																			
e_clear_sig_speed2																																			
e_clear_sig_speed3																																			
Type:	FUR - Functional Requirement																																		
Status:	draft	Stability:	stable																																
Target Release:	SEA SW GEN 3.0	Priority:	mandatory																																
Originator:	Technics (R&D)	Allocation:	SW EC-MUX																																
Verification Method:	Test	Verification Step:	module test, subit																																
Comments:																																			
Satisfies:																																			

EC-MUX-SIG:SSS: REQ:3504	Signal With Three Speed Aspects and Stop and Dark commands - execute SPEED commands in STOP / DARK												
Given the current element state is <State> When receiving the command <Command> from CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to <NewState> to OC-LS And SEA shall set the current element state to <ChangingToSpeedState> <table><tr><th>State</th><th>Command</th><th>NewState</th><th>ChangingToSpeedState</th></tr><tr><td>STOP</td><td>e_clear_sig_speed1</td><td>SPEED1</td><td>CHANGING_TO_SPEED1</td></tr><tr><td></td><td>e_clear_sig_</td><td></td><td>CHANGING_TO_</td></tr></table>		State	Command	NewState	ChangingToSpeedState	STOP	e_clear_sig_speed1	SPEED1	CHANGING_TO_SPEED1		e_clear_sig_		CHANGING_TO_
State	Command	NewState	ChangingToSpeedState										
STOP	e_clear_sig_speed1	SPEED1	CHANGING_TO_SPEED1										
	e_clear_sig_		CHANGING_TO_										

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State	Command	NewState	ChangingToSpeedState
STOP	speed2	SPEED2	SPEED2
STOP	e_clear_sig_speed3	SPEED3	CHANGING_TO_SPEED3
DARK	e_clear_sig_speed1	SPEED1	CHANGING_TO_SPEED1
DARK	e_clear_sig_speed2	SPEED2	CHANGING_TO_SPEED2
DARK	e_clear_sig_speed3	SPEED3	CHANGING_TO_SPEED3

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3505	Signal With Three Speed Aspects and Stop and Dark commands - reject STOP commands in STOP			
Given the current element state is STOP When receiving the command <Command> from CC Then SEA shall report the element state e_sig_stop_es with command reply flag true to CC And SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_set_to_stop_p</td></tr><tr><td>e_set_to_stop_2</td></tr></table>		Command	e_set_to_stop_p	e_set_to_stop_2
Command				
e_set_to_stop_p				
e_set_to_stop_2				
Type:	FUR - Functional Requirement			

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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3506	Signal With Three Speed Aspects and Stop and Dark commands - reject DARK commands in DARK					
Given the current element state is DARK When receiving the command <Command> from CC Then SEA shall report the element state e_sig_dark_es with command reply flag true to CC And SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_set_to_dark_p</td></tr><tr><td>e_set_to_dark_2</td></tr></table>				Command	e_set_to_dark_p	e_set_to_dark_2
Command						
e_set_to_dark_p						
e_set_to_dark_2						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

EC-MUX-SIG:SSS: REQ:3507	Signal With Three Speed Aspects and Stop and Dark commands - execute STOP commands in DARK
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Given the current element state is DARK

When receiving the command <Command> from CC

Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS

And SEA shall set the current element state to CHANGING_TO_STOP

Command
e_set_to_stop_p
e_set_to_stop_2

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

**EC-MUX-SIG:SSS:
REQ:3508**

**Signal With Three Speed Aspects and Stop and Dark commands -
execute DARK commands in STOP**

Given the current element state is STOP

When receiving the command <Command> from CC

Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS

And SEA shall set the current element state to CHANGING_TO_DARK

Command
e_set_to_dark_p
e_set_to_dark_2

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<table><tr><th>Command</th></tr><tr><td>2</td></tr></table>				Command	2
Command					
2					
Type:	FUR - Functional Requirement				
Status:	draft	Stability:	stable		
Target Release:	SEA SW GEN 3.0	Priority:	mandatory		
Originator:	Technics (R&D)	Allocation:	SW EC-MUX		
Verification Method:	Test	Verification Step:	module test, subit		
Comments:					
Satisfies:					

EC-MUX-SIG:SSS: REQ:3509	Signal With Three Speed Aspects and Stop and Dark commands - spontaneous transition between STOP / DARK														
Given the current element state is <State> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag false to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>State</th><th>NewState</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>STOP</td><td>DARK</td><td>e_sig_dark_es</td><td>SIG_DARK</td></tr><tr><td>DARK</td><td>STOP</td><td>e_sig_stop_es</td><td>SIG_STOP</td></tr></table>				State	NewState	CCState	DGPMMessage	STOP	DARK	e_sig_dark_es	SIG_DARK	DARK	STOP	e_sig_stop_es	SIG_STOP
State	NewState	CCState	DGPMMessage												
STOP	DARK	e_sig_dark_es	SIG_DARK												
DARK	STOP	e_sig_stop_es	SIG_STOP												
Type:	FUR - Functional Requirement														
Status:	draft	Stability:	stable												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory												
Originator:	Technics (R&D)	Allocation:	SW EC-MUX												
Verification Method:	Test	Verification Step:	module test, subit												
Comments:															

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Satisfies:	
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EC-MUX-SIG:SSS: REQ:3510	Signal With Three Speed Aspects and Stop and Dark commands - execute STOP commands in SPEEDx																
Given the current element state is <State> When receiving the command <Command> from CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS And SEA shall set the current element state to CHANGING_TO_STOP <table><tr><th>State</th><th>Command</th></tr><tr><td>SPEED1</td><td>e_set_to_stop_p</td></tr><tr><td>SPEED2</td><td>e_set_to_stop_p</td></tr><tr><td>SPEED3</td><td>e_set_to_stop_p</td></tr><tr><td>SPEED1</td><td>e_set_to_stop_2</td></tr><tr><td>SPEED2</td><td>e_set_to_stop_2</td></tr><tr><td>SPEED3</td><td>e_set_to_stop_2</td></tr></table>				State	Command	SPEED1	e_set_to_stop_p	SPEED2	e_set_to_stop_p	SPEED3	e_set_to_stop_p	SPEED1	e_set_to_stop_2	SPEED2	e_set_to_stop_2	SPEED3	e_set_to_stop_2
State	Command																
SPEED1	e_set_to_stop_p																
SPEED2	e_set_to_stop_p																
SPEED3	e_set_to_stop_p																
SPEED1	e_set_to_stop_2																
SPEED2	e_set_to_stop_2																
SPEED3	e_set_to_stop_2																
Type:	FUR - Functional Requirement																
Status:	draft	Stability:	stable														
Target Release:	SEA SW GEN 3.0	Priority:	mandatory														
Originator:	Technics (R&D)	Allocation:	SW EC-MUX														
Verification Method:	Test	Verification Step:	module test, subit														
Comments:																	
Satisfies:																	

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EC-MUX-SIG:SSS: REQ:3511	Signal With Three Speed Aspects and Stop and Dark commands - execute DARK commands in SPEEDx																
Given the current element state is <State> When receiving the command <Command> from CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to CHANGING_TO_DARK <table><tr><th>State</th><th>Command</th></tr><tr><td>SPEED1</td><td>e_set_to_dark_p</td></tr><tr><td>SPEED2</td><td>e_set_to_dark_p</td></tr><tr><td>SPEED3</td><td>e_set_to_dark_p</td></tr><tr><td>SPEED1</td><td>e_set_to_dark_2</td></tr><tr><td>SPEED2</td><td>e_set_to_dark_2</td></tr><tr><td>SPEED3</td><td>e_set_to_dark_2</td></tr></table>				State	Command	SPEED1	e_set_to_dark_p	SPEED2	e_set_to_dark_p	SPEED3	e_set_to_dark_p	SPEED1	e_set_to_dark_2	SPEED2	e_set_to_dark_2	SPEED3	e_set_to_dark_2
State	Command																
SPEED1	e_set_to_dark_p																
SPEED2	e_set_to_dark_p																
SPEED3	e_set_to_dark_p																
SPEED1	e_set_to_dark_2																
SPEED2	e_set_to_dark_2																
SPEED3	e_set_to_dark_2																
Type:	FUR - Functional Requirement																
Status:	draft	Stability:	stable														
Target Release:	SEA SW GEN 3.0	Priority:	mandatory														
Originator:	Technics (R&D)	Allocation:	SW EC-MUX														
Verification Method:	Test	Verification Step:	module test, subit														
Comments:																	
Satisfies:																	

EC-MUX-SIG:SSS: REQ:3512	Signal With Three Speed Aspects and Stop and Dark commands - reject SPEED commands in SPEEDx
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Given the current element state is <State>

When receiving the command <Command> from CC

Then SEA shall report the element state <CCState> with command reply flag true to CC

And SEA shall report the rejected command diagnosis to DGP

State	Command	CCState
SPEED1	e_clear_sig_speed1	e_sig_speed1_es
SPEED1	e_clear_sig_speed2	e_sig_speed1_es
SPEED1	e_clear_sig_speed3	e_sig_speed1_es
SPEED2	e_clear_sig_speed1	e_sig_speed2_es
SPEED2	e_clear_sig_speed2	e_sig_speed2_es
SPEED2	e_clear_sig_speed3	e_sig_speed2_es
SPEED3	e_clear_sig_speed1	e_sig_speed3_es
SPEED3	e_clear_sig_speed2	e_sig_speed3_es
SPEED3	e_clear_sig_speed3	e_sig_speed3_es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	CC shall not send the command SPEED, if the corresponding aspect is already given. CC shall not send any other SPEED command (downgrade / upgrade the speed) without STOP command between.		
Satisfies:			

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EC-MUX-SIG:SSS: REQ:3513	Signal With Three Speed Aspects and Stop and Dark commands - spontaneous transition from SPEEDx to STOP / DARK																														
Given the current element state is <State> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag false to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>State</th><th>NewState</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>SPEED1</td><td>STOP</td><td>e_sig_stop_es</td><td>SIG_STOP</td></tr><tr><td>SPEED1</td><td>DARK</td><td>e_sig_dark_es</td><td>SIG_DARK</td></tr><tr><td>SPEED2</td><td>STOP</td><td>e_sig_stop_es</td><td>SIG_STOP</td></tr><tr><td>SPEED2</td><td>DARK</td><td>e_sig_dark_es</td><td>SIG_DARK</td></tr><tr><td>SPEED3</td><td>STOP</td><td>e_sig_stop_es</td><td>SIG_STOP</td></tr><tr><td>SPEED3</td><td>DARK</td><td>e_sig_dark_es</td><td>SIG_DARK</td></tr></table>				State	NewState	CCState	DGPMMessage	SPEED1	STOP	e_sig_stop_es	SIG_STOP	SPEED1	DARK	e_sig_dark_es	SIG_DARK	SPEED2	STOP	e_sig_stop_es	SIG_STOP	SPEED2	DARK	e_sig_dark_es	SIG_DARK	SPEED3	STOP	e_sig_stop_es	SIG_STOP	SPEED3	DARK	e_sig_dark_es	SIG_DARK
State	NewState	CCState	DGPMMessage																												
SPEED1	STOP	e_sig_stop_es	SIG_STOP																												
SPEED1	DARK	e_sig_dark_es	SIG_DARK																												
SPEED2	STOP	e_sig_stop_es	SIG_STOP																												
SPEED2	DARK	e_sig_dark_es	SIG_DARK																												
SPEED3	STOP	e_sig_stop_es	SIG_STOP																												
SPEED3	DARK	e_sig_dark_es	SIG_DARK																												
Type:	FUR - Functional Requirement																														
Status:	draft	Stability:	stable																												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory																												
Originator:	Technics (R&D)	Allocation:	SW EC-MUX																												
Verification Method:	Test	Verification Step:	module test, subit																												
Comments:																															
Satisfies:																															

EC-MUX-SIG:SSS:	Signal With Three Speed Aspects and Stop and Dark commands -
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REQ:3514	CHANGING_TO_SPEEDn - transition from CHANGING_TO_SPEEDx to SPEEDx																		
Given the current element state is <ChangingToSpeedState> When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS within WaitForSpeed ms Then SEA shall set the current element state to <State> And SEA shall report the element state <CCState> with command reply flag true to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>ChangingToSpeedState</th><th>State</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>CHANGING_TO_SPEED1</td><td>SPEED1</td><td>e_sig_speed1_es</td><td>SIG_SPEED_1</td></tr><tr><td>CHANGING_TO_SPEED2</td><td>SPEED2</td><td>e_sig_speed2_es</td><td>SIG_SPEED_2</td></tr><tr><td>CHANGING_TO_SPEED3</td><td>SPEED3</td><td>e_sig_speed3_es</td><td>SIG_SPEED_3</td></tr></table>				ChangingToSpeedState	State	CCState	DGPMMessage	CHANGING_TO_SPEED1	SPEED1	e_sig_speed1_es	SIG_SPEED_1	CHANGING_TO_SPEED2	SPEED2	e_sig_speed2_es	SIG_SPEED_2	CHANGING_TO_SPEED3	SPEED3	e_sig_speed3_es	SIG_SPEED_3
ChangingToSpeedState	State	CCState	DGPMMessage																
CHANGING_TO_SPEED1	SPEED1	e_sig_speed1_es	SIG_SPEED_1																
CHANGING_TO_SPEED2	SPEED2	e_sig_speed2_es	SIG_SPEED_2																
CHANGING_TO_SPEED3	SPEED3	e_sig_speed3_es	SIG_SPEED_3																
Type:	FUR - Functional Requirement																		
Status:	draft	Stability:	stable																
Target Release:	SEA SW GEN 3.0	Priority:	mandatory																
Originator:	Technics (R&D)	Allocation:	SW EC-MUX																
Verification Method:	Test	Verification Step:	module test, subit																
Comments:	Successful SPEED command execution - commanded aspect reached.																		
Satisfies:																			

EC-MUX-SIG:SSS: REQ:3515	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_SPEEDn - reject repeated SPEED command in CHANGING_TO_SPEEDn
Given the current element state is CHANGING_TO_SPEEDn When receiving the command <Command> from CC within WaitForSpeed ms Then SEA shall report the rejected command diagnosis to DGP	

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<table border="1"> <thead> <tr> <th colspan="4">Command</th></tr> </thead> <tbody> <tr> <td colspan="4">e_clear_sig_speed1</td></tr> <tr> <td colspan="4">e_clear_sig_speed2</td></tr> <tr> <td colspan="4">e_clear_sig_speed3</td></tr> </tbody> </table>				Command				e_clear_sig_speed1				e_clear_sig_speed2				e_clear_sig_speed3			
Command																			
e_clear_sig_speed1																			
e_clear_sig_speed2																			
e_clear_sig_speed3																			
Type:	FUR - Functional Requirement																		
Status:	draft	Stability:	stable																
Target Release:	SEA SW GEN 3.0	Priority:	mandatory																
Originator:	Technics (R&D)	Allocation:	SW EC-MUX																
Verification Method:	Test	Verification Step:	module test, subit																
Comments:	CC shall not repeat the command SPEED. CC shall not switch to another SPEED aspect without STOP command between them. The WaitForSpeed timer shall not to be re-started due to the rejected command.																		
Satisfies:																			

EC-MUX-SIG:SSS: REQ:3516	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_SPEEDn - return to STOP / DARK									
Given the current element state is CHANGING_TO_SPEEDn And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS within WaitForSpeed ms Then SEA shall set the current element state to <State> And SEA shall report the element state <CCState> with command reply flag true to CC <table><tr><th>LastStable</th><th>State</th><th>CCState</th></tr><tr><td>STOP</td><td>STOP</td><td>e_sig_stop_es</td></tr><tr><td>DARK</td><td>DARK</td><td>e_sig_dark_es</td></tr></table>		LastStable	State	CCState	STOP	STOP	e_sig_stop_es	DARK	DARK	e_sig_dark_es
LastStable	State	CCState								
STOP	STOP	e_sig_stop_es								
DARK	DARK	e_sig_dark_es								

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	OC-LS has probably sent the message as a downgrade aspect for failed clear command. As it is not possible to forward the command corresponding to SPEEDx aspect before receiving a reply to the previous STOP, it is not necessary to continue waiting for a further reply and thus the CHANGING_TO_SPEEDx state shall be left.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3517	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_SPEEDn - transition to STOP / DARK														
Given the current element state is CHANGING_TO_SPEEDn And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS within WaitForSpeed ms Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag true to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>LastStable</th><th>NewState</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>STOP</td><td>DARK</td><td>e_sig_dark_ es</td><td>SIG_DARK</td></tr><tr><td>DARK</td><td>STOP</td><td>e_sig_stop_ es</td><td>SIG_STOP</td></tr></table>				LastStable	NewState	CCState	DGPMMessage	STOP	DARK	e_sig_dark_ es	SIG_DARK	DARK	STOP	e_sig_stop_ es	SIG_STOP
LastStable	NewState	CCState	DGPMMessage												
STOP	DARK	e_sig_dark_ es	SIG_DARK												
DARK	STOP	e_sig_stop_ es	SIG_STOP												
Type:	FUR - Functional Requirement														
Status:	draft	Stability:	stable												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory												
Originator:	Technics (R&D)	Allocation:	SW EC-MUX												
Verification Method:	Test	Verification Step:	module test, subit												

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Comments:	OC-LS has probably sent the message as a downgrade aspect for failed command. As it is not possible to forward the command corresponding to SPEEDx aspect before receiving a reply to the previous STOP, it is not necessary to continue waiting for a further reply and thus the CHANGING_TO_SPEEDx state shall be left. The diagnosis aspect message is sent only if the last stable state is different than the OC-LS reply message.
Satisfies:	

EC-MUX-SIG:SSS: REQ:3518	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_SPEEDn - timeout								
Given the current element state is CHANGING_TO_SPEEDn And the last stable element state is <LastStable> When the current state of the element has been CHANGING_TO_SPEEDn for WaitForSpeed ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x". to DGP And SEA shall set the current element state to <LastStable> And SEA shall report the element state <CCState> with command reply flag true to CC <table><tr><th>LastStable</th><th>CCState</th></tr><tr><td>STOP</td><td>e_sig_dark_es</td></tr><tr><td>DARK</td><td>e_sig_stop_es</td></tr></table>				LastStable	CCState	STOP	e_sig_dark_es	DARK	e_sig_stop_es
LastStable	CCState								
STOP	e_sig_dark_es								
DARK	e_sig_stop_es								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:	Regularly the state CHANGING_TO_SPEEDn will be left after WaitForSpeed ms. No safety reaction OC Disconnection is performed as this is not a safety-critical command. Instead, SEA shall generate a diagnosis and try to fallback to original STOP/DARK.								

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Satisfies:	
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EC-MUX-SIG:SSS: REQ:3519	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_SPEEDn - execute STOP command					
Given the current element state is CHANGING_TO_SPEEDn When receiving the command <Command> from CC within WaitForSpeed ms Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS And SEA shall set the current element state to CHANGING_TO_STOP <table><tr><th>Command</th></tr><tr><td>e_set_to_stop_p</td></tr><tr><td>e_set_to_stop_2</td></tr></table>				Command	e_set_to_stop_p	e_set_to_stop_2
Command						
e_set_to_stop_p						
e_set_to_stop_2						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:	If the STOP command reaches the OC-LS before the SPEEDx command is answered, then it is possible that OC-LS will reply to none of both commands, because the last commanded aspect is still given.					
Satisfies:						

EC-MUX-SIG:SSS: REQ:3520	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_SPEEDn - execute DARK command
Given the current element state is CHANGING_TO_SPEEDn When receiving the command <Command> from CC within WaitForSpeed ms	

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Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS
And SEA shall set the current element state to CHANGING_TO_DARK

Command
e_set_to_dark_p
e_set_to_dark_2

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	If the DARK command reaches the OC-LS before the SPEEDx command is answered, then it is possible that OC-LS will reply to none of both commands, because the last commanded aspect is still given.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3521	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_STOP - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state SPEEDx
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Given the current element state is CHANGING_TO_STOP
And the last stable element state is <LastStable>

When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS within WaitForSafeState ms

Then SEA shall set the current element state to <NewState>
And SEA shall report the element state <CCState> with command reply flag true to CC
And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP

LastStable	NewState	CCState	DGPMMessage
SPEED1	STOP	e_sig_stop_es	SIG_STOP

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LastStable	NewState	CCState	DGPMessage
SPEED1	DARK	e_sig_dark_es	SIG_DARK
SPEED2	STOP	e_sig_stop_es	SIG_STOP
SPEED2	DARK	e_sig_dark_es	SIG_DARK
SPEED3	STOP	e_sig_stop_es	SIG_STOP
SPEED3	DARK	e_sig_dark_es	SIG_DARK

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Successful stop command execution - commanded aspect / downgrade aspect reached.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3527	Signal With Three Speed Aspects and Stop and Dark commands - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state STOP / DARK
-------------------------------------	--

Given the current element state is CHANGING_TO_STOP
And the last stable element state is <LastStable>

When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS within WaitForSafeState ms

Then SEA shall set the current element state to <State>
And SEA shall report the element state <CCState> with command reply flag true to CC

LastStable	State	CCState
STOP	STOP	e_sig_stop_es

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LastStable	State	CCState
DARK	DARK	e_sig_dark_ es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	After receiving the STOP / DARK as reply to the STOP command, the state will be reported to the CC with command reply flag true.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3528	Signal With Three Speed Aspects and Stop and Dark commands - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state DARK / STOP														
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag true to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>LastStable</th><th>NewState</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>DARK</td><td>STOP</td><td>e_sig_stop_es</td><td>SIG_STOP</td></tr><tr><td>STOP</td><td>DARK</td><td>e_sig_dark_es</td><td>SIG_DARK</td></tr></table>				LastStable	NewState	CCState	DGPMMessage	DARK	STOP	e_sig_stop_es	SIG_STOP	STOP	DARK	e_sig_dark_es	SIG_DARK
LastStable	NewState	CCState	DGPMMessage												
DARK	STOP	e_sig_stop_es	SIG_STOP												
STOP	DARK	e_sig_dark_es	SIG_DARK												
Type:	FUR - Functional Requirement														
Status:	draft	Stability:	stable												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory												
Originator:	Technics (R&D)	Allocation:	SW EC-MUX												

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Verification Method:	Test	Verification Step:	module test, subit
Comments:	After receiving the STOP / DARK as reply to the STOP command, the state will be reported to the CC with command reply flag true; aspect diagnosis is sent also as the state is changed.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3522	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_STOP - reject STOP / SPEED commands in CHANGING_TO_STOP								
Given the current element state is CHANGING_TO_STOP When receiving the command <Command> from CC within WaitForSafeState ms Then SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_set_to_stop_2</td></tr><tr><td>e_set_to_stop_p</td></tr><tr><td>e_clear_sig_speed1</td></tr><tr><td>e_clear_sig_speed2</td></tr><tr><td>e_clear_sig_speed3</td></tr></table>				Command	e_set_to_stop_2	e_set_to_stop_p	e_clear_sig_speed1	e_clear_sig_speed2	e_clear_sig_speed3
Command									
e_set_to_stop_2									
e_set_to_stop_p									
e_clear_sig_speed1									
e_clear_sig_speed2									
e_clear_sig_speed3									
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:	CC shall not send a clear command if there is STOP command not answered yet. CC shall not repeat the command STOP. The WaitForSafeState timer shall not to be re-started due to the rejected command.								

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Satisfies:	
------------	--

EC-MUX-SIG:SSS: REQ:3523	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_STOP - execute DARK command					
Given the current element state is CHANGING_TO_STOP When receiving the command <Command> from CC within WaitForSafeState ms Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to CHANGING_TO_DARK <table><tr><th>Command</th></tr><tr><td>e_set_to_dark_p</td></tr><tr><td>e_set_to_dark_2</td></tr></table>				Command	e_set_to_dark_p	e_set_to_dark_2
Command						
e_set_to_dark_p						
e_set_to_dark_2						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:	If the DARK command reaches the OC-LS before the STOP command is answered, then it is possible that OC-LS will reply to none of both commands, because the last commanded aspect is still given. After reaching the state CHANGING_TO_DARK, the WaitForSafeState timer is restarted.					
Satisfies:						

EC-MUX-SIG:SSS: REQ:3524	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_STOP - timeout when OC-LS SPEEDx - OC Disconnection
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable>	

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When the current state of the element has been CHANGING_TO_STOP for WaitForSafeState ms

Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP

And SEA shall execute EC-MUX safety reaction OC Disconnection

And SEA shall set the current element state to UNDEFINED

And SEA shall report the element state e_undefined_es with command reply flag true to CC

And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP

And SEA shall report the no_connection diagnosis device fault message to DGP

LastStable
SPEED1
SPEED2
SPEED3

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Regularly the state CHANGING_TO_STOP will be left after WaitForSafeState ms. Safety reaction OC Disconnection is applicable only if the OC-LS cannot reach the STOP aspect and stays in SPEEDx. See also EC-MUX-SIG:SSS:REQ:3525.		
Satisfies:			

EC-MUX-SIG:SSS:REQ:3525	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_STOP - timeout when OC-LS DARK / STOP - reach DARK / STOP
--------------------------------	---

Given the current element state is CHANGING_TO_STOP

And the last stable element state is <LastStable>

When the current state of the element has been CHANGING_TO_STOP for WaitForSafeState ms

Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP

And SEA shall set the current element state to <LastStable> after WaitForSafeState ms

And SEA shall report the element state <CCState> with command reply flag true to CC

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LastStable	CCState
STOP	e_sig_stop_es
DARK	e_sig_dark_es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Regularly the state CHANGING_TO_STOP will be left after WaitForSafeState ms. OC-LS state can be STOP or DARK when a STOP command has been executed after a SPEED command and no reply has been received for either command - see also EC-MUX-SIG:SSS:REQ:3518 and EC-MUX-SIG:SSS:REQ:3519. This is an allowed scenario so no safety reaction OC Disconnection is performed. OC-LS state can be DARK after a STOP command has been executed. OC-LS can be STOP when a STOP command has been executed after a DARK command and no reply has been received for either command - see also EC-MUX-SIG:SSS:REQ:3533.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3526	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_STOP - ignore SPEED message				
Given the current element state is CHANGING_TO_STOP When receiving the LS Indicated Signal Aspect message with Code corresponding to <SpeedAspect> from OC-LS within WaitForSafeState ms Then SEA shall do nothing <table><tr><th>SpeedAspect</th></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr><tr><td>SPEED3</td></tr></table>		SpeedAspect	SPEED1	SPEED2	SPEED3
SpeedAspect					
SPEED1					
SPEED2					
SPEED3					

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	It is possible that OC-LS has sent the message corresponding to SPEED as reply to the previous SPEED command not answered yet. Waiting for the reply to the STOP command is continued by keeping the current state CHANGING_TO_STOP. After timer expiry, if the expected STOP reply is not received, the element state is changed to STOP / DARK or a Safety Reaction OC Disconnection is performed, depending on the last received OC-LS state.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3529	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_DARK - transition from CHANGING_TO_DARK to STOP / DARK - from OC-LS state SPEEDx																								
Given the current element state is CHANGING_TO_DARK And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag true to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>LastStable</th><th>NewState</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>SPEED1</td><td>STOP</td><td>e_sig_stop_ es</td><td>SIG_STOP</td></tr><tr><td>SPEED1</td><td>DARK</td><td>e_sig_dark_ es</td><td>SIG_DARK</td></tr><tr><td>SPEED2</td><td>STOP</td><td>e_sig_stop_ es</td><td>SIG_STOP</td></tr><tr><td>SPEED2</td><td>DARK</td><td>e_sig_dark_ es</td><td>SIG_DARK</td></tr><tr><td>SPEED3</td><td>STOP</td><td>e_sig_stop_ es</td><td>SIG_STOP</td></tr></table>		LastStable	NewState	CCState	DGPMMessage	SPEED1	STOP	e_sig_stop_ es	SIG_STOP	SPEED1	DARK	e_sig_dark_ es	SIG_DARK	SPEED2	STOP	e_sig_stop_ es	SIG_STOP	SPEED2	DARK	e_sig_dark_ es	SIG_DARK	SPEED3	STOP	e_sig_stop_ es	SIG_STOP
LastStable	NewState	CCState	DGPMMessage																						
SPEED1	STOP	e_sig_stop_ es	SIG_STOP																						
SPEED1	DARK	e_sig_dark_ es	SIG_DARK																						
SPEED2	STOP	e_sig_stop_ es	SIG_STOP																						
SPEED2	DARK	e_sig_dark_ es	SIG_DARK																						
SPEED3	STOP	e_sig_stop_ es	SIG_STOP																						

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LastStable	NewState	CCState	DGPMessage
SPEED3	DARK	e_sig_dark_es	SIG_DARK
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Successful stop command execution - commanded aspect / downgrade aspect reached.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3535	Signal With Three Speed Aspects and Stop and Dark commands - transition from CHANGING_TO_DARK to STOP / DARK - from OC-LS state STOP / DARK											
Given the current element state is CHANGING_TO_DARK And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to <State> And SEA shall report the element state <CCState> with command reply flag true to CC <table><tr><th>LastStable</th><th>State</th><th>CCState</th></tr><tr><td>STOP</td><td>STOP</td><td>e_sig_stop_es</td></tr><tr><td>DARK</td><td>DARK</td><td>e_sig_dark_es</td></tr></table>				LastStable	State	CCState	STOP	STOP	e_sig_stop_es	DARK	DARK	e_sig_dark_es
LastStable	State	CCState										
STOP	STOP	e_sig_stop_es										
DARK	DARK	e_sig_dark_es										
Type:	FUR - Functional Requirement											
Status:	draft	Stability:	stable									
Target Release:	SEA SW GEN 3.0	Priority:	mandatory									
Originator:	Technics (R&D)	Allocation:	SW EC-MUX									

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Verification Method:	Test	Verification Step:	module test, subit
Comments:	After receiving the STOP / DARK as reply to the STOP command, the state will be reported to the CC with command reply flag true.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3536	Signal With Three Speed Aspects and Stop and Dark commands - transition from CHANGING_TO_DARK to STOP / DARK - from OC-LS state DARK / STOP														
Given the current element state is CHANGING_TO_DARK And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to <NewState> from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to <NewState> And SEA shall report the element state <CCState> with command reply flag true to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>LastStable</th><th>NewState</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>DARK</td><td>STOP</td><td>e_sig_stop_es</td><td>SIG_STOP</td></tr><tr><td>STOP</td><td>DARK</td><td>e_sig_dark_es</td><td>SIG_DARK</td></tr></table>				LastStable	NewState	CCState	DGPMMessage	DARK	STOP	e_sig_stop_es	SIG_STOP	STOP	DARK	e_sig_dark_es	SIG_DARK
LastStable	NewState	CCState	DGPMMessage												
DARK	STOP	e_sig_stop_es	SIG_STOP												
STOP	DARK	e_sig_dark_es	SIG_DARK												
Type:	FUR - Functional Requirement														
Status:	draft	Stability:	stable												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory												
Originator:	Technics (R&D)	Allocation:	SW EC-MUX												
Verification Method:	Test	Verification Step:	module test, subit												
Comments:	After receiving the STOP / DARK as reply to the DARK command, the state will be reported to the CC with command reply flag true; aspect diagnosis is sent also as the state is changed.														
Satisfies:															

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EC-MUX-SIG:SSS: REQ:3530	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_DARK - reject DARK / SPEED commands in CHANGING_TO_DARK								
Given the current element state is CHANGING_TO_DARK When receiving the command <Command> from CC within WaitForSafeState ms Then SEA shall report the rejected command diagnosis to DGP <table><tr><th>Command</th></tr><tr><td>e_set_to_dark_2</td></tr><tr><td>e_set_to_dark_p</td></tr><tr><td>e_clear_sig_speed1</td></tr><tr><td>e_clear_sig_speed2</td></tr><tr><td>e_clear_sig_speed3</td></tr></table>				Command	e_set_to_dark_2	e_set_to_dark_p	e_clear_sig_speed1	e_clear_sig_speed2	e_clear_sig_speed3
Command									
e_set_to_dark_2									
e_set_to_dark_p									
e_clear_sig_speed1									
e_clear_sig_speed2									
e_clear_sig_speed3									
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:	CC shall not send a clear command if there is DARK command not answered yet. CC shall not repeat the command DARK. The WaitForSafeState timer shall not to be re-started due to the rejected command.								
Satisfies:									

EC-MUX-SIG:SSS: REQ:3531	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_DARK - execute STOP command
Given the current element state is CHANGING_TO_DARK	

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When receiving the command <Command> from CC within WaitForSafeState ms

Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to STOP to OC-LS

And SEA shall set the current element state to CHANGING_TO_STOP

Command
e_set_to_stop_p
e_set_to_stop_2

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	If the STOP command reaches the OC-LS before the DARK command is answered, then it is possible that OC-LS will reply to none of both commands, because the last commanded aspect is still given. After reaching the state CHANGING_TO_STOP, the WaitForSafeState timer is restarted.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3532	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_DARK - timeout when OC-LS SPEEDx - OC Disconnection
Given the current element state is CHANGING_TO_DARK And the last stable element state is <LastStable> When the current state of the element has been CHANGING_TO_DARK for WaitForSafeState ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP	

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<table><tr><td>LastStable</td></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr><tr><td>SPEED3</td></tr></table>				LastStable	SPEED1	SPEED2	SPEED3
LastStable							
SPEED1							
SPEED2							
SPEED3							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:	Regularly the state CHANGING_TO_DARK will be left after WaitForSafeState ms. Safety reaction OC Disconnection is applicable only if the OC-LS cannot reach the DARK aspect and stays in SPEEDx. See also EC-MUX-SIG:SSS:REQ:3533.						
Satisfies:							

EC-MUX-SIG:SSS: REQ:3533	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_DARK - timeout when OC-LS DARK - reach DARK		
Given the current element state is CHANGING_TO_DARK And the last stable element state is DARK When the current state of the element has been CHANGING_TO_DARK for WaitForSafeState ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP And SEA shall set the current element state to DARK And SEA shall report the element state e_sig_dark_es with command reply flag true to CC			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit

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Comments:	Regularly the state CHANGING_TO_DARK will be left after WaitForSafeState ms. OC-LS state can be DARK when a DARK command has been executed after a SPEED / STOP command and no reply has been received for either command - see also EC-MUX-SIG:SSS:REQ:3518 and EC-MUX-SIG:SSS:REQ:3520. This is an allowed scenario so no safety reaction OC Disconnection is performed.
Satisfies:	

EC-MUX-SIG:SSS: REQ:3560	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_DARK - timeout when OC-LS STOP - OC Disconnection		
Given the current element state is CHANGING_TO_DARK And the last stable element state is STOP When the current state of the element has been CHANGING_TO_DARK for WaitForSafeState ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP after WaitForSafeState ms after reaching CHANGING_TO_DARK And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Regularly the state CHANGING_TO_DARK will be left after WaitForSafeState ms. The safety reaction OC Disconnection is performed when OC-LS state is STOP and a DARK command has been executed and: - no reply has been received for the command; - after a SPEED command, no reply has been received for either commands.		
Satisfies:			

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EC-MUX-SIG:SSS: REQ:3534	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_DARK - ignore SPEED message						
Given the current element state is CHANGING_TO_DARK When receiving the LS Indicated Signal Aspect message with Code corresponding to <SpeedAspect> from OC-LS within WaitForSafeState ms Then SEA shall do nothing <table><tr><th>SpeedAspect</th></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr><tr><td>SPEED3</td></tr></table>				SpeedAspect	SPEED1	SPEED2	SPEED3
SpeedAspect							
SPEED1							
SPEED2							
SPEED3							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:	It is possible that OC-LS has sent the message corresponding to SPEED as reply to the previous SPEED command not answered yet. Waiting for the reply to the DARK command is continued by keeping the current state CHANGING_TO_DARK. After timer expiry, if the expected DARK reply is not received, the element state is changed to STOP / DARK or a Safety Reaction OC Disconnection is performed, depending on the last received OC-LS state.						
Satisfies:							

EC-MUX-SIG:SSS: REQ:3537	Signal With Three Speed Aspects and Stop and Dark commands - Disconnection on spontaneous SPEED message in STOP / DARK / SPEED
Given the current element state is <State>	
When receiving the LS Indicated Signal Aspect message with Code corresponding to <SpeedAspect> from OC-LS	

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Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: Spontaneous <SpeedAspect> reported for Signal x." to DGP

And SEA shall execute EC-MUX safety reaction OC Disconnection

And SEA shall set the current element state to UNDEFINED

And SEA shall report the element state e_undefined_es with command reply flag false to CC

And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP

And SEA shall report the no_connection diagnosis device fault message to DGP

State	SpeedAspect
STOP	SPEED1
STOP	SPEED2
STOP	SPEED3
DARK	SPEED1
DARK	SPEED2
DARK	SPEED3
SPEED1	SPEED2
SPEED1	SPEED3
SPEED2	SPEED1
SPEED2	SPEED3
SPEED3	SPEED1
SPEED3	SPEED2

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3538	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_SPEEDn - Disconnection on unexpected command reply
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Given the current element state is <ChangingToSpeedState>

When receiving the LS Indicated Signal Aspect message with Code corresponding to <SpeedMsg> from OC-LS within WaitForSpeed ms

Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: Cmd <SpeedCmd> replied by Msg <SpeedMsg> for Signal x." to DGP

And SEA shall execute EC-MUX safety reaction OC Disconnection

And SEA shall set the current element state to UNDEFINED

And SEA shall report the element state e_undefined_es with command reply flag true to CC

And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP

And SEA shall report the no_connection diagnosis device fault message to DGP

ChangingToSpeedState	SpeedMsg	SpeedCmd
CHANGING_TO_SPEED1	SPEED2	SPEED1
CHANGING_TO_SPEED1	SPEED3	SPEED1
CHANGING_TO_SPEED2	SPEED1	SPEED2
CHANGING_TO_SPEED2	SPEED3	SPEED2
CHANGING_TO_SPEED3	SPEED1	SPEED3
CHANGING_TO_SPEED3	SPEED2	SPEED3

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3539	Signal With Three Speed Aspects and Stop and Dark commands - Ignore repeated messages
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Given the current element state is <State>

When receiving the LS Indicated Signal Aspect message with Code corresponding to <SpeedAspect> from OC-LS

Then SEA shall do nothing

State	SpeedAspect
STOP	STOP
SPEED1	SPEED1
SPEED2	SPEED2
SPEED3	SPEED3
DARK	DARK

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

3.1.7.3 Connection - Signal With Three Speed Aspects and Stop and Dark commands

In order to control and supervise a Signal With Three Speed Aspects and Stop and Dark commands
As a SEA

I want to be able to supervise connections and when changes occur then

- on CC connection establishment report the current state to CC
- on CC connection loss execute the safe command DARK if the last command forwarded to OC-LS was SPEED1 / SPEED2 / SPEED3 / STOP and it was not answered by DARK
- on OC-LS connection loss take on the default state UNDEFINED
- on OC-LS connection establishment transition to DARK / STOP state as the OC-LS element state
- report the connection state changes to CC and DGP

For requirement language see Chapter 2.8 "Requirement Language".

CHANGING_TO_SPEEDn, CHANGING_TO_STOP and CHANGING_TO_DARK element states are not considered in the mapping between element state and CC State because when reaching these states,

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nothing is reported to CC.

WaitForSpeed and WaitForSafeState are configurable timers for receiving a command reply from OC-LS. WaitForSpeed is used when waiting for a reply to a SPEED command (e_clear_sig_speed1, e_clear_sig_speed2, e_clear_sig_speed3), its default value is 2000ms. WaitForSafeState is used when waiting for a reply to a STOP command (e_set_to_stop_2, e_set_to_stop_p) or to a DARK command (e_set_to_dark_2, e_set_to_dark_p), its default value is 1000ms.

Given a Signal With Three Speed Aspects and Stop and Dark commands

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a STOP command or a DARK command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_stop_es
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es
SPEED3	e_sig_speed3_es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12
SIG_STOP	13
SIG_SPEED_1	16
SIG_SPEED_2	17
SIG_SPEED_3	18

EC-MUX-SIG:SSS: REQ:3540	Signal With Three Speed Aspects and Stop and Dark commands - [Re-] Connecting to CC - OC-LS CONNECTED - General inquiry in stable states
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Given the current element state is <State>
And CC connection state is NOT_CONNECTED
And EC-MUX - OC-LS connection state is CONNECTED

When establishing the connection to CC

Then SEA shall report the element state <CCState> with command reply flag false to CC

State	CCState
STOP	e_sig_stop_ es
DARK	e_sig_dark_ es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	[Re-]Connecting in SPEEDx state is impossible. [Re-]Connecting in STOP is possible only if the disconnect happens in STOP and the OC-LS does not answer the generated safe command DARK - in this case the timer in CHANGING_TO_DARK shall timeout and state STOP shall be reached again.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3541	Signal With Three Speed Aspects and Stop and Dark commands - [Re-] Connecting to CC - OC-LS CONNECTED - General inquiry in CHANGING_TO_DARK - no message after CC disconnect
Given the current element state is CHANGING_TO_DARK And the last stable element state is <LastStable> And CC connection state is NOT_CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When establishing the connection to CC within WaitForSafeState ms Then SEA shall report the element state <CCState> with command reply flag false to CC	

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LastStable	CCState
STOP	e_sig_stop_es
DARK	e_sig_dark_es
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es
SPEED3	e_sig_speed3_es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Re-connecting to CC in CHANGING_TO_DARK state is impossible in real operation, as re-connecting will regularly take much longer time than the configured WaitForSafeState. Due to the execution of the safe command [Re-]Connecting in CHANGING_TO_DARK state is possible only with last forwarded command DARK, but depending on the previous events the last state reported to CC can be different. Following initial states are covered with this requirement: - CHANGING_TO_DARK with last stable element state SPEED1/SPEED2 /SPEED3 - CHANGING_TO_DARK with last stable element state DARK/STOP - CHANGING_TO_DARK with last stable element state DARK/STOP and SPEED1/SPEED2/SPEED3 received from OC-LS in CHANGING_TO_DARK - CHANGING_TO_DARK with last stable element state SPEED1/SPEED2 /SPEED3/DARK/STOP and SPEED1/SPEED2/SPEED3 received from OC-LS in CHANGING_TO_STOP - CHANGING_TO_DARK as a safe command reaction upon CC connection loss with last stable element state STOP		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3542	Signal With Three Speed Aspects and Stop and Dark commands - [Re-] Connecting to CC - OC-LS NOT_CONNECTED - General inquiry
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Given the current element state is UNDEFINED
And CC connection state is NOT_CONNECTED
And EC-MUX - OC-LS connection state is NOT_CONNECTED

When establishing the connection to CC

Then SEA shall report the element state e_undefined_es with command reply flag false to CC

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS:
REQ:3558 **Signal With Three Speed Aspects and Stop and Dark commands - Connection loss to CC - execute safe command in STOP**

Given the current element state is STOP
And CC connection state is CONNECTED

When losing the connection to CC

Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS

And SEA shall set the current element state to CHANGING_TO_DARK

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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EC-MUX-SIG:SSS: REQ:3543	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss to CC - execute safe command in SPEEDx						
Given the current element state is <State> And CC connection state is CONNECTED When losing the connection to CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to CHANGING_TO_DARK <table><tr><th>State</th></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr><tr><td>SPEED3</td></tr></table>				State	SPEED1	SPEED2	SPEED3
State							
SPEED1							
SPEED2							
SPEED3							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:							
Satisfies:							

EC-MUX-SIG:SSS: REQ:3544	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss to CC - do nothing in DARK / UNDEFINED
Given the current element state is <State> And CC connection state is CONNECTED When losing the connection to CC Then SEA shall do nothing	

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State
DARK
UNDEFINED

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3545	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss to CC - CHANGING_TO_SPEEDn		
Given the current element state is CHANGING_TO_SPEEDn And CC connection state is CONNECTED When losing the connection to CC within WaitForSpeed ms Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to CHANGING_TO_DARK			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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EC-MUX-SIG:SSS: REQ:3546	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss to CC - CHANGING_TO_STOP - execute safe command		
Given the current element state is CHANGING_TO_STOP And CC connection state is CONNECTED When losing the connection to CC within WaitForSafeState ms Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to CHANGING_TO_DARK			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3547	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss to CC - CHANGING_TO_DARK - do nothing		
Given the current element state is CHANGING_TO_DARK And CC connection state is CONNECTED When losing the connection to CC within WaitForSafeState ms Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			

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Satisfies:	
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EC-MUX-SIG:SSS: REQ:3501	Signal With Three Speed Aspects and Stop and Dark commands - [Re-] Connecting between EC-MUX and OC-LS in state STOP		
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED And OC-LS state is STOP When establishing the connection between EC-MUX and OC-LS Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: OC-LS connect with STOP for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3502	Signal With Three Speed Aspects and Stop and Dark commands - [Re-] Connecting between EC-MUX and OC-LS in state DARK		
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED And OC-LS state is DARK When establishing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to DARK And SEA shall report the element state e_sig_dark_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_DARK to DGP And SEA shall report the connection_ok diagnosis device fault repair message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3556	Signal With Three Speed Aspects and Stop and Dark commands - [Re-] Connecting between EC-MUX and OC-LS in state SPEEDx						
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED And OC-LS state is <SpeedAspect> When establishing the connection between EC-MUX and OC-LS Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: OC-LS connect with <SpeedAspect> for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection <table><tr><td>SpeedAspect</td></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr><tr><td>SPEED3</td></tr></table>				SpeedAspect	SPEED1	SPEED2	SPEED3
SpeedAspect							
SPEED1							
SPEED2							
SPEED3							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:							
Satisfies:							

EC-MUX-SIG:SSS:	Signal With Three Speed Aspects and Stop and Dark commands -
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REQ:3549	Connection loss between EC-MUX and OC-LS in regular states							
Given the current element state is <State> And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP <table><tr><td>State</td></tr><tr><td>STOP</td></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr><tr><td>SPEED3</td></tr></table>				State	STOP	SPEED1	SPEED2	SPEED3
State								
STOP								
SPEED1								
SPEED2								
SPEED3								
Type:	FUR - Functional Requirement							
Status:	draft	Stability:	stable					
Target Release:	SEA SW GEN 3.0	Priority:	mandatory					
Originator:	Technics (R&D)	Allocation:	SW EC-MUX					
Verification Method:	Test	Verification Step:	module test, subit					
Comments:								
Satisfies:								

EC-MUX-SIG:SSS: REQ:3550	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss between EC-MUX and OC-LS in DARK state
Given the current element state is DARK And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS	

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Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3551	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss between EC-MUX and OC-LS in CHANGING_TO_SPEEDn		
Given the current element state is CHANGING_TO_SPEEDn And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS within WaitForSpeed ms Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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EC-MUX-SIG:SSS: REQ:3552	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss between EC-MUX and OC-LS in CHANGING_TO_DARK		
Given the current element state is CHANGING_TO_DARK And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS within WaitForSafeState ms Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3557	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss between EC-MUX and OC-LS in CHANGING_TO_STOP		
Given the current element state is CHANGING_TO_STOP And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS within WaitForSafeState ms Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

3.1.7.4 Diagnosis - Signal With Three Speed Aspects and Stop and Dark commands

In order to provide the aspect diagnosis for a Signal With Three Speed Aspects and Stop and Dark commands

As a SEA

I want to be able to receive an aspect diagnosis query from DGP and answer it by the DGP diagnosis values corresponding to the last diagnosis aspect message reported to DGP for the queried Signal

For requirement language see Chapter 2.8 "Requirement Language".

Given a Signal With Three Speed Aspects and Stop and Dark commands

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a STOP command or a DARK command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_stop_es
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es
SPEED3	e_sig_speed3_es
DARK	e_sig_dark_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0

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SEA diagnosis aspect	DGP diagnosis value
SIG_ DARK	12
SIG_ STOP	13
SIG_ SPEED_ 1	16
SIG_ SPEED_ 2	17
SIG_ SPEED_ 3	18

EC-MUX-SIG:SSS: REQ:3553	Signal With Three Speed Aspects and Stop and Dark commands - aspect query - last stable state														
Given the last stable element state is <LastStable> When receiving an aspect query from DGP Then SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>LastStable</th><th>DGPMMessage</th></tr><tr><td>DARK</td><td>SIG_ DARK</td></tr><tr><td>STOP</td><td>SIG_ STOP</td></tr><tr><td>SPEED1</td><td>SIG_ SPEED_ 1</td></tr><tr><td>SPEED2</td><td>SIG_ SPEED_ 2</td></tr><tr><td>SPEED3</td><td>SIG_ SPEED_ 3</td></tr></table>				LastStable	DGPMMessage	DARK	SIG_ DARK	STOP	SIG_ STOP	SPEED1	SIG_ SPEED_ 1	SPEED2	SIG_ SPEED_ 2	SPEED3	SIG_ SPEED_ 3
LastStable	DGPMMessage														
DARK	SIG_ DARK														
STOP	SIG_ STOP														
SPEED1	SIG_ SPEED_ 1														
SPEED2	SIG_ SPEED_ 2														
SPEED3	SIG_ SPEED_ 3														
Type:	FUR - Functional Requirement														
Status:	draft	Stability:	stable												
Target Release:	SEA SW GEN 3.0	Priority:	mandatory												
Originator:	Technics (R&D)	Allocation:	SW EC-MUX												
Verification Method:	Test	Verification Step:	module test, subit												
Comments:															
Satisfies:															

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EC-MUX-SIG:SSS: REQ:3555	Signal With Three Speed Aspects and Stop and Dark commands - aspect query in UNDEFINED - OC-LS NOT_CONNECTED		
Given EC-MUX - OC-LS connection state is NOT_CONNECTED And the current element state is UNDEFINED When receiving an aspect query from DGP Then SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

3.1.8 Signal Repeater Combined on the mast of the superior PRSIG

3.1.8.1 Signal Repeater Combined

The Signal Repeater Combined on the mast of the superior Protection Signal in SEA is a configuration of the SEA Eulynx Signal with two speed aspects and Stop command. While the superior Protection Signal has aspect SPEED1, the behavior of the Signal Repeater Combined is the same as the behavior of Signal Repeater Standalone. When the Protection Signal loses SPEED1 aspect, the Signal Repeater Combined is commanded to DARK. This command is generated by EC-MUX directly.

The state machine of the SEA Eulynx Signal Repeater Combined on the mast of the superior Protection Signal is shown in the following figure:

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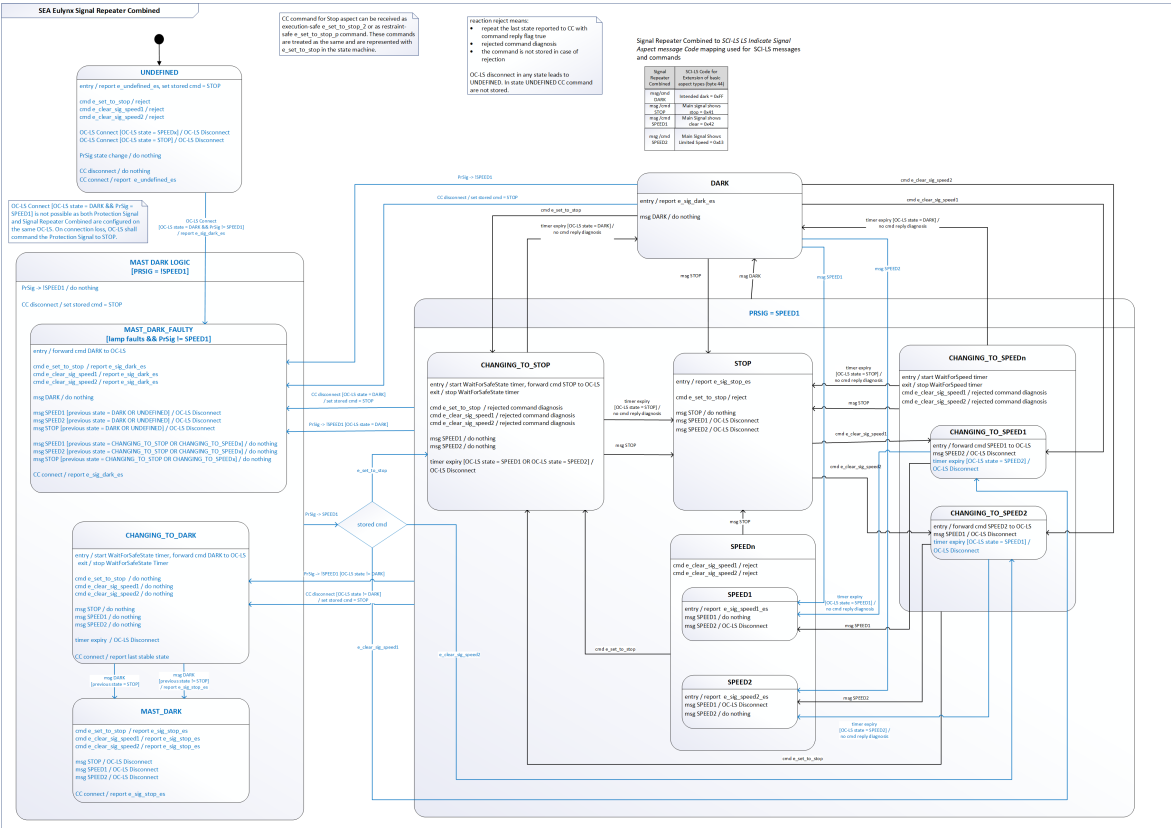


Figure 16: SEA Eulynx Signal Repeater Combined on the mast of the superior Protection Signal - State Machine

The operational and aspect diagnosis query requirements are split into two operational and diagnosis features:

- specific for Signal Repeater Combined element see in sub-chapter Operational - Signal Repeater Combined on the mast of the superior PRSIG and in sub-chapter Diagnosis - Signal Repeater Combined on the mast of the superior PRSIG
- common with Signal Repeater Standalone element in sub-chapter Operational - Signal With Two Speed Aspects and Stop command and in sub-chapter Diagnosis - Signal With Two Speed Aspects and Stop command

Used Cmd LS Indicate Signal Aspect and Msg LS Indicated Signal Aspect shall be addressed to the element identified by the Signal Repeater Combined operational identifier. CC command and state mappings to SCI-LS telegrams are defined in following tables.

CC Command	Mast Protection Signal aspect	Signal Repeater Combined	Command byte / parameter	Parameter value	OBB aspect
e_set_to_stop_2	Speed 1 /		44 = code for	Main Signal	Hauptsignal

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OR e_set_to_stop_p	Stop / Dark / Undefined	Stop	extension of basic aspect types	shows Stop = 0x41	zeigt Halt
e_clear_sig_speed1	Speed 1	Speed 1	44 = code for extension of basic aspect types	Main Signal shows Clear (1) = 0x42	Hauptsignal zeigt Frei
e_clear_sig_speed2	Speed 1	Speed 2	44 = code for extension of basic aspect types	Main Signal shows Limited Speed = 0x43	Hauptsignal zeigt Frei beschr.

**Table 43: Signal Repeater Combined - Mapping CC command (CC to EC-MUX) to SCI-LS Cmd
Indicate Signal Aspect (EC-MUX to OC-LS)**

The SCI-LS Cmd Indicate Signal Aspect irrelevant and relevant parameters for the Signal Repeater Combined are listed in the table *Eulynx Signal Mast configuration*.

Message byte / parameter	Parameter value	Mast Protection Signal aspect	Signal Repeater Combined	OBB aspect	CC State
44 = code for extension of basic aspect types	Main Signal shows Stop = 0x41	Speed 1	Stop	Hauptsignal zeigt Halt	e_sig_stop_es
44 = code for extension of basic aspect types	Main Signal shows Clear = 0x42	Speed 1	Speed 1	Hauptsignal zeigt Frei	e_sig_speed1_es
44 = code for extension of basic aspect types	Main Signal shows Limited Speed = 0x43	Speed 1	Speed 2	Hauptsignal zeigt Frei beschr.	e_sig_speed2_es
44 = code for extension of basic aspect types	Intended dark = 0xFF	Speed 1	Dark	(*)	e_sig_dark_es
44 = code for extension of basic aspect types	Intended dark = 0xFF	Stop / Dark / Undefined	Dark	(*)	e_sig_stop_es only if previous Signal Repeater Combined state is Dark / Speed 1 / Speed 2

Table 44: Signal Repeater Combined - Mapping SCI-LS Msg Indicated Signal Aspect (OC-LS to EC-MUX) to gen_elem_stat (EC-MUX to CC)

Remark: (*) The aspect defined in Signal aspect table [9] as "Intended dark" cannot be commanded, it can be reached only due to a lamp fault.

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For default values of the remaining parameters (byte 43, 45 - 60) of the Msg Indicated Signal Aspect see table for default values of the corresponding command parameters.

3.1.8.2 Operational - Signal Repeater Combined on the mast of the superior PRSIG

In order to control and supervise a Signal Repeater Combined on the mast of the superior Protection Signal

As a SEA

I want to be able to receive commands from CC, translate them into SCI-LS commands and forward them to OC-LS and to receive SCI-LS state messages from OC-LS, translate them into signal state changes and report them to CC.

For requirement language see Chapter 2.8 "Requirement Language".

CHANGING_TO_SPEEDn, CHANGING_TO_STOP and CHANGING_TO_DARK element states are not considered in the mapping between element state and CC State because when reaching these states, nothing is reported to CC.

WaitForSpeed and WaitForSafeState are configurable timers for receiving a command reply from OC-LS. WaitForSpeed is used when waiting for a reply to a SPEED command (e_clear_sig_speed1, e_clear_sig_speed2), its default value is 2000ms.

WaitForSafeState is used when waiting for a reply to a DARK command, its default value is 1000ms.

Given a Signal Repeater Combined on the mast of the superior PRSIG

And WaitForSafeState is the configured time for receiving a command reply for a DARK command from OC-LS

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_stop_es
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es
DARK	e_sig_dark_es
MAST_DARK_FAULTY	e_sig_dark_es
MAST_DARK	e_sig_stop_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

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SEA diagnosis aspect	DGP diagnosis value
SIG_ UNDEFINED	0
SIG_ DARK	12
SIG_ STOP	13
SIG_ SPEED_ 1	16
SIG_ SPEED_ 2	17

EC-MUX-SIG:SSS: REQ:3700	Signal Repeater Combined - Initial state		
When initializing the element			
Then SEA shall set the current element state to UNDEFINED			
And SEA shall set the stored command to STOP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3702	Signal Repeater Combined - UNDEFINED - Protection Signal state change - do nothing
Given the current element state is UNDEFINED When changing the Protection Signal state to <PrSigState> Then SEA shall do nothing	

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<table><tr><td>PrSigState</td></tr><tr><td>STOP</td></tr><tr><td>DARK</td></tr></table>				PrSigState	STOP	DARK
PrSigState						
STOP						
DARK						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test			
Comments:						
Satisfies:						

EC-MUX-SIG:SSS: REQ:3721	Signal Repeater Combined - DARK - spontaneous transition to SPEEDx											
Given the current element state is DARK When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS Then SEA shall set the current element state to <State> And SEA shall report the element state <CCState> with command reply flag false to CC And SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>State</th><th>CCState</th><th>DGPMMessage</th></tr><tr><td>SPEED1</td><td>e_sig_speed1_es</td><td>SIG_SPEED_1</td></tr><tr><td>SPEED2</td><td>e_sig_speed2_es</td><td>SIG_SPEED_2</td></tr></table>				State	CCState	DGPMMessage	SPEED1	e_sig_speed1_es	SIG_SPEED_1	SPEED2	e_sig_speed2_es	SIG_SPEED_2
State	CCState	DGPMMessage										
SPEED1	e_sig_speed1_es	SIG_SPEED_1										
SPEED2	e_sig_speed2_es	SIG_SPEED_2										
Type:	FUR - Functional Requirement											
Status:	draft	Stability:	stable									
Target Release:	SEA SW GEN 3.0	Priority:	mandatory									
Originator:	Technics (R&D)	Allocation:	SW EC-MUX									
Verification Method:	Test	Verification Step:	module test, subit									

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Comments:	
Satisfies:	

EC-MUX-SIG:SSS: REQ:3722	Signal Repeater Combined - DARK - transition to MAST_DARK_FAULTY						
Given the current element state is DARK When changing the Protection Signal state to <PrSigState> Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to MAST_DARK_FAULTY <table><tr><td>PrSigState</td></tr><tr><td>UNDEFINED</td></tr><tr><td>DARK</td></tr><tr><td>STOP</td></tr></table>				PrSigState	UNDEFINED	DARK	STOP
PrSigState							
UNDEFINED							
DARK							
STOP							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:							
Satisfies:							

EC-MUX-SIG:SSS: REQ:3726	Signal Repeater Combined - STOP - transition to CHANGING_TO_DARK
Given the current element state is STOP When changing the Protection Signal state to <PrSigState>	

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Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS

And SEA shall set the current element state to CHANGING_TO_DARK

PrSigState

UNDEFINED

DARK

STOP

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS:
REQ:3730

Signal Repeater Combined - transition from SPEEDx to CHANGING_TO_DARK

Given the current element state is <State>

When changing the Protection Signal state to <PrSigState>

Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS

And SEA shall set the current element state to CHANGING_TO_DARK

State	PrSigState
SPEED1	DARK
SPEED1	STOP
SPEED1	UNDEFINED
SPEED2	DARK
SPEED2	STOP
SPEED2	UNDEFINED

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Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3737	Signal Repeater Combined - last stable state DARK - transition from CHANGING_TO_SPEEDn to MAST_DARK_FAULTY						
Given the current element state is CHANGING_TO_SPEEDn And the last stable element state is DARK When changing the Protection Signal state to <PrSigState> Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to MAST_DARK_FAULTY <table><tr><th>PrSigState</th></tr><tr><td>UNDEFINED</td></tr><tr><td>DARK</td></tr><tr><td>STOP</td></tr></table>				PrSigState	UNDEFINED	DARK	STOP
PrSigState							
UNDEFINED							
DARK							
STOP							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:							
Satisfies:							

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EC-MUX-SIG:SSS: REQ:3738	Signal Repeater Combined - last stable state STOP - transition from CHANGING_TO_SPEEDn to CHANGING_TO_DARK						
Given the current element state is CHANGING_TO_SPEEDn And the last stable element state is STOP When changing the Protection Signal state to <PrSigState> Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to CHANGING_TO_DARK <table><tr><th>PrSigState</th></tr><tr><td>UNDEFINED</td></tr><tr><td>DARK</td></tr><tr><td>STOP</td></tr></table>				PrSigState	UNDEFINED	DARK	STOP
PrSigState							
UNDEFINED							
DARK							
STOP							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:							
Satisfies:							

EC-MUX-SIG:SSS: REQ:3769	Signal Repeater Combined - CHANGING_TO_SPEEDn - timeout with SPEEDx
Given the current element state is CHANGING_TO_SPEEDn And OC-LS state is <OcLsState> When the current state of the element has been CHANGING_TO_SPEEDn for WaitForSpeed ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP And SEA shall set the current element state to <OcLsState> And SEA shall report the element state <CCState> with command reply flag true to CC	

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OcLsState	CCState
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	This requirement is applicable in case the CHANGING_TO_SPEEDn state with last stable SPEED1 / SPEED2 is reached from CHANGING_TO_DARK state after the Protection Signal state changes to SPEED1. No safety reaction OC Disconnection is performed as this is a valid use case for the Signal Repeater Combined.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3770	Signal Repeater Combined - CHANGING_TO_SPEEDn - timeout with not matching SPEEDx				
Given the current element state is <ChangingToState> And OC-LS state is <OcLsState> When the current state of the element has been <ChangingToState> for WaitForSpeed ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall set the stored command to STOP And SEA shall report the no_connection diagnosis device fault message to DGP <table> <tr> <th>OcLsState</th><th>ChangingToState</th></tr> <tr> <td>SPEED2</td><td>CHANGING_TO_SPEED1</td></tr> </table>		OcLsState	ChangingToState	SPEED2	CHANGING_TO_SPEED1
OcLsState	ChangingToState				
SPEED2	CHANGING_TO_SPEED1				

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<table><tr><th>OcLsState</th><th>ChangingToState</th></tr><tr><td>SPEED1</td><td>CHANGING_TO_SPEED2</td></tr></table>		OcLsState	ChangingToState	SPEED1	CHANGING_TO_SPEED2
OcLsState	ChangingToState				
SPEED1	CHANGING_TO_SPEED2				
Type:	FUR - Functional Requirement				
Status:	draft	Stability:	stable		
Target Release:	SEA SW GEN 3.0	Priority:	mandatory		
Originator:	Technics (R&D)	Allocation:	SW EC-MUX		
Verification Method:	Test	Verification Step:	module test, subit		
Comments:					
Satisfies:					

EC-MUX-SIG:SSS: REQ:3745	Signal Repeater Combined - last stable state DARK - transition from CHANGING_TO_STOP to MAST_DARK_FAULTY						
Given the current element state is CHANGING_TO_STOP And the last stable element state is DARK When changing the Protection Signal state to <PrSigState> Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to MAST_DARK_FAULTY <table><tr><th>PrSigState</th></tr><tr><td>UNDEFINED</td></tr><tr><td>DARK</td></tr><tr><td>STOP</td></tr></table>				PrSigState	UNDEFINED	DARK	STOP
PrSigState							
UNDEFINED							
DARK							
STOP							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				

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Comments:	
Satisfies:	

EC-MUX-SIG:SSS: REQ:3746	Signal Repeater Combined - last stable state STOP / SPEEDx - transition from CHANGING_TO_STOP to CHANGING_TO_DARK																						
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable> When changing the Protection Signal state to <PrSigState> Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to CHANGING_TO_DARK <table><tr><th>LastStable</th><th>PrSigState</th></tr><tr><td>STOP</td><td>UNDEFINED</td></tr><tr><td>STOP</td><td>DARK</td></tr><tr><td>STOP</td><td>STOP</td></tr><tr><td>SPEED1</td><td>UNDEFINED</td></tr><tr><td>SPEED1</td><td>DARK</td></tr><tr><td>SPEED1</td><td>STOP</td></tr><tr><td>SPEED2</td><td>UNDEFINED</td></tr><tr><td>SPEED2</td><td>DARK</td></tr><tr><td>SPEED2</td><td>STOP</td></tr></table>				LastStable	PrSigState	STOP	UNDEFINED	STOP	DARK	STOP	STOP	SPEED1	UNDEFINED	SPEED1	DARK	SPEED1	STOP	SPEED2	UNDEFINED	SPEED2	DARK	SPEED2	STOP
LastStable	PrSigState																						
STOP	UNDEFINED																						
STOP	DARK																						
STOP	STOP																						
SPEED1	UNDEFINED																						
SPEED1	DARK																						
SPEED1	STOP																						
SPEED2	UNDEFINED																						
SPEED2	DARK																						
SPEED2	STOP																						
Type:	FUR - Functional Requirement																						
Status:	draft	Stability:	stable																				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory																				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX																				
Verification Method:	Test	Verification Step:	module test, subit																				
Comments:																							
Satisfies:																							

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EC-MUX-SIG:SSS: REQ:3703	Signal Repeater Combined - MAST_DARK_FAULTY - spontaneous message DARK - do nothing		
Given the current element state is MAST_DARK_FAULTY			
When receiving the LS Indicated Signal Aspect message with Code corresponding to DARK from OC-LS			
Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3704	Signal Repeater Combined - MAST_DARK_FAULTY - previous state DARK / UNDEFINED - spontaneous message SPEEDx / STOP - OC Disconnection			
Given the current element state is MAST_DARK_FAULTY When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS Then SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall set the stored command to STOP And SEA shall report the no_connection diagnosis device fault message to DGP <table><tr><th>State</th></tr><tr><td>STOP</td></tr><tr><td>SPEED1</td></tr></table>		State	STOP	SPEED1
State				
STOP				
SPEED1				

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State SPEED2			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	This requirement is applicable for the following cases: - MAST_DARK_FAULTY state is reached from the state UNDEFINED - MAST_DARK_FAULTY state is reached from the state DARK		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3705	Signal Repeater Combined - MAST_DARK_FAULTY - previous state CHANGING_TO_STOP / CHANGING_TO_SPEEDn - spontaneous message SPEEDx / STOP - do nothing						
Given the current element state is MAST_DARK_FAULTY When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS Then SEA shall do nothing <table><tr><td>State</td></tr><tr><td>STOP</td></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr></table>				State	STOP	SPEED1	SPEED2
State							
STOP							
SPEED1							
SPEED2							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				

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Comments:	This requirement is applicable for the following cases: - MAST_DARK_FAULTY state is reached from the state CHANGING_TO_STOP with the last stable state DARK - MAST_DARK_FAULTY state is reached from the state CHANGING_TO_SPEED1 / CHANGING_TO_SPEED2 with the last stable state DARK
Satisfies:	

EC-MUX-SIG:SSS: REQ:3706	Signal Repeater Combined - MAST_DARK_FAULTY - STOP / SPEEDx commands - report e_sig_dark_es												
Given the current element state is MAST_DARK_FAULTY When receiving the command <Command> from CC Then SEA shall report the element state e_sig_dark_es with command reply flag true to CC And SEA shall set the stored command to <StoredCommand> <table><tr><th>Command</th><th>StoredCommand</th></tr><tr><td>e_set_to_stop_p</td><td>STOP</td></tr><tr><td>e_set_to_stop_2</td><td>STOP</td></tr><tr><td>e_clear_sig_speed1</td><td>SPEED1</td></tr><tr><td>e_clear_sig_speed2</td><td>SPEED2</td></tr></table>				Command	StoredCommand	e_set_to_stop_p	STOP	e_set_to_stop_2	STOP	e_clear_sig_speed1	SPEED1	e_clear_sig_speed2	SPEED2
Command	StoredCommand												
e_set_to_stop_p	STOP												
e_set_to_stop_2	STOP												
e_clear_sig_speed1	SPEED1												
e_clear_sig_speed2	SPEED2												
Type:	FUR - Functional Requirement												
Status:	draft	Stability:	stable										
Target Release:	SEA SW GEN 3.0	Priority:	mandatory										
Originator:	Technics (R&D)	Allocation:	SW EC-MUX										
Verification Method:	Test	Verification Step:	module test, subit										
Comments:													
Satisfies:													

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EC-MUX-SIG:SSS: REQ:3707	Signal Repeater Combined - MAST_DARK_FAULTY / CHANGING_TO_DARK / MAST_DARK - Protection Signal state change - do nothing																						
Given the current element state is <State> When changing the Protection Signal state to <PrSigState> Then SEA shall do nothing <table><tr><th>State</th><th>PrSigState</th></tr><tr><td>MAST_DARK_FAULTY</td><td>UNDEFINED</td></tr><tr><td>MAST_DARK_FAULTY</td><td>DARK</td></tr><tr><td>MAST_DARK_FAULTY</td><td>STOP</td></tr><tr><td>CHANGING_TO_DARK</td><td>UNDEFINED</td></tr><tr><td>CHANGING_TO_DARK</td><td>DARK</td></tr><tr><td>CHANGING_TO_DARK</td><td>STOP</td></tr><tr><td>MAST_DARK</td><td>UNDEFINED</td></tr><tr><td>MAST_DARK</td><td>DARK</td></tr><tr><td>MAST_DARK</td><td>STOP</td></tr></table>				State	PrSigState	MAST_DARK_FAULTY	UNDEFINED	MAST_DARK_FAULTY	DARK	MAST_DARK_FAULTY	STOP	CHANGING_TO_DARK	UNDEFINED	CHANGING_TO_DARK	DARK	CHANGING_TO_DARK	STOP	MAST_DARK	UNDEFINED	MAST_DARK	DARK	MAST_DARK	STOP
State	PrSigState																						
MAST_DARK_FAULTY	UNDEFINED																						
MAST_DARK_FAULTY	DARK																						
MAST_DARK_FAULTY	STOP																						
CHANGING_TO_DARK	UNDEFINED																						
CHANGING_TO_DARK	DARK																						
CHANGING_TO_DARK	STOP																						
MAST_DARK	UNDEFINED																						
MAST_DARK	DARK																						
MAST_DARK	STOP																						
Type:	FUR - Functional Requirement																						
Status:	draft	Stability:	stable																				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory																				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX																				
Verification Method:	Test	Verification Step:	module test, subit																				
Comments:																							
Satisfies:																							

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EC-MUX-SIG:SSS: REQ:3708	Signal Repeater Combined - MAST_DARK_FAULTY / CHANGING_TO_DARK / MAST_DARK - Protection Signal state change to SPEED1 - forward last stored command																																										
Given the current element state is <State> And the last stored command is <Command> When changing the Protection Signal state to SPEED1 Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to <NewState> to OC-LS And SEA shall set the current element state to <ChangingToState>																																											
<table><tr><th>State</th><th>Command</th><th>NewState</th><th>ChangingToState</th></tr><tr><td>MAST_DARK_FAULTY</td><td>STOP</td><td>STOP</td><td>CHANGING_TO_STOP</td></tr><tr><td>MAST_DARK_FAULTY</td><td>SPEED1</td><td>SPEED1</td><td>CHANGING_TO_SPEED1</td></tr><tr><td>MAST_DARK_FAULTY</td><td>SPEED2</td><td>SPEED2</td><td>CHANGING_TO_SPEED2</td></tr><tr><td>CHANGING_TO_DARK</td><td>STOP</td><td>STOP</td><td>CHANGING_TO_STOP</td></tr><tr><td>CHANGING_TO_DARK</td><td>SPEED1</td><td>SPEED1</td><td>CHANGING_TO_SPEED1</td></tr><tr><td>CHANGING_TO_DARK</td><td>SPEED2</td><td>SPEED2</td><td>CHANGING_TO_SPEED2</td></tr><tr><td>MAST_DARK</td><td>STOP</td><td>STOP</td><td>CHANGING_TO_STOP</td></tr><tr><td>MAST_DARK</td><td>SPEED1</td><td>SPEED1</td><td>CHANGING_TO_SPEED1</td></tr><tr><td>MAST_DARK</td><td>SPEED2</td><td>SPEED2</td><td>CHANGING_TO_SPEED2</td></tr></table>				State	Command	NewState	ChangingToState	MAST_DARK_FAULTY	STOP	STOP	CHANGING_TO_STOP	MAST_DARK_FAULTY	SPEED1	SPEED1	CHANGING_TO_SPEED1	MAST_DARK_FAULTY	SPEED2	SPEED2	CHANGING_TO_SPEED2	CHANGING_TO_DARK	STOP	STOP	CHANGING_TO_STOP	CHANGING_TO_DARK	SPEED1	SPEED1	CHANGING_TO_SPEED1	CHANGING_TO_DARK	SPEED2	SPEED2	CHANGING_TO_SPEED2	MAST_DARK	STOP	STOP	CHANGING_TO_STOP	MAST_DARK	SPEED1	SPEED1	CHANGING_TO_SPEED1	MAST_DARK	SPEED2	SPEED2	CHANGING_TO_SPEED2
State	Command	NewState	ChangingToState																																								
MAST_DARK_FAULTY	STOP	STOP	CHANGING_TO_STOP																																								
MAST_DARK_FAULTY	SPEED1	SPEED1	CHANGING_TO_SPEED1																																								
MAST_DARK_FAULTY	SPEED2	SPEED2	CHANGING_TO_SPEED2																																								
CHANGING_TO_DARK	STOP	STOP	CHANGING_TO_STOP																																								
CHANGING_TO_DARK	SPEED1	SPEED1	CHANGING_TO_SPEED1																																								
CHANGING_TO_DARK	SPEED2	SPEED2	CHANGING_TO_SPEED2																																								
MAST_DARK	STOP	STOP	CHANGING_TO_STOP																																								
MAST_DARK	SPEED1	SPEED1	CHANGING_TO_SPEED1																																								
MAST_DARK	SPEED2	SPEED2	CHANGING_TO_SPEED2																																								
Type:		FUR - Functional Requirement																																									
Status:	draft	Stability:	stable																																								
Target Release:	SEA SW GEN 3.0	Priority:	mandatory																																								
Originator:	Technics (R&D)	Allocation:	SW EC-MUX																																								
Verification Method:	Test	Verification Step:	module test, subit																																								
Comments:																																											

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Satisfies:	
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EC-MUX-SIG:SSS: REQ:3709	Signal Repeater Combined - CHANGING_TO_DARK - spontaneous message SPEEDx / STOP - do nothing						
Given the current element state is CHANGING_TO_DARK When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS within WaitForSafeState ms Then SEA shall do nothing <table><tr><td>State</td></tr><tr><td>STOP</td></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr></table>				State	STOP	SPEED1	SPEED2
State							
STOP							
SPEED1							
SPEED2							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:							
Satisfies:							

EC-MUX-SIG:SSS: REQ:3710	Signal Repeater Combined - CHANGING_TO_DARK - STOP / SPEEDx commands - store command
Given the current element state is CHANGING_TO_DARK When receiving the command <Command> from CC within WaitForSafeState ms Then SEA shall set the stored command to <StoredCommand>	

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Command	StoredCommand
e_set_to_stop_p	STOP
e_set_to_stop_2	STOP
e_clear_sig_speed1	SPEED1
e_clear_sig_speed2	SPEED2

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3711	Signal Repeater Combined - CHANGING_TO_DARK - timeout - OC Disconnection		
Given the current element state is CHANGING_TO_DARK When the current state of the element has been CHANGING_TO_DARK for WaitForSafeState ms Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: No command reply within expected time for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall set the stored command to STOP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX

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Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3712	Signal Repeater Combined - CHANGING_TO_DARK - transition from CHANGING_TO_DARK to MAST_DARK - last stable SPEEDx					
Given the current element state is CHANGING_TO_DARK And the last stable element state is <LastStable> When receiving the LS Indicated Signal Aspect message with Code corresponding to DARK from OC-LS within WaitForSafeState ms Then SEA shall set the current element state to MAST_DARK And SEA shall report the element state e_sig_stop_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_STOP to DGP <table><tr><td>LastStable</td></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr></table>				LastStable	SPEED1	SPEED2
LastStable						
SPEED1						
SPEED2						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

EC-MUX-SIG:SSS: REQ:3713	Signal Repeater Combined - CHANGING_TO_DARK - transition from CHANGING_TO_DARK to MAST_DARK - last stable STOP
Given the current element state is CHANGING_TO_DARK And the last stable element state is STOP	

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When receiving the LS Indicated Signal Aspect message with Code corresponding to DARK from OC-LS within WaitForSafeState ms			
Then SEA shall set the current element state to MAST_DARK			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3715	Signal Repeater Combined - MAST_DARK - spontaneous message SPEEDx / STOP - OC Disconnection						
Given the current element state is MAST_DARK When receiving the LS Indicated Signal Aspect message with Code corresponding to <State> from OC-LS Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: Spontaneous <State> reported for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection And SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall set the stored command to STOP And SEA shall report the no_connection diagnosis device fault message to DGP							
<table><tr><th>State</th></tr><tr><td>STOP</td></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr></table>				State	STOP	SPEED1	SPEED2
State							
STOP							
SPEED1							
SPEED2							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				

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Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3716	Signal Repeater Combined - MAST_DARK - STOP / SPEEDx commands - report e_sig_stop_es												
Given the current element state is MAST_DARK When receiving the command <Command> from CC Then SEA shall report the element state e_sig_stop_es with command reply flag true to CC And SEA shall set the stored command to <State> <table><tr><th>Command</th><th>State</th></tr><tr><td>e_set_to_stop_p</td><td>STOP</td></tr><tr><td>e_set_to_stop_2</td><td>STOP</td></tr><tr><td>e_clear_sig_speed1</td><td>SPEED1</td></tr><tr><td>e_clear_sig_speed2</td><td>SPEED2</td></tr></table>				Command	State	e_set_to_stop_p	STOP	e_set_to_stop_2	STOP	e_clear_sig_speed1	SPEED1	e_clear_sig_speed2	SPEED2
Command	State												
e_set_to_stop_p	STOP												
e_set_to_stop_2	STOP												
e_clear_sig_speed1	SPEED1												
e_clear_sig_speed2	SPEED2												
Type:	FUR - Functional Requirement												
Status:	draft	Stability:	stable										
Target Release:	SEA SW GEN 3.0	Priority:	mandatory										
Originator:	Technics (R&D)	Allocation:	SW EC-MUX										
Verification Method:	Test	Verification Step:	module test, subit										
Comments:													
Satisfies:													

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3.1.8.3 Connection - Signal Repeater Combined on the mast of the superior PRSIG

In order to control and supervise a Signal Repeater Combined on the mast of the superior Protection Signal

As a SEA

I want to be able to supervise connections and when changes occur then

- on CC connection establishment report the current state to CC
- on CC connection loss execute the safe command DARK if the last command forwarded to OC-LS was SPEED1 / SPEED2 / STOP and it was not answered by DARK
- on OC-LS connection loss take on the default state UNDEFINED
- on OC-LS connection establishment transition to MAST_DARK_FAULTY state
- report the connection state changes to CC and DGP

For requirement language see Chapter 2.8 "Requirement Language".

CHANGING_TO_SPEEDn, CHANGING_TO_STOP, CHANGING_TO_DARK element states are not considered in the mapping between element state and CC State because when reaching these states, nothing is reported to CC.

WaitForSpeed and WaitForSafeState are configurable timers for receiving a command reply from OC-LS. WaitForSpeed is used when waiting for a reply to a SPEED command (e_clear_sig_speed1, e_clear_sig_speed2), its default value is 2000ms.

WaitForSafeState is used when waiting for a reply to a STOP command (e_set_to_stop_2, e_set_to_stop_p) or to a DARK command (e_set_to_dark_2, e_set_to_dark_p), its default value is 1000ms.

Given a Signal Repeater Combined on the mast of the superior PRSIG

And WaitForSpeed is the configured time for receiving a command reply for a SPEED command from OC-LS

And WaitForSafeState is the configured time for receiving a command reply for a STOP command or a DARK command from OC-LS

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_stop_es
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es
DARK	e_sig_dark_es
MAST_DARK_FAULTY	e_sig_dark_es
MAST_DARK	e_sig_stop_es
UNDEFINED	e_undefined_es

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And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_ UNDEFINED	0
SIG_ DARK	12
SIG_ STOP	13
SIG_ SPEED_ 1	16
SIG_ SPEED_ 2	17

EC-MUX-SIG:SSS: REQ:3750	Signal Repeater Combined - [Re-]Connecting to CC - General inquiry in UNDEFINED / MAST_DARK / MAST_DARK_FAULTY										
Given the current element state is <State> And CC connection state is NOT_CONNECTED When establishing the connection to CC Then SEA shall report the element state <CCState> with command reply flag false to CC <table><tr><th>State</th><th>CCState</th></tr><tr><td>UNDEFINED</td><td>e_undefined_ es</td></tr><tr><td>MAST_DARK</td><td>e_sig_stop_es</td></tr><tr><td>MAST_DARK_ FAULTY</td><td>e_sig_dark_es</td></tr></table>				State	CCState	UNDEFINED	e_undefined_ es	MAST_DARK	e_sig_stop_es	MAST_DARK_ FAULTY	e_sig_dark_es
State	CCState										
UNDEFINED	e_undefined_ es										
MAST_DARK	e_sig_stop_es										
MAST_DARK_ FAULTY	e_sig_dark_es										
Type:	FUR - Functional Requirement										
Status:	draft	Stability:	stable								
Target Release:	SEA SW GEN 3.0	Priority:	mandatory								
Originator:	Technics (R&D)	Allocation:	SW EC-MUX								
Verification Method:	Test	Verification Step:	module test, subit								
Comments:											
Satisfies:											

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EC-MUX-SIG:SSS: REQ:3751	Signal Repeater Combined - [Re-]Connecting to CC - OC-LS CONNECTED - General inquiry in CHANGING_TO_DARK - no message after CC disconnect										
Given the current element state is CHANGING_TO_DARK And the last stable element state is <LastStable> And CC connection state is NOT_CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When establishing the connection to CC within WaitForSafeState ms Then SEA shall report the element state <CCState> with command reply flag false to CC <table><tr><th>LastStable</th><th>CCState</th></tr><tr><td>STOP</td><td>e_sig_stop_es</td></tr><tr><td>SPEED1</td><td>e_sig_speed1_es</td></tr><tr><td>SPEED2</td><td>e_sig_speed2_es</td></tr></table>				LastStable	CCState	STOP	e_sig_stop_es	SPEED1	e_sig_speed1_es	SPEED2	e_sig_speed2_es
LastStable	CCState										
STOP	e_sig_stop_es										
SPEED1	e_sig_speed1_es										
SPEED2	e_sig_speed2_es										
Type:	FUR - Functional Requirement										
Status:	draft	Stability:	stable								
Target Release:	SEA SW GEN 3.0	Priority:	mandatory								
Originator:	Technics (R&D)	Allocation:	SW EC-MUX								
Verification Method:	Test	Verification Step:	module test, subit								
Comments:	Re-connecting to CC in CHANGING_TO_DARK state is impossible in real operation, as re-connecting will regularly take much longer time than the configured WaitForSafeState. Due to the execution of the safe command [Re-]Connecting in CHANGING_TO_DARK state is possible only with last forwarded command DARK, but depending on the previous events the last state reported to CC can be different. Following initial states are covered with this requirement: - CHANGING_TO_DARK with last stable element state SPEED1/SPEED2 - CHANGING_TO_DARK with last stable element state STOP - CHANGING_TO_DARK with last stable element state STOP and SPEED1 /SPEED2 received from OC-LS in CHANGING_TO_DARK - CHANGING_TO_DARK with last stable element state SPEED1/SPEED2 and STOP received from OC-LS in CHANGING_TO_DARK										

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Satisfies:	
------------	--

EC-MUX-SIG:SSS: REQ:3752	Signal Repeater Combined - Connection loss to CC in UNDEFINED - do nothing		
Given the current element state is UNDEFINED And CC connection state is CONNECTED When losing the connection to CC Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3753		Signal Repeater Combined - Connection loss to CC in MAST_DARK / MAST_DARK_FAULTY - store command STOP				
Given the current element state is <State> And CC connection state is CONNECTED When losing the connection to CC Then SEA shall set the stored command to STOP <table><tr><td>State</td></tr><tr><td>MAST_ DARK</td></tr><tr><td>MAST_ DARK_ FAULTY</td></tr></table>				State	MAST_ DARK	MAST_ DARK_ FAULTY
State						
MAST_ DARK						
MAST_ DARK_ FAULTY						
Type:		FUR - Functional Requirement				
Status:		draft	Stability: stable			

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3754	Signal Repeater Combined - Connection loss to CC in DARK - transition to MAST_DARK_FAULTY and store command STOP		
Given the current element state is DARK And CC connection state is CONNECTED When losing the connection to CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to MAST_DARK_FAULTY And SEA shall set the stored command to STOP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3755	Signal Repeater Combined - Connection loss to CC in CHANGING_TO_DARK - store command STOP
Given the current element state is CHANGING_TO_DARK And CC connection state is CONNECTED When losing the connection to CC within WaitForSafeState ms	

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Then SEA shall set the stored command to STOP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3756	Signal Repeater Combined - Connection loss to CC in STOP / SPEEDx - transition to CHANGING_TO_DARK and store command STOP						
Given the current element state is <State> And CC connection state is CONNECTED When losing the connection to CC Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to CHANGING_TO_DARK And SEA shall set the stored command to STOP <table><tr><th>State</th></tr><tr><td>STOP</td></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr></table>				State	STOP	SPEED1	SPEED2
State							
STOP							
SPEED1							
SPEED2							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:							
Satisfies:							

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EC-MUX-SIG:SSS: REQ:3757	Signal Repeater Combined - Connection loss to CC in CHANGING_TO_SPEEDn - last stable STOP - transition to CHANGING_TO_DARK and store command STOP		
Given the current element state is CHANGING_TO_SPEEDn And the last stable element state is STOP And CC connection state is CONNECTED When losing the connection to CC within WaitForSpeed ms Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to CHANGING_TO_DARK And SEA shall set the stored command to STOP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3758	Signal Repeater Combined - Connection loss to CC in CHANGING_TO_SPEEDn - last stable DARK - transition to MAST_DARK_FAULTY and store command STOP
Given the current element state is CHANGING_TO_SPEEDn And the last stable element state is DARK And CC connection state is CONNECTED When losing the connection to CC within WaitForSpeed ms Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to MAST_DARK_FAULTY And SEA shall set the stored command to STOP	
Type:	FUR - Functional Requirement

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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3759	Signal Repeater Combined - Connection loss to CC in CHANGING_TO_STOP - last stable not DARK - transition to CHANGING_TO_DARK and store command STOP						
Given the current element state is CHANGING_TO_STOP And the last stable element state is <LastStable> And CC connection state is CONNECTED When losing the connection to CC within WaitForSafeState ms Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to CHANGING_TO_DARK And SEA shall set the stored command to STOP <table><tr><td>LastStable</td></tr><tr><td>STOP</td></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr></table>				LastStable	STOP	SPEED1	SPEED2
LastStable							
STOP							
SPEED1							
SPEED2							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:							
Satisfies:							

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EC-MUX-SIG:SSS: REQ:3760	Signal Repeater Combined - Connection loss to CC in CHANGING_TO_STOP - last stable DARK - transition to MAST_DARK_FAULTY and store command STOP		
Given the current element state is CHANGING_TO_STOP And the last stable element state is DARK And CC connection state is CONNECTED When losing the connection to CC within WaitForSafeState ms Then SEA shall forward the command LS Indicate Signal Aspect with Code corresponding to DARK to OC-LS And SEA shall set the current element state to MAST_DARK_FAULTY And SEA shall set the stored command to STOP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3761	Signal Repeater Combined - [Re-]Connecting between EC-MUX and OC-LS in state UNDEFINED - OC-LS in DARK state - transition to MAST_DARK_FAULTY
Given the current element state is UNDEFINED And EC-MUX - OC-LS connection state is NOT_CONNECTED And OC-LS state is DARK When establishing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to MAST_DARK_FAULTY And SEA shall report the element state e_sig_dark_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_DARK to DGP And SEA shall report the connection_ok diagnosis device fault repair message to DGP	
Type:	FUR - Functional Requirement

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Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:	Note that the last stored command is STOP.		
Satisfies:			

EC-MUX-SIG:SSS: REQ:3763	Signal Repeater Combined - [Re-]Connecting between EC-MUX and OC-LS in state UNDEFINED - OC-LS in SPEEDn / STOP - OC Disconnection						
Given EC-MUX - OC-LS connection state is NOT_CONNECTED And the current element state is UNDEFINED And OC-LS state is <SpeedAspect> When establishing the connection between EC-MUX and OC-LS Then SEA shall report the plain text diagnosis message "LS-IF-ERROR: OC-LS connect with <SpeedAspect> for Signal x." to DGP And SEA shall execute EC-MUX safety reaction OC Disconnection <table><tr><th>SpeedAspect</th></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr><tr><td>STOP</td></tr></table>				SpeedAspect	SPEED1	SPEED2	STOP
SpeedAspect							
SPEED1							
SPEED2							
STOP							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:							
Satisfies:							

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EC-MUX-SIG:SSS: REQ:3762	Signal Repeater Combined - Connection loss between EC-MUX and OC-LS in regular states						
Given the current element state is <State> And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall set the stored command to STOP And SEA shall report the no_connection diagnosis device fault message to DGP <table><tr><th>State</th></tr><tr><td>STOP</td></tr><tr><td>SPEED1</td></tr><tr><td>SPEED2</td></tr></table>				State	STOP	SPEED1	SPEED2
State							
STOP							
SPEED1							
SPEED2							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				
Target Release:	SEA SW GEN 3.0	Priority:	mandatory				
Originator:	Technics (R&D)	Allocation:	SW EC-MUX				
Verification Method:	Test	Verification Step:	module test, subit				
Comments:							
Satisfies:							

EC-MUX-SIG:SSS: REQ:3764	Signal Repeater Combined - Connection loss between EC-MUX and OC-LS in CHANGING_TO_SPEEDn
Given the current element state is CHANGING_TO_SPEEDn And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS within WaitForSpeed ms	

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Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall set the stored command to STOP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3765	Signal Repeater Combined - Connection loss between EC-MUX and OC-LS in CHANGING_TO_STOP		
Given the current element state is CHANGING_TO_STOP And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS within WaitForSafeState ms Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall set the stored command to STOP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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EC-MUX-SIG:SSS: REQ:3767	Signal Repeater Combined - Connection loss between EC-MUX and OC-LS in CHANGING_TO_DARK		
Given the current element state is CHANGING_TO_DARK And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS within WaitForSafeState ms Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall set the stored command to STOP And SEA shall report the no_connection diagnosis device fault message to DGP			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3768	Signal Repeater Combined - Connection loss between EC-MUX and OC-LS in MAST_DARK state
Given the current element state is MAST_DARK And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag false to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall set the stored command to STOP And SEA shall report the no_connection diagnosis device fault message to DGP	
Type:	FUR - Functional Requirement

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3766	Signal Repeater Combined - Connection loss between EC-MUX and OC-LS in MAST_DARK_FAULTY / DARK					
Given the current element state is <State> And CC connection state is CONNECTED And EC-MUX - OC-LS connection state is CONNECTED When losing the connection between EC-MUX and OC-LS Then SEA shall set the current element state to UNDEFINED And SEA shall report the element state e_undefined_es with command reply flag true to CC And SEA shall report the diagnosis aspect message SIG_UNDEFINED to DGP And SEA shall set the stored command to STOP And SEA shall report the no_connection diagnosis device fault message to DGP <table><tr><th>State</th></tr><tr><td>DARK</td></tr><tr><td>MAST_DARK_FAULTY</td></tr></table>				State	DARK	MAST_DARK_FAULTY
State						
DARK						
MAST_DARK_FAULTY						
Type:	FUR - Functional Requirement					
Status:	draft	Stability:	stable			
Target Release:	SEA SW GEN 3.0	Priority:	mandatory			
Originator:	Technics (R&D)	Allocation:	SW EC-MUX			
Verification Method:	Test	Verification Step:	module test, subit			
Comments:						
Satisfies:						

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3.1.8.4 Diagnosis - Signal Repeater Combined on the mast of the superior PRSIG

In order to provide the aspect diagnosis for a Signal Repeater Combined on the mast of the superior PRSIG

As a SEA

I want to be able to receive an aspect diagnosis query from DGP and answer it by the DGP diagnosis values corresponding to the last diagnosis aspect message reported to DGP for the queried Signal

For requirement language see Chapter 2.8 "Requirement Language".

CHANGING_TO_SPEEDn, CHANGING_TO_STOP, CHANGING_TO_DARK element states are not considered in the mapping between element state and CC State because when reaching these states, nothing is reported to CC.

Given a Signal Repeater Combined on the mast of the superior PRSIG

And the mapping between element state and CC State is defined

State	CC State
STOP	e_sig_stop_es
SPEED1	e_sig_speed1_es
SPEED2	e_sig_speed2_es
DARK	e_sig_dark_es
MAST_DARK_FAULTY	e_sig_dark_es
MAST_DARK	e_sig_stop_es
UNDEFINED	e_undefined_es

And the mapping between SEA diagnosis aspect and DGP diagnosis value is defined

SEA diagnosis aspect	DGP diagnosis value
SIG_UNDEFINED	0
SIG_DARK	12
SIG_STOP	13
SIG_SPEED_1	16
SIG_SPEED_2	17

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EC-MUX-SIG:SSS: REQ:3771	Signal Repeater Combined - aspect query								
Given the last stable element state is <LastStable> When receiving an aspect query from DGP Then SEA shall report the diagnosis aspect message <DGPMMessage> to DGP <table><tr><th>LastStable</th><th>DGPMMessage</th></tr><tr><td>MAST_ DARK_ FAULTY</td><td>SIG_ DARK</td></tr><tr><td>MAST_ DARK</td><td>SIG_ STOP</td></tr></table>				LastStable	DGPMMessage	MAST_ DARK_ FAULTY	SIG_ DARK	MAST_ DARK	SIG_ STOP
LastStable	DGPMMessage								
MAST_ DARK_ FAULTY	SIG_ DARK								
MAST_ DARK	SIG_ STOP								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:									
Satisfies:									

3.1.9 Element Related Features

3.1.9.1 Syntax checks

Incoming message syntax checks are defined generically for all Eulynx elements in SRS SEA OBB [4] Chapter 3.3.4.1 “Incoming Message Syntax checks”.

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3.2 Signal Mast

3.2.1 Luminosity

3.2.1.1 Signal Mast - Luminosity

Eulynx Signal Mast Luminosity is in SEA realized via the component **:eulx_sig_luminosity** as shown in the following Figure.

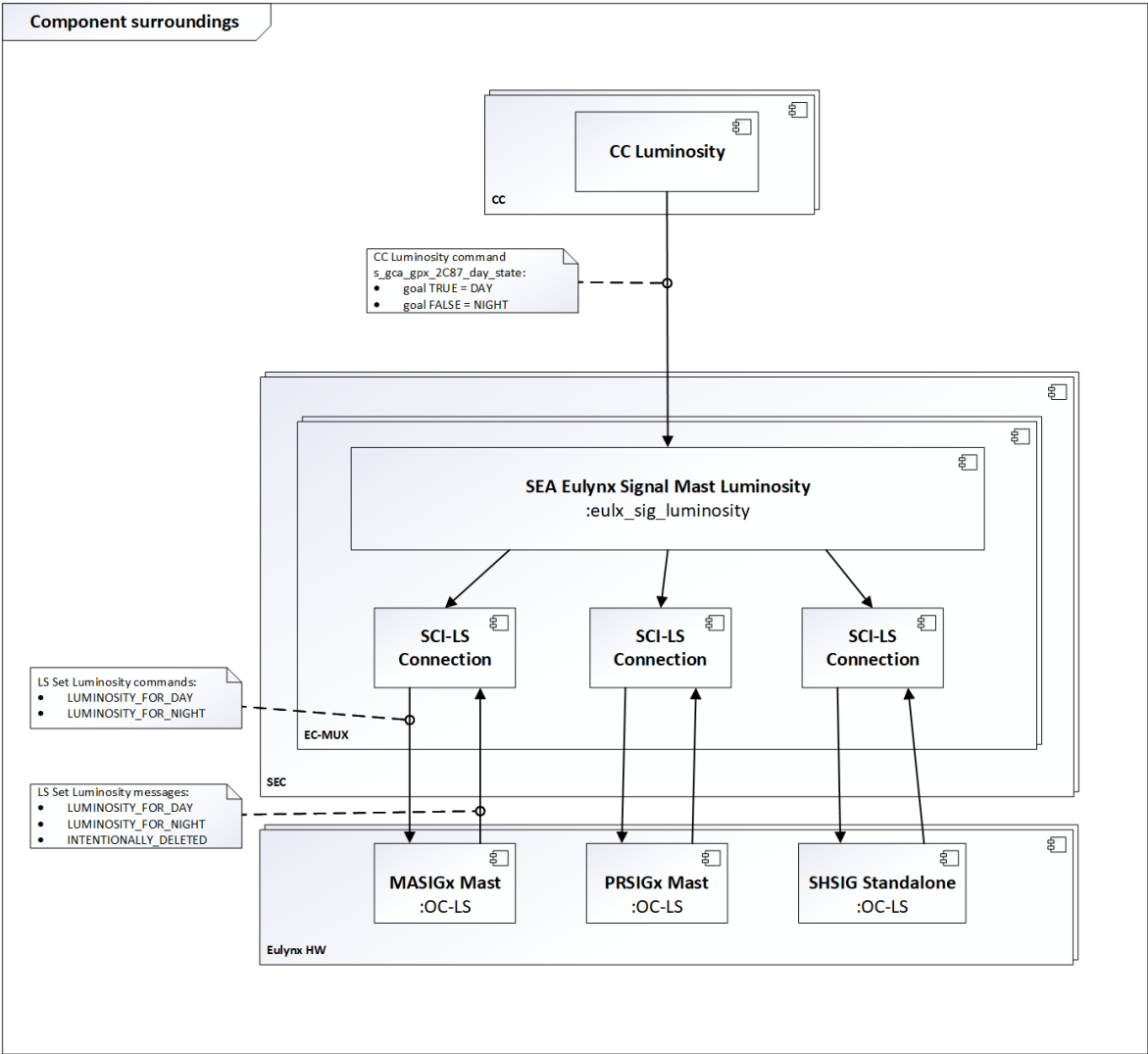


Figure 17: SEA Eulynx Signal Mast Luminosity - Surroundings

SEA Eulynx Signal Mast Luminosity can reach following regular luminosity states via CC command:

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- DAY
- NIGHT

SEA Eulynx Signal Mast Luminosity element states have no corresponding state in CC interface. For this reason, SEA does not report the element state message to CC and the diagnosis aspect message to DGP.

SEA Eulynx Signal Mast Luminosity element state shall depend only on the CC command, not on the OC-LS state: CC commands are always accepted and forwarded (if the connection to OC-LS is available) and the Signal Mast Luminosity state is immediately updated. For details see paragraphs “Command and Message” and “Connection Handling”.

The state machine of the SEA Eulynx Signal Mast Luminosity is shown in the following figure:

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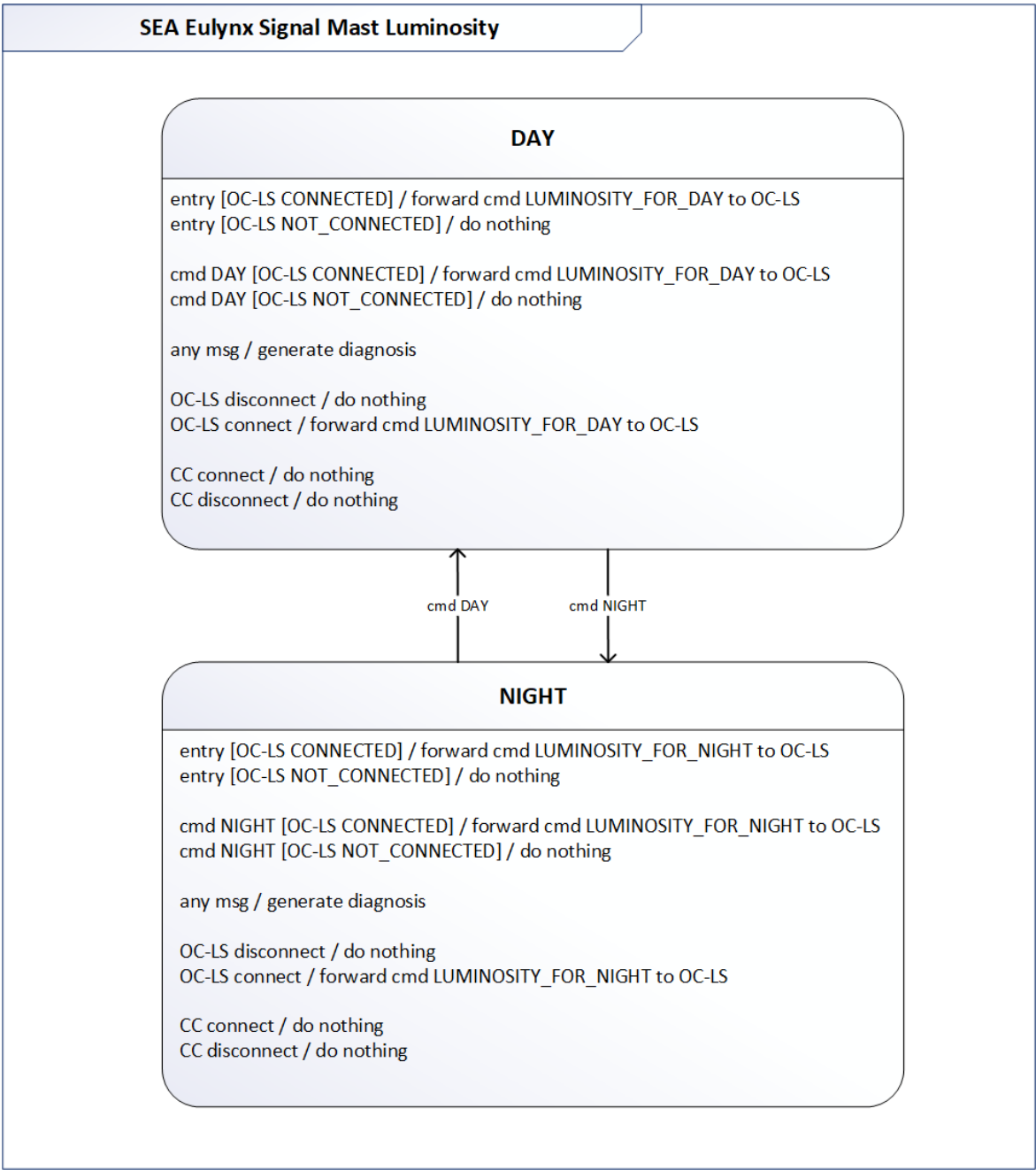


Figure 18: SEA Eulynx Signal Mast Luminosity - State Machine

Following internal element states are defined for the SEA Eulynx Signal Mast Luminosity:

Element State	Functionality	CC State
	initial state,	

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DAY	plain text diagnosis when msg received spontaneously / after cmd execution	none
NIGHT	plain text diagnosis when msg received spontaneously / after cmd execution	none

Table 45: SEA Eulynx Signal Mast Luminosity - Internal Element States**Connection handling****EC-MUX - OC-LS**

When losing the connection to OC-LS, SEA shall do nothing.

When (re-)establishing the connection to OC-LS, SEA shall forward the command corresponding to the current element state. to the OC-LS that is currently connecting to this EC-MUX.

CC - EC-MUX

When losing the connection to CC, SEA shall do nothing.

When (re-)establishing the connection to CC, SEA shall do nothing.

Command and Message

Used Cmd LS Set Luminosity and Msg LS Set Luminosity shall be addressed to the element identified by the OC-LS Signal Mast operational identifier. CC command and state mappings to SCI-LS telegrams are defined in the following tables.

Luminosity command s_gca_gpx_2C87_day_state received by EC-MUX from CC shall be distributed to each Signal Mast configured for this EC-MUX, i.e. each connected OC-LS (there is one OC-LS connection configured per Signal Mast).

Msg LS Set Luminosity message received from OC-LS does not change the state. SEA shall display a plain text diagnosis when the message is received spontaneously or after command execution.

Command Parameter	SCI-LS Command	Message byte / parameter	SCI-LS Parameter	SCI-LS Parameter value
TRUE	Cmd Set Luminosity	43 = Luminosity	0x01	LUMINOSITY_FOR_DAY
FALSE	Cmd Set Luminosity	43 = Luminosity	0x02	LUMINOSITY_FOR_NIGHT

Table 46: SEA Eulynx Signal Mast - Luminosity - Mapping CC command (CC to EC-MUX) to SCI-LS Cmd Set Luminosity (EC-MUX to OC-LS)

Message byte / parameter	Parameter value	SEA Eulynx Signal Mast Luminosity	CC State
43 = Luminosity	Luminosity for day = 0x01	DAY	none
	Luminosity for night =		

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43 = Luminosity	0x02	NIGHT	none
43 = Luminosity	Intentionally deleted = 0xFE	none	none

Table 47: SEA Eulynx Signal Mast - Luminosity - Mapping SCI-LS Msg Set Luminosity (OC-LS to EC-MUX) to SEA element state and gen_elem_stat (EC-MUX to CC)

In SCI-LS [7] and SCI-REQ-LS [8], there is no information about usage of the value 0xFE INTENTIONALLY_DELETED in Msg Set Luminosity.

3.2.1.2 Operational - Signal Mast - Luminosity

In order to control and supervise the signal mast luminosity

As a SEA

I want to be able to receive the Luminosity command from CC, translate it into SCI-LS Set Luminosity command and forward it to each connected OC-LS and to receive SCI-LS Set Luminosity messages from OC-LS and send the reported luminosity as plain text diagnosis to DGP

For requirement language see Chapter 2.8 "Requirement Language".

Given SEA with at least one configured Signal Mast

EC-MUX-SIG:SSS: REQ:3600	Signal Mast - Luminosity - Initial state		
When initializing the element			
Then SEA shall set the current Signal Mast luminosity state to DAY			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3601	Signal Mast - Luminosity - execute command
-------------------------------------	---

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When receiving the luminosity command with <Goal> from CC

Then SEA shall forward the command LS Set Luminosity with luminosity <Luminosity> to each connected OC-LS

And SEA shall set the current element state to <State>

Goal	Luminosity	State
TRUE	LUMINOSITY_ FOR_ DAY	DAY
FALSE	LUMINOSITY_ FOR_ NIGHT	NIGHT

Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3602	Signal Mast - Luminosity - LS Set Luminosity message - plain text diagnosis						
When receiving the LS Set Luminosity message with luminosity <Luminosity> from OC-LS Then SEA shall report the plain text diagnosis message "SCI-LS: LS Set Luminosity <Luminosity> message received for Signal Mast x" to DGP <table><tr><th>Luminosity</th></tr><tr><td>LUMINOSITY_ FOR_ DAY</td></tr><tr><td>LUMINOSITY_ FOR_ NIGHT</td></tr><tr><td>INTENTIONALLY_ DELETED</td></tr></table>				Luminosity	LUMINOSITY_ FOR_ DAY	LUMINOSITY_ FOR_ NIGHT	INTENTIONALLY_ DELETED
Luminosity							
LUMINOSITY_ FOR_ DAY							
LUMINOSITY_ FOR_ NIGHT							
INTENTIONALLY_ DELETED							
Type:	FUR - Functional Requirement						
Status:	draft	Stability:	stable				

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Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

3.2.1.3 Connection - Signal Mast - Luminosity

In order to control and supervise the signal mast luminosity

As a SEA

I want to be able to supervise connections and when changes occur then

- on CC connection establishment / connection loss do nothing
- on OC-LS connection establishment forward the last luminosity command received from CC
- on OC-LS connection loss do nothing

For requirement language see Chapter 2.8 "Requirement Language".

Given SEA with at least one configured Signal Mast

EC-MUX-SIG:SSS: REQ:3610	Signal Mast - Luminosity - [Re-]Connecting to CC - General inquiry		
Given CC connection state is NOT_CONNECTED			
When establishing the connection to CC			
Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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EC-MUX-SIG:SSS: REQ:3611	Signal Mast - Luminosity - Connection loss to CC - do nothing		
Given CC connection state is CONNECTED			
When losing the connection to CC			
Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3612	Signal Mast - Luminosity - OC-LS NOT_CONNECTED - command								
Given EC-MUX - OC-LS connection state is NOT_CONNECTED When receiving the luminosity command with <Goal> from CC Then SEA shall set the current element state to <State> <table><tr><th>Goal</th><th>State</th></tr><tr><td>TRUE</td><td>DAY</td></tr><tr><td>FALSE</td><td>NIGHT</td></tr></table>				Goal	State	TRUE	DAY	FALSE	NIGHT
Goal	State								
TRUE	DAY								
FALSE	NIGHT								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						

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Comments:	The luminosity command received from CC changes the state for the current element if its connection to EC-MUX is NOT_CONNECTED and the command is forwarded to each connected OC-LS as defined in the operational feature.
Satisfies:	

EC-MUX-SIG:SSS: REQ:3613	Signal Mast - Luminosity - [Re-]Connecting between EC-MUX and OC-LS								
Given EC-MUX - OC-LS connection state is NOT_CONNECTED And the current element state is <ElemState> When establishing the connection between EC-MUX and OC-LS Then SEA shall forward the command LS Set Luminosity with luminosity <Luminosity> to this OC-LS <table><tr><th>ElemState</th><th>Luminosity</th></tr><tr><td>DAY</td><td>LUMINOSITY_FOR_DAY</td></tr><tr><td>NIGHT</td><td>LUMINOSITY_FOR_NIGHT</td></tr></table>				ElemState	Luminosity	DAY	LUMINOSITY_FOR_DAY	NIGHT	LUMINOSITY_FOR_NIGHT
ElemState	Luminosity								
DAY	LUMINOSITY_FOR_DAY								
NIGHT	LUMINOSITY_FOR_NIGHT								
Type:	FUR - Functional Requirement								
Status:	draft	Stability:	stable						
Target Release:	SEA SW GEN 3.0	Priority:	mandatory						
Originator:	Technics (R&D)	Allocation:	SW EC-MUX						
Verification Method:	Test	Verification Step:	module test, subit						
Comments:									
Satisfies:									

EC-MUX-SIG:SSS: REQ:3614	Signal Mast - Luminosity - Connection loss between EC-MUX and OC-LS
Given the EC-MUX - OC-LS connection state is CONNECTED	

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When losing the connection between EC-MUX and OC-LS			
Then SEA shall do nothing			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

3.2.1.4 Diagnosis - Signal Mast - Luminosity

Diagnosis and query of current states/faults is not supported.

3.2.2 Command Handling

3.2.2.1 Signal Mast - Command and message handling

In order to control the signal mast command and message handling

As a SEA

I want to be able to:

- receive command from CC for a single signal element
 - translate it into SCI-LS Indicate Signal Aspect command changing only the aspect of this signal element while other signals on the mast stay unchanged
 - forward the SCI-LS Indicate Signal Aspect command to OC-LS
- and
- receive the SCI-LS Indicated Signal Aspect message from OC-LS
 - translate it into signal state change
 - report it to CC

A Signal Mast can have configured several signal elements, each signal element is represented by a separate element process in SEA, but all signals belonging to the same mast are configured on the same OC-LS.

Eulynx defines a LS Indicate Signal Aspect command and a LS Indicated Signal Aspect message that contain aspect values of all configured signal elements on the mast as byte values. Some byte values are shared by several signal elements on this mast.

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SEA needs to ensure that commanding an aspect change for one signal element does not influence other signal elements on the same mast; only the byte assigned to that signal element shall be updated with the commanded aspect while values for other bytes shall be set to last received values from the LS Indicated Signal Aspect message (in other words there shall be no aspect change commanded for other signal elements).

SEA needs to ensure that a byte value received in the LS Indicated Signal Aspect message is processed by all signal elements for which this value is configured as an aspect. Signal elements sharing the same byte could both be influenced depending on the received aspect, i.e. a Main Signal Mast configured with a Shunting Signal Combined share byte 43 of LS Indicate Signal Aspect command and LS Indicated Signal Aspect message and the message with aspect DARK - HEX value 0xFF - shall affect both Main Signal and Shunting Signal Combined.

For requirement language see Chapter 2.8 "Requirement Language".

Given SEA with at least one configured Signal Mast

EC-MUX-SIG:SSS: REQ:3901	Signal Mast - command executed for targeted signal element		
Given the Signal Mast with several signal elements configured			
When executing a command for aspect change of one signal element			
Then SEA shall ensure that aspects of other elements on the same Signal Mast stay unchanged			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

EC-MUX-SIG:SSS: REQ:3902	Signal Mast - LS Indicated Signal Aspect message handling
Given the Signal Mast with several signal elements configured	
When receiving a LS Indicated Signal Aspect message from OC-LS	

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Then SEA shall ensure that each signal element on the mast independently reacts to the change of the aspect Code received in the corresponding byte			
Type:	FUR - Functional Requirement		
Status:	draft	Stability:	stable
Target Release:	SEA SW GEN 3.0	Priority:	mandatory
Originator:	Technics (R&D)	Allocation:	SW EC-MUX
Verification Method:	Test	Verification Step:	module test, subit
Comments:			
Satisfies:			

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4 Requirements List

Identifier	Synopsis	Category	Status
EC-MUX-SIG:SSS:REQ:3021	Signal With One Speed Aspect and Stop command - [Re-] Connecting to CC - OC-LS CONNECTED - General inquiry in stable states	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3022	Signal With One Speed Aspect and Stop command - [Re-] Connecting to CC - OC-LS CONNECTED - General inquiry in CHANGING_TO_STOP state - no message after CC disconnect	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3024	Signal With One Speed Aspect and Stop command - [Re-] Connecting to CC - OC-LS NOT_CONNECTED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3025	Signal With One Speed Aspect and Stop command - Connection loss to CC - execute safe command in SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3035	Signal With One Speed Aspect and Stop command - Connection loss to CC - OC-LS NOT_CONNECTED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3027	Signal With One Speed Aspect and Stop command - Connection loss to CC - do nothing in STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3029	Signal With One Speed Aspect and Stop command - Connection loss to CC - execute safe command in DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3026	Signal With One Speed Aspect and Stop command - Connection loss to CC - CHANGING_TO_SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3028	Signal With One Speed Aspect and Stop command - Connection loss to CC - CHANGING_TO_STOP - do nothing	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3001	Signal With One Speed Aspect and Stop command - [Re-] Connecting between EC-MUX and OC-LS in state STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3002	Signal With One Speed Aspect and Stop command - [Re-] Connecting between EC-MUX and OC-LS in state DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3052	Signal With One Speed Aspect and Stop command - [Re-] Connecting between EC-MUX and OC-LS in state SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3030	Signal With One Speed Aspect and Stop command - Connection loss between EC-MUX and OC-LS in regular states	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:	Signal With One Speed Aspect and Stop command -	FUR -	

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Identifier	Synopsis	Category	Status
REQ:3031	Connection loss between EC-MUX and OC-LS in DARK state	Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3048	Signal With One Speed Aspect and Stop command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3032	Signal With One Speed Aspect and Stop command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3000	Signal With One Speed Aspect and Stop command - Initial state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3003	Signal With One Speed Aspect and Stop command - reject commands in UNDEFINED state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3004	Signal With One Speed Aspect and Stop command - execute e_clear_sig_speed1 command in STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3005	Signal With One Speed Aspect and Stop command - reject STOP commands in STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3011	Signal With One Speed Aspect and Stop command - execute STOP commands in DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3006	Signal With One Speed Aspect and Stop command - spontaneous transition between STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3007	Signal With One Speed Aspect and Stop command - execute STOP command in SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3008	Signal With One Speed Aspect and Stop command - reject e_clear_sig_speed1 command in SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3009	Signal With One Speed Aspect and Stop command - spontaneous transition from SPEED1 to STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3010	Signal With One Speed Aspect and Stop command - transition from CHANGING_TO_SPEED1 to SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3012	Signal With One Speed Aspect and Stop command - reject e_clear_sig_speed1 command in CHANGING_TO_SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3013	Signal With One Speed Aspect and Stop command - CHANGING_TO_SPEED1 - return to STOP / DARK	FUR - Functional Requirement	draft
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Identifier	Synopsis	Category	Status
EC-MUX-SIG:SSS:REQ:3014	Signal With One Speed Aspect and Stop command - CHANGING_TO_SPEED1 - transition to DARK / STOP	Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3019	Signal With One Speed Aspect and Stop command - CHANGING_TO_SPEED1 - execute STOP command	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3043	Signal With One Speed Aspect and Stop command - CHANGING_TO_SPEED1 - timeout	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3015	Signal With One Speed Aspect and Stop command - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3047	Signal With One Speed Aspect and Stop command - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state DARK / STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3044	Signal With One Speed Aspect and Stop command - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3016	Signal With One Speed Aspect and Stop command - reject commands in CHANGING_TO_STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3017	Signal With One Speed Aspect and Stop command - CHANGING_TO_STOP - ignore SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3045	Signal With One Speed Aspect and Stop command - CHANGING_TO_STOP - timeout when OC-LS SPEED1 - OC Disconnection	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3046	Signal With One Speed Aspect and Stop command - CHANGING_TO_STOP - timeout when OC-LS DARK / STOP - reach DARK / STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3020	Signal With One Speed Aspect and Stop command - Disconnection on spontaneous aspect message SPEED1 in STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3042	Signal With One Speed Aspect and Stop command - Ignore repeated messages	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3053	Signal With One Speed Aspect and Stop command - Receive unexpected DARK / SPEED commands from CC	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3050	Signal With One Speed Aspect and Stop command - aspect query - last stable state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3049	Signal With One Speed Aspect and Stop command - aspect query in UNDEFINED - OC-LS NOT_CONNECTED	FUR - Functional Requirement	draft
		FUR -	

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Identifier	Synopsis	Category	Status
EC-MUX-SIG:SSS:REQ:3300	Signal With Three Speed Aspects and Dark command - Initial state	Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3302	Signal With Three Speed Aspects and Dark command - reject commands in UNDEFINED state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3303	Signal With Three Speed Aspects and Dark command - execute SPEED command in DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3304	Signal With Three Speed Aspects and Dark command - reject DARK commands in DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3305	Signal With Three Speed Aspects and Dark command - execute DARK commands in SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3306	Signal With Three Speed Aspects and Dark command - reject SPEED command in SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3307	Signal With Three Speed Aspects and Dark command - spontaneous transition from SPEEDx to DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3308	Signal With Three Speed Aspects and Dark command - CHANGING_TO_SPEEDn - transition from CHANGING_TO_SPEEDx to SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3309	Signal With Three Speed Aspects and Dark command - CHANGING_TO_SPEEDn - reject repeated SPEED command in CHANGING_TO_SPEEDn	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3310	Signal With Three Speed Aspects and Dark command - CHANGING_TO_SPEEDn - return to DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3311	Signal With Three Speed Aspects and Dark command - CHANGING_TO_SPEEDn - timeout	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3312	Signal With Three Speed Aspects and Dark command - CHANGING_TO_SPEEDn - execute DARK command	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3313	Signal With Three Speed Aspects and Dark command - CHANGING_TO_DARK - transition from CHANGING_TO_DARK to DARK - from OC-LS state SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3314	Signal With Three Speed Aspects and Dark command - CHANGING_TO_DARK - reject commands in CHANGING_TO_DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3315	Signal With Three Speed Aspects and Dark command - CHANGING_TO_DARK - timeout when OC-LS SPEEDx - OC Disconnection	FUR - Functional Requirement	draft
		FUR -	

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Identifier	Synopsis	Category	Status
EC-MUX-SIG:SSS:REQ:3316	Signal With Three Speed Aspects and Dark command - CHANGING_TO_DARK - ignore SPEED message	Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3317	Signal With Three Speed Aspects and Dark command - transition from CHANGING_TO_DARK to DARK - from OC-LS state DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3336	Signal With Three Speed Aspects and Dark command - CHANGING_TO_DARK - timeout when OC-LS DARK - reach DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3318	Signal With Three Speed Aspects and Dark command - Disconnection on spontaneous SPEED message in DARK / SPEED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3319	Signal With Three Speed Aspects and Dark command - CHANGING_TO_SPEEDn - Disconnection on unexpected command reply	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3320	Signal With Three Speed Aspects and Dark command - Ignore repeated messages	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3338	Signal With Three Speed Aspects and Dark command - Receive unexpected STOP commands from CC	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3321	Signal With Three Speed Aspects and Dark command - [Re-] Connecting to CC - OC-LS CONNECTED - General inquiry in stable states	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3322	Signal With Three Speed Aspects and Dark command - [Re-] Connecting to CC - OC-LS CONNECTED - General inquiry in CHANGING_TO_DARK - no message after CC disconnect	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3323	Signal With Three Speed Aspects and Dark command - [Re-] Connecting to CC - OC-LS NOT_CONNECTED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3324	Signal With Three Speed Aspects and Dark command - Connection loss to CC - execute safe command in SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3325	Signal With Three Speed Aspects and Dark command - Connection loss to CC - do nothing in DARK / UNDEFINED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3326	Signal With Three Speed Aspects and Dark command - Connection loss to CC - CHANGING_TO_SPEEDn	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3327	Signal With Three Speed Aspects and Dark command - Connection loss to CC - CHANGING_TO_DARK - do nothing	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3301	Signal With Three Speed Aspects and Dark command - [Re-] Connecting between EC-MUX and OC-LS in state DARK	FUR - Functional Requirement	draft
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Identifier	Synopsis	Category	Status
EC-MUX-SIG:SSS:REQ:3339	Signal With Three Speed Aspects and Dark command - [Re-] Connecting between EC-MUX and OC-LS in state SPEEDx	Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3329	Signal With Three Speed Aspects and Dark command - Connection loss between EC-MUX and OC-LS in regular states	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3330	Signal With Three Speed Aspects and Dark command - Connection loss between EC-MUX and OC-LS in DARK state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3331	Signal With Three Speed Aspects and Dark command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_SPEEDn	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3332	Signal With Three Speed Aspects and Dark command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3333	Signal With Three Speed Aspects and Dark command - aspect query - last diagnosis aspect message	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3335	Signal With Three Speed Aspects and Dark command - aspect query in UNDEFINED - OC-LS NOT_CONNECTED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3615	Signal Mast - Luminosity - send Luminosity command while no OC-LS is configured	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3901	Signal Mast - command executed for targeted signal element	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3902	Signal Mast - LS Indicated Signal Aspect message handling	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3610	Signal Mast - Luminosity - [Re-]Connecting to CC - General inquiry	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3611	Signal Mast - Luminosity - Connection loss to CC - do nothing	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3612	Signal Mast - Luminosity - OC-LS NOT_CONNECTED - command	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3613	Signal Mast - Luminosity - [Re-]Connecting between EC-MUX and OC-LS	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3614	Signal Mast - Luminosity - Connection loss between EC-MUX and OC-LS	FUR - Functional Requirement	draft
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Identifier	Synopsis	Category	Status
EC-MUX-SIG:SSS: REQ:3600	Signal Mast - Luminosity - Initial state	Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3601	Signal Mast - Luminosity - execute command	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3602	Signal Mast - Luminosity - LS Set Luminosity message - plain text diagnosis	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3850	Shunting Signal Combined - aspect query - last stable state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3849	Shunting Signal Combined - aspect query in UNDEFINED - OC-LS NOT_CONNECTED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3800	Shunting Signal Combined - Initial state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3803	Shunting Signal Combined - reject commands in UNDEFINED state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3804	Shunting Signal Combined - execute e_clear_sig_speed1 command in STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3805	Shunting Signal Combined - reject STOP commands in STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3811	Shunting Signal Combined - execute STOP commands in DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3806	Shunting Signal Combined - spontaneous transition between STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3807	Shunting Signal Combined - execute STOP command in SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3808	Shunting Signal Combined - reject e_clear_sig_speed1 command in SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3809	Shunting Signal Combined - spontaneous transition from SPEED1 to STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3810	Shunting Signal Combined - transition from CHANGING_TO_SPEED1 to SPEED1	FUR - Functional Requirement	draft
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Identifier	Synopsis	Category	Status
EC-MUX-SIG:SSS:REQ:3812	Shunting Signal Combined - reject e_clear_sig_speed1 command in CHANGING_TO_SPEED1	Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3813	Shunting Signal Combined - CHANGING_TO_SPEED1 - return to STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3814	Shunting Signal Combined - CHANGING_TO_SPEED1 - transition to DARK / STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3819	Shunting Signal Combined - CHANGING_TO_SPEED1 - execute STOP command	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3843	Shunting Signal Combined - CHANGING_TO_SPEED1 - timeout	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3815	Shunting Signal Combined - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3847	Shunting Signal Combined - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state DARK / STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3844	Shunting Signal Combined - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3816	Shunting Signal Combined - reject commands in CHANGING_TO_STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3817	Shunting Signal Combined - CHANGING_TO_STOP - ignore SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3845	Shunting Signal Combined - CHANGING_TO_STOP - timeout when OC-LS SPEED1 - OC Disconnection	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3846	Shunting Signal Combined - CHANGING_TO_STOP - timeout when OC-LS DARK / STOP - reach DARK / STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3820	Shunting Signal Combined - Disconnection on spontaneous aspect message SPEED1 in STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3842	Shunting Signal Combined - Ignore repeated messages	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3853	Shunting Signal Combined - Receive unexpected DARK / SPEED commands from CC	FUR - Functional Requirement	draft
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Identifier	Synopsis	Category	Status
EC-MUX-SIG:SSS: REQ:3821	Shunting Signal Combined - [Re-]Connecting to CC - OC-LS CONNECTED - General inquiry in stable states	Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3822	Shunting Signal Combined - [Re-]Connecting to CC - OC-LS CONNECTED - General inquiry in CHANGING_TO_STOP state - no message after CC disconnect	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3824	Shunting Signal Combined - [Re-]Connecting to CC - OC-LS NOT_CONNECTED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3825	Shunting Signal Combined - Connection loss to CC - execute safe command in SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3835	Shunting Signal Combined - Connection loss to CC - OC-LS NOT_CONNECTED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3827	Shunting Signal Combined - Connection loss to CC - do nothing in STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3826	Shunting Signal Combined - Connection loss to CC - CHANGING_TO_SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3828	Shunting Signal Combined - Connection loss to CC - CHANGING_TO_STOP - do nothing	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3801	Shunting Signal Combined - [Re-]Connecting between EC- MUX and OC-LS in state STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3802	Shunting Signal Combined - [Re-]Connecting between EC- MUX and OC-LS in state DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3852	Shunting Signal Combined - [Re-]Connecting between EC- MUX and OC-LS in state SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3830	Shunting Signal Combined - Connection loss between EC- MUX and OC-LS in regular states	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3831	Shunting Signal Combined - Connection loss between EC- MUX and OC-LS in DARK state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3848	Shunting Signal Combined - Connection loss between EC- MUX and OC-LS in CHANGING_TO_SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3832	Shunting Signal Combined - Connection loss between EC- MUX and OC-LS in CHANGING_TO_STOP	FUR - Functional Requirement	draft
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Identifier	Synopsis	Category	Status
EC-MUX-SIG:SSS: REQ:3553	Signal With Three Speed Aspects and Stop and Dark commands - aspect query - last stable state	Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3555	Signal With Three Speed Aspects and Stop and Dark commands - aspect query in UNDEFINED - OC-LS NOT_CONNECTED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3540	Signal With Three Speed Aspects and Stop and Dark commands - [Re-]Connecting to CC - OC-LS CONNECTED - General inquiry in stable states	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3541	Signal With Three Speed Aspects and Stop and Dark commands - [Re-]Connecting to CC - OC-LS CONNECTED - General inquiry in CHANGING_TO_DARK - no message after CC disconnect	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3542	Signal With Three Speed Aspects and Stop and Dark commands - [Re-]Connecting to CC - OC-LS NOT_CONNECTED - General inquiry	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3558	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss to CC - execute safe command in STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3543	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss to CC - execute safe command in SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3544	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss to CC - do nothing in DARK / UNDEFINED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3545	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss to CC - CHANGING_TO_SPEEDn	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3546	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss to CC - CHANGING_TO_STOP - execute safe command	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3547	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss to CC - CHANGING_TO_DARK - do nothing	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3501	Signal With Three Speed Aspects and Stop and Dark commands - [Re-]Connecting between EC-MUX and OC-LS in state STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3502	Signal With Three Speed Aspects and Stop and Dark commands - [Re-]Connecting between EC-MUX and OC-LS in state DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3556	Signal With Three Speed Aspects and Stop and Dark commands - [Re-]Connecting between EC-MUX and OC-LS in state SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3549	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss between EC-MUX and OC-LS in regular states	FUR - Functional Requirement	draft

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Identifier	Synopsis	Category	Status
EC-MUX-SIG:SSS:REQ:3550	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss between EC-MUX and OC-LS in DARK state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3551	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss between EC-MUX and OC-LS in CHANGING_TO_SPEEDn	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3552	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss between EC-MUX and OC-LS in CHANGING_TO_DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3557	Signal With Three Speed Aspects and Stop and Dark commands - Connection loss between EC-MUX and OC-LS in CHANGING_TO_STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3500	Signal With Three Speed Aspects and Stop and Dark commands - Initial state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3503	Signal With Three Speed Aspects and Stop and Dark commands - reject commands in UNDEFINED state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3504	Signal With Three Speed Aspects and Stop and Dark commands - execute SPEED commands in STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3505	Signal With Three Speed Aspects and Stop and Dark commands - reject STOP commands in STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3506	Signal With Three Speed Aspects and Stop and Dark commands - reject DARK commands in DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3507	Signal With Three Speed Aspects and Stop and Dark commands - execute STOP commands in DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3508	Signal With Three Speed Aspects and Stop and Dark commands - execute DARK commands in STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3509	Signal With Three Speed Aspects and Stop and Dark commands - spontaneous transition between STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3510	Signal With Three Speed Aspects and Stop and Dark commands - execute STOP commands in SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3511	Signal With Three Speed Aspects and Stop and Dark commands - execute DARK commands in SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3512	Signal With Three Speed Aspects and Stop and Dark commands - reject SPEED commands in SPEEDx	FUR - Functional Requirement	draft

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Identifier	Synopsis	Category	Status
EC-MUX-SIG:SSS: REQ:3513	Signal With Three Speed Aspects and Stop and Dark commands - spontaneous transition from SPEEDx to STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3514	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_SPEEDn - transition from CHANGING_TO_SPEEDx to SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3515	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_SPEEDn - reject repeated SPEED command in CHANGING_TO_SPEEDn	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3516	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_SPEEDn - return to STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3517	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_SPEEDn - transition to STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3518	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_SPEEDn - timeout	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3519	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_SPEEDn - execute STOP command	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3520	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_SPEEDn - execute DARK command	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3521	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_STOP - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3527	Signal With Three Speed Aspects and Stop and Dark commands - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3528	Signal With Three Speed Aspects and Stop and Dark commands - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state DARK / STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3522	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_STOP - reject STOP / SPEED commands in CHANGING_TO_STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3523	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_STOP - execute DARK command	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3524	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_STOP - timeout when OC-LS SPEEDx - OC Disconnection	FUR - Functional Requirement	draft
	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_STOP - timeout when OC-LS	FUR - Functional	

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EC-MUX-SIG:SSS: REQ:3525	DARK / STOP - reach DARK / STOP	Requirement	draft
EC-MUX-SIG:SSS: REQ:3526	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_STOP - ignore SPEED message	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3529	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_DARK - transition from CHANGING_TO_DARK to STOP / DARK - from OC-LS state SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3535	Signal With Three Speed Aspects and Stop and Dark commands - transition from CHANGING_TO_DARK to STOP / DARK - from OC-LS state STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3536	Signal With Three Speed Aspects and Stop and Dark commands - transition from CHANGING_TO_DARK to STOP / DARK - from OC-LS state DARK / STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3530	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_DARK - reject DARK / SPEED commands in CHANGING_TO_DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3531	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_DARK - execute STOP command	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3532	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_DARK - timeout when OC-LS SPEEDx - OC Disconnection	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3533	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_DARK - timeout when OC-LS DARK - reach DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3560	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_DARK - timeout when OC-LS STOP - OC Disconnection	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3534	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_DARK - ignore SPEED message	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3537	Signal With Three Speed Aspects and Stop and Dark commands - Disconnection on spontaneous SPEED message in STOP / DARK / SPEED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3538	Signal With Three Speed Aspects and Stop and Dark commands - CHANGING_TO_SPEEDn - Disconnection on unexpected command reply	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3539	Signal With Three Speed Aspects and Stop and Dark commands - Ignore repeated messages	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3235	Signal With One Speed Aspect and Dark command - aspect query - last stable state	FUR - Functional Requirement	draft

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Identifier	Synopsis	Category	Status
EC-MUX-SIG:SSS:REQ:3237	Signal With One Speed Aspect and Dark command - aspect query in UNDEFINED - OC-LS NOT_CONNECTED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3222	Signal With One Speed Aspect and Dark command - [Re-] Connecting to CC - OC-LS CONNECTED - General inquiry in stable states	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3223	Signal With One Speed Aspect and Dark command - [Re-] Connecting to CC - OC-LS CONNECTED - General inquiry in CHANGING_TO_DARK state - no message after CC disconnect	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3224	Signal With One Speed Aspect and Dark command - [Re-] Connecting to CC - OC-LS NOT_CONNECTED - General inquiry	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3225	Signal With One Speed Aspect and Dark command - Connection loss to CC - execute safe command in SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3226	Signal With One Speed Aspect and Dark command - Connection loss to CC - do nothing in DARK / UNDEFINED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3227	Signal With One Speed Aspect and Dark command - Connection loss to CC - CHANGING_TO_SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3228	Signal With One Speed Aspect and Dark command - Connection loss to CC - CHANGING_TO_DARK - do nothing	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3201	Signal With One Speed Aspect and Dark command - [Re-] Connecting between EC-MUX and OC-LS in state DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3220	Signal With One Speed Aspect and Dark command - [Re-] Connecting between EC-MUX and OC-LS in state SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3230	Signal With One Speed Aspect and Dark command - Connection loss between EC-MUX and OC-LS in SPEED1 state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3231	Signal With One Speed Aspect and Dark command - Connection loss between EC-MUX and OC-LS in DARK state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3232	Signal With One Speed Aspect and Dark command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3233	Signal With One Speed Aspect and Dark command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_DARK	FUR - Functional Requirement	draft
		FUR - Functional	

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Identifier	Synopsis	Category	Status
EC-MUX-SIG:SSS:REQ:3200	Signal With One Speed Aspect and Dark command - Initial state	Requirement	draft
EC-MUX-SIG:SSS:REQ:3202	Signal With One Speed Aspect and Dark command - reject commands in UNDEFINED state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3203	Signal With One Speed Aspect and Dark command - execute e_clear_sig_speed1 command in DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3204	Signal With One Speed Aspect and Dark command - reject DARK commands in DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3205	Signal With One Speed Aspect and Dark command - execute DARK command in SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3206	Signal With One Speed Aspect and Dark command - reject e_clear_sig_speed1 command in SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3207	Signal With One Speed Aspect and Dark command - spontaneous transition from SPEED1 to DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3208	Signal With One Speed Aspect and Dark command - transition from CHANGING_TO_SPEED1 to SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3209	Signal With One Speed Aspect and Dark command - reject e_clear_sig_speed1 command in CHANGING_TO_SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3210	Signal With One Speed Aspect and Dark command - CHANGING_TO_SPEED1 - return to DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3211	Signal With One Speed Aspect and Dark command - CHANGING_TO_SPEED1 - execute DARK command	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3212	Signal With One Speed Aspect and Dark command - CHANGING_TO_SPEED1 - timeout	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3213	Signal With One Speed Aspect and Dark command - transition from CHANGING_TO_DARK to DARK - from OC-LS state SPEED1	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3214	Signal With One Speed Aspect and Dark command - transition from CHANGING_TO_DARK to DARK - from OC-LS state DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3215	Signal With One Speed Aspect and Dark command - reject commands in CHANGING_TO_DARK	FUR - Functional Requirement	draft
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EC-MUX-SIG:SSS:REQ:3216	Signal With One Speed Aspect and Dark command - CHANGING_TO_DARK - ignore SPEED1	Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3217	Signal With One Speed Aspect and Dark command - CHANGING_TO_DARK - timeout when OC-LS SPEED1 - OC Disconnection	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3218	Signal With One Speed Aspect and Dark command - CHANGING_TO_DARK - timeout when OC-LS DARK - reach DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3219	Signal With One Speed Aspect and Dark command - Disconnection on spontaneous aspect message SPEED1 in DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3221	Signal With One Speed Aspect and Dark command - Ignore repeated messages	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3238	Signal With One Speed Aspect and Dark command - Receive unexpected STOP / SPEED commands from CC	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3092	Signal With Three Speed Aspects and Stop command - Initial state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3095	Signal With Three Speed Aspects and Stop command - reject commands in UNDEFINED state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3096	Signal With Three Speed Aspects and Stop command - execute SPEED command in STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3097	Signal With Three Speed Aspects and Stop command - reject STOP commands in STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3018	Signal With Three Speed Aspects and Stop command - execute STOP commands in DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3098	Signal With Three Speed Aspects and Stop command - spontaneous transition between STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3099	Signal With Three Speed Aspects and Stop command - execute STOP commands in SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3100	Signal With Three Speed Aspects and Stop command - reject SPEED command in SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3101	Signal With Three Speed Aspects and Stop command - spontaneous transition from SPEEDx to STOP / DARK	FUR - Functional Requirement	draft
	Signal With Three Speed Aspects and Stop command -	FUR -	

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EC-MUX-SIG:SSS: REQ:3102	CHANGING_TO_SPEEDn - transition from CHANGING_TO_SPEEDx to SPEEDx	Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3103	Signal With Three Speed Aspects and Stop command - CHANGING_TO_SPEEDn - reject repeated SPEED command in CHANGING_TO_SPEEDn	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3104	Signal With Three Speed Aspects and Stop command - CHANGING_TO_SPEEDn - return to STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3105	Signal With Three Speed Aspects and Stop command - CHANGING_TO_SPEEDn - transition to STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3140	Signal With Three Speed Aspects and Stop command - CHANGING_TO_SPEEDn - timeout	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3143	Signal With Three Speed Aspects and Stop command - CHANGING_TO_SPEEDn - execute STOP command	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3107	Signal With Three Speed Aspects and Stop command - CHANGING_TO_STOP - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3111	Signal With Three Speed Aspects and Stop command - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3112	Signal With Three Speed Aspects and Stop command - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state DARK / STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3108	Signal With Three Speed Aspects and Stop command - CHANGING_TO_STOP - reject commands in CHANGING_TO_STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3141	Signal With Three Speed Aspects and Stop command - CHANGING_TO_STOP - timeout when OC-LS SPEEDx - OC Disconnection	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3142	Signal With Three Speed Aspects and Stop command - CHANGING_TO_STOP - timeout when OC-LS DARK / STOP - reach DARK / STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3110	Signal With Three Speed Aspects and Stop command - CHANGING_TO_STOP - ignore SPEED message	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3114	Signal With Three Speed Aspects and Stop command - Disconnection on spontaneous SPEED messages in STOP / DARK / SPEED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3106	Signal With Three Speed Aspects and Stop command - CHANGING_TO_SPEEDx - Disconnection on unexpected command reply	FUR - Functional Requirement	draft

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EC-MUX-SIG:SSS:REQ:3115	Signal With Three Speed Aspects and Stop command - Ignore repeated messages	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3146	Signal With Three Speed Aspects and Stop command - Receive unexpected DARK commands from CC	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3116	Signal With Three Speed Aspects and Stop command - [Re-] Connecting to CC - OC-LS CONNECTED - General inquiry in stable states	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3117	Signal With Three Speed Aspects and Stop command - [Re-] Connecting to CC - OC-LS CONNECTED - General inquiry in CHANGING_TO_STOP - no message after CC disconnect	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3139	Signal With Three Speed Aspects and Stop command - [Re-] Connecting to CC - OC-LS NOT_CONNECTED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3120	Signal With Three Speed Aspects and Stop command - Connection loss to CC - execute safe command in SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3118	Signal With Three Speed Aspects and Stop command - Connection loss to CC - do nothing in STOP / UNDEFINED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3150	Signal With Three Speed Aspects and Stop command - Connection loss to CC - execute safe command in DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3123	Signal With Three Speed Aspects and Stop command - Connection loss to CC - CHANGING_TO_SPEEDn	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3122	Signal With Three Speed Aspects and Stop command - Connection loss to CC - CHANGING_TO_STOP - do nothing	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3093	Signal With Three Speed Aspects and Stop command - [Re-] Connecting between EC-MUX and OC-LS in state STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3094	Signal With Three Speed Aspects and Stop command - [Re-] Connecting between EC-MUX and OC-LS in state DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3147	Signal With Three Speed Aspects and Stop command - [Re-] Connecting between EC-MUX and OC-LS in state SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3127	Signal With Three Speed Aspects and Stop command - Connection loss between EC-MUX and OC-LS in regular states	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3128	Signal With Three Speed Aspects and Stop command - Connection loss between EC-MUX and OC-LS in DARK state	FUR - Functional Requirement	draft

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EC-MUX-SIG:SSS: REQ:3129	Signal With Three Speed Aspects and Stop command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_SPEEDn	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3144	Signal With Three Speed Aspects and Stop command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3138	Signal With Three Speed Aspects and Stop command - aspect query - last stable state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3145	Signal With Three Speed Aspects and Stop command - aspect query in UNDEFINED - OC-LS NOT_CONNECTED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3750	Signal Repeater Combined - [Re-]Connecting to CC - General inquiry in UNDEFINED / MAST_DARK / MAST_DARK_FAULTY	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3751	Signal Repeater Combined - [Re-]Connecting to CC - OC-LS CONNECTED - General inquiry in CHANGING_TO_DARK - no message after CC disconnect	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3752	Signal Repeater Combined - Connection loss to CC in UNDEFINED - do nothing	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3753	Signal Repeater Combined - Connection loss to CC in MAST_DARK / MAST_DARK_FAULTY - store command STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3754	Signal Repeater Combined - Connection loss to CC in DARK - transition to MAST_DARK_FAULTY and store command STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3755	Signal Repeater Combined - Connection loss to CC in CHANGING_TO_DARK - store command STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3756	Signal Repeater Combined - Connection loss to CC in STOP / SPEEDx - transition to CHANGING_TO_DARK and store command STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3757	Signal Repeater Combined - Connection loss to CC in CHANGING_TO_SPEEDn - last stable STOP - transition to CHANGING_TO_DARK and store command STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3758	Signal Repeater Combined - Connection loss to CC in CHANGING_TO_SPEEDn - last stable DARK - transition to MAST_DARK_FAULTY and store command STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3759	Signal Repeater Combined - Connection loss to CC in CHANGING_TO_STOP - last stable not DARK - transition to CHANGING_TO_DARK and store command STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3760	Signal Repeater Combined - Connection loss to CC in CHANGING_TO_STOP - last stable DARK - transition to MAST_DARK_FAULTY and store command STOP	FUR - Functional Requirement	draft

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EC-MUX-SIG:SSS:REQ:3761	Signal Repeater Combined - [Re-]Connecting between EC-MUX and OC-LS in state UNDEFINED - OC-LS in DARK state - transition to MAST_DARK_FAULTY	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3763	Signal Repeater Combined - [Re-]Connecting between EC-MUX and OC-LS in state UNDEFINED - OC-LS in SPEEDn / STOP - OC Disconnection	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3762	Signal Repeater Combined - Connection loss between EC-MUX and OC-LS in regular states	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3764	Signal Repeater Combined - Connection loss between EC-MUX and OC-LS in CHANGING_TO_SPEEDn	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3765	Signal Repeater Combined - Connection loss between EC-MUX and OC-LS in CHANGING_TO_STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3767	Signal Repeater Combined - Connection loss between EC-MUX and OC-LS in CHANGING_TO_DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3768	Signal Repeater Combined - Connection loss between EC-MUX and OC-LS in MAST_DARK state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3766	Signal Repeater Combined - Connection loss between EC-MUX and OC-LS in MAST_DARK_FAULTY / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3771	Signal Repeater Combined - aspect query	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3700	Signal Repeater Combined - Initial state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3702	Signal Repeater Combined - UNDEFINED - Protection Signal state change - do nothing	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3721	Signal Repeater Combined - DARK - spontaneous transition to SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3722	Signal Repeater Combined - DARK - transition to MAST_DARK_FAULTY	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3726	Signal Repeater Combined - STOP - transition to CHANGING_TO_DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3730	Signal Repeater Combined - transition from SPEEDx to CHANGING_TO_DARK	FUR - Functional Requirement	draft

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EC-MUX-SIG:SSS: REQ:3737	Signal Repeater Combined - last stable state DARK - transition from CHANGING_TO_SPEEDn to MAST_DARK_FAULTY	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3738	Signal Repeater Combined - last stable state STOP - transition from CHANGING_TO_SPEEDn to CHANGING_TO_DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3769	Signal Repeater Combined - CHANGING_TO_SPEEDn - timeout with SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3770	Signal Repeater Combined - CHANGING_TO_SPEEDn - timeout with not matching SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3745	Signal Repeater Combined - last stable state DARK - transition from CHANGING_TO_STOP to MAST_DARK_FAULTY	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3746	Signal Repeater Combined - last stable state STOP / SPEEDx - transition from CHANGING_TO_STOP to CHANGING_TO_DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3703	Signal Repeater Combined - MAST_DARK_FAULTY - spontaneous message DARK - do nothing	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3704	Signal Repeater Combined - MAST_DARK_FAULTY - previous state DARK / UNDEFINED - spontaneous message SPEEDx / STOP - OC Disconnection	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3705	Signal Repeater Combined - MAST_DARK_FAULTY - previous state CHANGING_TO_STOP / CHANGING_TO_SPEEDn - spontaneous message SPEEDx / STOP - do nothing	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3706	Signal Repeater Combined - MAST_DARK_FAULTY - STOP / SPEEDx commands - report e_sig_dark_es	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3707	Signal Repeater Combined - MAST_DARK_FAULTY / CHANGING_TO_DARK / MAST_DARK - Protection Signal state change - do nothing	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3708	Signal Repeater Combined - MAST_DARK_FAULTY / CHANGING_TO_DARK / MAST_DARK - Protection Signal state change to SPEED1 - forward last stored command	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3709	Signal Repeater Combined - CHANGING_TO_DARK - spontaneous message SPEEDx / STOP - do nothing	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3710	Signal Repeater Combined - CHANGING_TO_DARK - STOP / SPEEDx commands - store command	FUR - Functional Requirement	draft
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EC-MUX-SIG:SSS:REQ:3711	Signal Repeater Combined - CHANGING_TO_DARK - timeout - OC Disconnection	Requirement	draft
EC-MUX-SIG:SSS:REQ:3712	Signal Repeater Combined - CHANGING_TO_DARK - transition from CHANGING_TO_DARK to MAST_DARK - last stable SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3713	Signal Repeater Combined - CHANGING_TO_DARK - transition from CHANGING_TO_DARK to MAST_DARK - last stable STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3715	Signal Repeater Combined - MAST_DARK - spontaneous message SPEEDx / STOP - OC Disconnection	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3716	Signal Repeater Combined - MAST_DARK - STOP / SPEEDx commands - report e_sig_stop_es	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3427	Signal With Two Speed Aspects and Stop command - [Re-] Connecting to CC - OC-LS CONNECTED - General inquiry in stable states	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3428	Signal With Two Speed Aspects and Stop command - [Re-] Connecting to CC - OC-LS CONNECTED - General inquiry in CHANGING_TO_STOP - no message after CC disconnect	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3429	Signal With Two Speed Aspects and Stop command - [Re-] Connecting to CC - OC-LS NOT_CONNECTED - General inquiry	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3430	Signal With Two Speed Aspects and Stop command - Connection loss to CC - execute safe command in SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3431	Signal With Two Speed Aspects and Stop command - Connection loss to CC - do nothing in STOP / UNDEFINED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3451	Signal With Two Speed Aspects and Stop command - Connection loss to CC - do nothing in DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3432	Signal With Two Speed Aspects and Stop command - Connection loss to CC - CHANGING_TO_SPEEDn	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3433	Signal With Two Speed Aspects and Stop command - Connection loss to CC - CHANGING_TO_STOP - do nothing	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3401	Signal With Two Speed Aspects and Stop command - [Re-] Connecting between EC-MUX and OC-LS in state STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3402	Signal With Two Speed Aspects and Stop command - [Re-] Connecting between EC-MUX and OC-LS in state DARK	FUR - Functional Requirement	draft
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EC-MUX-SIG:SSS: REQ:3434	Signal With Two Speed Aspects and Stop command - [Re-] Connecting between EC-MUX and OC-LS in state SPEEDx	Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3435	Signal With Two Speed Aspects and Stop command - Connection loss between EC-MUX and OC-LS in regular states	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3436	Signal With Two Speed Aspects and Stop command - Connection loss between EC-MUX and OC-LS in DARK state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3437	Signal With Two Speed Aspects and Stop command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_SPEEDn	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3438	Signal With Two Speed Aspects and Stop command - Connection loss between EC-MUX and OC-LS in CHANGING_TO_STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3440	Signal With Two Speed Aspects and Stop command - aspect query - last stable state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3442	Signal With Two Speed Aspects and Stop command - aspect query in UNDEFINED - OC-LS NOT_CONNECTED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3400	Signal With Two Speed Aspects and Stop command - Initial state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3450	Signal With Two Speed Aspects and Stop command - Disconnection on spontaneous SPEED message in DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3403	Signal With Two Speed Aspects and Stop command - reject commands in UNDEFINED state	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3404	Signal With Two Speed Aspects and Stop command - execute SPEED command in STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3405	Signal With Two Speed Aspects and Stop command - reject STOP commands in STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3439	Signal With Two Speed Aspects and Stop command - execute STOP commands in DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3406	Signal With Two Speed Aspects and Stop command - spontaneous transition between STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3407	Signal With Two Speed Aspects and Stop command - execute STOP commands in SPEEDx	FUR - Functional Requirement	draft
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EC-MUX-SIG:SSS: REQ:3408	Signal With Two Speed Aspects and Stop command - reject SPEED command in SPEEDx	Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3409	Signal With Two Speed Aspects and Stop command - spontaneous transition from SPEEDx to STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3410	Signal With Two Speed Aspects and Stop command - CHANGING_TO_SPEEDn - transition from CHANGING_TO_SPEEDx to SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3411	Signal With Two Speed Aspects and Stop command - CHANGING_TO_SPEEDn - reject repeated SPEED command in CHANGING_TO_SPEEDn	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3412	Signal With Two Speed Aspects and Stop command - CHANGING_TO_SPEEDn - return to STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3413	Signal With Two Speed Aspects and Stop command - CHANGING_TO_SPEEDn - transition to STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3414	Signal With Two Speed Aspects and Stop command - CHANGING_TO_SPEEDn - timeout	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3415	Signal With Two Speed Aspects and Stop command - CHANGING_TO_SPEEDn - execute STOP command	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3416	Signal With Two Speed Aspects and Stop command - CHANGING_TO_STOP - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state SPEEDx	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3417	Signal With Two Speed Aspects and Stop command - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state STOP / DARK	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3418	Signal With Two Speed Aspects and Stop command - transition from CHANGING_TO_STOP to STOP / DARK - from OC-LS state DARK / STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3419	Signal With Two Speed Aspects and Stop command - CHANGING_TO_STOP - reject commands in CHANGING_TO_STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3420	Signal With Two Speed Aspects and Stop command - CHANGING_TO_STOP - timeout when OC-LS SPEEDx - OC Disconnection	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3421	Signal With Two Speed Aspects and Stop command - CHANGING_TO_STOP - timeout when OC-LS DARK / STOP - reach DARK / STOP	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS: REQ:3422	Signal With Two Speed Aspects and Stop command - CHANGING_TO_STOP - ignore SPEED message	FUR - Functional Requirement	draft

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Identifier	Synopsis	Category	Status
EC-MUX-SIG:SSS:REQ:3423	Signal With Two Speed Aspects and Stop command - Disconnection on spontaneous SPEED message in STOP / SPEED	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3424	Signal With Two Speed Aspects and Stop command - CHANGING_TO_SPEEDx - Disconnection on unexpected command reply	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3425	Signal With Two Speed Aspects and Stop command - Ignore repeated messages	FUR - Functional Requirement	draft
EC-MUX-SIG:SSS:REQ:3426	Signal With Two Speed Aspects and Stop command - Receive unexpected DARK / SPEED commands from CC	FUR - Functional Requirement	draft

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